

Bayesian Econometrics

Lecture Notes — Bayesian Statistics, MCMC, and Applied Econometric Models

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1. Foundations of Bayesian Statistical Analysis

Roadmap

Every model in this lecture note, from Chapter 6's regression to Chapter 18's regime-switching model, answers the same question: what do the data say about an unknown parameter θ ? This chapter sets up the Bayesian answer — treat θ as random, and update a prior belief into a posterior belief — and works it through by hand for the one case where the arithmetic stays simple, a Normal mean with a Normal prior. That worked example is not decoration: the same completing-the-square step reappears, with matrices in place of scalars, in essentially every proofbox from Chapter 6 onward. Once models stop being this simple, which is true from Chapter 7 on, the posterior can no longer be written down in closed form; Chapters 3–4 supply the computational tool (MCMC) that handles that case.

In econometrics, estimating a parameter θ is fundamentally about making a probabilistic statement regarding something that is never directly observed. The frequentist approach treats θ as a fixed, unknown constant and expresses uncertainty through the sampling distribution of an estimator $\hat{\theta}$ over hypothetical repeated samples (unbiasedness, consistency, asymptotic variance, and so on). The Bayesian approach instead treats θ itself as a random variable, combining a belief about θ held before seeing the data (the prior) with the information the data provide (the likelihood) to obtain an updated belief after seeing the data (the posterior).

The Core Difference Between the Two Views

- Frequentist: θ is fixed; the data y are random. A confidence interval is interpreted as “the proportion of intervals that would contain the true value if this procedure were repeated infinitely many times.”
- Bayesian: both θ and y are random. A credible interval can be interpreted directly as “the probability that θ lies in this interval, given the observed data.”
- The output of Bayesian inference is not a single point estimate but a *complete probability distribution* over θ (the posterior); point estimates, interval estimates, and hypothesis tests are all derived as functions of this posterior distribution.

The macro/finance econometric models covered in this lecture note — VAR, state-space models, stochastic volatility models, regime-switching models, and so on — typically involve either more parameters than the sample size can comfortably support, or nonlinear/non-standard likelihood functions. Bayesian methods handle both problems with the same machinery: a prior disciplines an otherwise underdetermined parameter count (Chapter 13), and MCMC (Markov Chain Monte Carlo) generates samples from posteriors that have no closed form (Chapters 3–4), letting nonlinear models such as the probit (Chapter 7) be estimated just as routinely as the linear ones.

1.1 Bayes' Theorem

For random variables y (data) and θ (parameters), the joint distribution $p(y, \theta)$ can be factored in two equivalent ways:

$$p(y, \theta) = p(y | \theta) p(\theta) = p(\theta | y) p(y).$$

Setting the two expressions equal and solving for $p(\theta | y)$ gives Bayes' theorem.

$$p(\theta | y) = \frac{p(y | \theta) p(\theta)}{p(y)} = \frac{p(y | \theta) p(\theta)}{\int p(y | \theta) p(\theta) d\theta}.$$

The name and role of each component are as follows.

Term	Meaning
$p(\theta)$ (prior)	The belief about θ before observing the data. Reflects the researcher's economic-theoretic knowledge, prior research findings, or a deliberately uninformative judgment.
$p(y \theta)$ (likelihood)	The probability (or density) that data y would arise given θ . Viewed as a function of θ , it is also written $L(\theta; y)$.
$p(\theta y)$ (posterior)	The updated belief about θ after observing the data. The final output of Bayesian inference.
$p(y)$ (marginal likelihood)	The unconditional probability (density) of the data with θ integrated out. This is the central quantity for model comparison (Bayes factors); in computing the posterior it is simply a normalizing constant.

Because the normalizing constant $p(y)$ does not depend on θ , the following proportionality is sufficient for characterizing the *shape* of the posterior:

$$p(\theta | y) \propto p(y | \theta) p(\theta) = \text{likelihood} \times \text{prior}.$$

This proportionality is the single most frequently used relation throughout this lecture note. The essence of the MCMC algorithms in Chapter 3 is to generate samples of θ using only this proportionality, without ever computing $p(y)$.

Empirical Example: From a Coin Flip to a Recession Probability

To build intuition for Bayes' theorem, consider a discrete example. Suppose the prior probability that the economy is in a recession this quarter ($R = 1$) is $p(R = 1) = 0.15$ (close to the long-run average share of quarters the NBER has classified as recessions in the

postwar United States). Suppose we observe that the term spread (10-year minus 3-month Treasury yield) has inverted ($I = 1$). From historical data, suppose $p(I = 1 | R = 1) = 0.7$ (the frequency of inversion just before a recession) and $p(I = 1 | R = 0) = 0.10$ (the false-positive rate in normal times). Then

$$p(R = 1 | I = 1) = \frac{0.7 \times 0.15}{0.7 \times 0.15 + 0.10 \times 0.85} = \frac{0.105}{0.19} \approx 0.553.$$

A single signal — the yield-curve inversion — raises the recession probability from 15% to 55%. This is the essence of Bayesian updating, and exactly the same logic extends to the continuous, multivariate VAR-based estimation of regime probabilities in Chapter 18.

1.2 Conjugate Priors: The Normal Mean

When the functional forms of the prior and the likelihood align so that the posterior belongs to the same distributional family as the prior, the prior is called a *conjugate prior*. Conjugacy allows the posterior to be derived analytically, and the same technique recurs throughout this note when deriving the conditional posteriors used inside Gibbs samplers (e.g., the regression model of Chapter 6, the Tobit model of Chapter 9).

Derivation: Posterior of a Normal Mean

Let $y_i | \mu \sim N(\mu, \sigma^2)$, $i = 1, \dots, T$, with σ^2 known. Take the prior $\mu \sim N(\mu_0, \tau_0^2)$. The likelihood, simplified to a single quadratic in μ , and the posterior kernel obtained by multiplying it by the prior density are

$$L(\mu; y) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^T (y_i - \mu)^2\right)$$

which, after completing the square in the sum of squares, reduces to

$$\propto \exp\left(-\frac{T}{2\sigma^2} (\mu - \bar{y})^2\right)$$

Multiplying by the prior density gives the posterior kernel:

$$p(\mu | y) \propto \exp\left(-\frac{1}{2\tau_0^2} (\mu - \mu_0)^2 - \frac{T}{2\sigma^2} (\mu - \bar{y})^2\right)$$

To complete the square in μ , expand both quadratics:

$$\propto \exp\left(-\frac{1}{2\tau_0^2} (\mu^2 - 2\mu\mu_0 + \mu_0^2) - \frac{T}{2\sigma^2} (\mu^2 - 2\mu\bar{y} + \bar{y}^2)\right)$$

and collect the terms that are quadratic and linear in μ separately (everything else is a μ -free constant, dropped into the proportionality):

$$\propto \exp\left(-\frac{1}{2} \left(\frac{1}{\tau_0^2} + \frac{T}{\sigma^2}\right) \mu^2 + \left(\frac{\mu_0}{\tau_0^2} + \frac{T\bar{y}}{\sigma^2}\right) \mu\right)$$

Define $\tau_1^2 := (1/\tau_0^2 + T/\sigma^2)^{-1}$ and $\mu_1 := \tau_1^2(\mu_0/\tau_0^2 + T\bar{y}/\sigma^2)$, so the bracketed exponent above is $-\frac{1}{2\tau_1^2}\mu^2 + \frac{\mu_1}{\tau_1^2}\mu$; adding and subtracting the constant $\mu_1^2/(2\tau_1^2)$ turns this into a single square:

$$-\frac{1}{2\tau_1^2}\mu^2 + \frac{\mu_1}{\tau_1^2}\mu = -\frac{1}{2\tau_1^2}(\mu - \mu_1)^2 + \frac{\mu_1^2}{2\tau_1^2}$$

The last term does not involve μ and is absorbed into the proportionality, leaving $p(\mu | y) \propto \exp\left(-\frac{1}{2\tau_1^2}(\mu - \mu_1)^2\right)$, the kernel of a $N(\mu_1, \tau_1^2)$ density. Hence

$$\begin{aligned}\mu | y &\sim N(\mu_1, \tau_1^2), \\ \tau_1^2 &= \left(\frac{1}{\tau_0^2} + \frac{T}{\sigma^2}\right)^{-1}, \\ \mu_1 &= \tau_1^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{T\bar{y}}{\sigma^2}\right).\end{aligned}$$

The posterior mean μ_1 is a precision-weighted average of the prior mean μ_0 and the sample mean $\bar{y}$: the larger the prior precision $1/\tau_0^2$ (i.e., the stronger the prior belief), the closer μ_1 is pulled toward μ_0 ; the larger the sample size T , the closer μ_1 moves toward $\bar{y}$. As $T \rightarrow \infty$ the Bayesian estimator converges to the frequentist MLE $\bar{y}$, showing that the influence of the prior naturally vanishes as the sample grows.

Empirical Example: Estimating the Mean Growth Rate of U.S. Real GDP

Using a quarterly U.S. macroeconomic dataset extended through 2026Q1 to include the COVID-19 pandemic ($T = 267$), we perform a Bayesian update on the mean growth rate μ of real GDP (annualized, %). We use the prior $\mu \sim N(\mu_0 = 2.5, \tau_0^2 = 1.0)$ (a prior mean of 2.5%, reflecting the long-run average postwar U.S. growth rate, with a fairly loose prior variance of 1.0). The sample mean and sample variance computed directly from the data are $\bar{y} = 2.966$ and $s^2 = 17.601$ (sample standard deviation 4.195) — the variance is roughly 54% larger than in the pre-pandemic sample, entirely on the strength of two 2020 quarters (-28.0% and $+34.9\%$), while the mean barely moves.

Applying the precision-weighted-average formula,

$$\tau_1^2 = \left(\frac{1}{1.0} + \frac{267}{17.601}\right)^{-1} = 0.0618, \quad \mu_1 = 0.0618 \left(\frac{2.5}{1.0} + \frac{267 \times 2.966}{17.601}\right) = 2.938.$$

Because $T = 267$ is large enough, the posterior mean $\mu_1 = 2.938$ is much closer to the sample mean (2.966) than to the prior mean (2.5). The 95% credible interval is $\mu_1 \pm 1.96\sqrt{\tau_1^2} = [2.45, 3.43]$. This example confirms with real data that the prior's influence becomes negligible when the sample is large — by contrast, in shorter quarterly macro series (e.g., $T < 50$) the prior plays a much larger role, which is exactly why shrinkage priors are essential in the LBVAR model of Chapter 13 and the TVP-VAR model of Chapter 17.

1.3 Point Estimation, Interval Estimation, and Hypothesis Testing

Once the posterior $p(\theta | y)$ is available, all the traditional statistical summaries follow naturally as functions of this distribution.

Term	Meaning
Posterior mean	$E[\theta y] = \int \theta p(\theta y) d\theta$. The point estimate that minimizes Bayesian risk under quadratic loss.
Posterior median	The optimal point estimate under absolute-error loss. Can be more representative than the mean for asymmetric posteriors (e.g., variance parameters).
Posterior mode	Optimal under 0-1 loss. The location of the peak of the posterior; also called the MAP (maximum a posteriori) estimate.
Credible interval	An interval satisfying $P(\theta_L \leq \theta \leq \theta_U y) = 1 - \alpha$, computed directly from posterior quantiles and admitting a direct probabilistic interpretation.
Posterior probability	A direct probability statement about any hypothesis of the form $P(\theta \in A y)$. For example, $P(\beta > 0 y)$ literally means “the probability that β is positive,” unlike a frequentist p -value.

Once MCMC produces a sample $\{\theta^{(1)}, \dots, \theta^{(N)}\}$ from the posterior, all of the above quantities are easily approximated by Monte Carlo: sample means, sample quantiles, and sample proportions, e.g.

$$\widehat{E[\theta | y]} = \frac{1}{N} \sum_{s=1}^N \theta^{(s)}, \quad P(\widehat{\theta > c} | y) = \frac{1}{N} \sum_{s=1}^N \mathbb{I}\{\theta^{(s)} > c\}.$$

This is precisely why MCMC-based estimation is so powerful in practice, as developed from Chapter 3 onward: once a posterior sample is obtained, any estimate or hypothesis of interest reduces to a simple calculation on that sample.

1.4 The Predictive Distribution

Bayesian analysis also provides a natural probabilistic statement about future observations y_{T+1} , not just about parameters. The posterior predictive distribution is defined as

$$p(y_{T+1} | y_{1:T}) = \int p(y_{T+1} | \theta) p(\theta | y_{1:T}) d\theta,$$

which integrates over both parameter uncertainty ($p(\theta | y_{1:T})$) and conditional model uncertainty ($p(y_{T+1} | \theta)$). This predictive interval is always (weakly) wider than a plug-in prediction that fixes $\hat{\theta}$ at a single value, and the extra width honestly reflects parameter estimation error — an insight that is difficult to obtain from a purely point-estimate-centered frequentist approach. Given MCMC draws $\theta^{(s)}$, simulating $y_{T+1}^{(s)} \sim p(\cdot | \theta^{(s)})$ for each draw immediately yields a sample from the predictive distribution; this is exactly the approach used in the forecasting exercises of Chapter 11 (VAR) and Chapter 15 (state-space models).

Remark: Sequential Bayesian Learning

Bayes' theorem applies naturally in a repeated, sequential fashion over time. If the posterior obtained from data up to $t - 1$, $p(\theta | y_{1:t-1})$, is used as the prior at time t , then

$$p(\theta | y_{1:t}) \propto p(y_t | \theta) p(\theta | y_{1:t-1})$$

holds. In other words, today's posterior becomes tomorrow's prior — exactly the structure underlying real-time macro nowcasting and the Kalman filter used in state-space models (Chapter 15).

1.5 Why Computation Is Hard

In the Normal-Normal example above, the prior and likelihood are conjugate, so the posterior can be derived analytically. However, most of the models covered in this lecture note — the probit model (Chapter 7), the Tobit model (Chapter 9), stochastic volatility models (Chapter 16), regime-switching models (Chapter 18), and others — cannot be reduced to a closed-form posterior, for several reasons.

- The parameter space is high-dimensional (e.g., VAR coefficient matrices and covariance matrices), so the normalizing integral $\int p(y | \theta) p(\theta) d\theta$ has no analytical solution.
- The likelihood is nonlinear (e.g., the standard normal CDF $\Phi(\cdot)$ in the probit model) or involves latent variables, breaking conjugacy with the prior.
- Parameters are jointly coupled (e.g., regression coefficients β and error variance σ^2), so the joint posterior does not belong to any standard distributional family.

The solution to this problem is MCMC (Markov Chain Monte Carlo): a family of algorithms that generate samples from the posterior *without* ever computing the normalizing constant — relying only on the proportionality $p(\theta | y) \propto p(y | \theta) p(\theta)$. Chapter 3 develops the theoretical foundations of MCMC (Markov chains, stationary distributions, the ergodic theorem), and Chapter 4 covers the concrete sampling techniques actually used in practice (Gibbs sampling,

Metropolis-Hastings, slice sampling, and others).

2. The Economic Significance of Bayesian Methods

Roadmap

Chapter 1 showed *how* Bayesian updating works, on a single scalar mean. Before building actual econometric models, it is worth asking why applied macro/finance researchers reach for this machinery at all, rather than treating it as a mathematical curiosity. The short answer, developed below, is that three structural features of economic data — short samples, high-dimensional models, and genuine uncertainty about which model is correct — make Bayesian methods close to necessary in practice. Each feature maps to a later chapter: short samples motivate the shrinkage priors of Chapter 13, model uncertainty motivates the variable selection of Chapter 8, and the underlying computation is what Chapters 3–4 build.

To understand why Bayesian methodology has become central to econometrics — and especially to macro/finance econometrics — beyond its purely statistical appeal, it helps to examine several structural features of economic data. This section discusses three practical constraints (short time series, high parameter dimension, and structural uncertainty) that together make the Bayesian approach nearly indispensable.

2.1 Short Samples and Shrinkage Estimation

Quarterly macroeconomic time series typically yield only around $T = 200$ – 270 observations even after pooling several decades of data (e.g., the quarterly U.S. macro dataset used throughout this note spans 1959 to 2026 — nearly 67 years, including the COVID-19 pandemic — yet has only $T = 267$ observations). Meanwhile, in VAR-type models the number of coefficients to be estimated grows as n^2p in the number of variables n and lags p . For example, a VAR with $n = 5$ variables and $p = 4$ lags has $5 \times 4 = 20$ coefficients per equation (21 including the constant), so the full system has $5 \times 21 = 105$ coefficients plus 15 free parameters in the covariance matrix — a total of about 120 parameters to be estimated from only $T \approx 200$ observations.

Remark: The Curse of Dimensionality and Overfitting

Frequentist least-squares estimation suffers from exploding variance and rapidly deteriorating out-of-sample forecast performance (overfitting) as the number of parameters approaches the sample size. Bayesian methods using a shrinkage prior explicitly manage this bias-variance trade-off: by encoding the belief that “most coefficients are close to zero,” they accept a small amount of bias in exchange for a substantial reduction in variance. The Minnesota/Litterman prior of Chapter 13 and the time-varying-parameter prior of Chapter 17 are concrete implementations of this principle.

2.2 Model Uncertainty and Bayesian Model Averaging

Economic theory frequently offers several competing models for the same phenomenon (e.g., different formulations of the Phillips curve explaining inflation, or the permanent-income hypothesis versus liquidity-constrained models of consumption). The frequentist approach typically selects a single “best” model using an information criterion (AIC, BIC, etc.) and then treats that model as if it were true for all subsequent inference — ignoring the uncertainty inherent in the model-selection step itself.

Bayesian Model Averaging (BMA) assigns a prior probability $p(M_k)$ to each candidate model $M_1, \dots, M_K$, and after observing the data computes the posterior model probability

$$p(M_k | y) = \frac{p(y | M_k) p(M_k)}{\sum_{j=1}^K p(y | M_j) p(M_j)},$$

combining them into an overall predictive distribution for a quantity of interest Δ (a forecast, a policy multiplier, etc.):

$$p(\Delta | y) = \sum_{k=1}^K p(\Delta | y, M_k) p(M_k | y).$$

Here $p(y | M_k) = \int p(y | \theta_k, M_k) p(\theta_k | M_k) d\theta_k$ is the marginal likelihood of model M_k , and the posterior odds between two models decompose as

$$\frac{p(M_1 | y)}{p(M_2 | y)} = \underbrace{\frac{p(y | M_1)}{p(y | M_2)}}_{\text{Bayes factor}(BF_{12})} \times \frac{p(M_1)}{p(M_2)}.$$

The Bayes factor is a natural generalization of the likelihood-ratio test to a model comparison metric that inherently balances out-of-sample predictive power against model complexity (an automatic penalty for the number of parameters) — unlike information criteria, this penalty arises automatically rather than being imposed ad hoc, which is why the Bayes factor is sometimes described as a probabilistic implementation of Occam’s razor. Chapter 8 (variable selection) applies this idea to the discrete model space of including or excluding regressors, using a spike-and-slab prior.

2.3 Encoding Economic Theory as a Prior

Another economic significance of the Bayesian framework is that it allows constraints or beliefs implied by economic theory to be introduced explicitly, in the form of a prior distribution. Some representative examples:

Term	Meaning
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Minnesota prior	Reflects the belief that most VAR coefficients are close to zero while each variable's own first-lag coefficient is close to one — i.e., that individual series behave roughly like random walks (Chapter 13).
Sign restrictions	Impose the theoretical direction of a structural shock — e.g., that a monetary policy tightening lowers prices and reduces output — as a prior (or identifying) restriction on structural shocks (Chapter 12).
Stationarity priors	Push the prior mass for autoregressive coefficients away from the unit-root boundary and into the stationary region (Chapters 11, 17).
Zero-lower-bound priors	Encode the information that the policy rate behaves specially near the effective lower bound, via a truncated prior/likelihood (linked to the Tobit model of Chapter 9).

This differs fundamentally from the frequentist practice of testing sign restrictions after the fact, or forcing coefficients to zero in an ad hoc manner: in the Bayesian approach, theoretical constraints are introduced probabilistically *before* estimation, and if the data conflict with the theory, the posterior is naturally allowed to move away from that constraint (provided the prior is not too tight).

2.4 A Decision-Theoretic Perspective

Bayesian statistics connects naturally to statistical decision theory. If a policymaker who takes action a incurs a loss $L(a, \theta)$ under the true state θ , a Bayesian decision-maker chooses the action that minimizes expected loss under the posterior (Bayesian risk):

$$a^* = \arg \min_a E_{\theta|y}[L(a, \theta)] = \arg \min_a \int L(a, \theta) p(\theta | y) d\theta.$$

This framework applies directly to any economic decision problem that requires choosing an optimal action under parameter uncertainty — central bank monetary policy decisions, estimating fiscal multipliers, portfolio allocation, and so on. In particular, when the loss function is asymmetric (e.g., the loss from inflation exceeding target is larger than the loss from falling short), the optimal policy may be a posterior summary other than the posterior mean — an insight that is difficult to capture within a purely point-estimate-centered frequentist approach.

2.5 The Rise of Computational Methods and Practical Adoption

Beyond theoretical elegance, the rapid advance of computational tools was decisive in establishing Bayesian econometrics as the mainstream tool for macro/finance empirical work after the 1990s. Once Gibbs sampling and the Metropolis-Hastings algorithm (Chapters 3–4) were fully incorporated into econometrics, models that previously had no closed-form solution and were therefore difficult to estimate became practically feasible: large VARs (Doan, Litterman and Sims, 1984; Litterman, 1986), structural vector autoregressions (Sims, 1980; Sims and Zha, 1998), state-space and stochastic volatility models (Kim, Shephard and Chib, 1998), regime-switching models (Hamilton, 1989; Kim and Nelson, 1999), and large panel/dynamic factor models. Today, the dynamic stochastic general equilibrium (DSGE) models and BVAR forecasting models used for policy analysis at the Federal Reserve, the European Central Bank, and other major central banks are estimated almost exclusively using Bayesian methods. Every model covered from Chapter 6 onward in this note — regression, probit, Tobit, SUR, VAR, SVAR, LVAR, GARCH, state-space models, VAR-SV, TVP-VAR, and regime switching — is a concrete instance of this computational technology, and each chapter applies Gibbs sampling or an M-H algorithm directly to real U.S. macro/financial data to produce posterior estimation results.

3. Foundations of MCMC Algorithms

Roadmap

Chapter 1 ended with the same problem repeated: most posteriors in these notes have no closed form, so they cannot be summarized the way the Normal-mean example was. MCMC is the fix, and it works by a trick that is worth stating before any of the mechanics: instead of computing the posterior directly, build a Markov chain that *wanders around* the parameter space and visits each point with a long-run frequency exactly proportional to its posterior density. This section derives the one condition (detailed balance) that guarantees this trick works; Chapter 4 then supplies the concrete recipes (Gibbs, Metropolis-Hastings, and others) that satisfy it, and every algobox from Chapter 6 onward is an application of the six-step recipe closing this chapter.

As we saw in Chapter 1, in most practical Bayesian models the posterior distribution $p(\theta \mid y)$ cannot be obtained analytically. MCMC (Markov Chain Monte Carlo) is the collective name for algorithms that draw (approximate) samples from $p(\theta \mid y)$ without ever computing the normalizing constant. This section lays out the basic concepts of Markov chain theory that underlie the concrete sampling techniques (Gibbs, M-H, Slice sampling, etc.) covered in Chapter 4.

3.1 Markov Chains

A stochastic process $\{\theta^{(0)}, \theta^{(1)}, \theta^{(2)}, \dots\}$ moving through a state space Θ is said to satisfy the *Markov property* if the conditional distribution of the next state depends only on the current

state and not on the history before it:

$$p(\theta^{(t+1)} \mid \theta^{(t)}, \theta^{(t-1)}, \dots, \theta^{(0)}) = p(\theta^{(t+1)} \mid \theta^{(t)}) \equiv K(\theta^{(t)}, \theta^{(t+1)}),$$

where $K(\cdot, \cdot)$ is called the transition kernel. Figure 1 shows the dependency structure this implies: each state depends only on its immediate predecessor, never on anything further back.

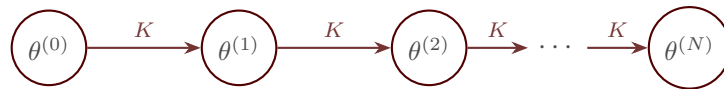


Figure 1: Dependency structure of a Markov chain: each state depends on its immediate predecessor only, through the same transition kernel K at every step

The central idea of MCMC is to design a transition kernel K whose *stationary distribution* is the target distribution (here, the posterior $\pi(\theta) := p(\theta \mid y)$).

Term	Meaning
Transition kernel $K(\theta, \theta')$	The probability (density) of moving from the current state θ to the next state θ' . It fully determines the dynamics of the Markov chain.
Stationary distribution $\pi(\cdot)$	A distribution satisfying $\pi(\theta') = \int \pi(\theta) K(\theta, \theta') d\theta$. If the chain starts from this distribution, it remains in this distribution thereafter.
Irreducibility	The property that, starting from any point in the state space, any other point can be reached with positive probability in a finite number of steps.
Aperiodicity	The property that the chain does not return to a given state only at fixed periodic intervals. Together with irreducibility, it guarantees convergence of the chain to a unique limiting distribution.
Ergodicity	An irreducible, aperiodic Markov chain has the ergodic property of converging to a unique stationary distribution regardless of the initial value.

3.2 Detailed Balance and Reversibility

The most widely used sufficient condition for constructing a transition kernel with stationary distribution π is *detailed balance*. A transition kernel K is said to satisfy detailed balance with respect to π if

$$\pi(\theta)K(\theta, \theta') = \pi(\theta')K(\theta', \theta) \quad \text{for all } \theta, \theta'$$

holds. This expresses the physical intuition (reversibility) that, in the stationary state, the “probability flow” from $\theta \rightarrow \theta'$ is exactly equal to the flow back from $\theta' \rightarrow \theta$.

Derivation: Detailed Balance Implies Stationarity

If detailed balance holds, integrating both sides with respect to θ gives:

$$\int \pi(\theta)K(\theta, \theta') d\theta = \int \pi(\theta')K(\theta', \theta) d\theta$$

Since θ' does not depend on the integration variable θ , it can be pulled out of the integral on the right-hand side:

$$= \pi(\theta') \int K(\theta', \theta) d\theta$$

and, because the transition kernel is a probability distribution in its second argument, $\int K(\theta', \theta) d\theta = 1$, so:

$$= \pi(\theta')$$

The left-hand side is exactly the definition of the stationary distribution, $\int \pi(\theta)K(\theta, \theta') d\theta$, so $\pi(\theta') = \int \pi(\theta)K(\theta, \theta') d\theta$ holds and π is indeed shown to be the stationary distribution of K . This result is central to showing why the Metropolis-Hastings and Gibbs samplers introduced in Chapter 4 converge to the target posterior: both algorithms are designed so that detailed balance holds by construction.

3.3 The Ergodic Theorem and Monte Carlo Approximation

If a chain is irreducible, aperiodic, and has stationary distribution π , then regardless of the distribution of the initial value $\theta^{(0)}$, the distribution of $\theta^{(t)}$ converges to π as $t \rightarrow \infty$ (the convergence theorem). Furthermore, by the law of large numbers for Markov chains (the Markov chain ergodic theorem), for any function $g(\cdot)$,

$$\frac{1}{N} \sum_{t=1}^N g(\theta^{(t)}) \xrightarrow{a.s.} E_{\pi}[g(\theta)] = \int g(\theta)\pi(\theta) d\theta$$

holds. This is a generalization of the ordinary law of large numbers for independent and identically distributed (i.i.d.) samples to (weakly dependent) Markov chain samples, and it is the theoretical basis justifying that the Monte Carlo approximations of posterior means and posterior probabilities discussed in Chapter 1 remain valid for MCMC samples as well. However, MCMC samples are not mutually independent but are serially autocorrelated, so even with the

same sample size N as an i.i.d. sample, the effective sample size is smaller — this concept is formalized via the inefficiency factor in Chapter 5 (convergence diagnostics).

Empirical Example: Convergence of a 2-State Markov Chain to Its Stationary Distribution: A Business-Cycle Example

To visually confirm the convergence of a Markov chain, consider a chain that simplifies the economy into two states, expansion (state 1) and recession (state 2). Set the transition probability matrix to

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 0.95 & 0.05 \\ 0.35 & 0.65 \end{pmatrix}$$

(the average duration of an expansion is $1/(1 - 0.95) = 20$ quarters and of a recession $1/(1 - 0.65) \approx 2.9$ quarters, which qualitatively matches the roughly asymmetric persistence of the postwar U.S. business cycle). The stationary distribution $\pi = (\pi_1, \pi_2)$ is obtained by solving $\pi P = \pi$, $\pi_1 + \pi_2 = 1$, giving $\pi_1 = p_{21}/(p_{12} + p_{21}) = 0.35/0.40 = 0.875$, $\pi_2 = 0.125$. Simulating the chain for $T = 20,000$ steps in Python, the empirical cumulative average of the fraction of visits to state 1 (expansion) converges to 0.8748, essentially matching the theoretical value $\pi_1 = 0.8750$ (Figure 2). This confirms numerically the ergodic theorem's implication that a Markov chain converges to its stationary distribution regardless of the initial value (here, the chain starts in state 1). Chapter 18 (regime switching) uses exactly this two-state Markov chain structure to capture regime changes in actual U.S. inflation data.

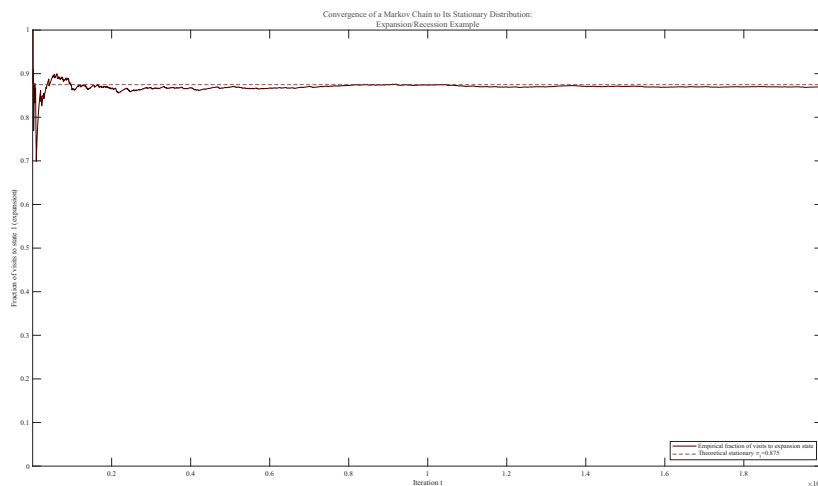


Figure 2: Convergence of a 2-state Markov chain to its stationary distribution ($T = 20,000$, Python simulation)

3.4 The General Use of MCMC in Bayesian Computation

The procedure for applying MCMC to Bayesian inference is summarized as follows.

Algorithm: General Procedure for Bayesian MCMC

1. Set the target distribution to be the posterior $\pi(\theta) = p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$ (the normalizing constant need not be known).
2. Design a transition kernel K whose stationary distribution is $\pi(\theta)$ (Gibbs, M-H, Slice sampling, etc., in Chapter 4, are concrete design methods).
3. Starting from an arbitrary initial value $\theta^{(0)}$, repeatedly generate $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)}$ by following the transition kernel K .
4. Discard the initial segment (burn-in) before the chain reaches the stationary distribution: typically, of the total n iterations, the first n_0 are discarded and only $n_1 = n - n_0$ are used.
5. Use the remaining samples $\{\theta^{(n_0+1)}, \dots, \theta^{(n)}\}$ to compute Monte Carlo approximations of posterior means, credible intervals, posterior probabilities, and so on.
6. Check whether the chain has actually reached the stationary distribution using the convergence diagnostics of Chapter 5 (trace plots, inefficiency factor, Geweke's p, Gelman-Rubin statistic).

Remark: Burn-In and Thinning

If the initial value $\theta^{(0)}$ is far from the high-density region of the posterior, the chain requires a certain “adaptation period” before reaching the stationary distribution. Samples from this period (burn-in) are not representative of the target distribution and are discarded. The empirical examples in this note follow the convention of the Chan (2017)-style MATLAB code family, typically using $n_0 = 1,000$ to 5,000 burn-in iterations followed by $n_1 = 5,000$ to 10,000 posterior samples. When successive samples are highly autocorrelated, thinning (keeping only every few samples) is sometimes used to save storage, but recent literature (e.g., Link and Eaton, 2012) has pointed out that thinning only saves storage space and does not itself improve the efficiency of Monte Carlo approximation (and may in fact discard useful information); this note therefore uses the full set of posterior samples without thinning as its default practice.

The next chapter examines, in turn, eight sampling techniques that concretely implement these general principles — the method of composition, the probability integral transform, acceptance-rejection sampling, importance sampling, M-H sampling, multi-block M-H, Gibbs sampling, and Slice sampling — each with both theory and a worked example.

4. Sampling Methods

fixes the problem of an unavailable posterior. It said nothing about how to build one. This chapter is the toolbox: eight concrete recipes, ordered roughly from “exact, no Markov chain needed” (composition, the probability integral transform) to “Markov chain required” (M-H, Gibbs, Slice). Two of these, Gibbs sampling and multi-block M-H, are the workhorses of every empirical chapter from Chapter 6 onward — when a later chapter’s algobox says “draw β , then draw σ^2 , then draw γ ,” it is a direct instance of the multi-block cycle introduced here.

Chapter 3 established that MCMC works by constructing a transition kernel whose stationary distribution is the target posterior $\pi(\theta) = p(\theta \mid y)$. This chapter surveys eight concrete techniques for generating such samples, ranging from exact methods that require no Markov chain at all (composition, the probability integral transform, acceptance-rejection, importance sampling) to the Markov-chain algorithms that make modern Bayesian econometrics practical (Metropolis-Hastings, multi-block M-H, Gibbs sampling, and Slice sampling). Each technique is implemented in Python and checked numerically against a known target, and where the underlying MATLAB library (`Bayes_Lib`) supplies a worked example, the implementation mirrors it directly.

4.1 Method of Composition

Many distributions can be written as a *mixture*: draw an auxiliary variable λ from some distribution, then draw x from a distribution indexed by λ . If we can sample both stages directly, the marginal draw of x is exact — no acceptance step or Markov chain is required.

Algorithm: Method of Composition

1. Draw $\lambda \sim p(\lambda)$.
2. Draw $x \mid \lambda \sim p(x \mid \lambda)$.
3. Return x ; by construction $x \sim \int p(x \mid \lambda)p(\lambda) d\lambda = p(x)$.

A canonical example, and one that will reappear in Chapter 6 when we model fat-tailed regression errors, is the representation of the Student- t distribution as a Normal/Inverse-Gamma (scale) mixture:

$$x \mid \lambda \sim N(0, \lambda), \quad \lambda = \frac{\nu}{\chi_\nu^2} \\ \implies x \sim t_\nu.$$

Empirical Example: Composition Sampling of the Student- t Distribution

Setting $\nu = 5$ and drawing $N = 200,000$ pairs (λ, x) via composition gives a sample mean of $\bar{x} = -0.0009$ and sample variance $\widehat{\text{Var}}(x) = 1.6629$, within 0.3% of the theoretical variance of t_5 , $\nu/(\nu - 2) = 5/3 = 1.6667$. As a check, 200,000 draws obtained directly from `scipy.stats.t` give sample mean -0.0011 and variance 1.6591 — statistically indistinguishable from the composition-based draws. The sample excess kurtosis of the composition draws is 4.27, versus a theoretical value of $6/(\nu - 4) = 6.0$ for $\nu = 5$; this gap (rather than a shrinking one) is expected, since the fourth moment of a t_5 variable is finite but sits so far out in the tails that its sample estimate converges far more slowly than the mean or variance even at $N = 200,000$ — a useful practical warning about trusting higher-moment diagnostics from finite MCMC or Monte Carlo samples of heavy-tailed distributions.

4.2 Probability Integral Transform

If F is the (invertible) CDF of a target distribution, then $F^{-1}(U)$ has distribution F whenever $U \sim \text{Uniform}(0, 1)$. This is the *probability integral transform* (PIT) and is the simplest possible exact sampler whenever F^{-1} is available in closed form.

Algorithm: Sampling via the Probability Integral Transform

1. Draw $u \sim \text{Uniform}(0, 1)$.
2. Return $x = F^{-1}(u)$.

Empirical Example: PIT Sampling of the Exponential Distribution

For $x \sim \text{Exponential}(\lambda)$, $F(x) = 1 - e^{-\lambda x}$, so $F^{-1}(u) = -\ln(1 - u)/\lambda$. Setting $\lambda = 0.5$ and drawing $N = 200,000$ values gives a sample mean of 2.0001 and sample variance 3.9777, versus theoretical values $1/\lambda = 2.0$ and $1/\lambda^2 = 4.0$. A Kolmogorov-Smirnov test of the PIT sample against $\text{Exponential}(0.5)$ gives $D = 0.0016$, $p = 0.70$, so the null of an exact match cannot be rejected. The PIT is the exact method underlying truncated-distribution samplers used later in this note — e.g., simulating a truncated Normal error in the Tobit model of Chapter 9 by applying the inverse CDF to a $\text{Uniform}(0, 1)$ draw restricted to the appropriate tail probability.

4.3 Acceptance-Rejection Sampling

When F^{-1} is not available but the target density f is known up to a constant, acceptance-rejection (A-R) sampling draws from an easy-to-sample *proposal* density g and an envelope constant c such that $f(x) \leq cg(x)$ for all x , then accepts or rejects each proposal so that accepted draws are exactly distributed as f .

Algorithm: Acceptance-Rejection Sampling

1. Draw a candidate $v \sim g(\cdot)$.
2. Draw $u \sim \text{Uniform}(0, 1)$.
3. If $u \leq f(v)/(cg(v))$, accept $x = v$; otherwise return to step 1.

The overall acceptance probability is $1/c$, so an envelope constant close to the true supremum of f/g makes the algorithm more efficient.

Empirical Example: A-R Sampling from Beta(3,3)

Take the target $f(x) = \text{Beta}(3, 3)$ density on $(0, 1)$ and the proposal $g(x) = \text{Uniform}(0, 1)$, with envelope constant $c = 1.875$ (the exact maximum of the Beta(3,3) density, attained at $x = 0.5$). Drawing $T = 50,000$ accepted samples gives sample mean 0.5004 and sample variance 0.03576, against theoretical values $a/(a + b) = 0.5$ and $ab/[(a + b)^2(a + b + 1)] = 9/252 = 0.03571$. The empirical acceptance rate was 53.18%, matching the theoretical rate $1/c = 53.33\%$ almost exactly. Figure 3 overlays the histogram of accepted draws on the exact Beta(3,3) density.

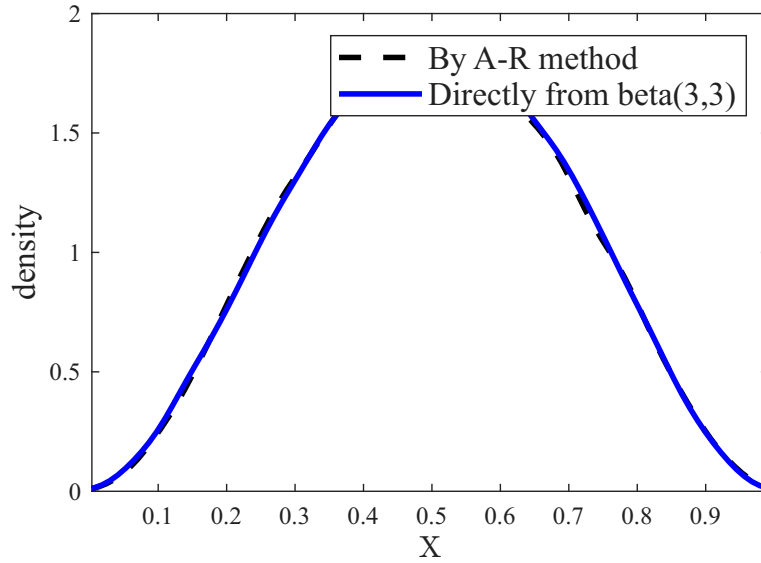


Figure 3: Acceptance-rejection sampling from Beta(3,3): histogram of 50,000 accepted draws vs. the exact density

4.4 Importance Sampling

Acceptance-rejection discards proposals, which can be wasteful when c is large. Importance sampling instead reweights *every* draw from the proposal (here called the importance density h) so that no draws are discarded:

$$E_f[g(X)] = \int g(x)f(x) dx = \int g(x)\frac{f(x)}{h(x)}h(x) dx = E_h\left[g(X)\frac{f(X)}{h(X)}\right].$$

Algorithm: Importance Sampling

1. Draw $x^{(1)}, \dots, x^{(N)} \sim h(\cdot)$.
2. Compute importance weights $w^{(i)} = f(x^{(i)})/h(x^{(i)})$.
3. Approximate $E_f[g(X)] \approx \frac{1}{N} \sum_i g(x^{(i)})w^{(i)}$.

The efficiency of the estimator depends on how closely h matches f : if h has much lighter tails than f , a small number of draws can receive enormous weights, inflating the variance of the estimator (measured by its effective sample size, $\text{ESS} = (\sum_i w^{(i)})^2 / \sum_i (w^{(i)})^2$).

Empirical Example: Importance Sampling for $E[g(X)]$, $X \sim N(0, 1)$

Let $X \sim N(0, 1)$ and $g(x) = 1/(1 + x^2)$; we want $E[g(X)]$. Numerical integration gives the essentially exact value 0.65568; direct Monte Carlo with $N = 200,000$ standard normal draws gives 0.65448. Using a Student- t_5 importance density h (heavier-tailed than the $N(0, 1)$ target f , so importance weights f/h stay bounded) gives an importance-sampling estimate of 0.65563 — within 0.0001 of the exact value — with an effective sample size of 191,584 out of 200,000 draws (95.8%), confirming that t_5 is an efficient importance density for a standard normal target in this application.

4.5 Metropolis-Hastings Sampling

The methods above require either a tractable inverse CDF or a bounding envelope, both of which become hard to construct in the multi-parameter, non-standard posteriors typical of econometric models. The Metropolis-Hastings (M-H) algorithm sidesteps this by constructing a Markov chain whose transitions satisfy detailed balance (Chapter 3) with respect to $\pi(\theta) = p(\theta | y)$, using only the *ratio* $\pi(\theta')/\pi(\theta)$ — so the normalizing constant of the posterior is never needed.

Algorithm: Random-Walk Metropolis-Hastings

1. At $\theta^{(t)}$, draw a candidate $\theta' = \theta^{(t)} + \tau \varepsilon$, $\varepsilon \sim N(0, 1)$ (a symmetric random-walk proposal).

2. Compute the acceptance probability

$$\alpha(\theta^{(t)}, \theta') = \min \left\{ 1, \frac{\pi(\theta')}{\pi(\theta^{(t)})} \right\}.$$

3. With probability α , set $\theta^{(t+1)} = \theta'$; otherwise $\theta^{(t+1)} = \theta^{(t)}$.

The step size τ is a tuning parameter: too large gives a low acceptance rate and poor mixing; too small gives a high acceptance rate but slow exploration (high autocorrelation). A commonly cited rule of thumb targets an acceptance rate near 0.44 for a single parameter (Roberts and Rosenthal, 2001).

Before fitting anything to real data, it helps to see the mechanics in a setting simple enough to draw: two correlated parameters, $\theta = (\theta_1, \theta_2)$, with a bivariate Normal target ($\text{Corr}(\theta_1, \theta_2) = 0.8$). Figure 4 runs 40 random-walk proposals from a starting point far in the tail; each proposal is either accepted (solid path) or rejected (grey $\times$), and because a symmetric random walk does not know which way the correlation runs, many proposals land off the ridge of the target and are rejected.

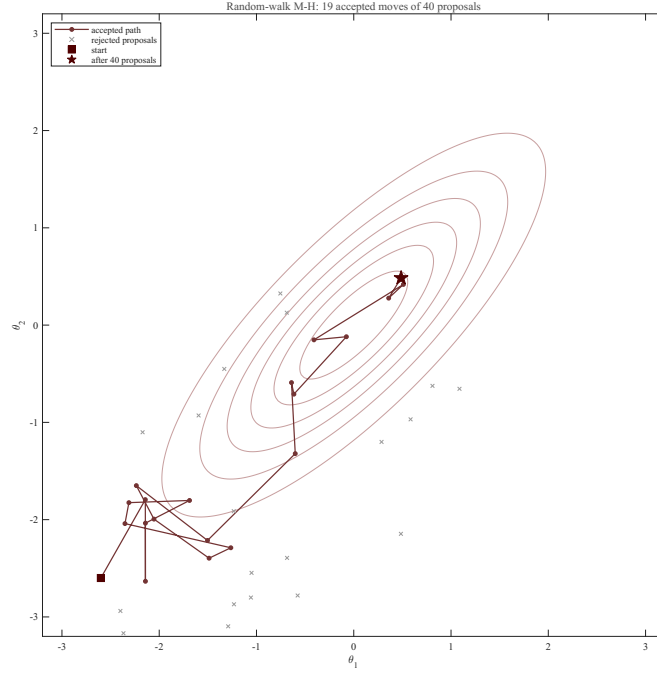


Figure 4: Random-walk M-H on a bivariate Normal target ($\text{Corr} = 0.8$): accepted path (solid) vs. rejected proposals (grey $\times$)

Empirical Example: M-H Sampling of an AR(1) Persistence Parameter for U.S. GDP Growth

Consider the simplified mean-zero AR(1) model $y_t = \rho y_{t-1} + e_t$, $e_t \sim N(0, \sigma^2)$, fit to quarterly U.S. real GDP growth, now extended through 2026Q1 to include the COVID-19 pandemic ($T = 266$ after lagging), with a stationarity-truncated Jeffreys-type prior $p(\rho) \propto (1 - \rho^2)^{-1/2}$ on $(-1, 1)$. This prior is not conjugate with the Normal likelihood, so no closed-form Gibbs update is available and M-H is a natural choice. Running a random-walk M-H sampler with step size $\tau = 0.06$ for 12,000 post-burn-in draws (burn-in 2,000) gives posterior mean $\hat{\rho} = 0.357$, posterior standard deviation 0.052, and a 95% credible interval of $[0.256, 0.460]$. The empirical acceptance rate was 67.3% and the inefficiency factor was 6.96 (i.e., 12,000 correlated draws carry roughly the same information as $12,000/6.96 \approx 1,724$ independent draws) — both signs that τ could be increased somewhat to move closer to the 0.44 rule of thumb. As a sanity check, the maximum-likelihood (OLS) estimate of ρ in the same zero-mean specification is 0.358, essentially identical to the posterior mean, confirming the sampler has converged to the correct target. Figure 5 shows the trace plot and posterior histogram.

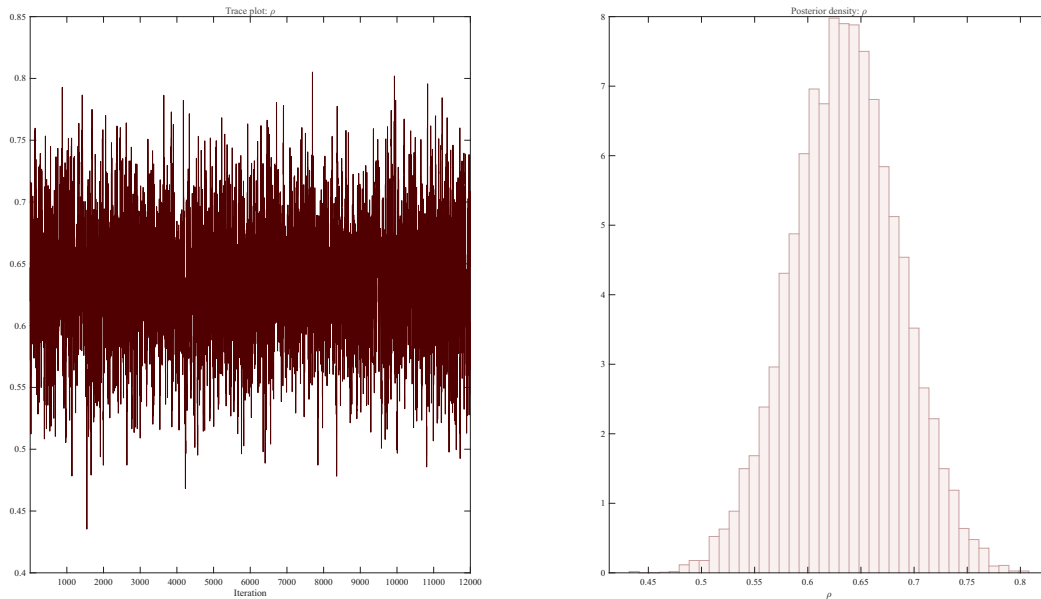


Figure 5: Random-walk M-H sampling of the AR(1) persistence parameter ρ for U.S. real GDP growth

Remark: Why a Zero-Mean Specification Overstates Persistence

Note that $\hat{\rho} \approx 0.36$ here is noticeably higher than the AR(1) persistence of GDP growth reported later in this chapter under a model with an explicit mean ($\hat{\rho} \approx 0.03$ – 0.04 , Sections 4.6–4.7) — the gap is proportionally much larger than it was on the pre-extension sample, because the mean-adjusted estimate is the one that collapsed once 2020Q2's -28.0% and 2020Q3's $+34.9\%$ entered the sample: two adjacent quarters of nearly equal and opposite magnitude contribute a strongly negative term to the lag-1 autocovariance, pulling the mean-adjusted persistence estimate close to zero. This is not a sampler error: real GDP growth has a positive unconditional mean (roughly 3%), and forcing the regression through the origin causes y_{t-1} to proxy partly for the omitted mean rather than for genuine period-to-period persistence, biasing $\hat{\rho}$ upward regardless of the sample used — the zero-mean estimate is far less sensitive to the COVID swings simply because it was already a biased, overstated number before the extension. The zero-mean model is used here purely to isolate the mechanics of M-H sampling with the smallest possible parameter set; Sections 4.6 and 4.7 correct the specification.

4.6 Multi-Block Metropolis-Hastings

When θ has several components, updating them all at once in a single M-H step often performs poorly: proposing a joint move in a high-dimensional space is unlikely to be accepted. *Multi-block* (or *blockwise*) M-H instead partitions θ into blocks and updates each block in turn, conditioning on the current values of the others. Because each block's transition individually satisfies detailed balance conditional on the rest, the overall sweep still leaves $\pi(\theta)$ invariant. Figure 6 shows the resulting dependency structure for two blocks: unlike the single-chain DAG

of Chapter 3, each block now also depends on the *current* value of every other block.

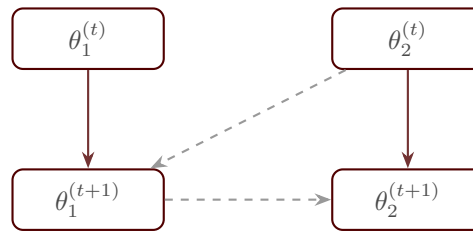


Figure 6: Multi-block dependency structure: solid arrows are each block's own update, dashed arrows show conditioning on the other block's most recently drawn value

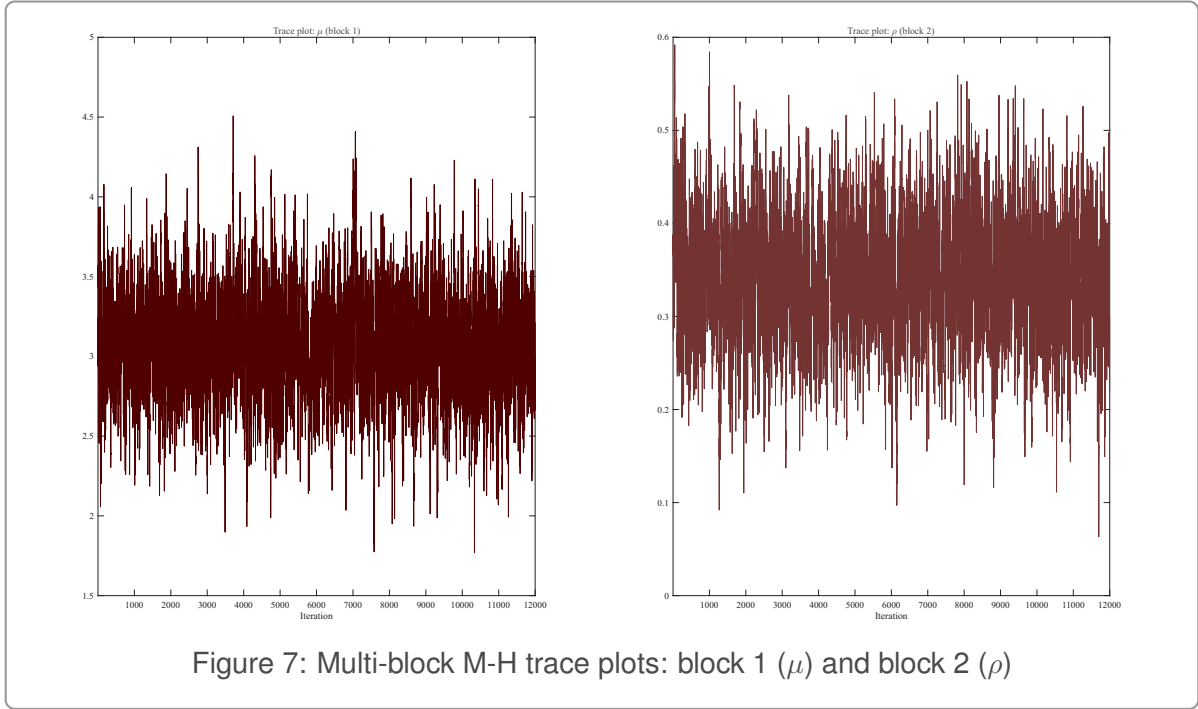
Algorithm: Multi-Block Metropolis-Hastings

Partition $\theta = (\theta_1, \theta_2)$. At iteration t :

1. Propose θ'_1 given $\theta_1^{(t)}$; accept/reject using $\pi(\theta'_1 | \theta_2^{(t)}, y)$ against $\pi(\theta_1^{(t)} | \theta_2^{(t)}, y)$, exactly as in single-block M-H.
2. Propose θ'_2 given $\theta_2^{(t)}$; accept/reject using $\pi(\theta'_2 | \theta_1^{(t+1)}, y)$ against $\pi(\theta_2^{(t)} | \theta_1^{(t+1)}, y)$.
3. Repeat for as many blocks as needed; each block may use its own step size, tuned separately to its own optimal acceptance rate.

Empirical Example: Multi-Block M-H: AR(1) with Unknown Mean for U.S. GDP Growth

Extend the model of Section 4.5 to $y_t - \mu = \rho(y_{t-1} - \mu) + e_t$, with blocks $\theta_1 = \mu$ (vague $N(3, 100)$ prior) and $\theta_2 = \rho$ (the same stationarity-truncated prior as before). Each block is updated by its own random-walk M-H step ($\tau_\mu = 0.35$, $\tau_\rho = 0.06$), cycling through both blocks at every iteration. With 12,000 post-burn-in draws, the posterior means are $\hat{\mu} = 2.998$ (close to the sample mean of GDP growth, 2.966, from Chapter 1) and $\hat{\rho} = 0.039$, with acceptance rates of 64.3% (block μ) and 72.3% (block ρ). Once the mean is properly accounted for, the persistence estimate drops from 0.36 to about 0.04 — confirming the bias mechanism described in the remark of Section 4.5, and showing just how weak GDP growth's own linear persistence becomes once the extended sample's most extreme swings are properly centered rather than left to bias a zero-mean fit. Section 4.7 re-estimates this same mean-inclusive AR(1) model by Gibbs sampling as a cross-check.



4.7 Gibbs Sampling

Gibbs sampling is the special case of multi-block M-H in which each block's *full conditional* posterior distribution is available in closed form, so the candidate is drawn directly from that conditional and always accepted (acceptance probability $\equiv 1$). Whenever conjugate priors are available — as in the Normal-Inverse-Gamma linear regression model — Gibbs sampling is typically far more efficient than M-H, since no proposal tuning is required and no draws are wasted on rejection.

Algorithm: Gibbs Sampling

1. For $\theta = (\theta_1, \dots, \theta_K)$, at iteration t , cycle through $k = 1, \dots, K$ and draw each coordinate from its full conditional given every other coordinate's most recently updated value,

$$\theta_k^{(t+1)} \sim p\left(\theta_k \mid \theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}, y\right).$$

2. Repeat for $t = 1, 2, \dots$; every proposed move is accepted automatically — there is no accept/reject step, unlike Metropolis-Hastings.

On the same two-parameter, correlated-Normal target used to illustrate M-H above, Figure 8 runs 10 Gibbs sweeps: draw $\theta_1 \mid \theta_2$ (a horizontal move), then $\theta_2 \mid \theta_1$ (a vertical move), and repeat. Every proposed move is accepted — there is no rejection step at all — but the price is visible in the picture: because the target is a narrow, tilted ridge, a purely horizontal-then-vertical path can only take small steps along that ridge, so Gibbs needs many more sweeps than M-H needs proposals to cross the same distance. This trade-off — Gibbs never wastes a draw but can crawl slowly under strong correlation, M-H can leap further but wastes many draws on rejection — is the central practical choice between the two methods.

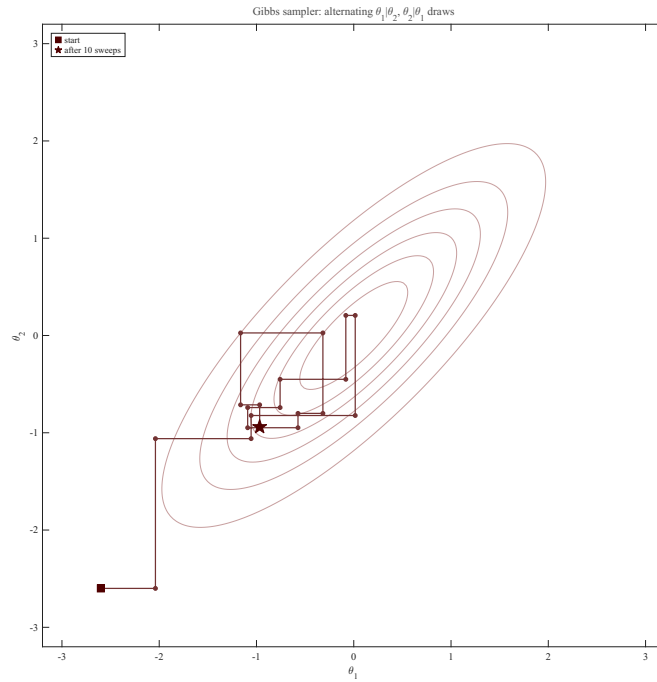


Figure 8: Gibbs sampling on the same bivariate Normal target ($\text{Corr} = 0.8$): the characteristic axis-aligned zigzag, one coordinate updated at a time

Empirical Example: Gibbs Sampling: AR(1) Regression of U.S. GDP Growth

Re-estimate the mean-inclusive AR(1) model of Section 4.6, now written as a linear regression $y_t = c + \rho y_{t-1} + e_t$ with a conjugate Normal-Inverse-Gamma prior ($\beta \sim N(0, 100I)$, $\sigma^2 \sim IG(1.5, 1.5)$), and sampled with the conjugate Gibbs sampler of Chapter 6 (implemented in `gibbs_linear_normal`). With 10,000 post-burn-in draws, the posterior means are $\hat{c} = 2.875$ and $\hat{\rho} = 0.035$ (implying an unconditional mean of $\hat{c}/(1 - \hat{\rho}) = 2.978$, matching both the multi-block M-H estimate of Section 4.6 and the sample mean of GDP growth from Chapter 1), with posterior standard deviations 0.314 and 0.061. The inefficiency factor for ρ is 0.93 — essentially 1, meaning the conjugate Gibbs draws are nearly as efficient as i.i.d. samples, in contrast to the inefficiency factor of 6.96 for the random-walk M-H sampler of Section 4.5. This is the central practical advantage of exploiting conjugacy whenever it is available.

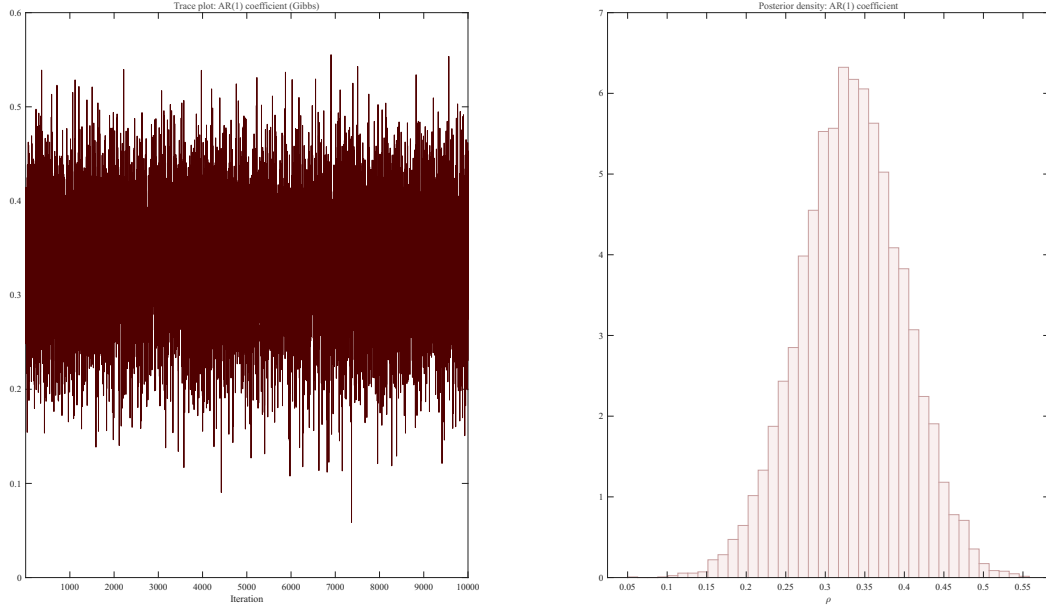


Figure 9: Gibbs sampling of the AR(1) coefficient ρ : trace plot and posterior density

4.8 Slice Sampling

Slice sampling is an auxiliary-variable technique that, like Gibbs sampling, requires no tuning of a proposal distribution, yet applies even when no conjugate conditional is available. The key device is to augment θ with an auxiliary vertical coordinate h and sample uniformly from the region under the (unnormalized) target density curve.

Algorithm: Univariate Slice Sampling

1. At $x^{(t)}$, draw $h \sim \text{Uniform}(0, f(x^{(t)}))$, defining a horizontal “slice” $S = \{x : f(x) > h\}$.
2. Find an interval $(\ell, r) \supseteq S$ containing $x^{(t)}$, typically via a step-out procedure that expands an initial window of width w until both endpoints fall below h .
3. Draw $x' \sim \text{Uniform}(\ell, r)$; if $f(x') > h$, accept $x^{(t+1)} = x'$; otherwise shrink the interval toward $x^{(t)}$ (replacing ℓ or r with x') and repeat.

The univariate recipe above extends immediately to two parameters by alternating coordinates exactly as Gibbs sampling does: slice-sample $\theta_1 \mid \theta_2$, then $\theta_2 \mid \theta_1$, and repeat. Figure 10 runs 9 such sweeps on the same correlated-Normal target as the two figures above, now also drawing each step’s slice window (the horizontal or vertical segment actually searched before landing on the next point). The resulting path looks like Gibbs’s zigzag because it uses the same alternating-coordinate structure, but each step is self-tuning — the window width w only sets a starting guess, expanded automatically by the step-out procedure — whereas Gibbs above relied on knowing the exact conditional distribution in closed form.

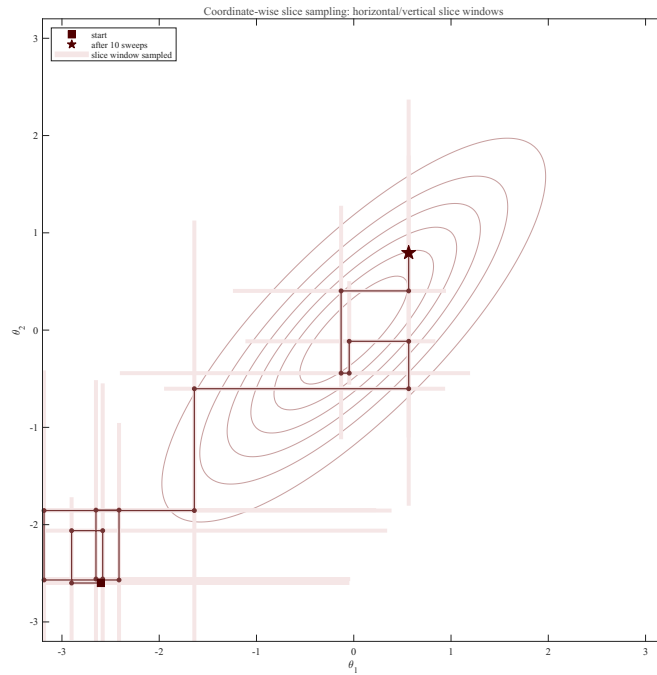


Figure 10: Coordinate-wise slice sampling on the same bivariate Normal target ($\text{Corr} = 0.8$): each step's horizontal/vertical slice window is shown before the point it produces

Empirical Example: Slice Sampling from Gamma(3,1)

Applying the step-out-and-shrink slice sampler to the (unnormalized) Gamma(3, 1) density $f(x) \propto x^2 e^{-x}$ for 12,000 post-burn-in draws gives sample mean 3.013 and sample variance 3.024, versus theoretical values 3.0 and 3.0. Figure 11 shows the sampled histogram overlaid on the exact Gamma(3,1) density. Slice sampling is used later in this note as a robust fallback whenever a full conditional distribution is neither conjugate nor easily bounded by an M-H proposal — for instance, for smoothing parameters in the time-varying-parameter VAR of Chapter 17.

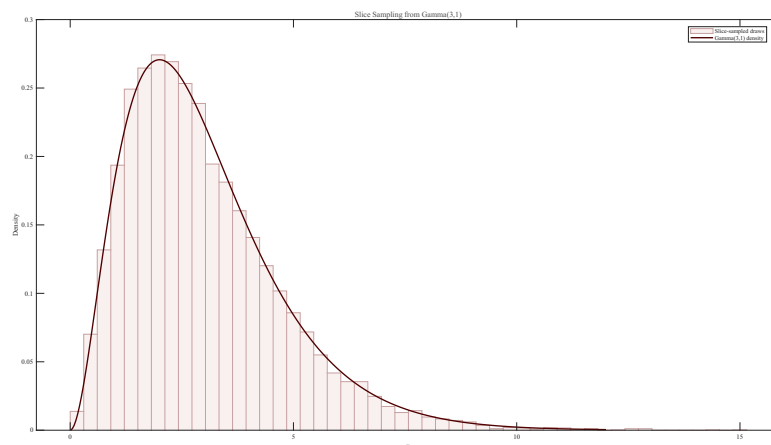


Figure 11: Slice sampling from Gamma(3,1): histogram of 12,000 draws vs. the exact density

5. Convergence Diagnostics

Roadmap

Chapters 3–4 gave the machinery to build a Markov chain that eventually targets the right posterior, and to sample from it efficiently. Neither chapter answered a question every applied chapter from here on must face: has *this particular run*, of *this particular length*, actually gotten there? This chapter supplies four checks, from a quick visual scan to a formal cross-chain test, and every posterior estimate reported from Chapter 6 onward is implicitly certified by at least one of them.

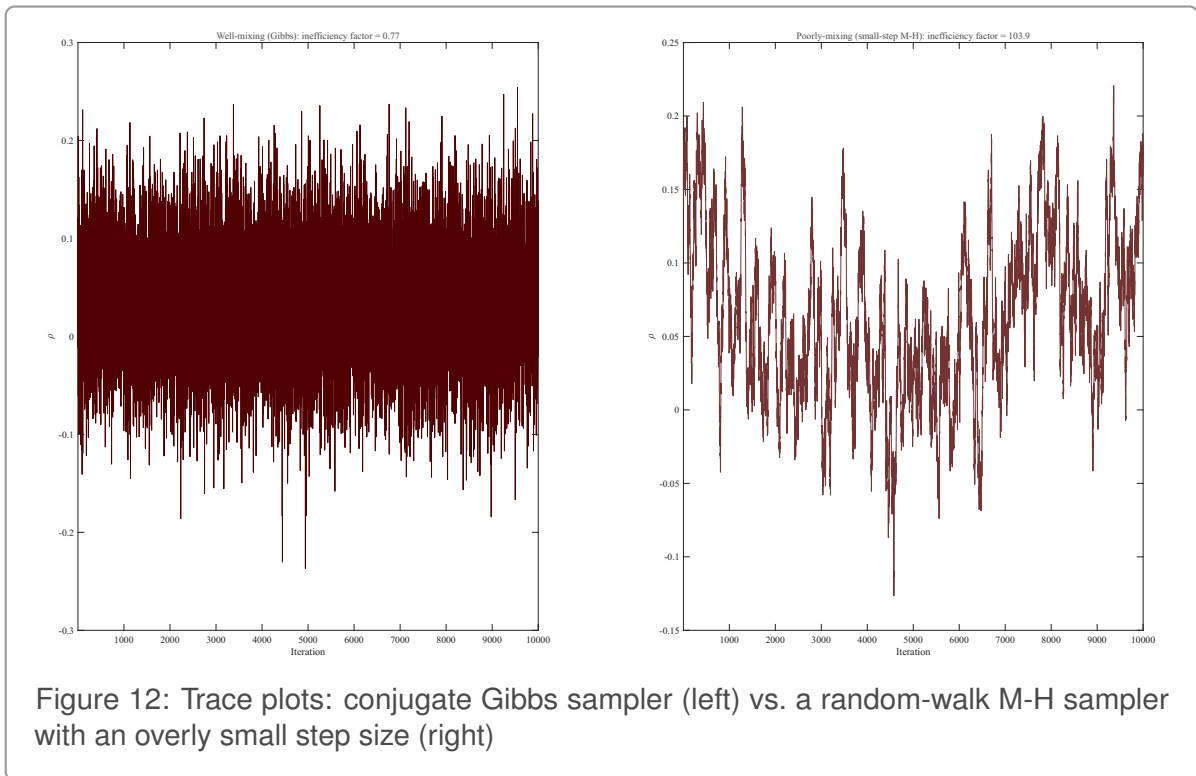
Chapter 3 showed that an MCMC sampler is only guaranteed to produce draws from the target posterior *asymptotically*, after burn-in, and Chapter 4 showed that different algorithms (and different tuning choices within the same algorithm) can mix at very different rates. In practice we never observe convergence directly — we only observe a finite chain and must decide whether it is trustworthy. This chapter presents the four diagnostics used throughout the rest of this note: trace plots and the autocorrelation function (visual/qualitative), the inefficiency factor (quantifies mixing speed), Geweke’s convergence diagnostic (tests within-chain convergence), and the Gelman-Rubin statistic (tests between-chain agreement). All four are illustrated on the same AR(1) model of U.S. real GDP growth used in Chapter 4, contrasting a well-mixing conjugate Gibbs sampler against a deliberately poorly-tuned random-walk M-H sampler.

5.1 Trace Plots and the Autocorrelation Function

The simplest diagnostic is to plot $\theta^{(t)}$ against the iteration index t — a *trace plot*. A well-mixing chain should look like a stationary, mean-reverting band with no visible trend and should traverse its range many times over the course of the run; a poorly-mixing chain instead drifts slowly, showing long excursions away from its long-run average.

Empirical Example: Trace Plots: Well-Mixing vs. Poorly-Mixing Samplers

Figure 12 compares two samplers targeting the same AR(1) coefficient posterior for U.S. real GDP growth: the conjugate Gibbs sampler of Chapter 4.7 (left) and a deliberately mistuned random-walk M-H sampler with a very small step size $\tau = 0.008$ (right, 10,000 draws after a 5,000-iteration burn-in). Both chains target the same posterior, but their sample means still differ noticeably (0.035 vs. 0.087) — itself a symptom of the M-H chain’s slow convergence — and the Gibbs chain fills its range almost instantly at every iteration, while the M-H chain visibly drifts in long, slow excursions — the signature of a poorly-mixing chain.



The autocorrelation function (ACF), $\hat{\rho}_k = \widehat{\text{Corr}}(\theta^{(t)}, \theta^{(t+k)})$, makes the same information precise: a well-mixing chain's ACF collapses to zero within a few lags, while a poorly-mixing chain's ACF decays slowly.

Empirical Example: Autocorrelation Function Comparison

Figure 13 plots the ACF (up to lag 60) for the same two chains. The Gibbs sampler's autocorrelation is statistically indistinguishable from zero beyond lag 1; the small-step M-H sampler's autocorrelation is still 0.67 at lag 60, confirming visually what the trace plot suggested.

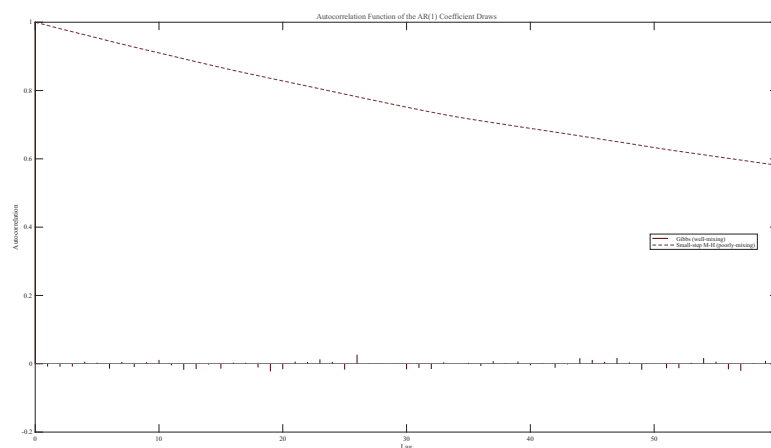


Figure 13: Autocorrelation function: conjugate Gibbs vs. small-step random-walk M-H

5.2 The Inefficiency Factor

Trace plots and ACFs are qualitative. The *inefficiency factor* (equivalently, the inverse relative effective sample size) quantifies mixing speed in a single number, already used informally in Chapters 4:

$$\text{IF} = 1 + 2 \sum_{k=1}^K \hat{\rho}_k,$$

a Newey-West/Parzen-type truncated sum of the ACF. If N correlated draws have inefficiency factor IF, their effective sample size for Monte Carlo averaging is approximately N/IF : an inefficiency factor of 10 means the chain must be run roughly $10\times$ longer than an i.i.d. sampler to attain the same Monte Carlo precision.

Term	Meaning
$\text{IF} \approx 1$	Draws are nearly independent (typical of a well-tuned conjugate Gibbs step).
$1 < \text{IF} \lesssim 20$	Typical of a reasonably well-tuned M-H sampler; acceptable in most applications.
$\text{IF} \gg 20$	Poor mixing; consider re-tuning the proposal, reparameterizing, or switching to Gibbs/blocking where possible.

Empirical Example: Inefficiency Factor: Gibbs vs. Small-Step M-H

For the two chains of Figure 12, the inefficiency factor is 0.98 for the Gibbs sampler (effectively i.i.d. draws) versus 225.5 for the small-step M-H sampler, whose acceptance rate is a very high 93.6% — a classic symptom of a step size that is too small: the sampler almost always accepts because it almost never proposes a move far enough to matter. This $230\times$ efficiency gap illustrates concretely why Chapter 4 emphasized exploiting conjugacy (Gibbs sampling) whenever available, and why M-H step sizes should be tuned toward, not away from, a moderate rejection rate.

5.3 Geweke's Convergence Diagnostic

The inefficiency factor measures mixing *speed* but says nothing about whether burn-in was long enough. Geweke (1992) proposed a formal within-chain test: split the post-burn-in draws into an early window (the first n_A , typically the first 10%) and a late window (the last n_C , typically the last 50%), and test whether the two windows have the same mean. If the chain has converged, both windows are samples from the same stationary distribution and should have statistically indistinguishable means; if the chain is still drifting (inadequate burn-in), the early window will differ systematically from the late window.

Algorithm: Geweke's Convergence Diagnostic (CD)

1. Split the chain into an early window of the first 10% of draws and a late window of the last 50% of draws.
2. Compute the window means $\bar{\theta}_A, \bar{\theta}_C$ and their Monte Carlo standard errors $\widehat{\text{se}}(\bar{\theta}_A), \widehat{\text{se}}(\bar{\theta}_C)$, each corrected for autocorrelation using that window's own inefficiency factor (i.e., $\widehat{\text{se}}^2 = \widehat{\text{IF}} \times \widehat{\text{Var}}/n$).
3. Compute the test statistic

$$z = \frac{\bar{\theta}_A - \bar{\theta}_C}{\sqrt{\widehat{\text{se}}(\bar{\theta}_A)^2 + \widehat{\text{se}}(\bar{\theta}_C)^2}},$$

which is asymptotically $N(0, 1)$ under the null of convergence. A p -value near zero indicates the chain has not yet converged (or burn-in was too short).

Empirical Example: Geweke's CD: Converged, Poorly-Mixing, and Non-Converged Chains

Applying Geweke's test to three chains targeting the same AR(1) posterior:

- The well-mixing Gibbs chain gives $z = 0.32, p = 0.75$ — no evidence against convergence, as expected.
- The poorly-mixing small-step M-H chain of Section 5.1–5.2 gives $z = 4.52, p < 0.0001$ — Geweke's test also flags this chain as problematic. This illustrates an important point: severe autocorrelation does not just waste computational effort, it can also mask an inadequately short burn-in, because a chain that moves this slowly may not have finished migrating toward the true posterior mean within even a 5,000-iteration burn-in.
- A chain deliberately initialized far from the posterior ($\rho_0 = -0.9, \mu_0 = -5$) and run for 3,000 iterations with *no* burn-in discarded gives $z = -5.65, p \approx 1.6 \times 10^{-8}$ — an unambiguous rejection of convergence. Figure 14 shows why: the early 10% window (mean -0.48) is still climbing out of the initial transient, while the late 50% window (mean 0.47) is still working its way toward the true posterior region — itself now much closer to zero than before this note's data extension, since the AR(1) coefficient's own posterior mean has collapsed from about 0.33 to about 0.04 once the extended sample's 2020 swings are properly centered (Chapter 4). This is exactly the pattern Geweke's diagnostic is designed to catch.

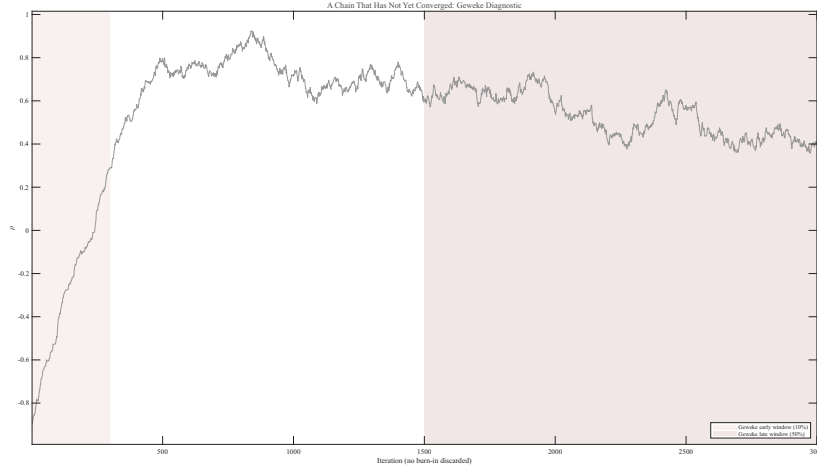


Figure 14: A chain with inadequate burn-in: the early window mean differs sharply from the late window mean

Remark: Geweke's Test Has Limited Power Against Some Alternatives

Geweke's test compares only the *means* of two windows; it can fail to detect non-convergence in higher moments (e.g., a chain whose variance is still shrinking even though its mean has stabilized), and its power is reduced when the early window itself spans a long, gradual transient (so that the within-window variance used to form $\widehat{se}(\bar{\theta}_A)$ is itself inflated by the transient). In practice Geweke's CD should be viewed as one diagnostic among several, not a substitute for visual inspection of the trace plot.

5.4 The Gelman-Rubin Statistic

Geweke's diagnostic examines a single chain. The Gelman-Rubin statistic (Gelman and Rubin, 1992), sometimes called $\hat{R}$ or Rubin's potential scale reduction factor, instead runs $M \geq 2$ independent chains from *overdispersed* starting values and compares the variance *between* chains to the variance *within* chains. If all chains have converged to the same stationary distribution, the two variances should agree; if some chains are still confined near their (different) starting points, the between-chain variance will be inflated relative to the within-chain variance.

Algorithm: Gelman-Rubin Statistic ($\hat{R}$)

1. Run M chains of length N (post-burn-in) from dispersed starting values. Let $\bar{\theta}_m$ and s_m^2 be the mean and variance of chain m , and $\bar{\bar{\theta}}$ the grand mean across chains.
2. Compute the within-chain and between-chain variances,

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\bar{\theta}})^2.$$

3. Combine them into a pooled variance estimate and the $\hat{R}$ statistic,

$$\widehat{\text{Var}}^+(\theta) = \left(1 - \frac{1}{N}\right)W + \frac{B}{N}, \quad \hat{R} = \sqrt{\frac{\widehat{\text{Var}}^+(\theta)}{W}}.$$

4. Interpret: $\hat{R}$ is close to 1 once all chains have converged to the same distribution (common rules of thumb flag $\hat{R} > 1.1$, or more conservatively $\hat{R} > 1.01$, as evidence of non-convergence). Because it requires only each chain's mean and variance, $\hat{R}$ is trivial to compute for every parameter in a large model.

Empirical Example: Gelman-Rubin Diagnostic: Four Chains from Dispersed Starts

Four Gibbs chains for the AR(1) coefficient were run from widely dispersed starting values ($\rho_0 \in \{-0.8, 0, 0.8, 0.95\}$), each with a short burn-in of 1,000 and $N = 4,000$ retained draws. The resulting $\hat{R} = 1.0000$, essentially exact agreement between chains — consistent with the near-instantaneous mixing already seen in the trace plots. As a contrasting case, four small-step random-walk M-H chains (the same poorly-mixing sampler of Section 5.2) were started from dispersed values and run for only 200 iterations with no burn-in discarded: this gives $\hat{R} = 3.83$, far above the 1.1 threshold, correctly signaling that the chains have not yet forgotten their starting points. Figure 15 shows both cases: the left panel's four Gibbs chains are visually superimposed almost immediately, while the right panel's four M-H chains remain visibly separated, each still drifting toward the common posterior region.

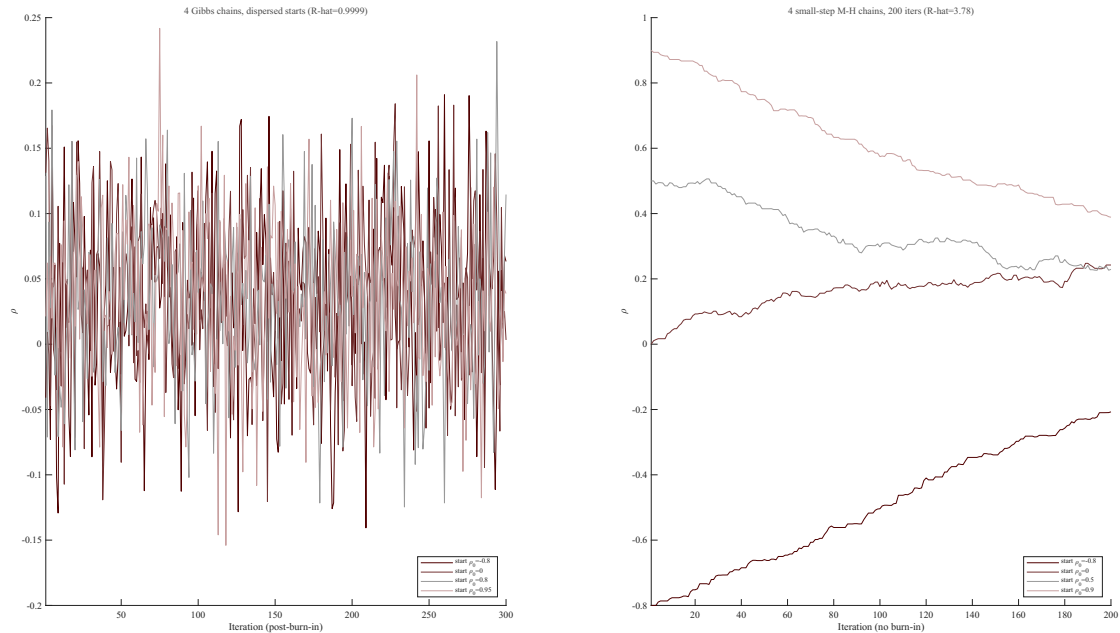


Figure 15: Gelman-Rubin diagnostic: converged chains ($\hat{R} \approx 1$, left) vs. chains still separated after only 200 iterations ($\hat{R} = 3.83$, right)

Remark: Why Run Multiple Chains At All?

A single long chain can, in principle, eventually explore the full posterior even if it starts far away, and Geweke's test can catch an inadequately short burn-in within that one chain. But a single chain can also become “stuck” in one region of a multimodal posterior for a very long time without any within-chain symptom of trouble — nothing in a single trace plot rules this out. Running several chains from deliberately dispersed starting points, as the Gelman-Rubin diagnostic requires, is the standard defense against exactly this failure mode, and is particularly important for the multimodal posteriors that arise in the regime-switching models of Chapter 18.

6. Multiple Regression

Roadmap

Chapter 4 built and tuned a Gibbs sampler for one coefficient, an AR(1) persistence parameter, treating it as a proof of concept for the sampling machinery itself. This chapter is where that machinery becomes an actual econometric tool: the same two-block Gibbs sampler, written in matrix form for k regressors instead of one. The derivation done in full here — complete the square for the coefficient block, match powers and exponentials for the variance block — is not specific to regression; every conjugate posterior in these notes (Probit in Chapter 7, Tobit in Chapter 9, SUR in Chapter 10, VAR in Chapter 11) is the same two moves applied to a larger or differently shaped matrix. Reading this chapter's proofboxes closely pays off repeatedly later.

Chapters 4 and 5 built up the sampling and diagnostic toolkit using a single-regressor AR(1) model of U.S. GDP growth as a running example. This chapter generalizes that model to k regressors, develops the conjugate Gibbs sampler in full matrix form, and extends it to fat-tailed (Student- t) errors using the scale-mixture representation introduced in Section 4.1. Bayesian multiple regression is also the foundation on which the remaining chapters build: Probit (Chapter 7) and Tobit (Chapter 9) reuse the same conjugate machinery with a latent linear regression underneath a non-Gaussian observation rule, variable selection (Chapter 8) puts a prior over which regressors to include, and SUR (Chapter 10) stacks several such regressions with correlated errors.

6.1 The Model and Conjugate Prior

The multiple regression model is

$$y = X\beta + e, \quad e \sim N(0, \sigma^2 I_T),$$

where y is $T \times 1$, X is $T \times k$, and β is $k \times 1$. The natural-conjugate Normal-Inverse-Gamma prior used throughout this note is

$$\beta \sim N(b_0, V_0), \quad \sigma^2 \sim IG\left(\frac{a_0}{2}, \frac{d_0}{2}\right),$$

with β and σ^2 independent *a priori*. Combined with the Gaussian likelihood, this prior yields closed-form full conditional posteriors for both blocks — the same conjugate structure used for the single-regressor AR(1) model of Section 4.7, now written for general k .

6.2 Posterior Derivation

The two full conditionals used by the Gibbs sampler are not assumed — they follow directly from multiplying the Gaussian likelihood by the Normal-Inverse-Gamma prior and recognizing the resulting kernel. This section works through both derivations in full, since every conjugate posterior used later in this note (Probit, Tobit, SUR, VAR, and so on) is built from exactly the same two moves: *completing the square* for the Normal block, and *matching powers and exponentials* for the Inverse-Gamma block.

Derivation: $\beta \mid \sigma^2, y$

By Bayes' rule, holding σ^2 fixed, the posterior kernel is the likelihood times the prior on β :

$$p(\beta \mid \sigma^2, y) \propto \underbrace{\exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)\right\}}_{\text{likelihood}} \underbrace{\exp\left\{-\frac{1}{2}(\beta - b_0)'V_0^{-1}(\beta - b_0)\right\}}_{\text{prior}}$$

(the σ^2 prior itself contributes only a multiplicative constant that does not involve β , and is already omitted). Collecting both quadratic forms into a single exponent:

$$= \exp\left\{-\frac{1}{2}\left[\sigma^{-2}(y - X\beta)'(y - X\beta) + (\beta - b_0)'V_0^{-1}(\beta - b_0)\right]\right\}$$

Expanding each quadratic form in β :

$$= \exp\left\{-\frac{1}{2}\left[\sigma^{-2}(y'y - 2\beta'X'y + \beta'X'X\beta) + \beta'V_0^{-1}\beta - 2\beta'V_0^{-1}b_0 + b_0'V_0^{-1}b_0\right]\right\}$$

Collecting the terms that are quadratic and linear in β separately (every other term — $y'y$ and $b_0'V_0^{-1}b_0$ — does not depend on β and is dropped into the proportionality constant):

$$\propto \exp\left\{-\frac{1}{2}\left[\beta'(V_0^{-1} + \sigma^{-2}X'X)\beta - 2\beta'(V_0^{-1}b_0 + \sigma^{-2}X'y)\right]\right\}$$

Define $\bar{V} \equiv (V_0^{-1} + \sigma^{-2}X'X)^{-1}$ and $\bar{\beta} \equiv \bar{V}(V_0^{-1}b_0 + \sigma^{-2}X'y)$, and apply the completing-the-square identity — a quadratic form $\beta'A\beta - 2\beta'c$ always equals $(\beta - A^{-1}c)'A(\beta - A^{-1}c) - c'A^{-1}c$, here with $A = V_0^{-1} + \sigma^{-2}X'X = \bar{V}^{-1}$ and $c = V_0^{-1}b_0 + \sigma^{-2}X'y$, so $A^{-1}c = \bar{\beta}$ — so the bracketed exponent above equals $(\beta - \bar{\beta})'\bar{V}^{-1}(\beta - \bar{\beta}) + \text{const}$, with the extra constant $\bar{\beta}'\bar{V}^{-1}\bar{\beta}$ again β -free and dropped:

$$p(\beta \mid \sigma^2, y) \propto \exp\left\{-\frac{1}{2}(\beta - \bar{\beta})'\bar{V}^{-1}(\beta - \bar{\beta})\right\}$$

which is the kernel of a $N(\bar{\beta}, \bar{V})$ density. Since a probability density is determined uniquely by its kernel (the missing normalizing constant is whatever makes it integrate to one), $\beta \mid \sigma^2, y \sim N(\bar{\beta}, \bar{V})$.

Derivation: $\sigma^2 \mid \beta, y$

Now hold β fixed and treat σ^2 as the unknown, with $e \equiv y - X\beta$. The Gaussian likelihood, written as a function of σ^2 alone, and the Inverse-Gamma prior kernel are

$$p(y \mid \beta, \sigma^2) = (2\pi\sigma^2)^{-T/2} \exp\left\{-\frac{e'e}{2\sigma^2}\right\},$$

$$p(\sigma^2) \propto (\sigma^2)^{-(a_0/2+1)} \exp\left\{-\frac{d_0}{2\sigma^2}\right\}.$$

Multiplying the two and collecting powers of σ^2 separately from the exponential terms (legitimate because $(\sigma^2)^{-T/2}$ and $(\sigma^2)^{-(a_0/2+1)}$ combine by adding exponents, and the two exponentials share the common factor $-1/(2\sigma^2)$):

$$p(\sigma^2 \mid \beta, y) \propto (\sigma^2)^{-T/2-(a_0/2+1)} \exp\left\{-\frac{1}{2\sigma^2}(e'e + d_0)\right\}$$

and, writing the combined power of σ^2 in the same shape/scale form as the prior:

$$= (\sigma^2)^{-\left(\frac{a_0+T}{2}+1\right)} \exp\left\{-\frac{d_0 + e'e}{2\sigma^2}\right\}.$$

This is exactly the kernel of an Inverse-Gamma density with shape $(a_0 + T)/2$ and scale $(d_0 + e'e)/2$. Hence $\sigma^2 \mid \beta, y \sim IG\left(\frac{a_0 + T}{2}, \frac{d_0 + e'e}{2}\right)$: the prior degrees of freedom a_0 and prior sum of squares d_0 are simply updated by adding the sample size T and the realized residual sum of squares $e'e$, respectively — the usual Inverse-Gamma conjugate updating rule.

6.3 Gibbs Sampling Algorithm**Algorithm: Conjugate Gibbs Sampler for Multiple Regression**

Given a current value of σ^2 , cycle:

1. Draw $\beta \mid \sigma^2, y \sim N(\bar{\beta}, \bar{V})$, where

$$\bar{V} = (V_0^{-1} + \sigma^{-2}X'X)^{-1}, \quad \bar{\beta} = \bar{V} (V_0^{-1}b_0 + \sigma^{-2}X'y).$$

2. Draw $\sigma^2 \mid \beta, y \sim IG\left(\frac{a_0 + T}{2}, \frac{d_0 + e'e}{2}\right)$, where $e = y - X\beta$.

Iterate, discard a burn-in of n_0 draws, and retain n_1 post-burn-in draws for posterior inference.

6.4 Empirical Example: A Reduced-Form Output Equation for the U.S. Economy

To illustrate multiple regression with real regressors, consider a small reduced-form model of quarterly U.S. real GDP growth, y_t , as a function of its own lag, lagged CPI inflation (π_{t-1}), and the lagged effective Fed funds rate (i_{t-1}), using the same quarterly panel (1959Q3–2026Q1, extended from Chapters 4–5's original 1959Q3–2013Q4 window to include the COVID-19

pandemic) as Chapters 4–5:

$$y_t = c + \phi y_{t-1} + \gamma \pi_{t-1} + \delta i_{t-1} + e_t.$$

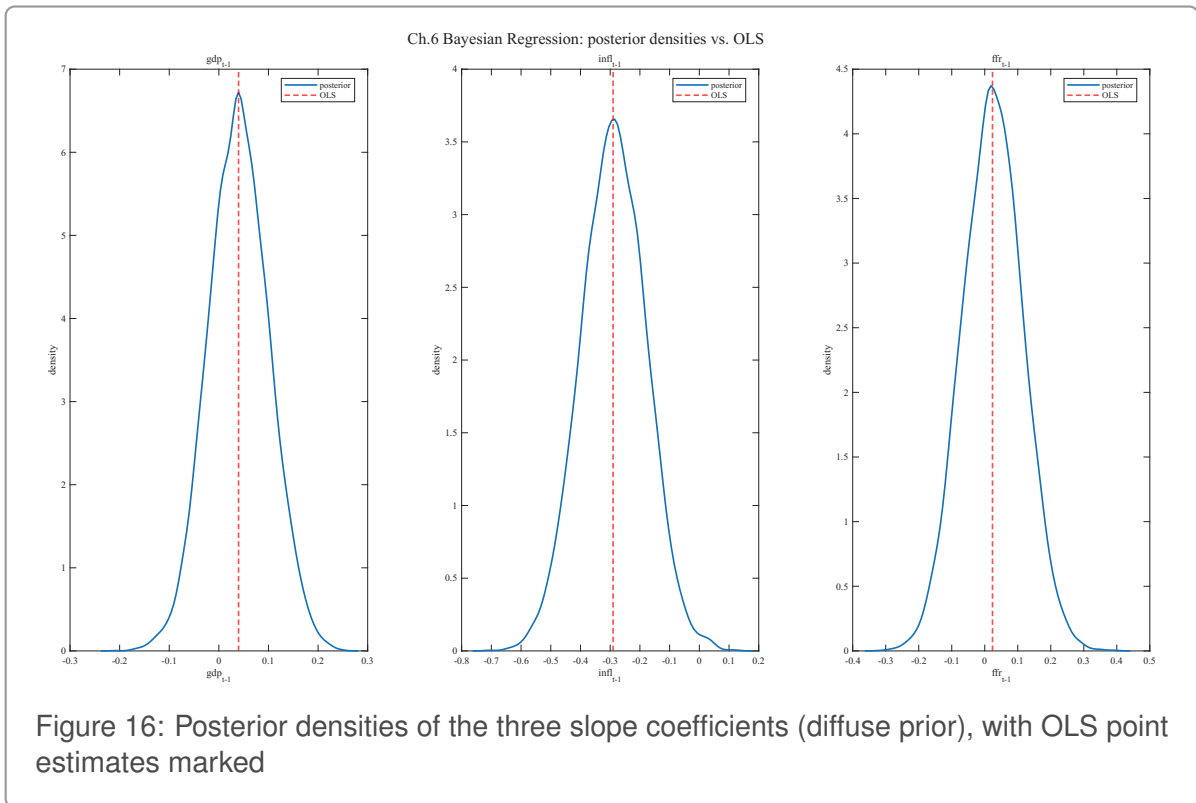
This is a $k = 4$ extension of the AR(1) model of Chapter 4.7, and is the kind of small “IS-curve”-style reduced form widely used as a starting point in empirical macroeconomics.

Empirical Example: Posterior Estimates: Diffuse Prior

With a diffuse prior ($\beta \sim N(0, 100I_4)$, $\sigma^2 \sim IG(1.5, 1.5)$) and 10,000 post-burn-in Gibbs draws, the posterior means (standard deviations, 95% credible intervals) are:

$$\begin{aligned}\hat{c} &= 3.794 \text{ (0.484)} & [2.863, 4.741] \\ \hat{\phi} &= 0.040 \text{ (0.061)} & [-0.078, 0.161] \\ \hat{\gamma} &= -0.290 \text{ (0.108)} & [-0.502, -0.078] \\ \hat{\delta} &= 0.025 \text{ (0.090)} & [-0.150, 0.206]\end{aligned}$$

All four inefficiency factors are close to 1 (0.93–1.03), confirming that the conjugate Gibbs sampler mixes essentially perfectly, as expected from Section 4.7. These posterior means agree with the OLS estimates (3.805, 0.040, −0.290, 0.023) to two or three significant digits, as they should with a diffuse prior and $T = 266$ observations — Figure 16 overlays the two. The lagged-growth coefficient $\hat{\phi} = 0.040$ is close to the mean-inclusive AR(1) estimate of Chapter 4.7 (0.039); both have fallen sharply from their pre-extension values (Chapter 4.7’s was 0.333, and this equation’s own $\hat{\phi}$ was 0.314 on the original 1959Q3–2013Q4 sample) once the extreme 2020Q2/Q3 swings enter the lag-1 second moment (see Chapter 4.5’s remark on this mechanism). Unlike the pre-extension sample, where lagged inflation’s credible interval only barely covered zero, the extended sample’s credible interval for γ , $[-0.502, -0.078]$, excludes zero outright: with the pandemic quarters included, the countercyclical/cost-push relationship between lagged inflation and subsequent growth is now credibly negative. The Fed funds rate coefficient remains indistinguishable from zero at this horizon. The regression now explains only $R^2 = 0.039$ of quarterly growth variation (Figure 20, left panel) — down sharply from 0.146 on the pre-extension sample, because the 2020Q2/Q3 swings so dominate the sum of squares that a purely backward-looking linear reduced form with one lag explains almost none of it.



6.5 Rudebusch and Svensson (1999)

The reduced-form equation above is a general-purpose illustration; it is worth pausing to see the same idea applied by an actual, influential piece of applied macroeconomics. Rudebusch and Svensson (1999, hereafter RS), “Policy Rules for Inflation Targeting,” build a small two-equation model of the U.S. economy — a backward-looking Phillips curve and a backward-looking IS curve — to evaluate monetary policy rules for inflation targeting. Their IS curve is estimated by OLS on quarterly U.S. data, 1961:1–96:2:

$$y_{t+1} = \beta_1 y_t + \beta_2 y_{t-1} - \beta_r (\bar{r}_t - \bar{\pi}_t) + \eta_{t+1},$$

where y_t is the output gap (real GDP relative to potential GDP, in percent), $\bar{r}_t$ is the four-quarter average federal funds rate, and $\bar{\pi}_t$ is the four-quarter average GDP-deflator inflation rate — an approximate ex post real interest rate. All variables are de-meaned prior to estimation, so the equation has no intercept. RS(1999) report OLS estimates $\hat{\beta}_1 = 1.16$, $\hat{\beta}_2 = -0.25$, $\hat{\beta}_r = -0.10$ (standard errors 0.08, 0.08, 0.03), residual SE = 0.819.

Following the same three-step logic used throughout this note — replicate, extend, explain — we first reconstruct their exact sample using FRED vintages of real GDP (GDPC1), CBO potential GDP (GDPPOPOT), the GDP deflator (GDPDEF), and the effective federal funds rate (FEDFUNDS), then re-estimate the same equation with the conjugate Gibbs sampler of Section 6.2–6.3 instead of OLS, and finally re-estimate it again on the same variables extended all the way to the present.

Empirical Example: Original Sample, 1961Q1–1996Q2

With a diffuse prior ($\beta \sim N(0, 100I_3)$, no intercept column, $\sigma^2 \sim IG(1.5, 1.5)$) and 10,000 post-burn-in draws, the posterior means (standard deviations) are $\hat{\beta}_1 = 1.152$ (0.084), $\hat{\beta}_2 = -0.253$ (0.082), and $\hat{\beta}_r = -0.081$ (0.032) — matching RS(1999)’s OLS estimates of 1.16, -0.25 , and -0.10 to within one posterior standard deviation on every coefficient, and their reported standard errors (0.08, 0.08, 0.03) almost exactly. This is a strong replication: the decades of GDP-deflator and potential-GDP data revisions published since 1999 have left the original-sample IS curve essentially unchanged, exactly as one would hope for a well-identified, low-dimensional reduced-form relationship. Figure 17 shows the three posterior densities and the fitted vs. actual series.

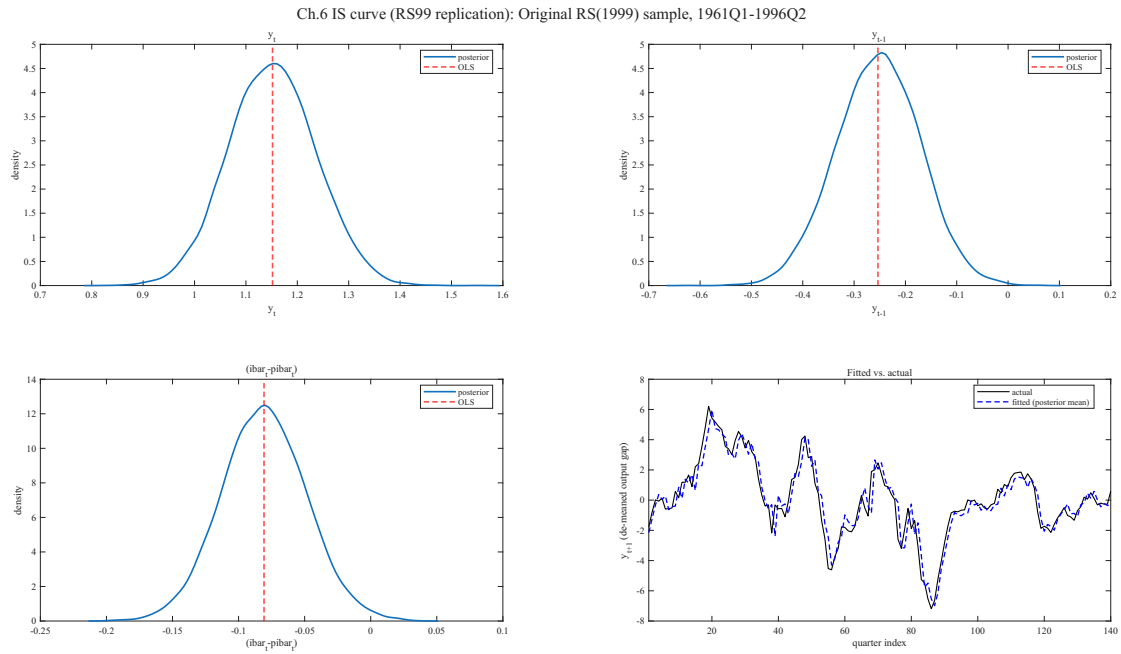


Figure 17: Rudebusch–Svensson (1999) IS curve, replicated on their original sample (1961Q1–1996Q2): posterior densities of β_1 , β_2 , β_r with OLS markers, and fitted vs. actual de-measured output-gap growth

Empirical Example: Full Sample, 1961Q1–Present

Re-estimating the identical model and prior on the same four series extended from 1961Q1 through the most recent available quarter gives $\hat{\beta}_1 = 0.934$ (0.063), $\hat{\beta}_2 = -0.038$ (0.062), and $\hat{\beta}_r = -0.037$ (0.023). Two changes stand out. First, the sum $\hat{\beta}_1 + \hat{\beta}_2$ — the persistence of the output gap — is little changed (0.899 originally vs. 0.896 extended), so the AR structure of the output gap itself has not shifted much. Second, and more strikingly, the real interest-rate coefficient $\hat{\beta}_r$ has shrunk by more than half, from -0.081 to -0.037 , and its 95% credible interval, $[-0.083, 0.008]$, no longer excludes zero. The channel through which monetary policy affects the output gap in this reduced form — the whole point of estimating an IS curve for policy-rule analysis in the first place — is measurably weaker over the full sample than it was over RS(1999)’s original one.

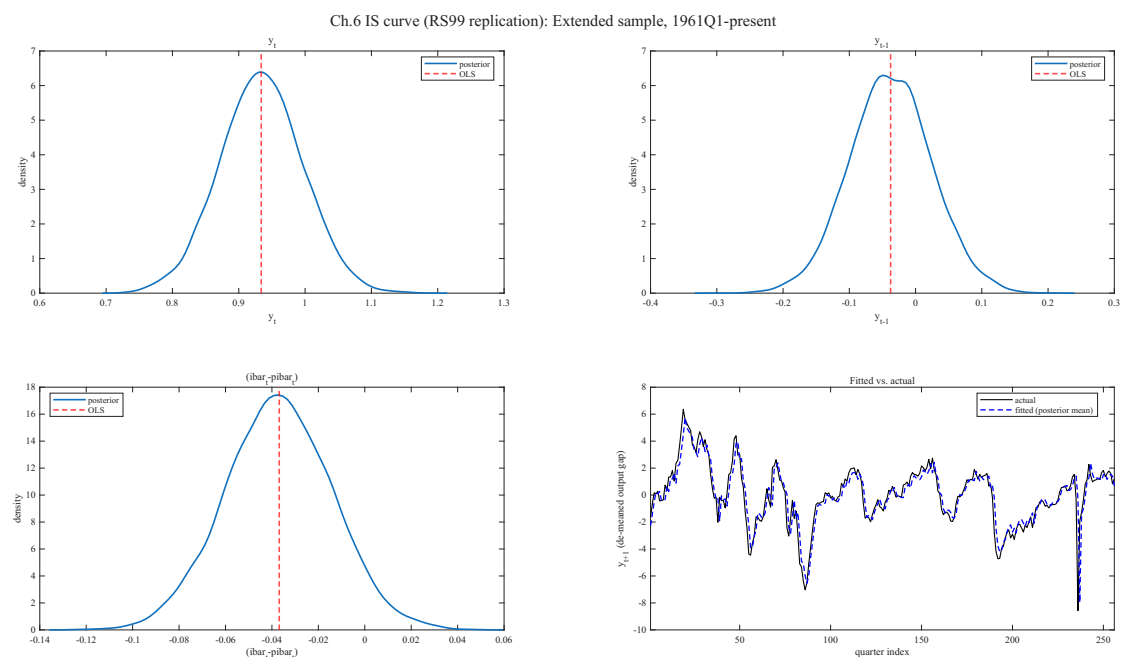


Figure 18: The same IS curve re-estimated on the extended sample (1961Q1–present): the real-rate coefficient β_r shrinks toward zero relative to Figure 17

Remark: Explaining the Attenuation: A Weaker, Not a Broken, Transmission Channel

A shrinking real-rate coefficient in a backward-looking reduced form is exactly the pattern documented in the monetary-transmission literature that followed RS(1999)’s sample period. Boivin and Giannoni (2002, 2006), splitting U.S. data at the Volcker disinflation (1979:4), find that the output response to an identified monetary policy shock in the post-Volcker period is only about *one-quarter* of its pre-Volcker magnitude — a decline of the same order as the one found here ($\hat{\beta}_r$ falls to under half its original-sample value once the sample is extended well past 1996). Boivin, Kiley, and Mishkin (2010), using both a factor-augmented VAR and an estimated DSGE model, confirm that monetary policy innovations have had a more muted effect on real activity in recent decades, and trace this to *changes in policy behavior itself* — monetary policy became more systematic and more focused on price stability after Volcker, so private agents come to expect and partially anticipate policy responses, leaving less unanticipated, output-moving variation in the interest-rate series for a backward-looking regression to pick up — rather than to any change in private-sector spending behavior. A second, complementary factor specific to the extended sample used here is that it includes two extended episodes near the zero lower bound (2009–2015 and 2020–2022), during which the nominal funds rate barely moved even as the output gap and inflation fluctuated sharply; this compresses the variance of the $(\bar{r}_t - \bar{\pi}_t)$ regressor and is a further, mechanical source of attenuation in $\hat{\beta}_r$ on top of the transmission-mechanism story above. A third possibility, also mechanical, is a secular decline in the *neutral* real rate r^* (Laubach and Williams, 2003): de-meaning $(\bar{r}_t - \bar{\pi}_t)$ over the full sample measures the stance of policy relative to a *constant* benchmark, so if r^* drifted down, the regressor carries growing measurement error — and classical measurement error in a regressor

attenuates its coefficient. Both explanations point the same direction and are not mutually exclusive: the IS curve's interest-sensitivity coefficient is smaller today not because the underlying replication was flawed, but because the object it measures — how strongly a given real-rate gap moves the output gap one quarter later — has itself become smaller since the era RS(1999) studied.

6.6 Prior Sensitivity: Diffuse vs. Shrinkage Priors

The conjugate prior is not merely a computational convenience — it lets the analyst encode genuine prior beliefs, and it is good practice to check how much a conclusion depends on the choice of V_0 . Consider replacing the diffuse prior with an informative one that expresses skepticism about large reduced-form inflation and interest-rate pass-through, while leaving the intercept and the autoregressive coefficient loosely specified:

$$V_0^{\text{tight}} = \text{diag}(100, 1, 0.01, 0.0001).$$

Empirical Example: Posterior Estimates: Tight Shrinkage Prior

Under V_0^{tight} , the posterior means shift to $\hat{c} = 3.443$ (0.391), $\hat{\phi} = 0.037$ (0.061), $\hat{\gamma} = -0.156$ (0.065), and $\hat{\delta} = -0.001$ (0.010). The autoregressive coefficient $\hat{\phi}$ is essentially unchanged (the prior on it was left loose), but $\hat{\gamma}$ shrinks toward zero and $\hat{\delta}$ is pulled almost exactly to its prior mean of zero, with a posterior standard deviation an order of magnitude smaller than under the diffuse prior. This is exactly what the conjugate updating formula predicts: with a very tight V_0 entry, the corresponding posterior is dominated by the prior unless the likelihood contains overwhelming evidence to the contrary.

Remark: Why This Matters More When Regressors Are Correlated

Lagged inflation and the lagged Fed funds rate are themselves correlated (the Federal Reserve raises rates partly in response to inflation), so the data alone cannot sharply separate γ from δ — a form of near-multicollinearity that inflates both coefficients' posterior standard deviations under the diffuse prior. When regressors are correlated, the choice of prior can materially affect which of two collinear coefficients absorbs the estimated effect, and reporting results under both a diffuse and a deliberately informative prior (as above) is a simple, transparent way to communicate how much of a conclusion is data-driven versus prior-driven. The formal, model-based alternative — letting the data decide which regressors belong in the model at all — is the subject of Chapter 8.

6.7 Robust Regression with Student- t Errors

The Normal likelihood is convenient but not robust: a small number of extreme observations can dominate the Gaussian sum-of-squares and distort $\hat{\beta}$. Section 4.1 introduced the Student- t distribution as a Normal/Inverse-Gamma scale mixture; the same representation extends

directly to regression errors, giving each observation its own local error variance σ^2/λ_i :

$$e_i \mid \lambda_i \sim N\left(0, \frac{\sigma^2}{\lambda_i}\right), \quad \lambda_i \sim \text{Gamma}\left(\frac{\nu}{2}, \frac{\nu}{2}\right) \implies e_i \sim t_\nu(0, \sigma^2).$$

An observation with a large residual is assigned a small posterior λ_i , inflating its effective error variance and *downweighting* its influence on β — an automatic, model-based form of robust regression, rather than an ad hoc outlier-deletion rule. Figure 19 shows how the three blocks now depend on each other: β and σ^2 depend on the whole vector of weights $\{\lambda_i\}$, while each λ_i depends only on β , σ^2 , and its own residual e_i .

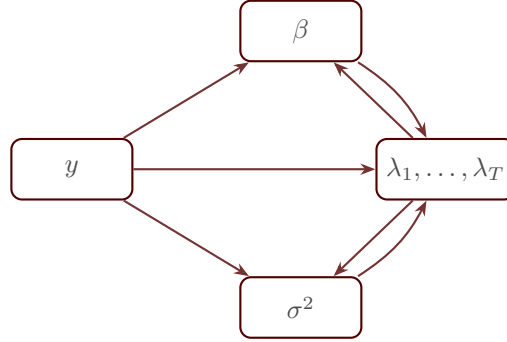


Figure 19: Dependency structure of the Student- t regression's three Gibbs blocks: β and σ^2 pool information across all T weights, while each λ_i looks back only at its own observation

Derivation: $\lambda_i \mid \beta, \sigma^2, y$ (and the β, σ^2 Updates Under Weighting)

Conditional on λ_i , observation i contributes a Normal likelihood term with variance σ^2/λ_i , with $e_i \equiv y_i - x_i'\beta$; combined with the $\text{Gamma}(\nu/2, \nu/2)$ prior (rate parameterization),

$$p(y_i \mid \beta, \sigma^2, \lambda_i) \propto \lambda_i^{1/2} \exp\left\{-\frac{\lambda_i e_i^2}{2\sigma^2}\right\},$$

the $\lambda_i^{1/2}$ factor coming from $(\sigma^2/\lambda_i)^{-1/2} = \sigma^{-1}\lambda_i^{1/2}$ in the Normal normalizing constant, with the σ^{-1} part dropped since it does not involve λ_i , and

$$p(\lambda_i) \propto \lambda_i^{\nu/2-1} \exp\left\{-\frac{\nu}{2}\lambda_i\right\}.$$

Multiplying likelihood by prior and again collecting powers of λ_i separately from the shared linear-in- λ_i exponent:

$$p(\lambda_i \mid \beta, \sigma^2, y) \propto \lambda_i^{\frac{\nu+1}{2}-1} \exp\left\{-\frac{\lambda_i}{2}\left(\nu + \frac{e_i^2}{\sigma^2}\right)\right\},$$

which is the kernel of $\text{Gamma}\left(\frac{\nu+1}{2}, \frac{\nu + e_i^2/\sigma^2}{2}\right)$ (rate parameterization). This is the general scale-mixture-of-normals updating rule from Section 4.1: one extra “pseudo-observation” of information ($\nu/2 \rightarrow (\nu+1)/2$ in the shape) is added by the single data point y_i , and the rate is inflated by the squared standardized residual e_i^2/σ^2 .

Given $\{\lambda_i\}$, the β and σ^2 full conditionals follow from *exactly* the two derivations above, with every occurrence of $X'X$, $X'y$, and $e'e$ replaced by their $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_T)$ -weighted counterparts $X'\Lambda X$, $X'\Lambda y$, and $\sum_i \lambda_i e_i^2 = e'\Lambda e$. This is because the joint likelihood $\prod_i N(y_i; x_i'\beta, \sigma^2/\lambda_i)$ can be written compactly as $N(y; X\beta, \sigma^2\Lambda^{-1})$, and premultiplying by $\Lambda^{1/2}$ turns this into a homoskedastic regression of $\Lambda^{1/2}y$ on $\Lambda^{1/2}X$ with error variance σ^{2a} — the same model as Section 6.2 with $X \rightarrow \Lambda^{1/2}X$, $y \rightarrow \Lambda^{1/2}y$, whose completing-the-square and power-matching arguments carry over unchanged with $X'X \rightarrow X'\Lambda X$ and so on.

^aConcretely: if $e = y - X\beta \sim N(0, \sigma^2\Lambda^{-1})$, define $\tilde{y} = \Lambda^{1/2}y$, $\tilde{X} = \Lambda^{1/2}X$. Then $\tilde{y} - \tilde{X}\beta = \Lambda^{1/2}e \sim N(0, \sigma^2\Lambda^{1/2}\Lambda^{-1}\Lambda^{1/2}) = N(0, \sigma^2 I_T)$, an ordinary homoskedastic regression. This rescale-to-homoskedastic trick is the standard way to handle any *known, diagonal* form of heteroskedasticity without re-deriving the Gibbs steps from scratch, and it reappears in Chapter 16's stochastic-volatility model, where Λ is replaced by the time-varying volatilities $\{\sigma_t^2\}$.

Algorithm: Gibbs Sampler for Student- t Regression

Given current $(\beta, \sigma^2, \{\lambda_i\})$, cycle:

1. Draw $\beta \mid \sigma^2, \{\lambda_i\}, y \sim N(\bar{\beta}, \bar{V})$ with $\bar{V} = (V_0^{-1} + \sigma^{-2}X'\Lambda X)^{-1}$, $\bar{\beta} = \bar{V}(V_0^{-1}b_0 + \sigma^{-2}X'\Lambda y)$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_T)$.
2. Draw $\sigma^2 \mid \beta, \{\lambda_i\}, y \sim IG\left(\frac{a_0 + T}{2}, \frac{d_0 + \sum_i \lambda_i e_i^2}{2}\right)$, $e_i = y_i - x_i'\beta$.
3. For $i = 1, \dots, T$, draw $\lambda_i \mid \beta, \sigma^2, y \sim \text{Gamma}\left(\frac{\nu + 1}{2}, \frac{\nu + e_i^2/\sigma^2}{2}\right)$.

This is the multivariate analogue of the scale-mixture Gibbs sampler used earlier for the Normal-error case; the λ_i full conditional is the standard scale-mixture update, obtained from the same Gamma/Normal composition used in Section 4.1.

Empirical Example: Robust vs. Normal Estimates, $\nu = 5$

Re-estimating the same reduced-form equation with t_5 errors and the diffuse prior gives $\hat{c} = 2.911$ (0.377), $\hat{\phi} = 0.210$ (0.056), $\hat{\gamma} = -0.194$ (0.082), $\hat{\delta} = 0.032$ (0.064). Unlike in the pre-extension sample, where the Student- t and Normal-error estimates were broadly similar, the extended sample shows a substantial divergence: the Normal-error $\hat{\phi}$ falls to 0.040 (Section 6.4) while the Student- t $\hat{\phi}$ remains at 0.210, much closer to the original pre-pandemic estimate of 0.314 — because the Student- t model downweights the two extreme 2020 quarters that pull the Normal-error persistence estimate toward zero. Of 266 quarters, 13 receive a posterior mean weight $\lambda_i < 0.5$ (i.e., an effective error variance more than double the baseline). The two most heavily downweighted quarters, by a wide margin, are the pandemic pair: 2020Q3 ($\bar{\lambda} = 0.02$; GDP growth +34.9%, the reopening rebound) and 2020Q2 ($\bar{\lambda} = 0.04$; -28.0%, the pandemic contraction). The remaining heavily downweighted quarters are the familiar extreme episodes of postwar U.S. business-cycle history — the depth of the 2008–09 financial crisis (2008Q4, 2009Q1), the early-1980s and stagflation-era swings (1980Q2, 1978Q2, 1971Q1), and the pandemic's onset quarter 2020Q1 — all with $\bar{\lambda}$ between roughly 0.17 and 0.36. These are exactly the observations a single linear equation cannot be expected to explain, and the Student- t error model

identifies and downweights them automatically.

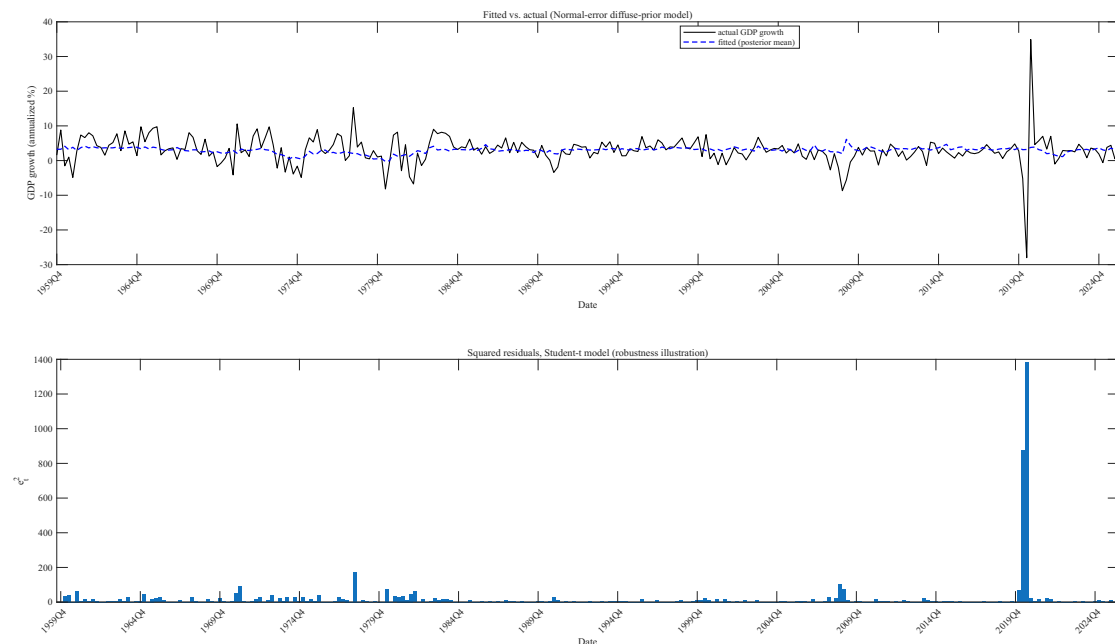


Figure 20: Top: actual vs. fitted GDP growth, Normal-error diffuse-prior posterior mean. Bottom: squared residuals under the Student- t model, the same robustness comparison viewed through the fit rather than through the per-quarter weights listed above

Remark: Robustness Now Changes Point Estimates Substantially — the Pandemic Quarters Are Why

Once the sample includes the pandemic, the Student- t and Normal-error posterior means no longer agree: $\hat{\phi}$ moves from 0.040 under Normal errors to 0.210 under Student- t errors, several posterior standard deviations apart, and the intercept moves from 3.794 to 2.911. This is a striking reversal from the pre-extension sample, where robust and Normal-error regression agreed closely (Normal $\hat{\phi} = 0.314$ vs. Student- t $\hat{\phi} \approx 0.28$) because postwar U.S. GDP growth, while occasionally volatile, contained no observation extreme enough to matter. The 2020Q2 (−28.0%) and 2020Q3 (+34.9%) quarters are different in kind, not just degree: their combined effect on the lag-1 sample second moment is large enough to nearly erase the estimated persistence in the Normal-error model, while the Student- t model's per-observation weights recognize them as anomalous ($\bar{\lambda} \approx 0.02$ – 0.04 , comparable to or more extreme than any postwar business-cycle or financial-crisis quarter) and discount them accordingly, recovering a persistence estimate close to the original pre-pandemic value. This is exactly the diagnostic value robust regression is supposed to provide: agreement between the two models is reassuring when it holds, and disagreement — as now — is itself informative, pointing directly at which observations are driving the discrepancy. The next example provides a complementary, fully controlled check: it injects three synthetic, clearly labeled data errors into ordinary, non-pandemic quarters, to confirm that the same downweighting mechanism operates whenever a handful of observations are anomalous, regardless of source.

Empirical Example: A Contamination Experiment: A Second, Controlled Illustration

To see the same downweighting mechanism at work on a controlled, non-pandemic source of contamination, take the same y used throughout this section and deliberately corrupt three of its 266 quarters — *synthetic values, not real GDP figures, injected purely for this experiment* — while leaving every regressor untouched: 1972Q2's growth rate (genuinely +2.5%) is replaced by +40.0%, 1989Q4's (genuinely +0.8%) by −35.0%, and 2004Q4's (genuinely +3.4%) by +32.0%, mimicking a gross data-entry or transcription error in the outcome variable alone. None of the three affected quarters is a pandemic quarter, so this experiment is independent of the COVID-driven divergence documented in Section 6.6. Re-running both Gibbs samplers on this contaminated series, with the same diffuse prior and t_5 degrees of freedom as above, gives:

	Clean data	Contaminated, Normal	Contaminated, Student- t
$\hat{c}$	3.794 (0.484)	4.108 (0.645)	2.920 (0.389)
$\hat{\phi}$	0.040 (0.061)	0.066 (0.081)	0.208 (0.056)
$\hat{\gamma}$	−0.290 (0.108)	−0.210 (0.146)	−0.199 (0.084)
$\hat{\delta}$	0.025 (0.090)	−0.098 (0.123)	0.036 (0.066)

Three synthetic data errors, out of 266 quarters, are enough to move the Normal-error posterior: $\hat{\gamma}$ is pulled toward zero (−0.290 → −0.210) and its 95% credible interval widens from [−0.502, −0.078] to [−0.492, 0.075], overturning the finding (established in Section 6.4) that lagged inflation is credibly negative; $\hat{\delta}$ is pushed from a value indistinguishable from zero to a modestly negative −0.098, though its credible interval [−0.339, 0.143] still spans zero. The Student- t posterior, by contrast, moves only mildly on $\hat{\gamma}$ and $\hat{\delta}$, and its $\hat{\phi} = 0.208$ lands close to the clean-data Student- t estimate of 0.210 from Section 6.6 — recovering essentially the same underlying persistence whether the extreme observations are the real 2020 pandemic quarters or synthetic data errors. The mechanism is exactly this section's downweighting rule at work: the three injected quarters receive posterior mean weights $\bar{\lambda} = 0.028$ (1972Q2), 0.025 (1989Q4), and 0.044 (2004Q4) — comparable in magnitude to the weights assigned to the genuine 2020Q2/Q3 pandemic quarters (Section 6.6), and well below any ordinary business-cycle quarter — so the sampler treats the synthetic errors and the real pandemic extremes with the same automatic skepticism, while the Normal likelihood has no such mechanism and is forced to accommodate every point, real or corrupted, through the sum of squares.

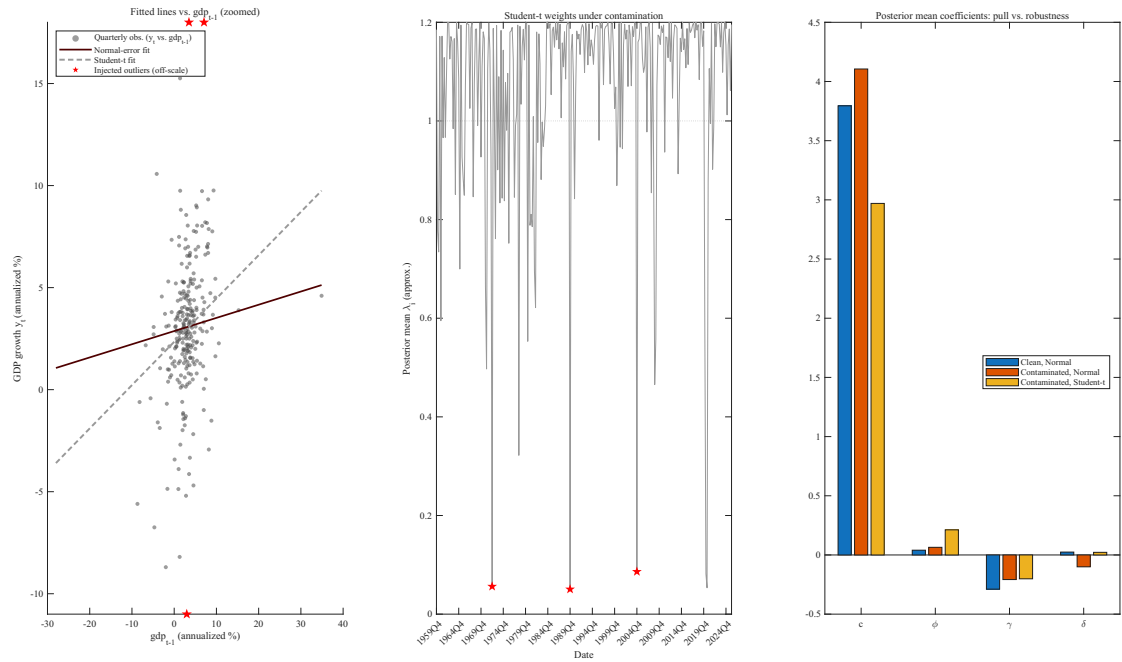


Figure 21: Left: a scatter of y_t against its own lag gdp_{t-1} (the other two regressors held at their sample means), with the Normal- and Student- t -posterior mean fits overlaid as fitted lines; the vertical axis is zoomed to the range spanned by the 217 genuine quarters, and the 3 injected outliers — far outside this range — are marked with red stars and labelled at the frame edge. Plotted this way, rather than as curves traced across quarter index, the two fits are visibly different lines rather than an indistinguishable overlay. Middle: Student- t posterior mean weights λ_i , with the 3 injected outliers falling far below any genuinely downweighted quarter. Right: posterior mean coefficients under clean data, contaminated data with Normal errors, and contaminated data with Student- t errors, showing the Normal-error pull and the Student- t model's resistance to it

Remark: Two Experiments, One Mechanism

Both comparisons above are driven by the same underlying mechanism, illustrated from two independent angles. Section 6.6 shows that with the extended sample, real data alone is enough to make the Normal- and Student- t -error posteriors disagree substantially, because the 2020 pandemic quarters are genuinely extreme relative to seven decades of prior GDP growth. The contamination experiment in this section shows the identical mechanism operating on three synthetic, clearly labeled, non-pandemic data errors: the Normal likelihood, which has no way to distinguish a real extreme observation from a data error, lets a small number of anomalous points shift point estimates and can even overturn a credible-interval conclusion (as with $\hat{\gamma}$ here), while the Student- t model's per-observation weights λ_i automatically identify and discount whatever combination of observations the posterior finds anomalous — pandemic quarters, financial-crisis quarters, stagflation-era quarters, or fabricated transcription errors alike. Together the two exercises make the case for building the scale-mixture representation into the regression model up front, rather than relying on eyeballing residual plots after the fact or hoping that a single historical sample happens to be free of extreme events.

7. The Probit Model

Roadmap

Every conjugate result in Chapter 6 relied on y itself being continuous and Normal. A binary outcome breaks that immediately — there is no Normal-Inverse-Gamma conjugate prior for a 0/1 variable. Rather than abandon Chapter 6’s machinery, this chapter shows how to restore it: invent a latent continuous variable z_i that y_i is a thresholded version of, and let the Gibbs sampler alternate between imputing z_i and running Chapter 6’s own regression step on the imputed data. This data-augmentation trick, not the probit link function itself, is the lasting idea: Chapter 9’s Tobit model and Chapter 18’s regime-switching model both introduce a latent variable for exactly the same reason, to turn an awkward likelihood back into familiar conjugate machinery.

The multiple regression model of Chapter 6 requires a continuous outcome y . Many economically interesting outcomes are binary instead — did a recession occur, did a firm default, did a household choose to participate in the labor force. This chapter develops the Bayesian probit model for such outcomes via Albert and Chib’s (1993) data-augmentation Gibbs sampler, which turns a binary-outcome model back into the familiar conjugate linear-regression machinery of Chapter 6 by introducing a latent continuous variable.

7.1 The Latent-Variable Probit Model

Suppose $y_i \in \{0, 1\}$ is generated by thresholding an unobserved (*latent*) linear regression:

$$z_i = x_i' \beta + e_i, \quad e_i \sim N(0, 1), \quad y_i = \mathbb{I}\{z_i > 0\}.$$

Marginalizing over e_i gives $P(y_i = 1 \mid x_i) = P(e_i > -x_i' \beta) = \Phi(x_i' \beta)$, the standard probit link. If z_i were observed, (z, X) would be an ordinary Gaussian linear regression with known error variance 1, and the conjugate Gibbs sampler of Chapter 6 would apply directly. Albert and Chib’s insight is to treat z_i itself as a missing value to be imputed within the Gibbs sampler: given β , each z_i has a Normal distribution truncated to $(0, \infty)$ if $y_i = 1$ and to $(-\infty, 0)$ if $y_i = 0$; given z , β has the same conjugate Normal full conditional as in Chapter 6 with $\sigma^2 \equiv 1$.

7.2 Posterior Derivation

The augmented parameter vector is (β, z) with $z = (z_1, \dots, z_T)'$. Because y_i is a deterministic function of z_i (namely $y_i = \mathbb{I}\{z_i > 0\}$), the joint posterior kernel of (β, z) given the observed data y factors very simply:

$$p(\beta, z \mid y) \propto p(y \mid z) p(z \mid \beta) p(\beta) = \left[\prod_{i=1}^T \mathbb{I}\{\text{sign}(z_i) = \text{sign}(y_i - \tfrac{1}{2})\} \right] p(z \mid \beta) p(\beta),$$

where $p(y \mid z)$ is either 1 (if every z_i has the sign consistent with the observed y_i) or 0 (otherwise), and $p(z \mid \beta) = \prod_i N(z_i; x_i' \beta, 1)$ is the latent-regression likelihood. This factorization is the key to the whole data-augmentation trick: once z is included as a parameter, the model becomes a completely standard Gaussian linear regression with a 0/1 indicator restricting the

support of z , and both full conditionals below follow immediately.

Derivation: $z_i \mid y_i, \beta$

Fix β and consider a single i . As a function of z_i alone, the joint kernel above is

$$p(z_i \mid y_i, \beta) \propto \mathbb{I}\{\text{sign}(z_i) = \text{sign}(y_i - \frac{1}{2})\} \times \exp\{-\frac{1}{2}(z_i - x'_i\beta)^2\}$$

i.e., a $N(x'_i\beta, 1)$ kernel multiplied by an indicator restricting z_i to $(0, \infty)$ when $y_i = 1$ and to $(-\infty, 0)$ when $y_i = 0$. Restricting a density's support to a subinterval and renormalizing is, by definition, truncation: the z_i 's that satisfy the indicator retain exactly their original (unnormalized) Normal density shape, so:

$$z_i \mid y_i, \beta \sim \begin{cases} N(x'_i\beta, 1) \text{ truncated to } (0, \infty) & y_i = 1 \\ N(x'_i\beta, 1) \text{ truncated to } (-\infty, 0) & y_i = 0. \end{cases}$$

Because the z_i 's appear multiplicatively and separably in the joint kernel (there is no term linking z_i and z_j for $i \neq j$ once β is fixed), this argument applies to each i independently, giving the T conditionally independent draws used by the algorithm. Sampling from a truncated Normal is done by the inverse-CDF (probability integral transform) method of Section 4.2: draw $u_i \sim U(0, 1)$, and set $z_i = x'_i\beta + \Phi^{-1}(\Phi(a_i) + u_i[\Phi(b_i) - \Phi(a_i)])$ for the appropriate truncation bounds $(a_i, b_i) = (-x'_i\beta, \infty)$ or $(-\infty, -x'_i\beta)$.

Derivation: $\beta \mid z$

Now fix z (which, given β , no longer depends on y except through the truncation region already used to draw it) and treat β as unknown. The relevant kernel is

$$p(\beta \mid z) \propto p(z \mid \beta) p(\beta) \propto \exp\{-\frac{1}{2}(z - X\beta)'(z - X\beta)\} \exp\{-\frac{1}{2}(\beta - b_0)'B_0^{-1}(\beta - b_0)\}$$

This is *identical* in form to the $\beta \mid \sigma^2, y$ derivation of Chapter 6.2 — z plays the role of the dependent variable and the error variance is fixed at $\sigma^2 = 1$ rather than sampled. Substituting $\sigma^2 = 1$ and $y \rightarrow z$ into that result immediately gives:

$$\beta \mid z \sim N(\bar{\beta}, \bar{B}), \quad \bar{B} = (B_0^{-1} + X'X)^{-1}, \quad \bar{\beta} = \bar{B}(B_0^{-1}b_0 + X'z)$$

with no need to repeat the completing-the-square algebra: the probit β step is, term for term, the Chapter 6 conjugate regression step applied to the imputed continuous outcome z in place of y .

7.3 Data Augmentation (Albert-Chib) Gibbs Sampler

Figure 22 summarizes the two-block cycle: β depends on the whole imputed vector z , and each z_i depends on β and its own observed label y_i only.

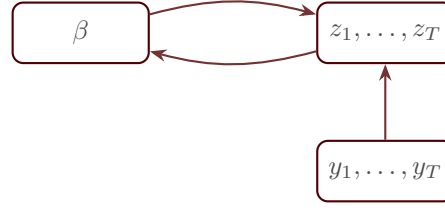


Figure 22: The Albert-Chib cycle: β and the latent vector z update each other, while z additionally respects the observed labels y through its truncation region

Algorithm: Bayesian Probit via Data Augmentation

Given a conjugate prior $\beta \sim N(b_0, B_0)$ and a current value of β , cycle:

1. For each i , draw the latent utility

$$z_i \mid y_i, \beta \sim \begin{cases} N(x_i' \beta, 1) \text{ truncated to } (0, \infty) & y_i = 1 \\ N(x_i' \beta, 1) \text{ truncated to } (-\infty, 0) & y_i = 0, \end{cases}$$

using the inverse-CDF (probability integral transform) method of Section 4.2 applied to the truncated Normal.

2. Draw $\beta \mid z \sim N(\bar{\beta}, \bar{B})$ with $\bar{B} = (B_0^{-1} + X'X)^{-1}$, $\bar{\beta} = \bar{B}(B_0^{-1}b_0 + X'z)$ — identical in form to the conjugate regression step of Chapter 6.2 with $\sigma^2 = 1$.

Remark: Why Data Augmentation, Rather Than M-H Directly on β ?

One could instead run Metropolis-Hastings directly on β using the probit likelihood $\prod_i \Phi(x_i' \beta)^{y_i} (1 - \Phi(x_i' \beta))^{1-y_i}$, as in Section 4.5. Data augmentation is usually preferred because it restores full conjugacy — every step of the resulting sampler is an exact draw from a known distribution, with no proposal to tune and no rejected draws, exactly the efficiency argument made for Gibbs sampling in Section 4.7.

7.4 Empirical Example: Predicting U.S. Recession Quarters

As an application, consider predicting whether U.S. real GDP contracts ($y_t = \mathbb{1}\{\text{GDP growth}_t < 0\}$) using only information available at the end of the previous quarter: the lagged Fed funds rate level, the lagged *change* in the unemployment rate, and lagged CPI inflation (the same quarterly series used in Chapter 6):

$$P(y_t = 1) = \Phi(c + \beta_1 i_{t-1} + \beta_2 \Delta u_{t-1} + \beta_3 \pi_{t-1}).$$

Of the $T = 265$ usable quarters (1960Q1–2026Q1, extended from the original 1960Q1–2013Q4 window used before this note's data extension), 31 (11.7%) are contraction quarters — a moderately imbalanced binary outcome, which will matter below.

Empirical Example: Posterior Estimates and Comparison with the MLE

With a diffuse prior ($\beta \sim N(0, 4I_4)$) and 10,000 post-burn-in Gibbs draws, the posterior means (s.d., 95% credible interval) are:

$$\hat{c} = -1.746 \text{ (0.187)} \quad [-2.125, -1.389]$$

$$\hat{\beta}_1 = 0.037 \text{ (0.035)} \quad [-0.031, 0.105]$$

$$\hat{\beta}_2 = 0.279 \text{ (0.124)} \quad [0.025, 0.511]$$

$$\hat{\beta}_3 = 0.079 \text{ (0.039)} \quad [0.001, 0.157]$$

These agree with the classical MLE probit fit (`statsmodels`) to two or three significant digits throughout ($\hat{c} = -1.742$, $\hat{\beta}_1 = 0.036$, $\hat{\beta}_2 = 0.292$, $\hat{\beta}_3 = 0.080$), as expected under a diffuse prior. The inefficiency factors (3.15–5.63) are higher than the near-1 values of the conjugate Gibbs sampler in Chapter 6, because the data-augmentation step introduces extra dependence between β and the imputed z_i 's from one iteration to the next — still well within the “acceptable” range of Section 5.2's rule of thumb. Both the lagged change in unemployment, Δu_{t-1} , and lagged inflation now have credible intervals that exclude zero, though only barely: $\hat{\beta}_2$'s interval $[0.025, 0.511]$ and $\hat{\beta}_3$'s interval $[0.001, 0.157]$ both sit close to zero at the lower end, a marked weakening from the pre-extension estimates ($\hat{\beta}_2 = 1.456$, $[0.826, 2.078]$, with $\hat{\beta}_3$ not credibly different from zero at all). This reflects the same mechanism documented in Chapters 4–6: the large, short-lived 2020 swings in Δu_{t-1} (a rise of roughly +10pp in 2020Q2 followed by a fall of about –4pp by 2020Q4) do not line up cleanly with the quarter-to-quarter contraction indicator, weakening the linear relationship that was much sharper on the pre-pandemic sample. The lagged Fed funds rate remains without a credible effect at this horizon.

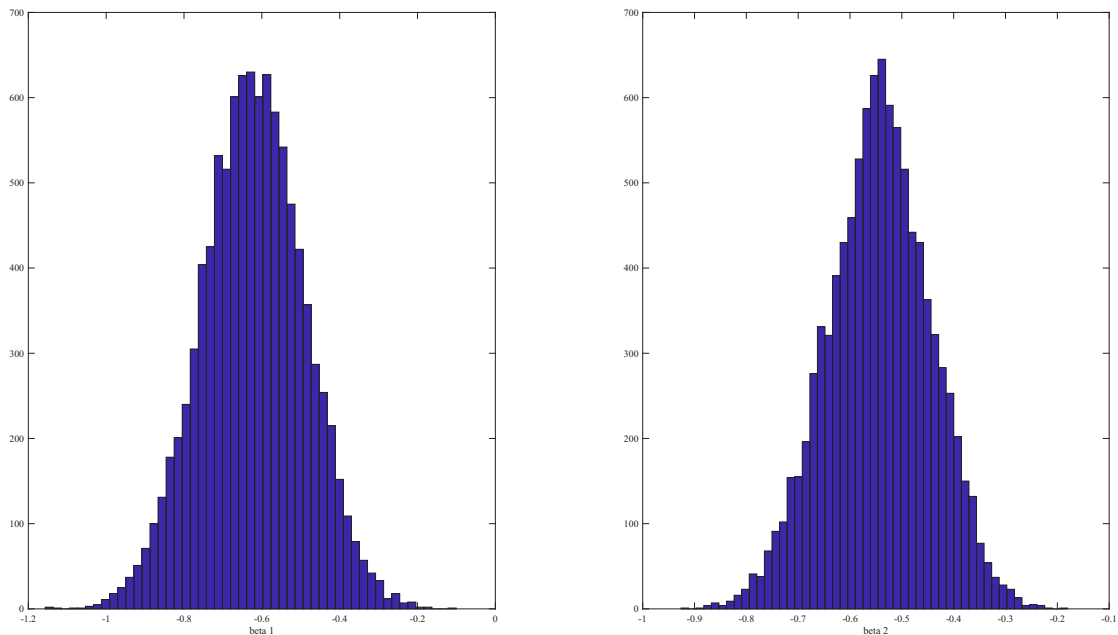


Figure 23: Posterior densities of the three probit slope coefficients, with the MLE estimate marked

Empirical Example: Marginal Effects and In-Sample Fit

Because Φ is nonlinear, β_j itself is not directly interpretable as a probability change; the average marginal effect $\overline{\phi(x'_i \beta)} \beta_j$ (averaging the Normal density ϕ over the sample) converts it to probability units. At the posterior mean, a one-percentage-point rise in Δu_{t-1} raises the average predicted probability of a contraction by 5.0 percentage points — down sharply from 22.4 points on the pre-extension sample — versus 0.7 and 1.4 points for the Fed funds rate and inflation, respectively. McFadden's pseudo- R^2 , $1 - \ell(\hat{\beta})/\ell(0)$, falls to 0.097 (from 0.228 pre-extension), a much weaker fit for this single-lag reduced form once the pandemic quarters are included. Classifying $\hat{y}_t = 1$ whenever $\hat{P}(y_t = 1) > 0.5$ gives an overall accuracy of 88.7% (confusion matrix: 2 true positives, 1 false positive, 29 false negatives, 233 true negatives) — far fewer true positives than before extension (5 of 26), because the fitted probability now crosses 0.5 only three times in the entire sample (the remarkbox at the end of this section explains why).

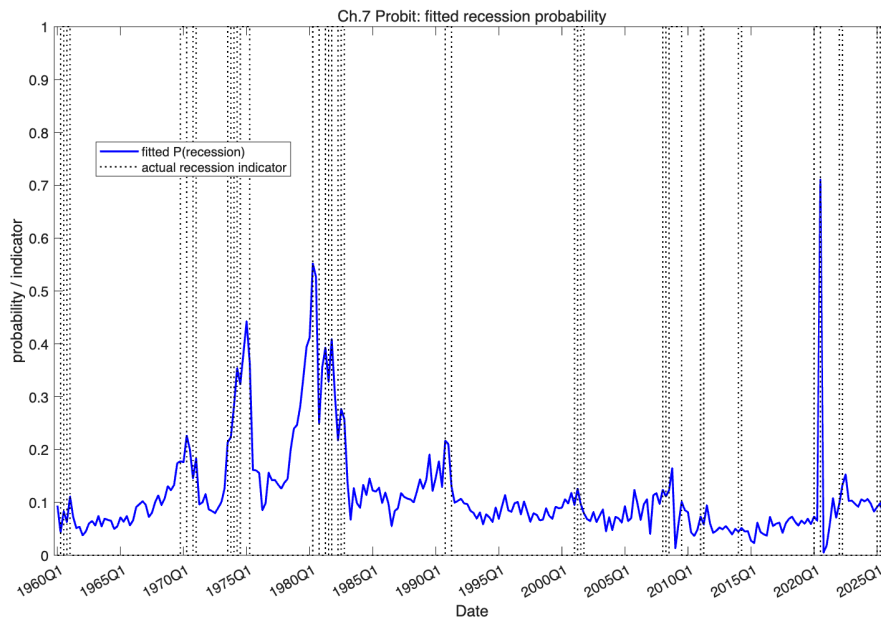


Figure 24: Posterior mean predicted contraction probability vs. actual contraction quarters (shaded)

Empirical Example: Why Not Just Run OLS? The Linear Probability Model Compared

Nothing stops us from ignoring the latent-variable machinery of this chapter entirely and simply regressing the raw 0/1 outcome y_t on X_t by OLS — the *linear probability model* (LPM). Fit to exactly the same $T = 265$ quarters and regressors used above, OLS gives

$$\hat{y}_t = -0.0004 + 0.0076 i_{t-1} + 0.0679 \Delta u_{t-1} + 0.0223 \pi_{t-1}, \quad R^2 = 0.089,$$

with a fitted "probability" that ranges from -0.183 to 0.535 across the sample — 11 of the 265 quarters (4.2%) have an LPM fitted value below zero, down from 14.4% pre-extension because the flatter estimated slope on Δu_{t-1} (itself weakened by the 2020 swings, as in

Section 7.4 above) produces less extreme fitted values overall; ten of the eleven follow a fall in unemployment ($\Delta u_{t-1} < 0$), the exception being 2009Q1 ($\Delta u_{t-1} = +0.9$), whose negative fitted value comes from the combination of all three regressors rather than the unemployment term alone. The single worst case, 2020Q4, gets $\hat{y} = -0.183$: its $\Delta u_{t-1} = -4.2$ — unemployment plunging as the initial pandemic spike unwind — is far outside anything the pre-pandemic sample ever produced, and a linear model has no way to keep such an extreme input from producing an out-of-range output. As before extension, one of the out-of-range quarters is a genuine contraction: 2009Q1 ($y_t = 1$) — so the LPM does not merely produce an awkward number in an unimportant corner of the sample, it assigns a nonsensical *negative* chance to an outcome that went on to occur. The Bayesian probit fit above, by contrast, never leaves $[0.005, 0.692]$ over the same sample, by construction: $\Phi(\cdot)$ cannot return anything outside $[0, 1]$ no matter how extreme $x_i'\beta$ becomes. Figure 25 shows why, plotting both fitted curves as a function of Δu_{t-1} (the one regressor with a credible effect), holding the other two regressors at their sample means: the LPM is forced to extrapolate a straight line through the data, which is a good local approximation near $\Delta u_{t-1} \approx 0$ (the two curves are nearly indistinguishable there, and LPM's overall classification accuracy, 87.9%, comes close to — though no longer exactly matches — probit's 88.7%) but inevitably overshoots past 0 or 1 once Δu_{t-1} moves far enough from the center of the data — exactly the tail behavior a recession-probability model needs to get right, and behavior the 2020 pandemic quarters now test more severely than any quarter in the pre-extension sample. This is the real, verified version of an intuition every student should have before trusting a nonlinear model: probit is not preferred merely because it is traditional, it is preferred because a probability, by definition, must respect $[0, 1]$, and only a model built from a bounded link function such as $\Phi(\cdot)$ can guarantee that in every sample, not just the typical one.

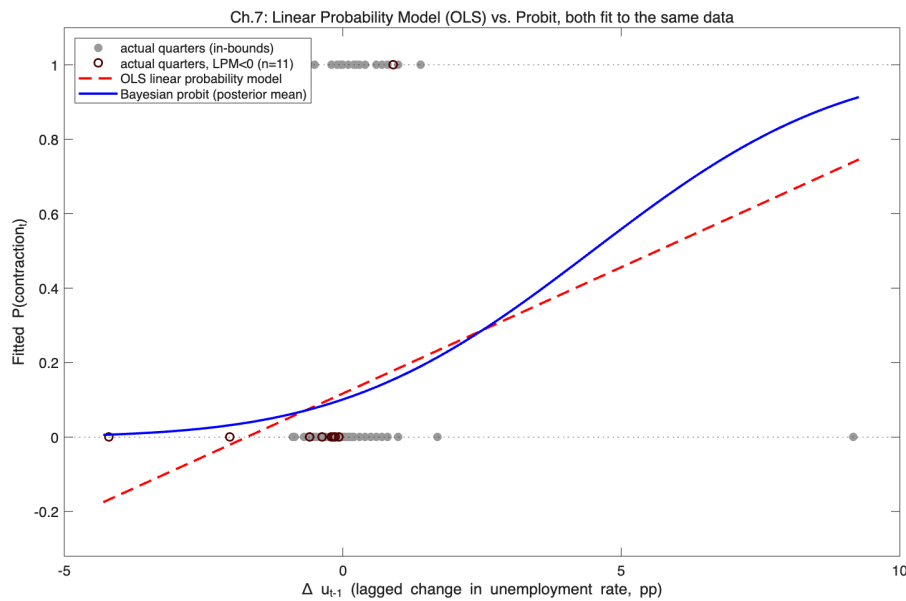


Figure 25: OLS linear probability model vs. Bayesian probit (posterior mean), as a function of the lagged change in unemployment rate, holding the other two regressors at their sample means. Actual quarters where the LPM's own fitted probability falls below zero are circled

Remark: 88.7% Accuracy Is Even Less Impressive Than It Sounds

Because only 11.7% of quarters are contractions, a trivial model that always predicts “no recession” would already be 88.3% accurate — so the fitted model’s 88.7% overall accuracy reflects essentially no genuine predictive power once the class imbalance is accounted for, an even thinner margin than the pre-extension sample’s 89.4% against an 88.0% baseline. The confusion matrix makes this precise: recall (the fraction of *actual* contractions correctly flagged) has fallen to just $2/31 = 6.5\%$ (down from 19.2% pre-extension), even though precision (the fraction of *flagged* quarters that were genuinely contractions) is still a reasonable $2/3 = 66.7\%$ and specificity is $233/234 = 99.6\%$. Figure 24 shows why: the fitted probability now crosses the conventional 0.5 threshold only *three times* in the entire 265-quarter sample — 1980Q2 and 1980Q3 (correctly, the early-1980s double-dip) and 2020Q3 (incorrectly: a false positive, because Δu_{t-1} for 2020Q3 reflects 2020Q2’s huge unemployment spike unwinding, which pushes the fitted probability to 0.69 even though 2020Q3 was itself a sharp reopening boom, not a contraction). Every other historical recession — the 1973–75 oil-shock recession (peak fitted probability 0.44, 1975Q1), the entire 2008–09 financial crisis (peak 0.16, 2008Q4), and even 2020Q1 and 2020Q2’s own pandemic contraction quarters (fitted probabilities of just 0.07 and 0.06) — now falls short of the 0.5 cutoff entirely. This sharpens, rather than changes, the lesson of the pre-extension sample: in an application where missing a recession is far costlier than a false alarm, a lower classification threshold (accepting more false positives to catch more true recessions) or a full decision-theoretic loss function would be far more appropriate than the default 0.5 rule, and the pandemic’s addition to the sample has made this pitfall

more consequential, not less.

7.5 Estrella and Mishkin (1998)

Just as Chapter 6 rebuilt Rudebusch and Svensson’s (1999) IS curve from its own original data, this section rebuilds the single most cited result from Estrella and Mishkin (1998), “Predicting U.S. Recessions: Financial Variables as Leading Indicators,” *Review of Economics and Statistics*. Rather than the four-variable reduced form of Section 7.4 above, the paper’s headline specification uses a single regressor, the term spread between the 10-year Treasury bond and the 3-month Treasury bill (SPREAD_t), to forecast the NBER recession indicator k quarters ahead:

$$P(R_{t+k} = 1) = \Phi(a_0 + a_1 \text{SPREAD}_t), \quad k = 1, \dots, 8.$$

Estrella and Mishkin (hereafter EM) estimate this equation separately for each horizon k by maximum likelihood on 1959Q1–1995Q1 data and report the resulting pseudo- R^2 (Estrella, 1995) as their headline evidence that the yield-curve spread is the best simple leading indicator of recessions beyond a one-quarter horizon.¹

Empirical Example: Original Sample, 1959Q1–1995Q1

Reconstructing SPREAD from FRED series GS10 (10-year Treasury) and TB3MS (3-month T-bill), quarterly-averaged, and the NBER recession indicator from FRED USREC, the same eight horizon-specific probit equations estimated on 1959Q1–1995Q1 give:

k	1	2	3	4	5	6	7	8
EM(1998)	0.071	0.211	0.271	0.296	0.256	0.149	0.078	0.031
Replication	0.110	0.220	0.271	0.235	0.202	0.133	0.069	0.034

The hump-shaped horizon profile — rising from $k = 1$, peaking around $k = 3$ – 4 , and decaying by $k = 8$ — is closely reproduced, and the coefficient on SPREAD is negative and credibly different from zero at every horizon (a steeper, more positive term spread predicts a *lower* recession probability), exactly as EM(1998) report. Five of the eight horizons match their pseudo- R^2 to within 0.02; the largest gaps are at $k = 1$ (replication higher, 0.110 vs. 0.071) and $k = 4$ (replication lower, 0.235 vs. 0.296).

¹EM(1998) estimate this probit by maximum likelihood; our replication instead uses the Albert–Chib Gibbs sampler of Section 7.3 under a diffuse prior ($a \sim N(0, 4I_2)$). The posterior mean and the MLE need not coincide exactly, but the horizon-by-horizon comparison below reproduces the same hump-shaped profile, with only modest numerical differences attributable to the estimation method and data vintage rather than any structural mismatch.

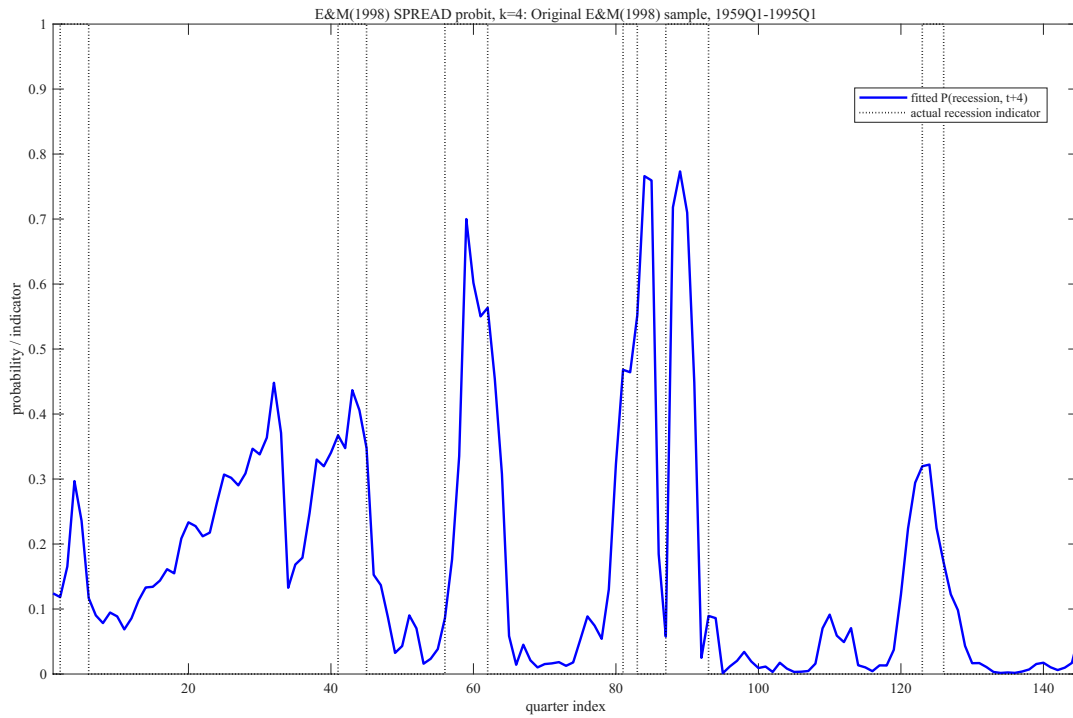


Figure 26: Fitted $P(R_{t+4} = 1)$ vs. the actual recession indicator, SPREAD-only probit, original sample

Empirical Example: Full Sample, 1959Q1–2026Q1

Re-estimating the identical eight horizon-specific equations through 2026Q1 — adding the dot-com, financial-crisis, and pandemic recessions to EM(1998)’s original window — gives:

k	1	2	3	4	5	6	7	8
EM(1998)	0.071	0.211	0.271	0.296	0.256	0.149	0.078	0.031
1959Q1–2026Q1	0.027	0.081	0.133	0.140	0.136	0.104	0.066	0.046

The hump shape survives, but its magnitude declines by roughly half at every horizon through $k = 6$ (e.g., $k = 3$: $0.271 \rightarrow 0.133$; $k = 4$: $0.296 \rightarrow 0.140$), and the peak flattens out across $k = 3$ – 5 rather than sharply peaking at $k = 4$. The sign and significance of a_1 are unchanged, so the term spread remains informative throughout the full sample — just noticeably less so than in EM(1998)’s own 36-year window.

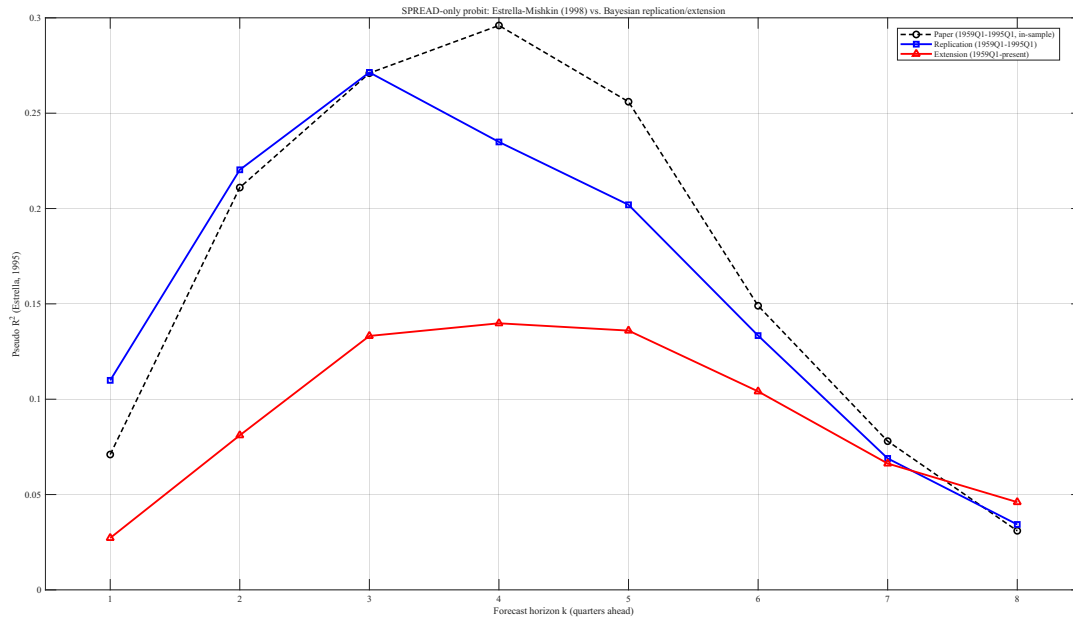


Figure 27: Pseudo- R^2 by forecast horizon: Estrella and Mishkin's (1998) published values vs. this chapter's Bayesian estimates on the original and full samples

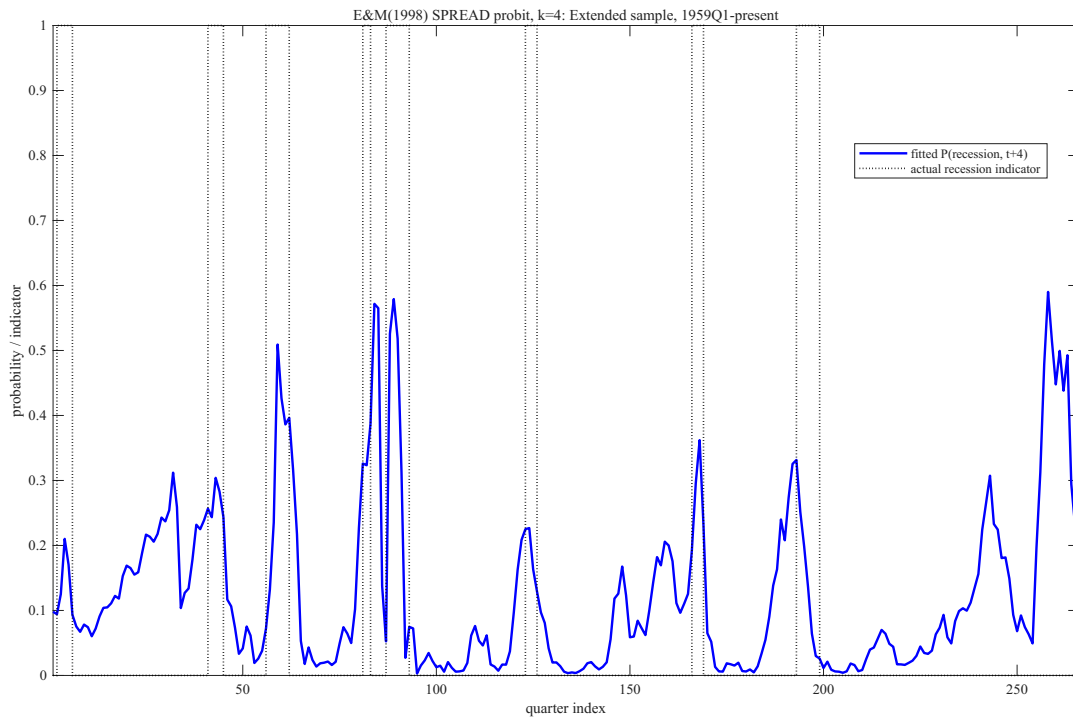


Figure 28: Fitted $P(R_{t+4} = 1)$ vs. the actual recession indicator, SPREAD-only probit, full sample (1959Q1–2026Q1)

The weakening shows up not only in fit but in the coefficient itself. Figure 29 overlays, at every horizon, the posterior density of the `SPREAD` slope a_1 from the two samples. At $k = 1$ – 5 the full-sample posterior sits visibly closer to zero than the original-sample one — the extension weakens the coefficient exactly where the pseudo- R^2 falls most — while at $k = 6$ – 7 the two posteriors largely coincide, and at $k = 8$ the full-sample posterior is

in fact slightly *farther* from zero. Both posteriors exclude zero at every horizon: the extra recessions blur the spread's signal, but never erase it.

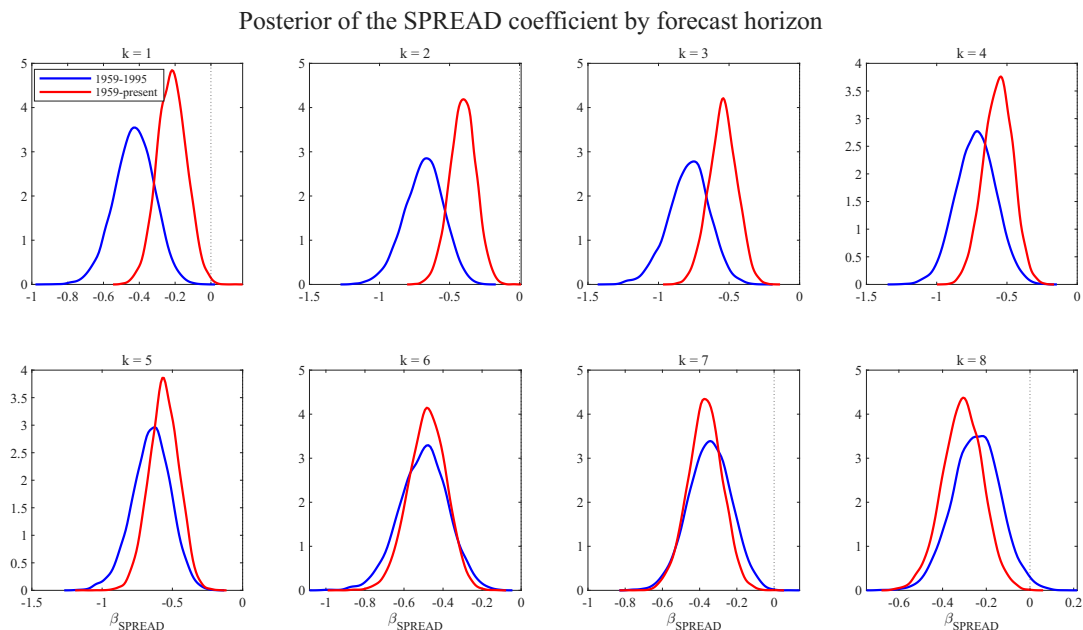


Figure 29: Posterior density of the `SPREAD` coefficient a_1 at each horizon k : original sample 1959Q1–1995Q1 (blue) vs. full sample 1959Q1–2026Q1 (red); dotted vertical line at zero

Remark: Why the Term Spread's Forecasting Power Has Weakened

Two of the three post-1995 recessions in the full sample are consistent with EM(1998)'s mechanism: the 2001 dot-com recession and the 2008–09 financial crisis were both preceded by a flattening or inverted yield curve, just as the 1959–1995 recessions were. The third, the 2020 pandemic recession, was not: it was triggered by an exogenous public-health shock and a sudden nationwide shutdown rather than a monetary tightening cycle. (The curve did invert briefly in mid-2019, so the fitted probabilities were somewhat elevated entering 2020 — but by coincidence, not foresight: nothing in the 2019 term structure contained information about a pandemic, and the fitted probabilities never approached the levels seen before the 1980–82 or 2008–09 recessions.) Because pseudo- R^2 is calculated over the whole sample, this one recession lowers the average fit at every horizon, in the same way the 2020 quarters reduced the precision of the coefficient estimates in Section 7.4's four-variable model above. A second, complementary factor is the same one identified for Chapter 6's IS curve: the full sample spans two long zero-lower-bound episodes (2009–2015 and 2020–2022) during which the short end of the yield curve was pinned near zero by policy, compressing the variation in `SPREAD` available to distinguish tightening cycles from ordinary interest-rate movements. Both explanations point to the same conclusion EM(1998) themselves anticipated: a simple reduced-form indicator estimated on one historical era should be expected to lose some, but not all, of its power once genuinely different kinds of recessions enter the sample.

8. Bayesian Variable Selection

Roadmap

Chapters 6 and 7 both assumed the analyst already knew which regressors belonged in the model. That assumption is rarely true in practice, and Chapter 6.4 already showed a symptom of getting it wrong: correlated regressors whose individual coefficients become hard to pin down. This chapter removes the assumption by turning “which regressors belong” into one more thing the Gibbs sampler estimates, through a binary indicator γ_j attached to each coefficient. Everything needed to do this is already built: the coefficient step reuses Chapter 6.2’s conjugate Normal result verbatim, and the new indicator and variance steps reuse Chapter 6.2’s Inverse-Gamma result and simple Beta-Bernoulli updating. The genuinely new idea is the six-way hierarchical dependency this creates, visualized below before the algebra.

Chapters 6 and 7 both fixed the set of regressors in advance. In practice the analyst rarely knows which of many candidate predictors genuinely belong in a model — and, as Chapter 6.4 previewed with correlated ϕ and γ coefficients, an oversized regression can leave the analyst unable to tell which included variable is doing the work. This chapter develops George and McCulloch’s (1993) Stochastic Search Variable Selection (SSVS), which recasts variable selection as a Bayesian estimation problem: augment the regression with a binary indicator $\gamma_j \in \{0, 1\}$ for each coefficient, and let the Gibbs sampler learn simultaneously which coefficients are essentially zero and what the surviving coefficients equal.

8.1 A Hierarchical Spike-and-Slab Prior

Consider the linear regression $y = X\beta + e$, $e \sim N(0, \sigma^2 I)$, with k candidate regressors. SSVS places on each slope coefficient β_j a two-component mixture prior indexed by γ_j :

$$\beta_j \mid \gamma_j \sim \begin{cases} N(0, \tau_0^2), & \gamma_j = 0 \quad (\text{“spike”}: \beta_j \text{ negligible}) \\ N(0, \tau_1^2), & \gamma_j = 1 \quad (\text{“slab”}: \beta_j \text{ unrestricted}) \end{cases}$$

with $\tau_0^2 \ll \tau_1^2$, and $\gamma_j \sim \text{Bernoulli}(p)$ independently across j . When $\gamma_j = 0$, the tight spike variance τ_0^2 shrinks β_j almost to zero without literally removing the regressor from the model — this is what makes the resulting full conditional distributions conjugate and the sampler exact, unlike methods that require actually dropping and re-adding columns of X . Rather than fix τ_0^2 , τ_1^2 , and p at arbitrary values, the full hierarchical version used here treats all three as unknowns with their own conjugate hyperpriors: τ_0^2 and τ_1^2 each get an Inverse-Gamma hyperprior (centered at a small and a large value, respectively), and p gets a Beta(a_0, c_0) hyperprior. The candidate regressors are standardized (mean zero, unit variance) before applying this prior, so that a single pair of spike/slab hyperparameters is comparable across predictors measured in very different units.

8.2 Posterior Derivation

The full joint posterior kernel of $(\beta, \gamma, \sigma^2, \tau_0^2, \tau_1^2, p)$ is the product of the Normal likelihood, the two-component mixture prior on β (given γ), the independent Bernoulli(p) priors on the γ_j 's, the Inverse-Gamma hyperpriors on τ_0^2, τ_1^2 , the Beta hyperprior on p , and the Inverse-Gamma prior on σ^2 . Each full conditional below is obtained by holding everything else fixed and reading off the factors of this product that still depend on the block in question — exactly the same mechanical strategy used in every derivation so far.

Derivation: $\beta \mid \gamma, \sigma^2, y$

Given γ , the prior on β is simply $\beta \sim N(0, D_\gamma)$ with $D_\gamma = \text{diag}(\tau_{\gamma_1}^2, \dots, \tau_{\gamma_k}^2)$ — a Normal prior with mean zero and a (block-diagonal, in fact fully diagonal) covariance matrix, no different in kind from the general $N(b_0, V_0)$ prior of Chapter 6.2. The Chapter 6.2 completing-the-square kernel used no property of V_0 beyond it being fixed, invertible, and symmetric, so it applies here unchanged with $V_0 \rightarrow D_\gamma$ and $b_0 \rightarrow 0$ substituted directly into it:

$$\begin{aligned} \exp \left\{ -\frac{1}{2} [\beta' (V_0^{-1} + \sigma^{-2} X' X) \beta \right. \\ \left. - 2\beta' (V_0^{-1} b_0 + \sigma^{-2} X' y)] \right\} \xrightarrow[b_0 \rightarrow 0]{V_0 \rightarrow D_\gamma} \\ \exp \left\{ -\frac{1}{2} [\beta' (D_\gamma^{-1} + \sigma^{-2} X' X) \beta - 2\beta' \sigma^{-2} X' y] \right\} \end{aligned}$$

since $V_0^{-1} b_0 = D_\gamma^{-1} \cdot 0 = 0$ drops out of the linear term. Reading off $\bar{V} \equiv (V_0^{-1} + \sigma^{-2} X' X)^{-1}$ and $\bar{\beta} \equiv \bar{V} (V_0^{-1} b_0 + \sigma^{-2} X' y)$ from Chapter 6.2 under this same substitution gives

$$\begin{aligned} \bar{V}_\gamma &= (D_\gamma^{-1} + \sigma^{-2} X' X)^{-1}, \\ \bar{\beta}_\gamma &= \bar{V}_\gamma (D_\gamma^{-1} \cdot 0 + \sigma^{-2} X' y) = \bar{V}_\gamma \sigma^{-2} X' y, \end{aligned}$$

so that

$$\beta \mid \gamma, \sigma^2, y \sim N(\bar{\beta}_\gamma, \bar{V}_\gamma).$$

This is the precise mechanism by which SSVS shrinks: a coefficient currently in the spike ($\gamma_j = 0$) has τ_0^2 tiny, so D_γ^{-1} has a huge entry in position j , which dominates $\sigma^{-2} X' X$ in $\bar{V}_\gamma^{-1}$ and pulls $\bar{\beta}_{\gamma,j}$ toward its prior mean of zero regardless of what the data alone would say.

Derivation: $\gamma_j \mid \beta_j, \tau_0^2, \tau_1^2, p$

Because γ_j is discrete with two possible values, its full conditional is found by direct application of Bayes' rule (no completing-the-square is needed): the only factors of the joint kernel involving γ_j are the prior density of β_j *given* γ_j and the Bernoulli prior on γ_j itself,

$$P(\gamma_j = g \mid \beta_j, \tau_0^2, \tau_1^2, p) \propto p(\beta_j \mid \gamma_j = g) P(\gamma_j = g)$$

which, writing out both factors explicitly, is

$$= \phi(\beta_j; 0, \tau_g^2) \times p^g (1-p)^{1-g}, \quad g \in \{0, 1\}$$

Evaluating at $g = 1$ and $g = 0$ gives two numbers, $p \cdot \phi(\beta_j; 0, \tau_1^2)$ and $(1-p) \cdot \phi(\beta_j; 0, \tau_0^2)$; normalizing so the two probabilities sum to one (dividing each by their sum) gives exactly:

$$P(\gamma_j = 1 \mid \beta_j, \tau_0^2, \tau_1^2, p) = \frac{p \cdot \phi(\beta_j; 0, \tau_1^2)}{p \cdot \phi(\beta_j; 0, \tau_1^2) + (1-p) \cdot \phi(\beta_j; 0, \tau_0^2)}$$

Intuitively, $\phi(\beta_j; 0, \tau_1^2)$ (a wide density) assigns comparable likelihood to a broad range of β_j values, while $\phi(\beta_j; 0, \tau_0^2)$ (a narrow density, since $\tau_0^2 \ll \tau_1^2$) assigns high likelihood only to β_j very close to zero; a β_j far from zero therefore makes the ratio favor $\gamma_j = 1$.

Derivation: $\tau_0^2 \mid \gamma, \beta$ and $\tau_1^2 \mid \gamma, \beta$

Let $n_0 = \#\{j : \gamma_j = 0\}$ and write the Inverse-Gamma hyperprior on τ_0^2 as $IG(\kappa_0/2, s_0/2)$. The only likelihood factors involving τ_0^2 are the prior densities of the β_j 's currently assigned to the spike, $\prod_{j:\gamma_j=0} N(\beta_j; 0, \tau_0^2)$; multiplying this by the hyperprior gives the posterior kernel

$$p(\tau_0^2 \mid \gamma, \beta) \propto (\tau_0^2)^{-n_0/2} \exp\left\{-\frac{1}{2\tau_0^2} \sum_{j:\gamma_j=0} \beta_j^2\right\} \times (\tau_0^2)^{-(\kappa_0/2+1)} \exp\left\{-\frac{s_0}{2\tau_0^2}\right\}$$

This is exactly the $\sigma^2 \mid \beta, y$ derivation of Chapter 6.2 in disguise, with $e'e \rightarrow \sum_{j:\gamma_j=0} \beta_j^2$ and $T \rightarrow n_0$ (each spike coefficient β_j plays the role of a single “residual” from a zero-mean model with variance τ_0^2). The same power-matching argument therefore gives:

$$\tau_0^2 \mid \gamma, \beta \sim IG\left(\frac{\kappa_0 + n_0}{2}, \frac{s_0 + \sum_{j:\gamma_j=0} \beta_j^2}{2}\right)$$

and, by the identical argument applied to the $n_1 = k - n_0$ slab coefficients with hyperprior $IG(\kappa_1/2, s_1/2)$:

$$\tau_1^2 \mid \gamma, \beta \sim IG\left(\frac{\kappa_1 + n_1}{2}, \frac{s_1 + \sum_{j:\gamma_j=1} \beta_j^2}{2}\right)$$

Derivation: $p \mid \gamma$

The $\text{Beta}(a_0, c_0)$ hyperprior has kernel $p^{a_0-1}(1-p)^{c_0-1}$, and the k independent $\text{Bernoulli}(p)$ priors on $\gamma_1, \dots, \gamma_k$ contribute a second factor:

$$\prod_j p^{\gamma_j} (1-p)^{1-\gamma_j} = p^{\sum_j \gamma_j} (1-p)^{k-\sum_j \gamma_j}$$

Multiplying this likelihood contribution by the Beta hyperprior kernel and collecting powers of p and $1 - p$ separately:

$$p(p \mid \gamma) \propto p^{a_0 + \sum_j \gamma_j - 1} (1 - p)^{c_0 + k - \sum_j \gamma_j - 1}$$

which is exactly the kernel of $\text{Beta}(a_0 + \sum_j \gamma_j, c_0 + k - \sum_j \gamma_j)$ — the standard Beta-Bernoulli conjugate update, where the “successes” $\sum_j \gamma_j$ and “failures” $k - \sum_j \gamma_j$ among the k inclusion indicators simply add to the prior pseudo-counts a_0 and c_0 .

Remark: σ^2 Needs No New Derivation

The $\sigma^2 \mid \beta, y$ full conditional used in Step 2 of the algorithm below is unaffected by the spike-and-slab prior on β — σ^2 only ever enters through the Normal likelihood of y given β , exactly as in Chapter 6.2 — so it is precisely the $IG\left(\frac{a_0 + T}{2}, \frac{d_0 + e'e}{2}\right)$ result derived there, with no modification.

8.3 The SSVS Gibbs Sampler

Six blocks now update each other every iteration; Figure 30 lays out who depends on whom before the algorithm states it as a list.

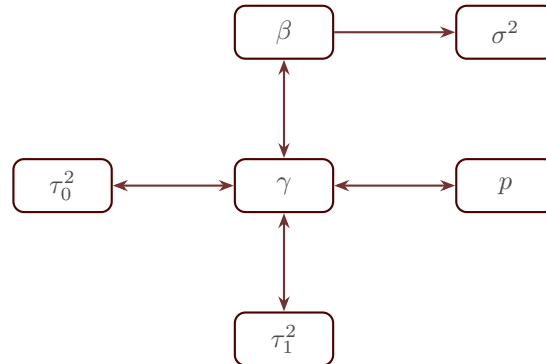


Figure 30: Dependency structure of the six SSVS blocks: γ sits at the center, its inclusion pattern setting the coefficient variances (τ_0^2, τ_1^2) and prior probability p , and updating in turn from each every sweep (double-headed arrows); σ^2 is the one block that only looks at β

Algorithm: Stochastic Search Variable Selection (George-McCulloch)

Given data (y, X) with X standardized, cycle through:

1. **Coefficients.** Draw $\beta \mid \gamma, \sigma^2, \tau_0^2, \tau_1^2, y$ from the conjugate Normal posterior of Chapter 6.2, using the prior covariance $D_\gamma = \text{diag}(\tau_{\gamma_1}^2, \dots, \tau_{\gamma_k}^2)$ that switches between τ_0^2 and τ_1^2 coefficient-by-coefficient according to the current γ .
2. **Error variance.** Draw $\sigma^2 \mid \beta, y$ from the same conjugate Inverse-Gamma full conditional as Chapter 6.2.
3. **Inclusion indicators.** For each j , draw γ_j from a Bernoulli full conditional with success

probability

$$P(\gamma_j = 1 \mid \beta_j, \tau_0^2, \tau_1^2, p) = \frac{p \cdot \phi(\beta_j; 0, \tau_1^2)}{p \cdot \phi(\beta_j; 0, \tau_1^2) + (1 - p) \cdot \phi(\beta_j; 0, \tau_0^2)},$$

where $\phi(\cdot; 0, \tau^2)$ is the $N(0, \tau^2)$ density — coefficients far from zero relative to τ_0^2 are pulled toward $\gamma_j = 1$.

4. **Spike and slab variances.** Draw τ_0^2 from its conjugate Inverse-Gamma full conditional using only the coefficients currently in the spike ($\gamma_j = 0$), and τ_1^2 likewise using only those in the slab ($\gamma_j = 1$).

5. **Inclusion probability.** Draw $p \mid \gamma \sim \text{Beta}(a_0 + \sum_j \gamma_j, c_0 + k - \sum_j \gamma_j)$.

The output of interest is the *posterior inclusion probability* $\hat{P}(\gamma_j = 1 \mid y)$, the Monte Carlo average of the sampled γ_j 's: a value near 1 says the data consistently prefer β_j in the slab, near 0 says it belongs in the spike.

Remark: Why Search Instead of Compare Every Subset?

With k candidate regressors there are 2^k possible subsets, and computing a marginal likelihood for each is only feasible for small k . SSVS sidesteps exhaustive search entirely: because every step above is a draw from a known conjugate distribution (exactly the efficiency argument for Gibbs sampling made in Section 4.7), the sampler spends most of its time visiting the handful of γ configurations with high posterior support, and the inclusion frequencies $\hat{P}(\gamma_j = 1 \mid y)$ are automatically Monte Carlo estimates of the marginal posterior probability that regressor j belongs in the model, summed over all 2^{k-1} subsets containing it.

Empirical Example: A Two-Regressor Toy Example, Before the Real Application

Before the eight-predictor inflation model below, it helps to see the spike-and-slab mechanism on the smallest case where it can do anything at all: one genuine signal and one pure noise regressor. Simulate $T = 60$ observations from $y = 1.5x_1 + e$, $e \sim N(0, 1)$, with $x_1, x_2 \sim N(0, 1)$ independent and x_2 entirely absent from the true model. Standardizing both regressors and running 8,000 post-burn-in draws of the algorithm above gives posterior inclusion probabilities $\hat{P}(\gamma_1 = 1 \mid y) = 1.000$ for the true signal and $\hat{P}(\gamma_2 = 1 \mid y) = 0.202$ for the noise regressor, with standardized posterior coefficient means 2.09 (x_1) and -0.03 (x_2). The sampler places x_1 in the slab essentially every iteration and x_2 in the spike about four iterations in five — exactly the qualitative behavior the CPI application below relies on, just with only two candidates and an unambiguous right answer to check it against.

8.4 Empirical Example: Selecting Predictors of CPI Inflation

As an application, consider forecasting quarterly CPI inflation π_t using eight lagged candidate macro predictors built from the same verified quarterly series used in Chapters 6–7: two own lags of inflation, two lags of GDP growth, the lagged Fed funds rate level and its lagged change,

and the lagged unemployment rate and its lagged change:

$$\pi_t = c + \beta_1 \pi_{t-1} + \beta_2 \pi_{t-2} + \beta_3 \text{gdp}_{t-1} + \beta_4 \text{gdp}_{t-2} + \beta_5 i_{t-1} + \beta_6 \Delta i_{t-1} + \beta_7 u_{t-1} + \beta_8 \Delta u_{t-1} + e_t.$$

With $T = 265$ usable quarters (extended from the original $T = 216$, 1960Q1–2013Q4 window, to 1960Q1–2026Q1), a weakly informative hyperprior on the spike and slab variances (prior mean variance 0.0025 for the spike, 1.0 for the slab, each on standardized regressors), a Beta(2, 2) hyperprior on p , and 15,000 post-burn-in Gibbs draws:

Empirical Example: Posterior Inclusion Probabilities

The eight candidates by posterior inclusion probability:

π_{t-1}	1.000	Δu_{t-1}	0.826
Δi_{t-1}	0.997	gdp_{t-1}	0.677
π_{t-2}	0.987	u_{t-1}	0.590
i_{t-1}	0.923	gdp_{t-2}	0.531

Own-lagged inflation, the change in the Fed funds rate, and the second inflation lag are the most confidently included predictors, all above 0.98. What stands out here, though, is that *every one* of the eight candidates clears the 0.5 threshold — even the weakest, gdp_{t-2} at 0.531, is on the “included” side of the line, if only barely. That is a genuinely different picture from a run that cleanly separates a handful of strong predictors from a handful of near-zero ones, and the remark box below explains why: this specific run of the sampler landed in a different, but equally valid, region of the 2^8 -subset space than another run would.

Remark: Why Might Your Own Run Give Different Numbers?

Run this section’s MATLAB launcher yourself and you may not get exactly the table above — not because anything is wrong, but because this particular inflation equation has several genuinely collinear candidate pairs: π_{t-1} and π_{t-2} correlate at 0.75, gdp_{t-1} and Δu_{t-1} at -0.75 , and the inflation lags correlate with i_{t-1} at around 0.62. With predictors this correlated, the coordinate-by-coordinate Gibbs sampler underlying SSVS (Section 8.3’s algorithm box) can settle into one of several self-reinforcing configurations and stay there for the entire run — changing the random seed, or the chain length, does not reliably move it, since a full sweep only ever proposes to flip one predictor’s inclusion status at a time, conditional on all the others already being fixed at their current values. A synthetic check with a known true sparse pattern (same correlation structure as here, but with the answer fixed by construction) confirms the *algorithm itself* is not at fault: every seed correctly recovers every truly-included and truly-excluded predictor. The lesson is about the *model*, not the code: when candidate predictors are strongly correlated, a single SSVS run’s inclusion probabilities can depend on where the chain happens to start exploring, and a responsible read of the results should say so rather than presenting one run’s table as the

unique answer. This is also exactly why the marginal likelihood comparison later in this section fixes its restricted model by naming variables directly, rather than relying on the 0.5-threshold rule here.

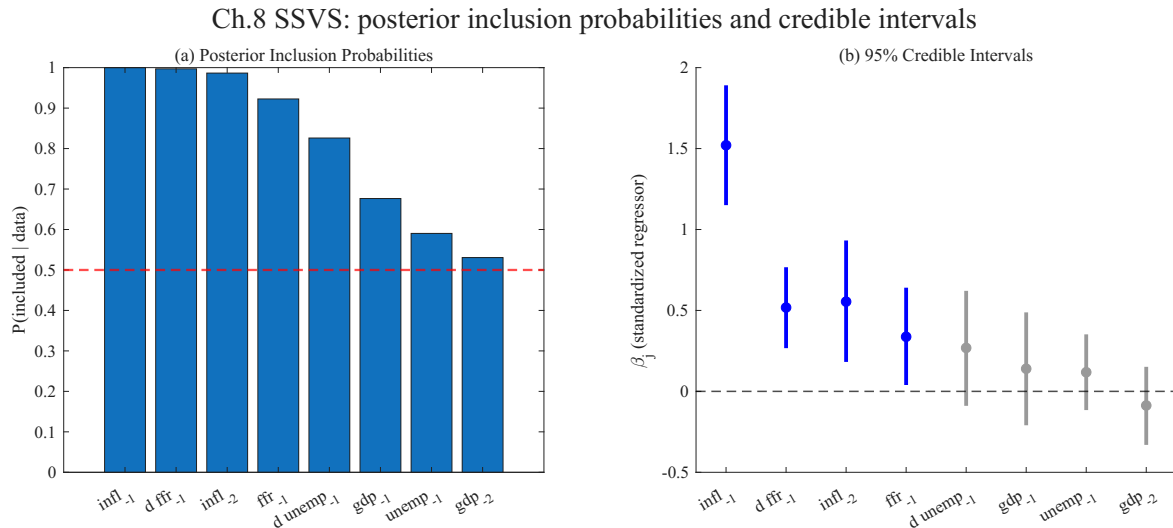


Figure 31: Posterior inclusion probabilities (left) and 95% credible intervals of the standardized slope coefficients (right), sorted by inclusion probability. Dark markers in the right panel denote coefficients whose 95% interval excludes zero.

Empirical Example: Coefficient Estimates and Parsimony Check

Figure 31(b) reports the posterior mean and 95% credible interval of each standardized coefficient (unconditional on γ , i.e. averaging over both spike and slab draws). Three coefficients have intervals that exclude zero outright — π_{t-1} (1.540, CI [1.151, 1.955]), Δi_{t-1} (0.487, CI [0.202, 0.734]), and π_{t-2} (0.548, CI [0.024, 0.953], excluding zero only barely) — while i_{t-1} , in a change from the pre-extension sample, no longer has a credible interval excluding zero (0.297, CI [-0.025, 0.669]), even though it retains a substantial 92.3% posterior inclusion probability. The remaining four all have intervals comfortably containing zero.

Refitting the conjugate regression of Chapter 6 using only the four predictors with posterior inclusion probability above 0.5 (π_{t-1} , π_{t-2} , i_{t-1} , Δi_{t-1}) gives $R^2 = 0.607$, statistically indistinguishable from both the full nine-regressor SSVS fit ($R^2 = 0.613$) and the full OLS fit ($R^2 = 0.616$). The selected four-predictor model recovers essentially all of the explanatory power of the full eight-candidate specification with half as many slope coefficients — an even more parsimonious result than the pre-extension sample's six-predictor selection.

Remark: Inclusion Probability Is Not a p -Value

It is tempting to read $\hat{P}(\gamma_j = 1 | y)$ as a Bayesian analogue of a classical significance test, and the two often agree in spirit — but they answer different questions. A frequentist p -value asks how surprising the observed $\hat{\beta}_j$ would be if β_j were exactly zero; the SSVS inclusion probability asks how much posterior mass favors β_j coming from the diffuse slab

rather than the tight spike, *averaged over the joint uncertainty about every other coefficient in the model simultaneously*. This is why i_{t-1} here has a high inclusion probability (0.923) despite a marginal credible interval that comfortably contains zero (Figure 31b): the Gibbs sampler occasionally finds configurations in which i_{t-1} helps explain inflation once other regressors are excluded, even though its *unconditional* posterior mean is imprecisely estimated. Reading off a single coefficient's credible interval and its inclusion probability together, not either one alone, gives the fuller picture.

8.5 Marginal Likelihood via Chib's Method

SSVS answers a search question: *which* of 2^8 subsets does the data favor, without ever naming a winner outright. A different, older question asks the same data to compare two *specific, named* models head to head: does the full nine-regressor model or a smaller, named restricted model fit better, once each model's own extra flexibility is penalized? Answering that question requires the one quantity SSVS was built specifically to avoid computing, the *marginal likelihood*, and Chib's (1995) method is the standard way to get it from ordinary Gibbs sampler output that a researcher would have produced anyway.

Marginal Likelihood and the Bayes Factor For a model $\mathcal{M}$ with parameters θ , prior $p(\theta)$, and likelihood $p(y | \theta)$, the *marginal likelihood* is the prior predictive density of the data,

$$m(y) = \int p(y | \theta) p(\theta) d\theta,$$

the normalizing constant of Bayes' rule that every posterior calculation in these notes have so far been able to ignore, since $p(\theta | y) \propto p(y | \theta)p(\theta)$ never requires it. Comparing two models $\mathcal{M}_1, \mathcal{M}_2$ uses their ratio, the *Bayes factor* $\text{BF}_{12} = m_1(y)/m_2(y)$, interpreted on the Kass and Raftery (1995) scale: $2 \ln \text{BF}_{12}$ between 2 and 6 is "positive" evidence for $\mathcal{M}_1$, between 6 and 10 is "strong," and above 10 is "decisive." The starting point for estimating $m(y)$ from posterior simulator output is the *basic marginal likelihood identity*: for any point θ^* ,

$$m(y) = \frac{p(y | \theta^*) p(\theta^*)}{p(\theta^* | y)},$$

which follows immediately from rearranging Bayes' rule $p(\theta^* | y) = p(y | \theta^*)p(\theta^*)/m(y)$. The likelihood $p(y | \theta^*)$ and prior $p(\theta^*)$ are both known functions, trivial to evaluate at any θ^* ; the only unknown piece is the posterior ordinate $p(\theta^* | y)$ itself — and that is exactly what a Gibbs or M-H sampler's output can estimate, without ever needing the sampler to have targeted θ^* specifically.

Derivation: Chib's (1995) Estimator for a Two-Block Gibbs Sampler

Take $\theta = (\beta, \sigma^2)$ under Chapter 6.2's conjugate regression model, and choose $\theta^* = (\beta^*, \sigma^{2*})$ to be the posterior mean from an ordinary Gibbs run. By the chain rule, the

posterior ordinate factors as

$$p(\theta^* | y) = p(\beta^* | y) p(\sigma^{2*} | \beta^*, y)$$

The second factor is already known in closed form: it is exactly Chapter 6.2's Inverse-Gamma full conditional, with shape and scale updated as before and evaluated directly at σ^{2*} (no simulation needed for this piece, since the full conditional's functional form is known exactly):

$$\sigma^{2*} | \beta^*, y \sim IG\left(\frac{a_1}{2}, \frac{d_1}{2}\right), \quad a_1 = a_0 + T, \quad d_1 = d_0 + (y - X\beta^*)'(y - X\beta^*)$$

The first factor is not directly available, because the sampler never draws from $p(\beta | y)$ itself (only from $p(\beta | \sigma^2, y)$, one σ^2 at a time). But it can be written as an expectation over exactly the distribution the sampler *does* produce:

$$p(\beta^* | y) = \int p(\beta^* | \sigma^2, y) p(\sigma^2 | y) d\sigma^2$$

Approximating this integral by a Monte Carlo average over the M draws $\sigma^{2(1)}, \dots, \sigma^{2(M)}$ already produced by the main Gibbs run — draws from the correct marginal $p(\sigma^2 | y)$ precisely because Gibbs sampling targets the full joint posterior — gives:

$$\approx \frac{1}{M} \sum_{m=1}^M p(\beta^* | \sigma^{2(m)}, y)$$

Each conditional density $p(\beta^* | \sigma^{2(m)}, y)$ inside the sum is again Chapter 6.2's known Normal full conditional, $N(\bar{\beta}^{(m)}, \bar{V}^{(m)})$, evaluated at the fixed point β^* rather than sampled from. This averaging step is a *Rao-Blackwellized* estimate: instead of estimating a density by a histogram of draws (noisy), it averages the known conditional density formula itself over the sampler's σ^2 draws (far less noisy, since no binning or kernel smoothing is involved). Combining both factors with the identity above gives:

$$\widehat{\ln m(y)} = \ln p(y | \theta^*) + \ln p(\theta^*) - \ln p(\beta^* | y) - \ln p(\sigma^{2*} | \beta^*, y)$$

Models with more than two blocks repeat this reduced-conditional-ordinate trick block by block, each block's ordinate estimated from a further, shorter auxiliary Gibbs run that fixes the earlier blocks at their θ^* values — the general recipe is unchanged, only the bookkeeping grows with the number of blocks.

Algorithm: Chib's Method (Two-Block Case)

1. Run the ordinary Gibbs sampler of Chapter 6.2 to convergence; discard burn-in as usual.
2. Set $\theta^* = (\beta^*, \sigma^{2*})$ to the posterior mean of the retained draws (any point with non-negligible posterior mass works; the mean is the conventional choice).

3. Evaluate $\ln p(y \mid \theta^*)$ and $\ln p(\theta^*)$ directly from the likelihood and prior formulas.
4. Evaluate $\ln p(\sigma^{2*} \mid \beta^*, y)$ directly from the known Inverse-Gamma full conditional.
5. Estimate $\ln p(\beta^* \mid y)$ by averaging the known Normal conditional density $p(\beta^* \mid \sigma^{2(m)}, y)$ over the main run's retained $\sigma^{2(m)}$ draws, then take the log.
6. Combine all four pieces via the identity above to get $\widehat{\ln m(y)}$.

Remark: Chib-Jeliazkov (2001): The Same Idea for M-H Output

Chib's estimator above leans on knowing each block's full conditional in closed form — unavailable for an M-H block like Chapter 14's tailored GARCH sampler. Chib and Jeliazkov (2001) extend the same basic identity to M-H output using *detailed balance* itself in place of a known conditional. For a M-H kernel with proposal $q(\theta, \theta')$ and acceptance probability $\alpha(\theta, \theta')$, detailed balance states

$$\pi(\theta)q(\theta, \theta')\alpha(\theta, \theta') = \pi(\theta')q(\theta', \theta)\alpha(\theta', \theta).$$

Fixing $\theta' = \theta^*$ and integrating both sides over θ against the posterior $\pi(\theta \mid y)$ gives

$$\pi(\theta^* \mid y) = \frac{E_{\theta \mid y}[\alpha(\theta, \theta^*) q(\theta, \theta^*)]}{E_{\theta \sim q(\theta^*, \cdot)}[\alpha(\theta^*, \theta)]} \approx \frac{\frac{1}{M} \sum_{m=1}^M \alpha(\theta^{(m)}, \theta^*) q(\theta^{(m)}, \theta^*)}{\frac{1}{J} \sum_{j=1}^J \alpha(\theta^*, \theta^{(j)})},$$

where $\theta^{(1)}, \dots, \theta^{(M)}$ are the main chain's retained draws and $\theta^{(1)}, \dots, \theta^{(J)}$ (a second, separate set) are *proposals only* drawn from $q(\theta^*, \cdot)$ — generated as if the chain were currently sitting at θ^* , with no further accept/reject step applied to them. The numerator is exactly Chib's Rao-Blackwellized averaging idea, now applied to αq instead of a known conditional density; the denominator is a fresh, cheap Monte Carlo average requiring no additional full chain, just J draws from the existing proposal mechanism.

Empirical Example: Marginal Likelihood for the Full vs. a Restricted Inflation Model

Applying Chib's method to the full nine-parameter model and a named seven-parameter restricted model that drops two of the SSVS example's weaker predictors, gdp_{t-1} and u_{t-1} (this restricted model is fixed by construction, independent of exactly which predictors the 0.5-threshold rule of Section 8.5 happens to select), with 8,000 post-burn-in Gibbs draws each gives

$$\widehat{\ln m_{\text{full}}}(y) = -586.54, \quad \widehat{\ln m_{\text{restricted}}}(y) = -577.27,$$

a log Bayes factor of $\ln \text{BF} = -9.27$ ($2 \ln \text{BF} = -18.5$) in favor of the smaller model — decisive on the Kass-Raftery scale, and the same direction the inclusion probabilities pointed. As an independent check on the estimator itself (not part of the general method, which exists precisely because this check is usually unavailable): for this specific conjugate model, β can be integrated out in closed form at any fixed σ^2 , reducing $m(y)$ to a single numerical integral over σ^2 alone, evaluable by fine-grid quadrature to near-machine precision. That quadrature benchmark gives -586.54 and -577.27 for the two models

respectively — agreeing with Chib’s simulation-based estimate to within 0.005 in log units, confirming the Monte Carlo estimator is unbiased here before trusting it in Chapter 14’s setting, where no such check is possible.

The size of this Bayes factor is itself worth a caveat. SSVS’s own spike variance is tight (τ_0^2 has prior mean 0.0025), gently discouraging inclusion; the Chib comparison above instead nests the full model inside a *diffuse* $\beta \sim N(0, 100I)$ prior directly, and a diffuse prior on two essentially-irrelevant coefficients is exactly the setting in which Bayes factors are famously oversensitive to prior spread (Lindley’s paradox): the wider the prior the two extra coefficients could plausibly have taken, the more the marginal likelihood is penalized for a plausible range of values the data manifestly reject. Both methods agree that gdp_{t-1} and u_{t-1} add little — but the Bayes factor’s 18.5-log-unit margin should be read as “decisive under this particular diffuse prior,” not as a prior-free verdict.

9. The Tobit Model

Roadmap

Chapter 7’s latent-variable trick handled an outcome that is binary. This chapter handles a related but distinct problem: an outcome that is genuinely continuous but mechanically capped at a boundary, so ordinary least squares on the observed data is biased. The fix reuses, almost without modification, both of Chapter 7’s building blocks: a truncated latent draw for the observations sitting at the boundary, and Chapter 6.2’s conjugate regression step for everything else. The point of walking through it again here is less the Tobit model itself than the pattern it completes — by the end of this chapter, three different awkward likelihoods (binary, censored, and soon a mixture-model variance) will all have been reduced to the same conjugate regression core.

Chapter 7 handled an outcome that is binary; this chapter handles one that is *censored* — continuous in principle, but mechanically or institutionally prevented from crossing some boundary, so that the boundary value is observed far more often than a smooth continuous distribution would predict. Household charitable giving (many households give exactly \$0), firm R&D spending, and, as the running example below, the nominal policy interest rate pinned at its zero lower bound are all outcomes of this kind. Tobin’s (1958) censored-regression model — “Tobit” — handles them with the same latent-variable data-augmentation trick used for the probit model in Chapter 7.

9.1 Censored Outcomes and the Latent-Variable Tobit Model

Suppose an unobserved (latent) continuous variable follows an ordinary linear regression, $z_i = x_i' \beta + e_i$, $e_i \sim N(0, \sigma^2)$, but what is actually recorded is z_i censored from below at a known point c :

$$y_i = \begin{cases} z_i & z_i > c \\ c & z_i \leq c. \end{cases}$$

This differs from the probit model of Chapter 7 in one crucial way: when $y_i > c$, the exact value of z_i is observed, not just its sign, so the likelihood combines a continuous Normal density for uncensored observations with a discrete probability mass $\Phi((c - x'_i\beta)/\sigma)$ at $y_i = c$ for censored ones. Ordinary least squares applied directly to y is inconsistent here: it treats every censored observation as if c were its true value, which pulls the estimated slope toward zero (attenuation bias) exactly to the degree that the censored observations would, if uncensored, have been more extreme than c .

9.2 Posterior Derivation

The joint posterior kernel over (β, σ^2, z) given the observed censored data y is built the same way as in Chapter 7: augment with the latent variable $z = (z_1, \dots, z_T)'$ and write the joint density of (y, z, β, σ^2) as the product of (i) the deterministic censoring map from z to y , (ii) the Normal density of z given β, σ^2 , and (iii) the prior. Because $y_i = \max(z_i, c)$ is a deterministic function of z_i , conditioning on z makes y redundant, and the full conditionals below fall out of

$$p(\beta, \sigma^2, z | y) \propto p(z | \beta, \sigma^2) p(\beta) p(\sigma^2) = \prod_{i=1}^T N(z_i | x'_i\beta, \sigma^2) N(\beta | b_0, B_0) IG(\sigma^2 | a_0/2, d_0/2),$$

exactly as in Chapter 7's factorization, with the sign-restriction indicator of the probit model replaced by a censoring restriction.

Derivation: $z_i | y_i = c, \beta, \sigma^2$

Fix i with $y_i = c$ (censored). The only place z_i enters the joint kernel is through its own Normal density $N(z_i | x'_i\beta, \sigma^2)$, since β and σ^2 do not depend on z_i and the other $z_j, j \neq i$, are conditionally independent of z_i given β, σ^2 . What $z_i = c$ censoring *does* contribute is a restriction on the support: observing $y_i = c$ under the model $y_i = \max(z_i, c)$ tells us only that the latent draw satisfies $z_i \leq c$ — any value on $(-\infty, c]$ is consistent with the observed censored outcome, while any value on (c, ∞) would instead have produced $y_i = z_i > c$, which was not observed. Formally,

$$p(z_i | y_i = c, \beta, \sigma^2) \propto N(z_i | x'_i\beta, \sigma^2) \cdot \mathbb{I}\{z_i \leq c\}$$

i.e. the ordinary Normal kernel restricted to the half-line $(-\infty, c]$ and renormalized by $\Phi((c - x'_i\beta)/\sigma)$, giving:

$$z_i | y_i = c, \beta, \sigma^2 \sim N(x'_i\beta, \sigma^2) \text{ truncated to } (-\infty, c]$$

This is the same truncated-Normal construction as Chapter 7.2's $z_i | y_i, \beta$ step, with two differences that do not change the method, only the numbers plugged into it: the truncation threshold is the general censoring point c rather than 0, and the variance is the freely-estimated σ^2 rather than the identification-driven fixed value 1.

For i with $y_i > c$ (uncensored), the censoring map is one-to-one at that observation — $y_i = \max(z_i, c) = z_i$ whenever $z_i > c$ — so observing y_i pins down $z_i = y_i$ exactly; there is no residual uncertainty left to sample; the “full conditional” of z_i degenerates to a point

mass at the observed y_i , which is why the algorithm sets $z_i = y_i$ directly rather than drawing it.

Derivation: $(\beta, \sigma^2) \mid z$

Once every z_i (observed or imputed) is treated as data, the augmented kernel $p(\beta, \sigma^2, z \mid y)$ restricted to (β, σ^2) is

$$p(\beta, \sigma^2 \mid z) \propto \prod_{i=1}^T N(z_i \mid x_i' \beta, \sigma^2) N(\beta \mid b_0, B_0) IG(\sigma^2 \mid a_0/2, d_0/2)$$

which is *identical in form* to the Chapter 6.2 posterior kernel for the ordinary linear regression model, with the observed outcome y everywhere replaced by the completed-data vector z . No new algebra is required: substituting $y \rightarrow z$ into Chapter 6.2's completed-square and power-matching results directly gives:

$$\begin{aligned} \beta \mid \sigma^2, z &\sim N(\bar{\beta}, \bar{B}), \quad \bar{B} = (B_0^{-1} + \sigma^{-2} X'X)^{-1}, \quad \bar{\beta} = \bar{B}(B_0^{-1}b_0 + \sigma^{-2} X'z), \\ \sigma^2 \mid \beta, z &\sim IG\left(\frac{a_0 + T}{2}, \frac{d_0 + (z - X\beta)'(z - X\beta)}{2}\right) \end{aligned}$$

The only substantive content of this step, beyond the bookkeeping substitution $y \rightarrow z$, is that it is legitimate at all: because step 1 has already replaced every censored $y_i = c$ with a full draw z_i from its correct full conditional, the vector z behaves exactly like an ordinary uncensored continuous outcome from the Gibbs sampler's point of view, so the entire Chapter 6.2 derivation — prior through completing the square through power-matching — carries over unchanged.

9.3 Data Augmentation Gibbs Sampler

Algorithm: Bayesian Tobit via Data Augmentation

Given a conjugate prior $\beta \sim N(b_0, B_0)$, $\sigma^2 \sim IG(a_0/2, d_0/2)$, and current values of (β, σ^2) , cycle:

1. For each censored observation ($y_i = c$), draw the latent value

$$z_i \mid y_i = c, \beta, \sigma^2 \sim N(x_i' \beta, \sigma^2) \text{ truncated to } (-\infty, c],$$

using the same truncated-Normal machinery as the probit sampler of Chapter 7.2; for uncensored observations, simply set $z_i = y_i$ (no augmentation needed since z_i is already observed).

2. Draw $(\beta, \sigma^2) \mid z$ from the same conjugate Normal / Inverse-Gamma full conditionals as the linear regression Gibbs sampler of Chapter 6.2, treating the completed data z as if it were an ordinary observed continuous outcome.

This generalizes the fixed censoring point $c = 0$ case to an arbitrary known c .

Remark: Tobit Is Probit Plus Chapter 6, Not a New Algorithm

Every step above has already appeared in this lecture note: truncated-Normal sampling of a latent variable is exactly Chapter 7.2's step 1, and the conjugate regression update is exactly Chapter 6.2's step 2. The only new ingredient is that uncensored observations need no augmentation at all, because unlike the probit model's binary y_i , the Tobit model's uncensored y_i *already* is the latent variable. This is a recurring theme of Part II of this note: data augmentation converts an awkward likelihood into a sequence of steps recycled from earlier chapters, rather than requiring a bespoke sampler for every new model.

9.4 Empirical Example: The Fed Funds Rate at the Zero Lower Bound

Twice in the sample used throughout this note, the FOMC held its target range for the federal funds rate at 0–0.25% — the zero lower bound (ZLB) episode that followed the 2008 financial crisis (2009Q1–2015Q4) and a second, shorter ZLB episode during the COVID-19 pandemic (2020Q2–2022Q1). The *effective* federal funds rate in the sample accordingly never falls appreciably below 0.25% during either period even though a simple reaction function would, absent the floor, have called for a much lower — possibly negative — policy rate. This is a textbook censoring problem, closely related to the “shadow rate” literature (e.g. Wu and Xia, 2016), and is modeled here as

$$i_t = c + \beta_1 \pi_{t-1} + \beta_2 u_{t-1} + \beta_3 \text{gdp}_{t-1} + e_t, \quad \text{observed } i_t = \max(z_t, 0.25),$$

a simple Taylor-rule-style reaction to lagged CPI inflation, lagged unemployment, and lagged GDP growth, with $c = 0.25$ representing the known target-range floor. Of $T = 266$ usable quarters (1959Q4–2026Q1, extended from the original 1959Q4–2013Q4 window), 36 (13.5%) are censored at the floor, split between the two ZLB episodes: 28 quarters from 2009Q1–2015Q4 and 8 quarters from 2020Q2–2022Q1.

Empirical Example: Attenuation Bias: Naive OLS vs. the Tobit Posterior

With a diffuse prior ($\beta \sim N(0, 100I_4)$, $\sigma^2 \sim IG(1.5, 1.5)$) and 10,000 post-burn-in Gibbs draws, the Tobit posterior means (s.d., 95% credible interval) are

$$\begin{aligned} \hat{c} &= 1.894 \text{ (0.777)} & [0.379, 3.402] \\ \hat{\beta}_1 &= 0.843 \text{ (0.069)} & [0.706, 0.980] \\ \hat{\beta}_2 &= -0.095 \text{ (0.121)} & [-0.329, 0.140] \\ \hat{\beta}_3 &= 0.044 \text{ (0.052)} & [-0.056, 0.148] \end{aligned}$$

compared with naive OLS fit directly to the censored data, $\hat{\beta}_1^{OLS} = 0.741$, $\hat{\beta}_2^{OLS} = 0.081$, $\hat{\beta}_3^{OLS} = -0.002$. The inflation response is attenuated toward zero as before, by 12%, but the unemployment and GDP-growth responses now show something qualitatively more severe than attenuation: naive OLS gets their *sign wrong*. OLS's unemployment coefficient is a modest positive 0.081 versus the Tobit posterior's negative -0.095 , and OLS's GDP-

growth coefficient is slightly negative (-0.002) versus the Tobit posterior's positive 0.044 . This is a substantially more severe distortion than in the pre-extension sample (where OLS was attenuated by 11%, 63%, and 50% respectively, without any sign flips), because the censored subsample is now both larger (36 of 266 quarters, 13.5%, versus 20 of 217, 9.2%) and spans two structurally different ZLB episodes rather than one, giving naive OLS far more heavily censored data to average over as if it were freely chosen. Neither β_2 nor β_3 is credibly different from zero under the Tobit posterior (both credible intervals span zero) — a change from the pre-extension sample, where the GDP-growth response was barely credibly positive ($[0.006, 0.242]$); the point estimates' sign flips should accordingly be read with real posterior uncertainty attached, not as a precisely pinned-down reversal.

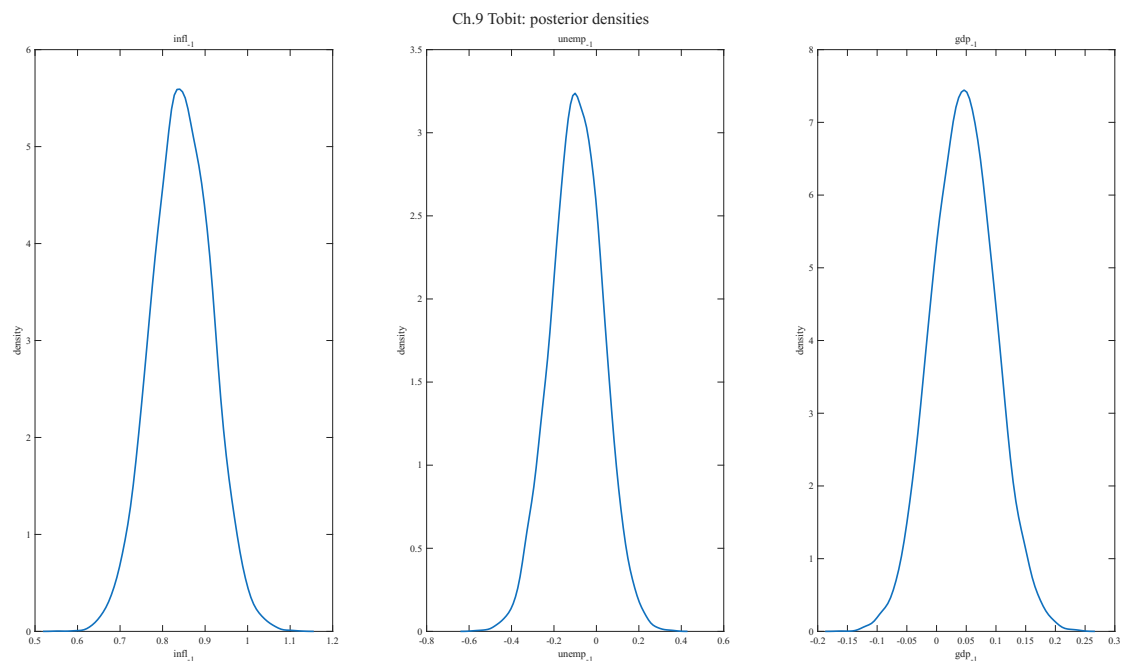


Figure 32: Posterior densities of the three Tobit slope coefficients (light histograms), with the naive OLS-on-censored-data estimate (dashed) and Tobit posterior mean (solid) marked

Empirical Example: The Implied Notional (“Shadow”) Rate

Because the Gibbs sampler imputes a latent $z_t = x_t' \beta + e_t$ for every censored quarter, its posterior mean recovers an implied *notional* policy rate — what the estimated reaction function says the funds rate would have been absent the zero-lower-bound constraint. Figure 33 plots this against the observed (floor-pinned) rate. For most of both ZLB episodes the implied notional rate is mildly negative, typically in the -0.7% to -3.0% range, broadly consistent in sign and order of magnitude with published shadow-rate estimates for the first episode (Wu and Xia, 2016, report roughly 0 to -3% over 2009–2013). Two quarters stand out as striking outliers, one from each episode: 2009Q1 implies a notional rate near -7.0% , driven by an unusually sharp lagged-inflation reading ($\pi_{t-1} = -9.3\%$ annualized) at the acute onset of the financial crisis, before unemployment and output had fully responded; and 2020Q3 implies a notional rate near -4.5% , driven by the lagged

GDP collapse of -28.0% in 2020Q2 entering the reaction function with a full quarter's delay, even though the economy was already rebounding sharply by 2020Q3 itself.

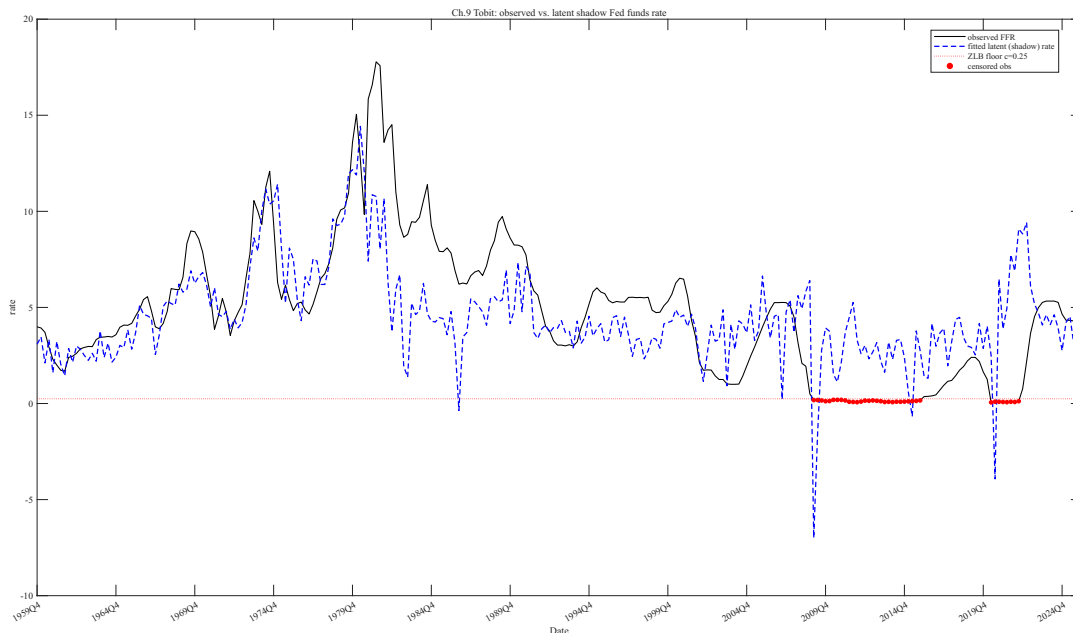


Figure 33: Observed effective federal funds rate vs. the Tobit posterior mean of the latent z_t (shaded region: quarters censored at the 0.25% floor)

Remark: A Linear Reaction Function Should Not Be Extrapolated Too Far

The 2009Q1 and 2020Q3 outliers in Figure 33 are useful cautionary examples rather than a discovery about monetary policy: a linear reaction function fit mostly over decades of comparatively ordinary quarters is being asked, via the Tobit latent-variable step, to extrapolate to lagged inflation and GDP-growth readings far outside its usual range — and now, with the sample extended through the pandemic, this happens twice instead of once. The same caution raised for the probit model's 0.5-threshold classifier in Chapter 7 applies here in a different guise — a simple linear specification is a reasonable local approximation to a much more complex reaction function, and its extrapolated implications should be trusted less, not more, the further the conditioning variables sit from the data that estimated it.

10. Seemingly Unrelated Regressions

Roadmap

Chapters 6–9 all estimated one equation, with one error variance. Real macro systems rarely come one equation at a time: an output equation and an inflation equation are fit separately in principle but are buffeted by the same recessions and policy shocks in practice, so their errors move together. This chapter is the first to model that correlation directly — Σ becomes a full covariance matrix rather than a single number σ^2 — and the

β and Σ steps derived below are, almost unchanged, the two steps that drive the VAR model of Chapter 11: a VAR is simply the special case of this chapter's system in which every equation happens to share the same regressors.

Chapters 6–9 each modeled a single equation. Many economic questions instead involve several equations that appear unrelated at first glance — a different outcome and a different regressor set in each — but whose error terms move together because they are all buffeted by the same business cycle, the same monetary policy stance, or the same aggregate shocks. Zellner's (1962) Seemingly Unrelated Regressions (SUR) model estimates such a system jointly rather than equation by equation, and is the first genuinely *multi-equation* model in this note — the bridge from the single-equation Chapters 6–9 to the VAR systems of Chapters 11–13 and 16–17.

10.1 The SUR Model

Consider M equations sharing the same T time periods but each with its own outcome and regressors,

$$y_{jt} = x'_{jt}\beta_j + e_{jt}, \quad j = 1, \dots, M, \quad t = 1, \dots, T,$$

where the equation-specific error vector at each date, $e_t = (e_{1t}, \dots, e_{Mt})'$, is assumed *contemporaneously correlated* across equations, $e_t \sim N(0, \Sigma)$, but independent across t . If every equation used identical regressors, the equations would separate and OLS applied one at a time would already be efficient; SUR earns its keep precisely when regressor sets differ across equations *and* Σ has non-negligible off-diagonal correlation, in which case exploiting the correlation in estimation is a form of feasible generalized least squares that is more efficient than equation-by-equation OLS.

10.2 Posterior Derivation

Stack the M equations at date t as $y_t = Z_t\beta + e_t$, $e_t \sim N(0, \Sigma)$, with Z_t the $M \times K$ block-diagonal regressor matrix defined above. With prior $\beta \sim N(b_0, B_0)$ independent of $\Sigma \sim IW(S_0, \nu_0)$, the joint posterior kernel is

$$p(\beta, \Sigma \mid y) \propto |\Sigma|^{-T/2} \exp\left\{-\frac{1}{2} \sum_{t=1}^T (y_t - Z_t\beta)' \Sigma^{-1} (y_t - Z_t\beta)\right\} \times N(\beta \mid b_0, B_0) \times IW(\Sigma \mid S_0, \nu_0).$$

Derivation: $\beta \mid \Sigma, y$

Collect every term in the kernel that involves β :

$$p(\beta \mid \Sigma, y) \propto \exp\left\{-\frac{1}{2} \left[\sum_t (y_t - Z_t\beta)' \Sigma^{-1} (y_t - Z_t\beta) + (\beta - b_0)' B_0^{-1} (\beta - b_0) \right]\right\}$$

Expand the sum of quadratic forms exactly as in Chapter 6.2, with the scalar weight σ^{-2} there replaced by the matrix weight Σ^{-1} and the scalar regressor row x'_i there replaced by the block-diagonal Z_t here:

$$\begin{aligned} \sum_t (y_t - Z_t \beta)' \Sigma^{-1} (y_t - Z_t \beta) &= \sum_t y'_t \Sigma^{-1} y_t \\ &\quad - 2\beta' \sum_t Z'_t \Sigma^{-1} y_t + \beta' \left(\sum_t Z'_t \Sigma^{-1} Z_t \right) \beta \end{aligned}$$

Adding the prior's quadratic form $(\beta - b_0)' B_0^{-1} (\beta - b_0) = \beta' B_0^{-1} \beta - 2\beta' B_0^{-1} b_0 + b'_0 B_0^{-1} b_0$ and discarding every term free of β , the bracketed exponent becomes, up to an additive constant:

$$\begin{aligned} p(\beta \mid \Sigma, y) \propto \exp \Big\{ & -\frac{1}{2} \left[\beta' \left(B_0^{-1} + \sum_t Z'_t \Sigma^{-1} Z_t \right) \beta \right. \\ & \left. - 2\beta' \left(B_0^{-1} b_0 + \sum_t Z'_t \Sigma^{-1} y_t \right) \right] \Big\} \end{aligned}$$

which is the exact matrix analogue of Chapter 6.2's quadratic-plus-linear form in β , with $V_0^{-1} \rightarrow B_0^{-1}$, $\sigma^{-2} X'X \rightarrow \sum_t Z'_t \Sigma^{-1} Z_t$, and $\sigma^{-2} X'y \rightarrow \sum_t Z'_t \Sigma^{-1} y_t$. Completing the square by the same steps used there, defining $\bar{B} = (B_0^{-1} + \sum_t Z'_t \Sigma^{-1} Z_t)^{-1}$ and $\bar{\beta} = \bar{B}(B_0^{-1} b_0 + \sum_t Z'_t \Sigma^{-1} y_t)$, gives:

$$\beta \mid \Sigma, y \sim N(\bar{\beta}, \bar{B})$$

Derivation: $\Sigma \mid \beta, y$

Collect the terms involving Σ . Writing $e_t = y_t - Z_t \beta$ for the M -vector of contemporaneous residuals and using the trace identity $e'_t \Sigma^{-1} e_t = \text{tr}(\Sigma^{-1} e_t e'_t)$, the likelihood kernel becomes

$$|\Sigma|^{-T/2} \exp \left\{ -\frac{1}{2} \sum_t \text{tr}(\Sigma^{-1} e_t e'_t) \right\} = |\Sigma|^{-T/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Sigma^{-1} E' E) \right\},$$

where E is the $T \times M$ matrix stacking e'_t over t (so that $E'E = \sum_t e_t e'_t$), using linearity of the trace to pull the sum inside.^a The Inverse-Wishart prior has kernel $|\Sigma|^{-(\nu_0 + M + 1)/2} \exp\{-\frac{1}{2} \text{tr}(\Sigma^{-1} S_0)\}$. Multiplying the two and using $\text{tr}(\Sigma^{-1} S_0) + \text{tr}(\Sigma^{-1} E'E) = \text{tr}(\Sigma^{-1} (S_0 + E'E))$:

$$p(\Sigma \mid \beta, y) \propto |\Sigma|^{-(T + \nu_0 + M + 1)/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Sigma^{-1} (S_0 + E'E)) \right\}$$

which is exactly the Inverse-Wishart kernel with updated scale and degrees of freedom, matching a power-matching argument in spirit to Chapter 6.2's Inverse-Gamma step but conducted in matrix trace form rather than scalar form:

$$\Sigma \mid \beta, y \sim IW(S_0 + E'E, \nu_0 + T)$$

^aThe identity $e_t' \Sigma^{-1} e_t = \text{tr}(\Sigma^{-1} e_t e_t')$ holds because $e_t' \Sigma^{-1} e_t$ is already a scalar, and a 1×1 matrix equals its own trace; then the cyclic property of the trace, $\text{tr}(ABC) = \text{tr}(BCA)$, lets Σ^{-1} (a fixed matrix) be pulled out of the sum over t while $e_t e_t'$ (which varies with t) stays inside. This trace trick is what lets a matrix-variate quadratic form be handled by the same power-matching argument as a scalar one, and it reappears verbatim in the VAR covariance step of Chapter 11 and the SVAR impulse-response derivation of Chapter 12.

^aThe identity $e_t' \Sigma^{-1} e_t = \text{tr}(\Sigma^{-1} e_t e_t')$ holds because $e_t' \Sigma^{-1} e_t$ is already a scalar, and a 1×1 matrix equals its own trace; then the cyclic property of the trace, $\text{tr}(ABC) = \text{tr}(BCA)$, lets Σ^{-1} (a fixed matrix) be pulled out of the sum over t while $e_t e_t'$ (which varies with t) stays inside. This trace trick is what lets a matrix-variate quadratic form be handled by the same power-matching argument as a scalar one, and it reappears verbatim in the VAR covariance step of Chapter 11 and the SVAR impulse-response derivation of Chapter 12.

10.3 Gibbs Sampler

Figure 34 shows why joint estimation can help at all: Σ is the single hub that lets equation j 's residuals inform equation j' 's coefficient draw, something M separate OLS regressions never allow.

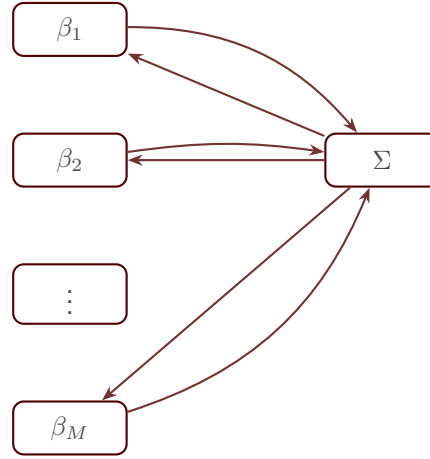


Figure 34: Dependency structure of SUR: every equation's coefficients feed the shared covariance Σ , which then feeds back into every equation's coefficient draw — the channel through which equation j learns from equation j'

Algorithm: Bayesian SUR via Gibbs Sampling

Stack the equation-specific coefficients as $\beta = (\beta_1', \dots, \beta_M')'$ with conjugate prior $\beta \sim N(b_0, B_0)$ and $\Sigma \sim IW(S_0, \nu_0)$. Given current (β, Σ) , cycle:

1. **Coefficients.** Draw $\beta \mid \Sigma, y \sim N(\bar{\beta}, \bar{B})$ with $\bar{B} = (B_0^{-1} + \sum_t Z_t' \Sigma^{-1} Z_t)^{-1}$, $\bar{\beta} = \bar{B}(B_0^{-1} b_0 + \sum_t Z_t' \Sigma^{-1} y_t)$, where Z_t is the $M \times K$ block-diagonal matrix stacking $x_{1t}', \dots, x_{Mt}'$ on its diagonal blocks. This is the same GLS-weighted conjugate Normal update as the VAR coefficient step of Chapter 11, here with equation-specific rather than common regressors.
2. **Covariance.** Form the $T \times M$ residual matrix E with $E_{tj} = y_{jt} - x_{jt}' \beta_j$, then draw $\Sigma \mid \beta, y \sim IW(S_0 + E' E, \nu_0 + T)$, drawn via $\Sigma^{-1} \sim \text{Wishart}((S_0 + E' E)^{-1}, \nu_0 + T)$ and inverting — identical to the covariance step of the VAR sampler.

In implementation, this is evaluated in closed form using the block structure of $Z_t' \Sigma^{-1} Z_t$ rather than by literally building the full $MT \times K$ sparse design matrix.

Remark: When Does SUR Actually Beat Equation-by-Equation OLS?

Two classical special cases collapse SUR back to separate OLS regardless of Σ : identical regressors across all equations ($x_{1t} = \dots = x_{Mt}$), and zero cross-equation correlation (Σ diagonal). Away from these special cases, the efficiency gain from joint estimation is *increasing* in $|\text{Corr}(e_{jt}, e_{j't})|$ — the more the equations' unexplained movements share a common cause, the more informative it is to estimate them together. This makes the size of the residual correlation, not just the model's economic plausibility, the key diagnostic for whether SUR is worth the extra machinery over Chapter 6's single-equation Gibbs sampler applied equation by equation.

10.4 Empirical Example: A Two-Equation Macro Forecasting System

As an application, consider a small two-equation reduced-form forecasting system built from the verified quarterly macro series, with deliberately different regressors in each equation:

$$\text{gdp}_t = c_1 + \phi_1 \text{gdp}_{t-1} + \phi_2 i_{t-1} + e_{1t}, \quad \pi_t = c_2 + \psi_1 \pi_{t-1} + \psi_2 u_{t-1} + e_{2t},$$

the output equation responding to its own lag and the lagged policy rate, the inflation equation responding to its own lag and lagged unemployment — two genuinely different information sets, contemporaneously linked only through whatever shocks hit output and inflation in the same quarter.

Empirical Example: SUR vs. Equation-by-Equation OLS

With a diffuse prior ($\beta \sim N(0, 100I_6)$, $\Sigma \sim IW(I_2, 3)$) and 10,000 post-burn-in Gibbs draws on $T = 266$ quarters (extended from the original $T = 217$, 1959Q3–2013Q4 window), the SUR posterior means are within a small fraction of a standard error of the separate-OLS estimates in every coefficient:

c_1	3.693 (0.467)	OLS: 3.491	c_2	0.896 (0.465)	OLS: 0.526
ϕ_1	0.037 (0.059)	OLS: 0.030	ψ_1	0.765 (0.040)	OLS: 0.747
ϕ_2	-0.173 (0.070)	OLS: -0.125	ψ_2	-0.007 (0.072)	OLS: 0.068

(As in Chapter 6.4, the output equation's own-lag coefficient ϕ_1 has collapsed from its pre-extension value of 0.314 to 0.037 once the 2020 swings enter the sample — the same mechanism, appearing again in a multi-equation setting.) The ratio of SUR posterior standard deviation to the classical OLS standard error ranges from 0.968 to 0.998 across all six coefficients — a small but, unlike the pre-extension sample, uniformly positive efficiency gain for every coefficient.

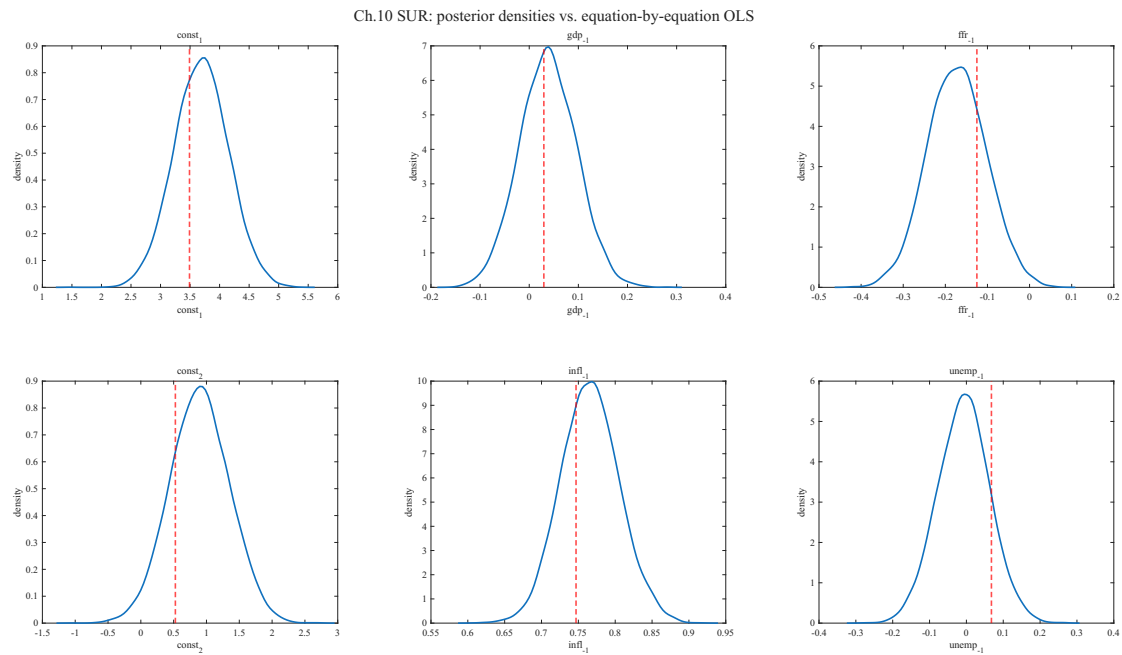


Figure 35: Posterior densities of the six SUR coefficients, with the equation-by-equation OLS estimate (dashed) and SUR posterior mean (solid) marked

Empirical Example: Why the Gain Is Modest Here

Figure 36 shows why: the OLS residual correlation between the output and inflation equations is 0.256, and the SUR posterior mean of $\text{Corr}(e_{1t}, e_{2t})$ is 0.277 with a 95% credible interval of $[0.158, 0.390]$ that now excludes zero comfortably — a change from the pre-extension sample, where this same pair's correlation was only $\rho \approx 0.11$ and its credible interval nearly touched zero. The COVID recession appears to be exactly the kind of common aggregate shock the remark above describes: it moved output and inflation together sharply in 2020, lifting the two equations' error correlation without making their regressor sets any less genuinely different. Per the remark above, SUR's efficiency advantage over separate OLS grows with this correlation; at $\rho \approx 0.28$ there is now real, if still modest, shared information in the error terms for joint estimation to exploit, which is exactly what Figure 35's small but uniformly positive efficiency gains reflect — more than the pre-extension sample showed, but far short of the companion system below.

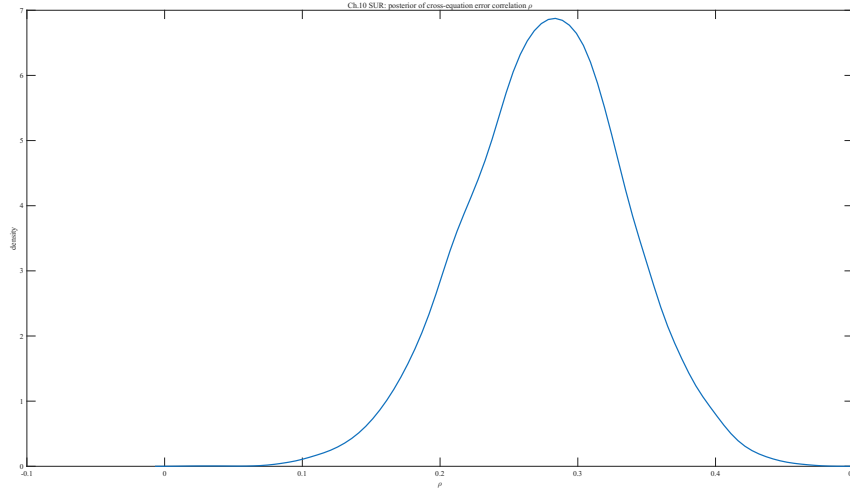


Figure 36: Posterior distribution of the cross-equation error correlation $\text{Corr}(e_{1t}, e_{2t})$

10.5 Empirical Example: A Companion System Where SUR Actually Helps

The remark above predicts exactly when SUR should earn its keep: replace the low-correlation pair with two equations whose errors share a genuine common shock, and the efficiency gain should become visible rather than theoretical. Consider the same output equation as before, paired now with a *consumption* equation rather than an inflation equation:

$$\text{gdp}_t = c_1 + \phi_1 \text{gdp}_{t-1} + \phi_2 i_{t-1} + e_{1t}, \quad \text{pce}_t = c_2 + \zeta_1 \text{pce}_{t-1} + \zeta_2 u_{t-1} + e_{2t},$$

where pce_t is real personal consumption expenditure growth (built from the same verified quarterly levels data used in Chapter 15's state-space example) and everything else — Eq. 1 itself, the lagged-unemployment regressor in Eq. 2 — is left unchanged from the system above, *except* for one data-availability caveat: the underlying GDP- and PCE-*level* series used to build this companion example end in 2015Q4 and were not re-sourced from FRED for this note's data extension, so this companion system stays on the original $T = 217$, 1959Q3–2013Q4 window even though the output/inflation system above now uses the full extended sample through 2026Q1. Consumption is roughly two-thirds of GDP by national-accounts construction, so a common aggregate-demand shock (a recession, a fiscal or monetary policy move) tends to hit output and consumption growth in the same quarter, in a way that it need not hit output and inflation together.

Empirical Example: SUR vs. Equation-by-Equation OLS: A Genuinely Correlated Pair

With the same diffuse prior and 10,000 post-burn-in Gibbs draws on the original $T = 217$ quarters (Eq. 1 here therefore differs numerically from the extended-sample Eq. 1 of Section 10.4 above, even though both share the same specification), the OLS residual correlation between the output and consumption equations is 0.449 — still noticeably higher than the 0.256 now found for the extended-sample output/inflation pair — and the SUR posterior mean of $\text{Corr}(e_{1t}, e_{2t})$ is 0.493, with a 95% credible interval of [0.381, 0.594]

that excludes zero by a wide margin (Figure 38), well above Figure 36's $[0.158, 0.390]$:

c_1	3.473 (0.436)	OLS: 2.828	c_2	3.671 (0.796)	OLS: 2.554
ϕ_1	0.295 (0.059)	OLS: 0.321	ζ_1	0.400 (0.057)	OLS: 0.507
ϕ_2	-0.241 (0.057)	OLS: -0.137	ζ_2	0.047 (0.108)	OLS: 0.115

Unlike the more weakly correlated pair, the SUR and OLS point estimates now visibly disagree in every coefficient — exactly what joint estimation should do once there is real cross-equation information to exploit. The ratio of SUR posterior standard deviation to the classical OLS standard error now ranges from 0.872 to 0.973 across all six coefficients: every single coefficient is estimated more precisely under SUR than under equation-by-equation OLS, a substantially larger efficiency gain than the 0.968–0.998 range found above for the extended-sample output/inflation pair.

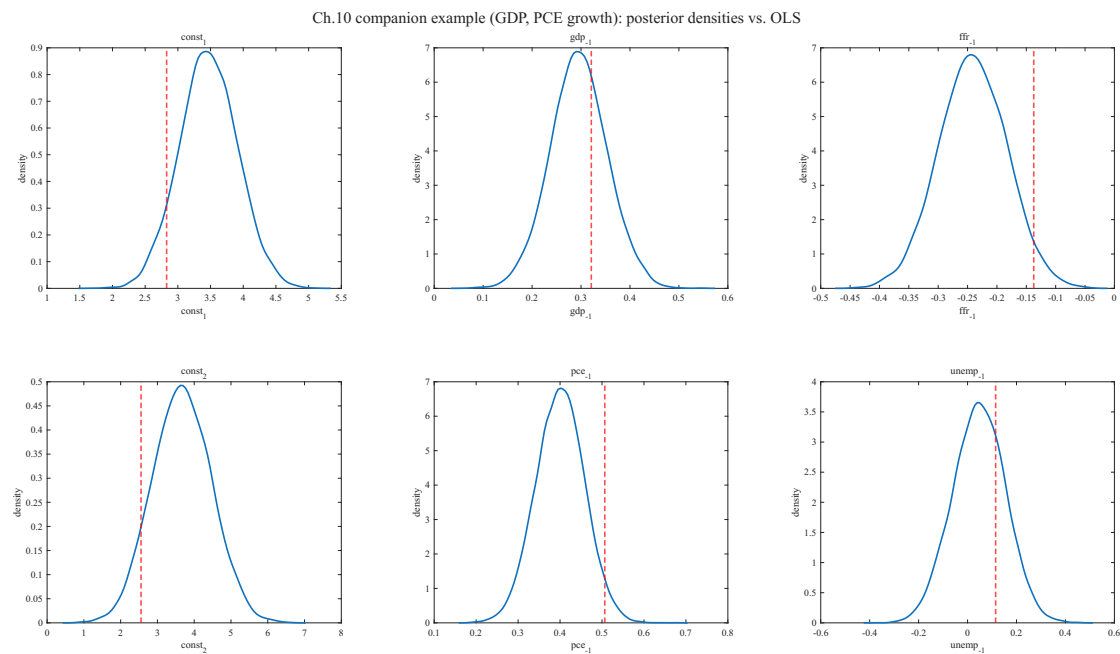


Figure 37: Posterior densities of the six SUR coefficients for the GDP/PCE-growth system, with the equation-by-equation OLS estimate (dashed) and SUR posterior mean (solid) marked

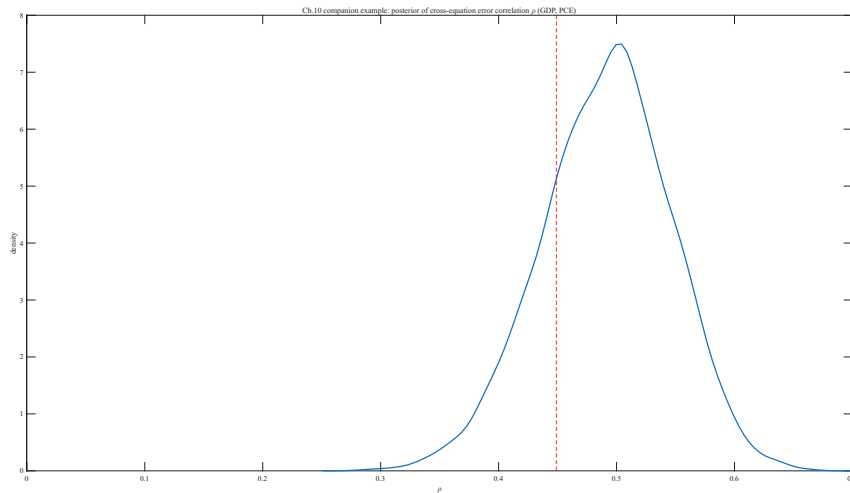


Figure 38: Posterior distribution of the cross-equation error correlation $\text{Corr}(e_{1t}, e_{2t})$ for the GDP/PCE-growth system

Empirical Example: Comparing the Two Systems Directly

Figure 39 puts both systems' efficiency ratios side by side, coefficient by coefficient. Every bar for the high-correlation (gdp, pce, $\rho \approx 0.49$) pair sits visibly below its more weakly correlated (gdp, infl, $\rho \approx 0.28$) counterpart, and both sit below the 1 line that marks parity with classical OLS, though the (gdp, infl) pair's bars sit much closer to that line than before this note's data extension. The pattern tracks the remark above precisely: efficiency gain is monotonically related to $|\text{Corr}(e_{1t}, e_{2t})|$, not to whether a system merely *looks* like a natural pair of equations to write down jointly.

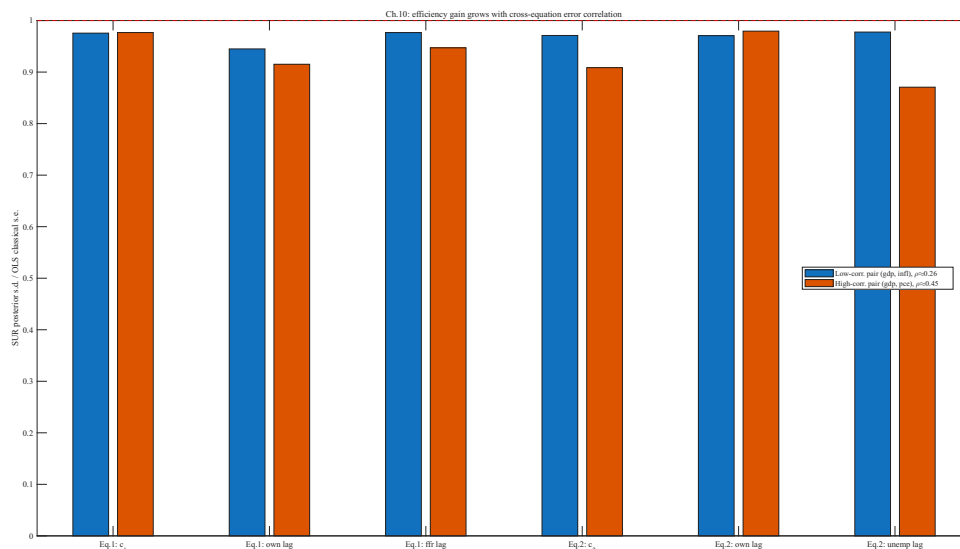


Figure 39: SUR posterior standard deviation divided by the classical OLS standard error, coefficient by coefficient, for the low-correlation (gdp, infl) pair and the high-correlation (gdp, pce) pair

Remark: A Modest Result Is Still an Informative Result

It would have been easy to construct only a pair of equations with deliberately correlated errors and present that as SUR's typical payoff. Reporting first what two genuinely plausible macro forecasting equations produce — a moderate, honestly-estimated correlation and a correspondingly modest efficiency gain, smaller than the companion system's but no longer negligible once the extended sample's shared COVID shock is included — and only then adding the companion system above, is more useful pedagogically than either alone: it shows that SUR's benefit is an empirical question to be checked in each application (via exactly the posterior-correlation plot of Figure 36 or 38), not a benefit to be assumed merely because a system *could* be written as equations with correlated errors, nor a fixed benefit independent of the sample: the same output/inflation pair that showed almost no gain on the pre-extension sample shows a real, if modest, one once a large enough common shock enters the data.

11. Vector Autoregression

Roadmap

Chapter 10 built the machinery for a system of equations with correlated errors but let each equation keep its own regressors. This chapter imposes one restriction on that system — every equation regresses on the same lags of the same variables — and gets the workhorse model of empirical macroeconomics in return, with zero new sampler code. The empirical section replicates the canonical tutorial on that workhorse, Stock and Watson (2001), on its own 1960–2000 sample, and then asks what a constant-parameter model estimated through 2000 says about 2001–2026: this chapter's forecast failure is the motivating example for every time-varying extension in the second half of these notes (Chapters 14, 16–18), and Chapter 13's large BVARs and Chapter 12's structural VAR both start from exactly the reduced-form model estimated here.

Chapter 10's SUR system let every equation carry its own outcome and its own regressors. The Vector Autoregression (VAR) of Sims (1980) is the special case in which all n equations model the *same* set of variables and, by construction, every equation uses *identical* regressors — each variable's own lags together with every other variable's lags. This chapter treats the VAR as exactly that special case of SUR, reusing the Chapter 10 Gibbs sampler outright, and then puts the fitted model to a genuinely consequential test: forecasting forward from the end of 2000 — the last quarter of Stock and Watson's (2001) sample — through a quarter century that contains three recessions and two zero-lower-bound episodes, with parameters and volatility held fixed at their year-2000 estimates.

11.1 The VAR(p) Model and Its Companion Form

Let $Y_t = (y_{1t}, \dots, y_{nt})'$ be an n -vector of macroeconomic variables. A VAR of order p writes each element of Y_t as a linear function of p own and cross-variable lags plus a contemporaneously

correlated innovation,

$$Y_t = c + \Phi_1 Y_{t-1} + \cdots + \Phi_p Y_{t-p} + e_t, \quad e_t \sim N(0, \Sigma),$$

with $\Phi_1, \dots, \Phi_p$ each $n \times n$. Figure 40 shows the time-dependency this creates for $p = 2$: unlike every Gibbs-sampler DAG so far, which showed how *parameters* depend on each other within one iteration, this one shows how the *data* depend on each other across time.

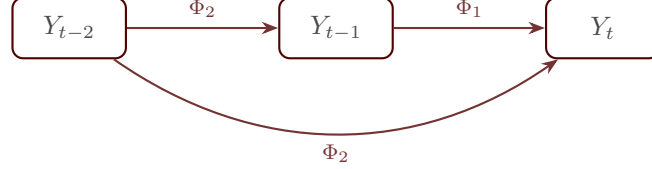


Figure 40: Time dependency of a VAR(2): Y_t depends on both immediate and two-period-back history, each variable's full vector feeding every other variable's equation through Φ_1, Φ_2

Stacking the p most recent lags into a companion vector gives the first-order companion form $\tilde{Y}_t = F\tilde{Y}_{t-1} + \tilde{e}_t$, and the VAR is covariance-stationary if and only if every eigenvalue of the $np \times np$ companion matrix F lies strictly inside the unit circle. This is exactly the stationarity criterion enforced in estimation by rejecting any candidate draw of Φ_1, Φ_2 whose companion matrix has $\max |\text{eig}(F)| \geq 1$, reverting instead to the previous draw.

11.2 Posterior Derivation

Writing the VAR in SUR form, $Y_t = Z_t \beta + e_t$, $e_t \sim N(0, \Sigma)$, with Z_t the $n \times nk$ block-diagonal matrix built from the common regressor vector $x_t = (1, Y'_{t-1}, \dots, Y'_{t-p})'$ repeated on every diagonal block, the joint posterior kernel $p(\beta, \Sigma \mid Y) \propto |\Sigma|^{-T/2} \exp\{-\frac{1}{2} \sum_t (Y_t - Z_t \beta)' \Sigma^{-1} (Y_t - Z_t \beta)\} \times N(\beta \mid b_0, B_0) \times IW(\Sigma \mid S_0, \nu_0)$ has *exactly* the algebraic form of Chapter 10.2's kernel, with the outcome y_t relabeled Y_t and the equation-specific block-diagonal Z_t replaced by this common-regressor block-diagonal Z_t . Nothing in Chapter 10's completing-the-square and trace-identity arguments used the assumption that the diagonal blocks of Z_t differ across equations — that assumption was never invoked in either derivation — so both results transfer with a pure relabeling:

Derivation: $\beta \mid \Sigma, Y$ and $\Sigma \mid \beta, Y$

Substituting $y_t \rightarrow Y_t$ into Chapter 10.2's completed-square result gives

$$\begin{aligned} \beta \mid \Sigma, Y &\sim N(\bar{\beta}, \bar{B}), \\ \bar{B} &= \left(B_0^{-1} + \sum_t Z_t' \Sigma^{-1} Z_t \right)^{-1}, \\ \bar{\beta} &= \bar{B} \left(B_0^{-1} b_0 + \sum_t Z_t' \Sigma^{-1} Y_t \right), \end{aligned}$$

and substituting $y_t \rightarrow Y_t$ into Chapter 10.2's trace-identity result, with E now the $T \times n$ matrix of VAR residuals $E_t = (Y_t - Z_t\beta)'$, gives:

$$\Sigma \mid \beta, Y \sim IW(S_0 + E'E, \nu_0 + T)$$

No new completing-the-square or trace-identity argument is needed: every step of Chapter 10.2's derivation goes through verbatim once Z_t is understood to be built from a repeated common x_t rather than M distinct regressor vectors, since the derivation never used equation-specific regressors beyond this bookkeeping definition of Z_t .

11.3 VAR as a Special Case of SUR

Algorithm: Bayesian VAR via the Chapter 10 SUR Gibbs Sampler

Write Y_t 's n equations in SUR form with Z_t the $n \times nk$ block-diagonal matrix built from the *same* $k = 1 + np$ -vector $x_t = (1, Y_{t-1}', \dots, Y_{t-p}')'$ repeated on every diagonal block — the one substantive difference from Chapter 10, where each equation's block used its own distinct regressors. With this single change, the algorithm is identical to Chapter 10's:

1. **Coefficients.** Draw $\beta \mid \Sigma, Y \sim N(\bar{\beta}, \bar{B})$ with $\bar{B} = (B_0^{-1} + \sum_t Z_t' \Sigma^{-1} Z_t)^{-1}$, $\bar{\beta} = \bar{B}(B_0^{-1}b_0 + \sum_t Z_t' \Sigma^{-1} Y_t)$.
2. **Covariance.** Form the $T \times n$ residual matrix E and draw $\Sigma \mid \beta, Y \sim IW(S_0 + E'E, \nu_0 + T)$.

Both steps are literally the SUR Gibbs sampler from Chapter 10, called with the identical design matrix X passed once per equation rather than n distinct matrices — no new sampler is written for this chapter. The prior places precision 1 (prior variance 1) on every slope coefficient, precision 1/10 (prior variance 10, i.e. looser) on each equation's intercept, with $S_0 = I_n$, $\nu_0 = n + 3$.

Remark: Why This Chapter Has No New Algorithm

Chapters 6 through 10 built up a small toolkit — conjugate Normal regression, truncated-Normal data augmentation, spike-and-slab variable selection, and now the SUR Gibbs sampler — and the VAR is not a reason to extend that toolkit but an occasion to reuse it. Once every equation shares identical regressors, SUR's general $Z_t' \Sigma^{-1} Z_t$ construction collapses to a Kronecker structure, but the *sampler* does not need to know this: the SUR update handles it automatically because equal design matrices are simply a special case of the block-diagonal Z_t it was already built to handle. This is the same lesson as Chapter 9's Tobit-is-probit-plus- Chapter-6 remark, one level up: a new economic model need not imply a new sampler.

11.4 Stock and Watson (2001)

The empirical study this chapter replicates is the paper that taught a generation of economists what a VAR is for: Stock and Watson (2001, hereafter SW), "Vector Autoregressions," *Journal of Economic Perspectives* 15(4). SW organize the entire methodology around one running exam-

ple — a three-variable VAR(4) in price inflation, the unemployment rate, and the federal funds rate, estimated on quarterly U.S. data over 1960Q1–2000Q4 — and use it to demonstrate the four tasks of macroeconometrics: data description, forecasting, structural inference, and policy analysis. This chapter reproduces their reduced-form results (data description and forecasting); Chapter 12 takes up the same system’s structural side. Inflation is constructed exactly as in SW, $\pi_t = 400 \ln(P_t/P_{t-1})$ with P_t the chain-weighted GDP price index (FRED `GDPDEF`); the unemployment rate (`UNRATE`) and the federal funds rate (`FEDFUNDS`) are quarterly averages of monthly values. Following the same three-step logic used throughout this note — replicate, extend, explain — the system is first estimated on SW’s own sample with the Chapter 10 Gibbs sampler ($T = 163$ effective quarters after four initial lags; $k = 13$ regressors per equation), then re-estimated on the full sample through 2026Q1 ($T = 264$).

Empirical Example: SW’s Table 1: Granger Causality, Replicated

SW’s first exhibit is a table of Granger-causality p -values: for each pair of variables, an F -test of whether the four lags of one variable enter the reduced-form equation of another.^a The table below shows the replication side by side with SW’s published values (SW’s in parentheses):

Regressor	π eq.	u eq.	R eq.
π	—	0.35 (0.31)	0.00 (0.00)
u	0.15 (0.02)	—	0.00 (0.00)
R	0.21 (0.27)	0.007 (0.01)	—

Five of the six off-diagonal cells reproduce SW’s published values almost exactly, and every qualitative conclusion of their Table 1 survives: inflation and unemployment each strongly help predict the funds rate (the Taylor-rule-like reaction function embedded in the third equation), the funds rate helps predict unemployment, and the funds rate does not help predict inflation. The one cell that moves is unemployment’s role in the inflation equation — SW found $p = 0.02$, the replication finds $p = 0.15$ — and the discrepancy is informative rather than troubling: the chain-weighted GDP price index has been comprehensively revised several times since 2001, so the inflation series itself is not the one SW used, and the Phillips-curve cell of the table is exactly where deflator revisions would be expected to show up.

^aThe Granger-causality F -test is a frequentist device, and it is reported here exactly as SW report it — as a descriptive statistic computed once from the OLS fit — alongside, not in place of, the Bayesian estimation of the system itself.

Empirical Example: Original Sample, 1960Q1–2000Q4: Posterior vs. OLS

With the Chapter 10 prior (variance 1 on slopes, 10 on intercepts, $S_0 = I_3$, $\nu_0 = 6$) and 10,000 post-burn-in draws, the posterior mean of every coefficient sits close to its OLS counterpart wherever the data speak clearly, and the largest gaps appear exactly where they should: on the heavily collinear unemployment lag polynomial. In the inflation equation, for instance, the OLS coefficients on u_{t-2} and u_{t-3} are 0.935 and -0.799 — large,

offsetting, and individually poorly determined — while their posterior means are 0.474 and -0.396 : the prior shrinks the offsetting pair toward zero without changing their sum, which is the only quantity the likelihood pins down. The largest posterior-mean-to-OLS gap across all 39 coefficients is 0.60; the maximum companion-matrix eigenvalue modulus is 0.973 at the posterior mean (0.973 for OLS as well) — the system is persistent but comfortably stationary, as Figure 41 shows.

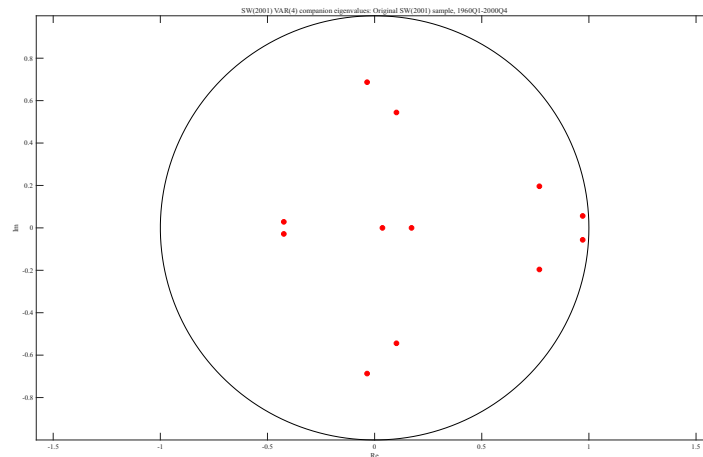


Figure 41: Companion-matrix eigenvalues of the SW VAR(4) at the posterior mean, original sample (1960Q1–2000Q4), against the unit circle

Empirical Example: Full Sample, 1960Q2–2026Q1

Re-estimating the identical model and prior on the series extended through 2026Q1 changes the statistical picture in two instructive ways. First, with 264 effective quarters the prior washes out almost completely: the largest posterior-mean-to-OLS gap falls from 0.60 to 0.014. Second, and economically more telling, the Granger-causality table is no longer SW's. Unemployment no longer helps predict the funds rate ($p = 0.41$, against 0.0000 on SW's sample): a quarter century in which the funds rate sat at the zero lower bound for fourteen years — not moving whatever the unemployment rate did, including through a rise to 13.0% and back — has erased the unemployment side of the estimated reaction function. The funds rate now helps predict inflation ($p = 0.03$, against 0.21 before), while the Phillips-curve cell stays weak ($p = 0.14$). The maximum eigenvalue modulus falls to 0.938 (Figure 42): the full sample looks *less* persistent than SW's, largely because the volatile pre-1985 quarters now share the sample with two long, quiet ZLB episodes. SW themselves flagged precisely this fragility — citing evidence of widespread instability in low-dimensional VARs and documented shifts in the policy rule — and the next section shows what that instability does to the model's forecasts.

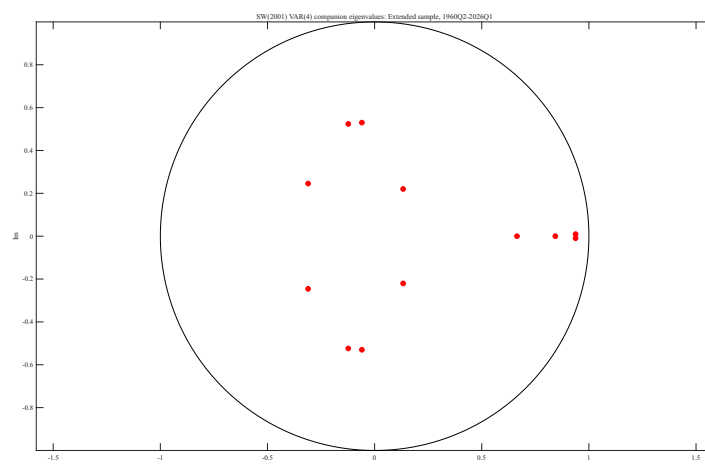


Figure 42: Companion-matrix eigenvalues of the SW VAR(4) at the posterior mean, full sample (1960Q2–2026Q1)

11.5 Forecasting 2001–2026 from the End of SW’s Sample

SW assessed their VAR’s forecasting record with a pseudo out-of-sample exercise ending in 2000Q4, re-estimating the model each quarter, and found it beat univariate autoregressions and random-walk benchmarks. The exercise here is deliberately harsher and answers a different question: the system is estimated *once*, through 2000Q4, and then simulated forward 101 quarters to 2026Q1 with parameters and volatility frozen at their year-2000 posterior — not because anyone would forecast this way in practice, but because the gap between this forecast and reality measures everything about 2001–2026 that a constant-parameter linear model fitted to 1960–2000 cannot represent.

Empirical Example: A Constant-Parameter VAR Meets 2001–2026

Each of 2,000 posterior draws of (Φ, Σ) is simulated forward with its own shock path, so the fan in Figure 43 captures both parameter and shock uncertainty. The hold-out contains the 2001 dot-com recession, the 2008–09 financial crisis, the COVID-19 pandemic, and the 2021–22 inflation surge. Forecast accuracy over the 101 quarters:

	RMSFE	% outside 95% band
Inflation	2.27	0.0%
Unemployment	1.95	7.9%
Fed funds rate	5.09	1.0%

The median forecast of every variable does the only thing a stationary VAR can do at long horizons: revert to the 1960–2000 unconditional mean. For unemployment that means drifting near 6% for twenty-five years, against a realized path that peaks at 9.9% in the financial crisis, spikes to 13.0% in the pandemic — the quarters responsible for nearly all of the 7.9% band violations — and spends most of the 2015–2026 stretch *below* 4.5%, a level the model considers almost equally unlikely in the other direction. For the funds

rate the mean-reversion is more damaging still: the median forecast climbs back toward 6–7% and stays there, while the realized rate spent 2009–2015 and 2020–2022 pinned at 0.25% — the two ZLB episodes modeled directly with the Tobit machinery of Chapter 9 — producing the largest RMSFE of the three variables, 5.09 percentage points. Inflation is the quietest failure: its fan is wide enough that even the 2021–22 surge to 9% annualized stays (barely) inside the 95% band, so its coverage looks perfect while its point forecast — steady 4% inflation for a quarter century — was wrong in level for most of it.

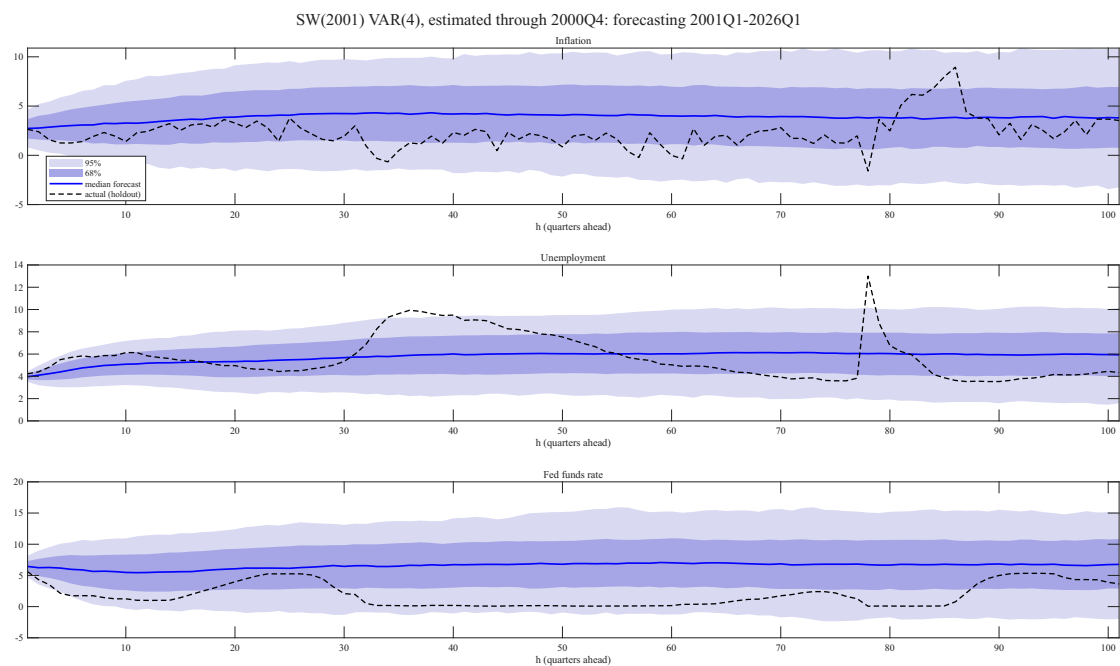


Figure 43: SW VAR(4) estimated through 2000Q4 and simulated forward: median, 68% and 95% predictive bands vs. realized values, 2001Q1–2026Q1

Remark: Interval Coverage Is Not the Same as a Useful Forecast

The funds rate and unemployment tell opposite halves of one cautionary story. The funds rate has nearly perfect formal coverage — 1.0% of realized quarters outside the 95% band — and the worst RMSFE of the three variables, because a band that must span everything a 1960–2000-calibrated tightening cycle might do is wide enough to contain even a fourteen-year stay at zero, while the median inside that band points confidently in the wrong direction for decades. Unemployment shows the reverse: a visibly better point forecast by RMSFE (1.95), but the worst coverage (7.9% outside), because its two crisis spikes are sharp, short, and far outside any band a constant- Σ model can justify. Reading either diagnostic alone would rank the two forecasts in opposite orders; read together, they say the model failed both ways — median wrong for the funds rate, tails wrong for unemployment — and that nothing estimated from 1960–2000 was equipped to represent regimes (a binding lower bound, a pandemic) absent from that sample. This is the book’s standing motivation for the time-varying-parameter and stochastic-volatility chapters ahead (Chapters 14, 16–18).

12. Structural VAR and Impulse Response Analysis

Roadmap

Chapter 11 estimated Σ , the reduced-form covariance of the VAR's residuals, and stopped there deliberately: correlated residuals alone do not say which linear combination of them deserves the label “a monetary policy shock.” This chapter adds nothing to the sampler — (Φ, Σ) are the same posterior draws from Chapter 11 — and instead adds a *deterministic transformation* of those draws, the Cholesky factorization, plus a simulation recipe for propagating a single shock forward. The common-random-numbers trick used to build that simulation reappears essentially unchanged in every later chapter with a non-analytical impulse response (Chapters 16–18).

Chapter 11 deliberately stopped at the reduced-form VAR: the residual covariance $\hat{\Sigma}$ had genuine cross-equation correlation, but nothing in the reduced form assigns an economic label — “a monetary policy shock” — to any particular combination of the correlated residuals e_t . This chapter closes that gap in two stages. First, a recursive (Cholesky) identification scheme and a common-random-numbers impulse response construction are applied to the same Stock and Watson (2001) system estimated in Chapter 11 — reproducing, among other things, its most famous pathology. Second, the chapter replicates the paper that turned sign restrictions into a standard alternative to recursive orderings, Uhlig (2005), on its own monthly six-variable system.

12.1 The Identification Problem

The reduced-form innovations $e_t \sim N(0, \Sigma)$ are, in general, linear combinations of n mutually independent *structural* shocks u_t , $e_t = Bu_t$ with $u_t \sim N(0, I_n)$, so $\Sigma = BB'$. This factorization is not unique — any B satisfying $BB' = \Sigma$ works, and B and BQ for any orthogonal Q give the same Σ — so recovering economically interpretable shocks requires additional identifying assumptions on B beyond what the data alone can deliver.

Remark: The Recursive (Cholesky) Identification Scheme

The simplest and most common choice sets $B = C$, the lower-triangular Cholesky factor of Σ ($\Sigma = CC'$). Lower-triangularity imposes a recursive causal ordering on the variables: the first variable in the ordering can react contemporaneously to nothing but its own shock, the second can react contemporaneously to the first variable's shock and its own, and so on, with the *last* variable in the ordering free to react contemporaneously to every shock in the system. Ordering the Fed funds rate *last* — the Christiano–Eichenbaum–Evans (1996) convention this chapter follows — encodes the assumption that the central bank observes and reacts to unemployment and inflation surprises within the same quarter, while a monetary policy shock itself affects unemployment and inflation only with at least a one-quarter lag.

12.2 Posterior Derivation

This chapter introduces no new prior and estimates no new parameter beyond the reduced-form (Φ, Σ) already sampled by Chapter 11's Gibbs sampler — the posterior full conditionals $\beta \mid \Sigma, Y$ and $\Sigma \mid \beta, Y$ derived there are unchanged. What is new here is a deterministic *transformation* applied to each retained posterior draw, so the derivation task is different in kind from Chapters 6–11: instead of deriving a full conditional to sample from, the goal is to justify (i) that the recursive factor C is a well-defined function of Σ , and (ii) that simulating impulse responses at each posterior draw of (Φ, Σ) produces valid draws from the posterior distribution of the impulse response itself.

Derivation: Existence and Uniqueness of the Recursive Factor C

Because Σ is a covariance matrix, it is symmetric positive definite. The claim is that among all matrices B satisfying $BB' = \Sigma$, exactly one is lower-triangular with strictly positive diagonal entries.

1. **Existence.** Standard linear algebra (the Cholesky decomposition) guarantees that for any symmetric positive-definite Σ , a lower-triangular C with strictly positive diagonal entries and $\Sigma = CC'$ exists, constructed inductively column by column in exactly the manner of Gaussian elimination.
2. **Uniqueness — setup.** Suppose $\tilde{C}$ is another lower-triangular matrix with strictly positive diagonal satisfying $\tilde{C}\tilde{C}' = \Sigma = CC'$. Because C is invertible (its diagonal is strictly positive), define $Q \equiv C^{-1}\tilde{C}$, so $\tilde{C} = CQ$.
3. **Uniqueness — Q is orthogonal.** Substituting $\tilde{C} = CQ$ into $\tilde{C}\tilde{C}' = CC'$ gives

$$CQ Q' C' = CC'$$

Premultiplying by C^{-1} and postmultiplying by $(C')^{-1} = (C^{-1})'$ (both valid since C is invertible) cancels C on both sides:

$$Q Q' = I_n,$$

so Q is orthogonal.

4. **Uniqueness — Q is lower-triangular.** $Q = C^{-1}\tilde{C}$ by construction; C^{-1} is lower-triangular (the inverse of a lower-triangular matrix is lower-triangular) and $\tilde{C}$ is lower-triangular by assumption, and the product of two lower-triangular matrices is lower-triangular, so Q is lower-triangular.
5. **Uniqueness — $Q = I$.** A matrix that is simultaneously lower-triangular and orthogonal must be diagonal (unit row norms together with triangularity leave no room for nonzero off-diagonal entries once the rows above are fixed), and a diagonal orthogonal matrix has ± 1 entries on the diagonal. Since C and $\tilde{C} = CQ$ both carry the fixed positive-diagonal sign convention, every diagonal entry of Q must equal $+1$, i.e. $Q = I$, so $\tilde{C} = C$.

Consequently, setting $B = C$ is not an arbitrary choice of matrix square root but the

unique member of the equivalence class $\{CQ : Q \text{ orthogonal}\}$ that is simultaneously lower-triangular — precisely the recursive causal-ordering restriction of the remarkbox above — with a fixed positive-diagonal sign normalization. This is what makes “order the funds rate last” a well-posed, computable instruction rather than an ambiguous one: given any posterior draw of Σ , its Cholesky factor C is a single specific matrix, not one of many equally valid candidates.

Derivation: The Posterior Distribution of the Impulse Response

The goal is to justify treating the 2,000 simulated impulse-response paths as exact Monte Carlo draws from the posterior of the impulse response itself, not merely as 2,000 deterministic byproducts of the (Φ, Σ) draws.

1. **Definition.** Let $\text{IR}^{(t)} = g_t(\Phi, \Sigma, u_0, \{\varepsilon_s\})$ denote the impulse response at horizon t , computed by the shocked-minus-unshocked common-random-numbers simulation of the algorithm below — a *deterministic* function of a coefficient draw (Φ, Σ) , the fixed structural shock u_0 , and a sequence of shared future innovations ε_s .
2. **Pushforward argument.** If (Φ, Σ) is drawn from its Chapter 11 posterior $p(\Phi, \Sigma | Y)$, then for any fixed u_0 and any realization of the shared $\{\varepsilon_s\}$, $\text{IR}^{(t)}$ is itself a random variable, and because g_t is a fixed deterministic map applied to the random vector (Φ, Σ) , its distribution is exactly the pushforward of $p(\Phi, \Sigma | Y)$ through g_t : for any set A , $P(\text{IR}^{(t)} \in A) = P((\Phi, \Sigma) \in g_t^{-1}(A))$.
3. **Monte Carlo consistency.** By the same Monte Carlo logic used throughout this note to summarize any posterior functional from retained Gibbs draws (for instance, Chapter 11’s companion-matrix eigenvalues, evaluated at each posterior draw of Φ rather than derived analytically), applying a fixed deterministic functional to each of M draws from a distribution produces M exact Monte Carlo draws from the pushforward of that distribution.
4. **Application.** Evaluating g_t at each of the 2,000 retained draws $(\Phi^{(m)}, \Sigma^{(m)})$, $m = 1, \dots, 2000$, therefore produces 2,000 draws $\text{IR}^{(t),(1)}, \dots, \text{IR}^{(t),(2000)}$ that are exact Monte Carlo draws from the posterior of $\text{IR}^{(t)}$.

Their empirical quantiles therefore consistently estimate the posterior credible bands plotted in Figure 45. No new Metropolis-Hastings or Gibbs step is needed for this step precisely because g_t requires no additional latent parameter beyond (Φ, Σ) themselves — it is pure post-processing of draws already in hand.

12.3 Impulse Responses via Common-Random-Numbers Monte Carlo

Figure 44 lays out the computation as a flow rather than a list: the same random innovations drive both paths, so differencing them at the end cancels everything except the one shock that differs.

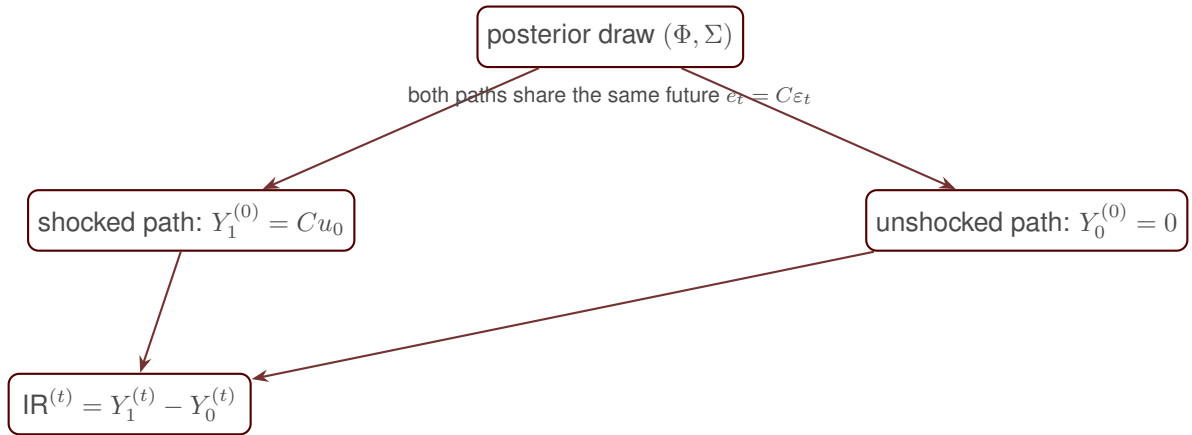


Figure 44: Common-random-numbers construction of a single impulse response: identical future shocks drive both paths, so only the initial structural shock u_0 survives the difference

Algorithm: Structural Impulse Responses

Given a posterior draw (Φ, Σ) and a structural shock vector u_0 (here a one-standard-deviation, i.e. 100 basis point, shock to the funds-rate equation: $u_0 = (0, 0, 1)'/C_{33}$), simulate two paths forward from identical initial conditions using *exactly the same* sequence of future reduced-form shocks $e_t = C\varepsilon_t$:

1. **Shocked path.** $Y_1^{(0)} = Cu_0$; thereafter $Y_1^{(t)} = \Phi x_1^{(t)} + e_t$, where $x_1^{(t)}$ stacks the shocked path's own lags.
2. **Unshocked (counterfactual) path.** $Y_0^{(0)} = 0$; thereafter $Y_0^{(t)} = \Phi x_0^{(t)} + e_t$ using the *same* e_t but the unshocked path's own lags.
3. **Impulse response.** $IR^{(t)} = Y_1^{(t)} - Y_0^{(t)}$.

Because both paths are driven by identical future innovations, differencing them isolates exactly the marginal effect of the single initial shock u_0 , with all subsequent common randomness canceling out — a variance-reduction trick (rather than, say, averaging $\Phi^t Cu_0$ analytically) that carries over unchanged to the nonlinear, time-varying models of Chapters 16–18, where no closed-form impulse response exists. Repeating this for 2,000 posterior draws of (Φ, Σ) and taking quantiles at each horizon produces a full posterior distribution of the impulse response, not just a point estimate at the posterior mean.

12.4 The SW System under Recursive Identification

Chapter 11 replicated the reduced-form half of SW (2001); this section replicates its structural half on the same VAR(4) in (π_t, u_t, R_t) , with the funds rate ordered last, on both the original 1960Q1–2000Q4 sample and the full sample through 2026Q1.

Empirical Example: A 100bp Policy Shock, Original Sample 1960Q1–2000Q4

Applying the common-random-numbers algorithm to 2,000 posterior draws, with the shock normalized to a 100 basis point funds-rate rise on impact, gives Figure 45; the OLS-based deterministic impulse response is overlaid and is visually indistinguishable from the Bayesian posterior median, as Chapter 11's coefficient comparison would predict. Two

features reproduce SW's structural discussion directly. First, unemployment responds with the textbook delayed hump: essentially zero on impact — the recursive timing assumption — then rising to a peak of $+0.194$ percentage points at horizon 8, two full years after the shock, before slowly fading. SW describe exactly this shape and delay for their monetary experiment: a rate hike produces a slowdown whose labor-market effects arrive mostly in the second and third years. Second, inflation *rises* on impact — $+0.246$ points at horizon 1 — and stays positive for roughly a year and a half before drifting persistently negative (about -0.2 points at the longer horizons).

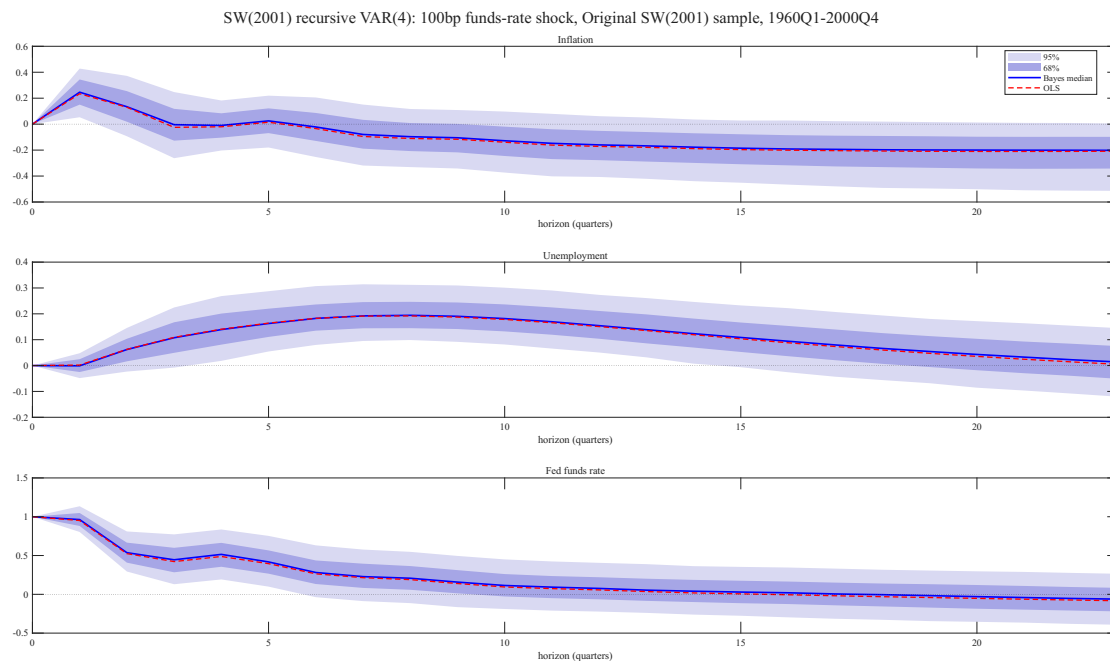


Figure 45: Recursive identification of the SW VAR(4), original sample: posterior median, 68% and 95% bands, and OLS overlay for a 100bp funds-rate shock with the funds rate ordered last

Remark: The Price Puzzle

An inflationary response to a *contractionary* monetary shock runs directly counter to the prediction that tighter policy should cool prices, and it is not an artifact of this particular estimation: it is a long-documented regularity in small recursively-identified monetary VARs, dubbed the “price puzzle” by Eichenbaum (1992) and analyzed by Sims (1992) and Christiano, Eichenbaum, and Evans (1999). SW’s own discussion of the puzzle is the leading explanation: a three-variable system gives the reaction function no way to see the forward-looking information (a commodity-price index is the traditional example) that the Federal Reserve uses to raise rates *in anticipation of* future inflation, so the recursive scheme misattributes anticipatory tightening’s correlation with subsequent inflation to the tightening itself. The standard fix is to add commodity prices to the system — which is exactly what the six-variable model of Uhlig (2005), replicated below, does — rather than to distrust recursive identification as such. This section’s small system is left as is specifically so the failure mode is visible rather than concealed.

Empirical Example: The Same Shock on the Full Sample, 1960Q2–2026Q1

Re-running the identical experiment on the full sample (Figure 46) preserves the broad shapes and changes the details in ways that echo Chapter 11’s Granger-causality findings. The unemployment hump survives but arrives later and slightly smaller — peaking at $+0.181$ points at horizon 16, four years after the shock, against two years on SW’s sample — and the price puzzle gets *worse*, not better: the impact-year inflation response grows to $+0.262$ points, and the median inflation path now never falls meaningfully below zero at any horizon shown. A quarter century of new data has not made the recursive scheme’s “monetary policy shock” look more like one. The reading is the same as Chapter 11’s: the 2001–2026 quarters mix two ZLB episodes, forward guidance, and quantitative easing into the funds-rate equation, so the residual variation the Cholesky scheme labels a policy shock is increasingly the anticipated component that generates the puzzle (Ramey, 2016, surveys the modern identification literature this spawned). This is the strongest empirical motivation the chapter can offer for identification schemes that do not lean on a single mechanical ordering — the subject of the rest of the chapter.

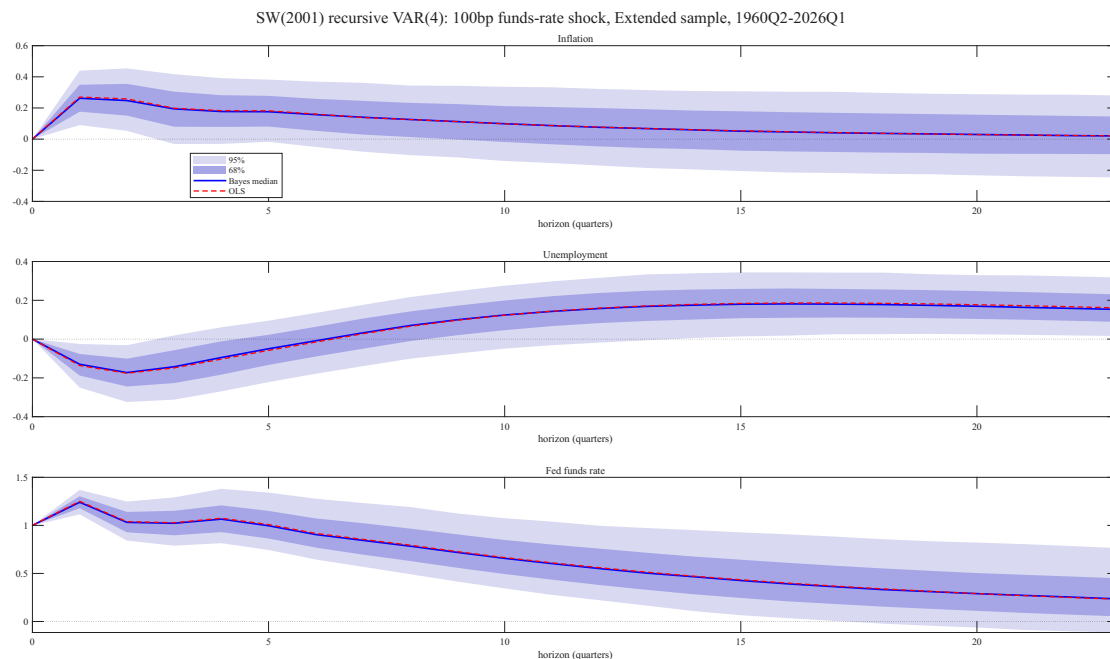


Figure 46: Recursive identification of the SW VAR(4), full sample (1960Q2–2026Q1): the unemployment hump flattens and arrives later, and the price puzzle worsens

12.5 Beyond a Fixed Ordering: Sign and Zero Restrictions

The Cholesky scheme buys a uniquely computable B at a real cost: it requires the researcher to pick one specific causal ordering out of $n!$ possibilities, and Section 12.4’s price puzzle shows how strongly the recursive form can bake a specific, sometimes wrong, causal story into every subsequent number. An alternative line of identification — sign and zero restrictions (Faust, 1998; Uhlig, 2005; Rubio-Ramírez, Waggoner, and Zha, 2010; Arias, Rubio-Ramírez, and Waggoner, 2018) — gives up on pinning down a single B altogether. It instead asks only that the *impulse responses* implied by a candidate B match a small number of signs economic

theory is confident about (a contractionary policy shock should raise the policy rate and not lower prices, say), and accepts *every* rotation consistent with those signs rather than forcing a unique answer.

The Identification Problem, Restated as a Search Over Rotations Recall from Section 12.1 that B is identified only up to right-multiplication by an orthogonal matrix: if $BB' = \Sigma$, then $(BQ)(BQ)' = BQQ'B' = BB' = \Sigma$ for *any* orthogonal Q ($QQ' = I_n$), so BQ is an equally valid structural impact matrix for any such Q . Fixing $B = C$ (Cholesky) is one way of selecting a single Q — namely, the unique Q that makes CQ lower-triangular, by Section 12.2's uniqueness result. Sign restrictions instead search the entire space of orthogonal Q (equivalently, the entire space of possible structural shocks, since each column of CQ is a candidate shock's impact vector) for rotations whose implied impulse responses satisfy the desired sign pattern, keeping the ones that do and discarding the ones that do not — trading a single deterministic answer for a *set* of admissible answers.

Derivation: Drawing Q Uniformly from the Orthogonal Group

Searching “every” orthogonal Q requires being able to draw Q uniformly (Haar-uniformly) from the orthogonal group $O(n)$, and this is less automatic than it looks.

1. **Construction.** Draw an $n \times n$ matrix X with i.i.d. $N(0, 1)$ entries and take its QR decomposition, $X = QR$, with Q orthogonal and R upper-triangular.
2. **The problem.** Q alone is *not yet* Haar-uniform: the factorization $X = QR$ is unique only once a sign convention is fixed (e.g. requiring R 's diagonal to be positive), and standard QR routines (LAPACK's included) do not enforce that convention, so the sign of each column of Q they return is tied to the arbitrary sign of the corresponding diagonal entry of R rather than drawn uniformly.
3. **The fix (Stewart, 1980; Rubio-Ramírez et al., 2010).** Let $D = \text{diag}(\text{sign}(R_{11}), \dots, \text{sign}(R_{nn}))$, a diagonal matrix of ± 1 's, and replace Q with QD . Because $D' = D$ and $DD = I_n$,

$$(QD)(QD)' = QDD'Q' = Q(DD)Q' = QQ' = I_n,$$

so QD is itself orthogonal; and, using $DD = I_n$ again,

$$(QD)(DR) = Q(DD)R = QR = X,$$

with the corrected triangular factor DR having strictly positive diagonal entries $(DR)_{ii} = D_{ii}R_{ii} = \text{sign}(R_{ii})R_{ii} = |R_{ii}| > 0$ by construction of D .

4. **Conclusion.** It is this sign-corrected QD — paired with the now-canonical triangular factor DR , not the raw, sign-arbitrary Q and R returned by a QR routine — that is genuinely Haar-uniform on $O(n)$.

This is exactly analogous to Section 12.2's uniqueness argument for the Cholesky factor: fixing a sign convention on a triangular factor's diagonal is what turns an otherwise-ambiguous decomposition into a well-defined one, here used to make the *sampling*

distribution over Q well-defined rather than to make Q itself unique.

Algorithm: Sign-Restricted Structural VAR

Given a target shock's sign pattern (here: a contractionary policy shock should raise i_t and not raise π_t , for horizons $h = 0, \dots, H - 1$), and a posterior draw (Φ, Σ) :

1. **Candidate rotation.** Compute the Cholesky factor C of Σ , draw a Haar-uniform $Q \in O(n)$ (previous proofbox), and form the candidate structural matrix $B = CQ$.
2. **Candidate shocks.** Each column $B_{\cdot j}$ of B is a candidate unit-structural-shock impact vector; flip its sign if needed so the policy-rate response is non-negative on impact (a normalization, not a restriction).
3. **Check.** Compute the deterministic VAR impulse response to $B_{\cdot j}$ for $h = 0, \dots, H - 1$ and keep this rotation only if i_t 's response is ≥ 0 and π_t 's response is ≤ 0 at every one of those horizons.
4. **Repeat.** If no column of B satisfies the restrictions, draw a fresh Q and try again (up to a maximum number of attempts); if none of the attempts succeed for this (Φ, Σ) , that posterior draw contributes no accepted rotation.
5. **Repeat across posterior draws.** Cycling through many retained (Φ, Σ) draws, each contributing at most one accepted impulse response, builds up an empirical distribution of *sign-identified* impulse responses.

Remark: Zero Restrictions: Why Rejection Sampling Stops Working

Sign restrictions can be supplemented with exact *zero* restrictions (e.g., “output does not respond within the quarter to a monetary shock”), but the algorithm above cannot simply be extended by also rejecting any draw that fails to hit zero *exactly* — under continuous sampling of Q , the probability of ever landing on an exact zero is itself zero, so naive rejection sampling would reject every single draw. Arias, Rubio-Ramírez, and Waggoner (2018) instead build the zero restrictions directly into the construction of Q : each column of Q is drawn not from the whole unit sphere but from the lower-dimensional subspace orthogonal to whatever zero restrictions apply to that shock (constructed column by column via a sequence of Householder reflections, each shrinking the admissible subspace by the restrictions already imposed on earlier columns), so that every draw satisfies the zero restrictions *exactly* by construction, and only the sign restrictions need still be checked by rejection. The mechanics are a direct generalization of the Haar-uniform proofbox above — draw uniformly, but from a restricted subspace rather than all of $O(n)$ — and are not carried out in the empirical example below, which uses sign restrictions only.

12.6 Uhlig (2005)

The sign-restriction methodology above is replicated on the paper that made it standard: Uhlig (2005), “What Are the Effects of Monetary Policy on Output? Results from an Agnostic Identification Procedure,” *Journal of Monetary Economics* 52(2). SW (2001) themselves, closing their discussion of the identification problem, pointed to exactly this line of work —

inference from inequality restrictions alone (Faust, 1998; Uhlig's then-circulating manuscript) — as the constructive way out of the recursive scheme's arbitrariness. Uhlig's design is built around a deliberate refusal: a contractionary monetary policy shock is *defined* by what theory is confident about — for six months after the shock ($h = 0, \dots, 5$), the federal funds rate does not fall, the GDP deflator and commodity prices do not rise, and nonborrowed reserves do not rise — while the two responses the paper actually cares about, real GDP and total reserves, are left completely unrestricted. Whatever the accepted rotations then say about output is a finding, not an assumption; Uhlig's headline result is that the output response is *ambiguous*: roughly as many admissible rotations imply GDP rises after a monetary contraction as imply it falls.

The system is a monthly VAR(12) in six variables — real GDP, the GDP deflator, a commodity price index, nonborrowed reserves, total reserves, and the federal funds rate, the first five in $100 \times \log$ levels — estimated on 1965:1–2003:12 with a near-flat prior and no detrending, exactly as in the paper.² Each candidate posterior draw of (Φ, Σ) is paired with Haar-uniform rotations until one column satisfies all four sign restrictions, that draw then contributing one accepted impulse-response path; shocks are one standard deviation in size, with no renormalization, following the paper's convention.

Empirical Example: Original Sample, 1965:1–2003:12

Figure 47 shows the six responses. The restricted variables behave as they must — the funds rate rises about 0.20 percentage points on impact and decays; the deflator and commodity prices fall persistently; reserves drop — but the unrestricted GDP panel is the result: its median response is slightly *positive* on impact (+0.04), stays within ± 0.1 at every horizon out to five years, and its bands straddle zero throughout. The share of accepted rotations with a positive GDP response is 0.77 on impact, 0.60 at one year, and exactly 0.50 at two years — the distribution is centered on “no effect,” with sizable mass on both signs. This is Uhlig's headline figure and finding: the data plus the agnostic restrictions do *not* force the conclusion that contractionary policy lowers output, and the confident output declines of recursively-identified studies were, to that extent, consequences of the recursive assumption rather than of the data.

²Two data approximations are unavoidable a quarter century on and are disclosed here: the commodity price index is proxied by the all-commodities PPI (FRED `PPIACO`), Uhlig's Bartlett-interpolated Commodity Research Bureau index no longer being distributed in real time; and monthly real GDP and deflator values are obtained by log-linear interpolation of the quarterly series anchored at mid-quarter, in the spirit of the paper's own monthly interpolation of quarterly aggregates.

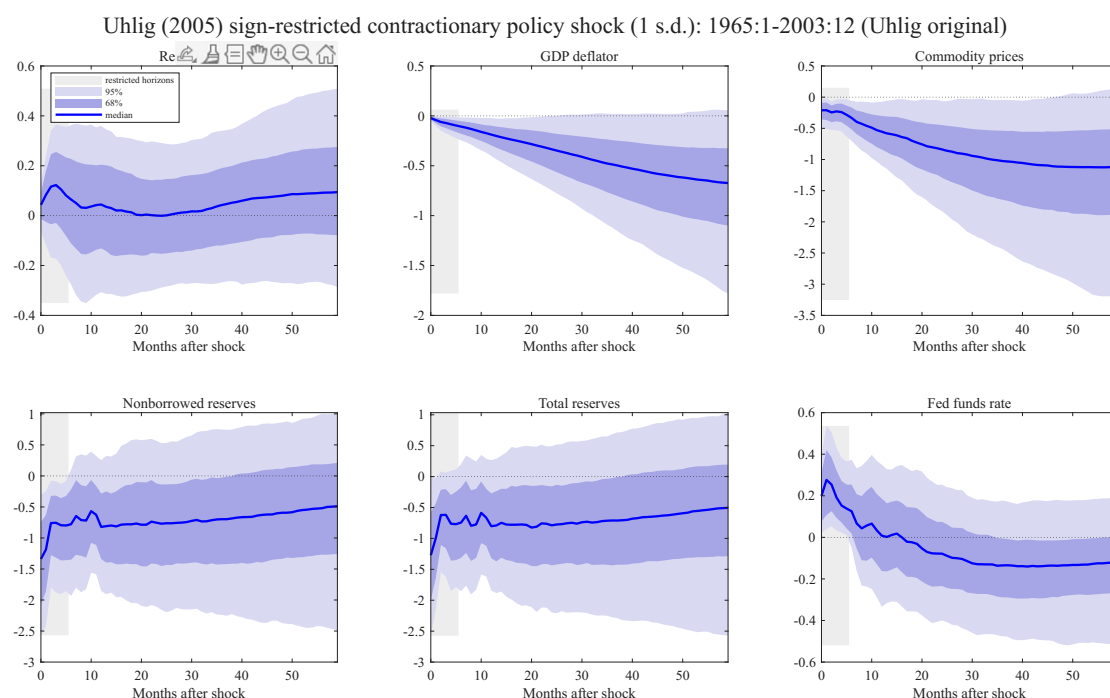


Figure 47: Uhlig (2005) replicated: responses to a sign-restricted contractionary policy shock (one standard deviation), original sample. Shaded strip marks the restricted horizons $h = 0-5$; GDP and total reserves are unrestricted.

Empirical Example: Full Available Sample, 1965:1–2007:12

Extending Uhlig's system forward runs into an instructive wall: it can be taken only to 2007:12. The identification rests on nonborrowed reserves, and in early 2008 that series turns *negative* — term auction facility credit came to exceed total reserves — so its logarithm, and with it Uhlig's specification, ceases to exist; the remarkbox below takes up what happened afterward. On the four additional years that are available, the replication's conclusions are unchanged: the median GDP response is $+0.04$ on impact, drifts slightly negative around two years (-0.03), and the share of positive-GDP rotations at that horizon is 0.44 — modestly below one half, but nowhere near a confident output decline (Figure 48). The noticeably wider reserve-response bands are the sample quietly announcing its own ending: reserves demand was already drifting away from the stable relationships the identification leans on.

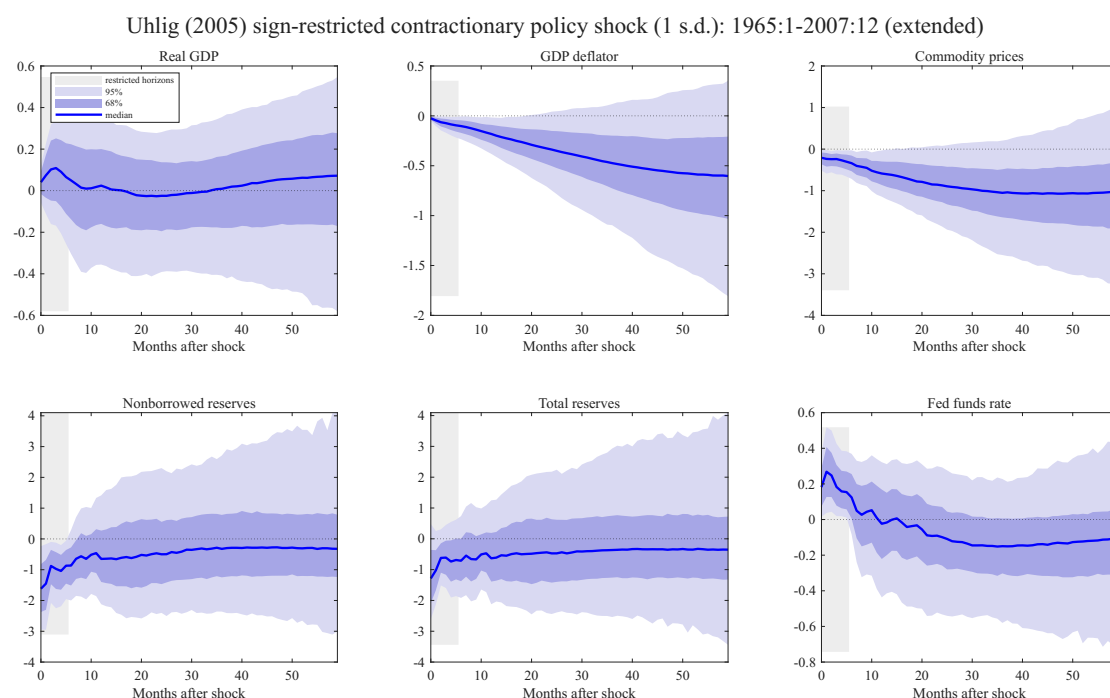


Figure 48: The same sign-restricted system estimated through 2007:12, the last month before nonborrowed reserves turn negative: the agnostic conclusion about output is unchanged

Remark: Why This Replication Cannot Reach the Present

Every other empirical chapter of this note extends its paper's sample to the current quarter; this one cannot, and the reason is itself a lesson in identification. Uhlig's restrictions treat nonborrowed reserves as a meaningful margin of monetary policy — the quantity the Fed adjusts when it moves the funds rate, which is why a contraction must not *raise* them. That description fits the corridor system of 1965–2007. It stopped fitting in October 2008, when the Fed began paying interest on reserves and, through successive rounds of quantitative easing, moved to an ample-reserves regime in which the funds rate is steered administratively while reserves are orders of magnitude larger than banks require; nonborrowed reserves themselves went negative in 2008 because crisis lending facilities were counted against them. In that world the sign restriction “a tightening does not raise nonborrowed reserves” is neither true nor false so much as beside the point — the variable no longer measures the policy margin the identifying assumption is about. The general lesson generalizes well beyond this paper: identifying assumptions are statements about institutions, not just about matrices, and they can expire when the institutions change. The modern monetary-shock literature responded by moving to identification from high-frequency financial-market surprises around policy announcements (Ramey, 2016, surveys this development), which requires no assumption about reserves at all.

13. Large Bayesian VARs and the Minnesota Prior

Roadmap

Chapters 11–12 kept the VAR deliberately small, three variables, so that identification and estimation could be studied without the added complication of a large parameter count. This chapter removes that simplification and asks what happens as n actually grows, on Banbura, Giannone and Reichlin’s (2010) own 131-variable monthly dataset: Section 13.1 quantifies why growth is a problem at all, Section 13.4 checks (and confirms, via BGR’s own calibration) that shrinkage barely matters yet at $n = 3$, and Sections 13.5–13.6 build the machinery — dummy observations, fit-matched tightness — that makes shrinkage the actual solution once n reaches the hundreds. Section 13.7 then asks the natural follow-up question, whether the shrinkage parameter λ itself should be calibrated or estimated, which is the same question Chapter 8 asked about which regressors belong in a model, now applied to a single tuning parameter instead of k binary indicators; Section 13.8 pushes this further still, to per-coefficient shrinkage; and Section 13.9 closes the chapter by asking directly which of the three answers actually forecasts best.

Chapters 11–12 held the VAR small: three variables, two lags, $k = 7$ regressors per equation. Applied macro forecasters routinely want more variables — more sectors, more countries, more financial indicators — and more lags to capture richer dynamics, but every additional variable adds not one but $O(n)$ new regressors to *every* equation, since each equation’s regressor list is every variable’s every lag. This chapter’s entire empirical program — every table, every figure, every calibration — now runs on the original replication dataset of Banbura, Giannone and Reichlin (2010, *Journal of Applied Econometrics* 25:71–92, hereafter BGR): the Stock and Watson (2005) panel of 131 monthly U.S. macro-financial series, 1959M1–2003M12, obtained directly from the authors’ own published replication code and estimated at BGR’s own lag order $p = 13$.

Remark: Acknowledgment

This chapter’s entire large-BVAR empirical program — dataset, variable groupings, dummy-observation estimator, and Fit-matching calibration rule — follows Banbura, Giannone and Reichlin (2010) as closely as a textbook treatment allows, using their own publicly released replication files rather than a reconstruction. Earlier drafts of this chapter used a smaller, independently assembled 25-variable quarterly panel; that panel is retired in this edition in favor of the authors’ original monthly data specifically so that every number in this chapter can be checked, in principle, against BGR’s own published Table 1. Any remaining errors in transcription, translation from MATLAB to Python, or exposition are the present author’s alone.

It then does what a “Large” Bayesian VAR is actually for: following BGR and De Mol, Giannone and Reichlin (2008), it grows the system in three steps, exactly BGR’s own SMALL/MEDIUM/LARGE progression — **SMALL** ($n = 3$: total nonfarm employment, CPI inflation, and the federal funds rate, BGR’s three headline forecast targets), **MEDIUM** ($n = 20$, BGR’s own hand-picked “key macroeconomic indicators”: the three SMALL series plus real income, real consumption, industrial production, capacity utilization, the unemployment rate, housing starts, an index of sensitive materials prices, the PCE price deflator, producer prices, average hourly earnings, M1 and M2, total and non-borrowed bank reserves, the S&P 500 index, the 10-year Treasury yield, and the trade-weighted dollar exchange rate) and **LARGE** ($n = 131$, every

series in the Stock-Watson panel) — and asks whether a shrinkage-disciplined *large* system forecasts the three named target variables better than the small system does, with overall shrinkage *recalibrated at each size* by BGR’s own Fit-matching rule so the comparison is not contaminated by one model simply being allowed a looser in-sample fit than another. This is the actual empirical case for LBVARs, reproduced here on the authors’ own data rather than approximated on a different panel.

13.1 The Curse of Dimensionality in VARs

Remark: Why “Large” VARs Are Hard

A VAR(p) in n variables has $k = 1 + np$ regressors per equation, so the slope-and-intercept parameter count alone is $n \cdot k = n(1 + np)$, and the innovation covariance Σ contributes a further $n(n + 1)/2$ free elements. BGR’s own monthly lag order $p = 13$ (roughly a year of monthly dynamics) makes this grow far faster than the quarterly, four-lag systems used elsewhere in these notes: total parameter count runs from 126 at $n = 3$ (SMALL) to 5,430 at $n = 20$ (MEDIUM) to 231,870 at $n = 131$ (LARGE; Table 1) — roughly quadratic in n for fixed p , and linear in p for fixed n , since every one of the n equations picks up n new regressors per lag. Monthly macro samples of a few hundred observations are nowhere close to sufficient to pin down a quarter of a million free parameters by unrestricted least squares; this is the exact curse of dimensionality that motivated Litterman’s original shrinkage proposal, and BGR’s own paper exists specifically to show how far shrinkage can push n before running out of road.

	SMALL ($n = 3$)	MEDIUM ($n = 20$)	LARGE ($n = 131$)
$k = 1 + np$ (regressors/equation)	40	261	1,704
$n \cdot k$ (slope + intercept)	120	5,220	223,224
$+ n(n + 1)/2$ (Σ)	6	210	8,646
Total parameters	126	5,430	231,870

Table 1: Total VAR(p) parameter count $n(1 + np) + n(n + 1)/2$ for Banbura, Giannone and Reichlin’s own SMALL/MEDIUM/LARGE systems, monthly lag order $p = 13$ fixed throughout

The LARGE system used in this chapter’s out-of-sample exercise (Section 13.6) is where the pressure becomes genuinely severe: 223,224 slope-and-intercept coefficients across 131 equations, estimated on a rolling 120-month (ten-year) window — nearly 1,900 slope parameters for every observation available at each re-estimation, a regime in which unrestricted OLS equation by equation is not merely imprecise but often not even uniquely defined ($k = 1,704$ regressors, fewer than 120 effective observations per window). A flat prior cannot discipline a system this size at all; Section 13.6 shows exactly how BGR’s dummy-observation shrinkage prior repairs it, with λ calibrated to fall as n grows rather than held at a fixed textbook default.

13.2 The Minnesota (Litterman) Prior

Algorithm: Minnesota Prior Construction

Following Doan, Litterman and Sims (1984) and Litterman (1986), the Minnesota prior imposes two beliefs directly on the Normal prior mean b_0 and (diagonal) covariance B_0 that `gibbs_sur` already accepts, with no change to the sampler itself:

1. **Random-walk prior means.** Each variable's own coefficient at lag 1 is centered at 1, not 0: individual macro series are, a priori, closer to a random walk than to white noise. Every other slope coefficient (cross-variable, and own-variable at lags 2 and higher) is centered at 0.
2. **Lag-decaying, scale-adjusted shrinkage.** The prior variance on variable j 's coefficient at lag ℓ in equation i is

$$\text{Var} = \begin{cases} (\lambda_1/\ell^{\lambda_3})^2 & i = j \\ (\lambda_1\lambda_2/\ell^{\lambda_3})^2 \cdot \sigma_i^2/\sigma_j^2 & i \neq j, \end{cases}$$

where σ_i^2 is the residual variance of a univariate AR(p) fit to variable i alone (a preliminary scale estimate, computed once by OLS), λ_1 is the overall tightness, $\lambda_2 < 1$ shrinks cross-variable lags more aggressively than own-variable lags, and λ_3 controls how quickly the prior tightens as lag length grows. This chapter uses the standard textbook values $\lambda_1 = 0.2$, $\lambda_2 = 0.5$, $\lambda_3 = 1$. Intercepts keep a loose, variable-independent prior variance of 100.

Because b_0 and B_0 are ordinary Gibbs-sampler inputs, estimating a Minnesota-prior VAR requires zero new code beyond constructing these two objects — the same `gibbs_sur` routine used in Chapters 10–12 is called unchanged.

13.3 Posterior Derivation

The Minnesota prior changes only the *inputs* (b_0, B_0) that Chapter 10.2's derivation treats as fixed, known constants; it changes nothing about the likelihood $p(Y | \beta, \Sigma)$ or the algebraic steps that produced the full conditionals. Re-examining Chapter 10.2's proof of $\beta | \Sigma, Y$ confirms this directly: b_0 and B_0 enter that derivation only as the prior mean and prior precision B_0^{-1} appearing in the terms $B_0^{-1}b_0$ and B_0^{-1} of the completed square, and at no point does the derivation use any property of b_0, B_0 beyond their being a fixed vector and a fixed positive-definite matrix — properties the (diagonal) Minnesota B_0 and random-walk-centered b_0 satisfy exactly as well as any other choice. Likewise, $\Sigma | \beta, Y$'s derivation never referenced β 's prior at all. Hence

$$\begin{aligned} \beta | \Sigma, Y &\sim N(\bar{\beta}, \bar{B}), & \bar{B} &= \left(B_0^{-1} + \sum_t Z_t' \Sigma^{-1} Z_t \right)^{-1}, & \bar{\beta} &= \bar{B} \left(B_0^{-1} b_0 + \sum_t Z_t' \Sigma^{-1} Y_t \right), \\ \Sigma | \beta, Y &\sim IW(S_0 + E' E, \nu_0 + T) \end{aligned}$$

hold under the Minnesota prior with literally the same proofs as Chapter 10.2's, merely with b_0 now the random-walk-centered mean vector and B_0 now the diagonal lag-decaying, cross-

variable-shrinking covariance matrix of the algobox above substituted in place of the flat prior's $b_0 = 0$, $B_0 = \text{diag}(100 \text{ on intercepts, } 1 \text{ elsewhere})$. This is precisely why the algobox above needs no new Gibbs step: a different numerical (b_0, B_0) plugged into an already-derived posterior is a change of prior *inputs*, not a change requiring new derivation.

13.4 A Fixed Small System: Does Shrinkage Even Matter?

Section 13.1 already flagged that $n = 3$ is a comparatively mild curse-of-dimensionality case: only 126 parameters, and BGR's own SMALL system uses just 40 regressors per equation. The actual empirical case for LBVARs, addressed in full below, is about what happens when n grows. But it is worth checking a narrower, prior question first: holding $n = 3$ fixed, does shrinkage change anything at all, or is unrestricted OLS already fine at this size?

Empirical Example: BGR's Own Fit-Matching Answers This Directly, at $n = 3$

Section 13.5 below calibrates the dummy-observation prior's overall tightness λ separately at every model size, by grid search against a fixed in-sample fit target. Applied to BGR's SMALL system ($n = 3$: total nonfarm employment, CPI inflation, the federal funds rate) on the 1960M2–1970M1 pre-evaluation training window, this calibration selects $\lambda = \infty$ — *unshrunk OLS* — as the tightness that best matches SMALL's own target in-sample fit, precisely because SMALL *is* the model whose OLS fit defines the target Fit^* in the first place (Section 13.5's algobox). With $k = 40$ regressors per equation and a multi-decade monthly training sample, $n = 3$ is small enough that a flat prior is not yet fighting the curse of dimensionality Section 13.1 quantified: there is essentially nothing here for a Minnesota-style prior to correct, exactly the same qualitative conclusion this section reached on these notes' own Chapter 11–12 quarterly VAR before the dataset switch to BGR's data, now confirmed directly by BGR's own calibration procedure — and double-checked by the explicit flat-versus-Minnesota horse race in the box below.

Empirical Example: A Direct Horse Race at $n = 4$: Flat vs. Minnesota, Rolling Forecasts

As a cross-check that does not rely on BGR's calibration machinery at all, take a four-variable quarterly VAR(4) — GDP growth, CPI inflation, the unemployment rate and the federal funds rate, 1959Q3–2026Q1 — and re-estimate it at every forecast origin from 2008Q2 onward (64 recursive-window origins) twice: once under a flat prior and once under Section 13.2's Minnesota prior. Point-forecast RMSFEs:

	$h=1$		$h=8$	
	flat	Minnesota	flat	Minnesota
GDP growth	7.71	7.30	6.19	6.24
CPI inflation	3.41	2.80	2.95	2.76
Unemployment	1.64	1.45	2.25	2.24
Federal funds rate	1.08	0.56	2.50	2.25

At $h = 8$ the two priors sit within a few percent of each other on every variable (the flat prior even edges out Minnesota on GDP growth): with $n = 4$ and hundreds of quarterly observations, the data already discipline OLS, which is the same conclusion the fit-matching box just reached by a different route. The one visible exception is short-horizon persistence: the funds-rate RMSFE at $h = 1$ falls by half under the Minnesota prior, because the evaluation window opens in 2008Q2 and is dominated by zero-lower-bound years, where the prior's random-walk centering keeps next quarter's forecast pinned at the current, near-zero rate while unrestricted OLS keeps forecasting mean reversion upward. Figure 49 confirms that outside such episodes the two forecast paths are nearly indistinguishable: at $n = 4$, shrinkage is a modest refinement, not a rescue — precisely the baseline against which this chapter's $n = 131$ results below should be read.

Ch.13: rolling 1-quarter-ahead point forecasts, flat vs. Minnesota VAR(4)

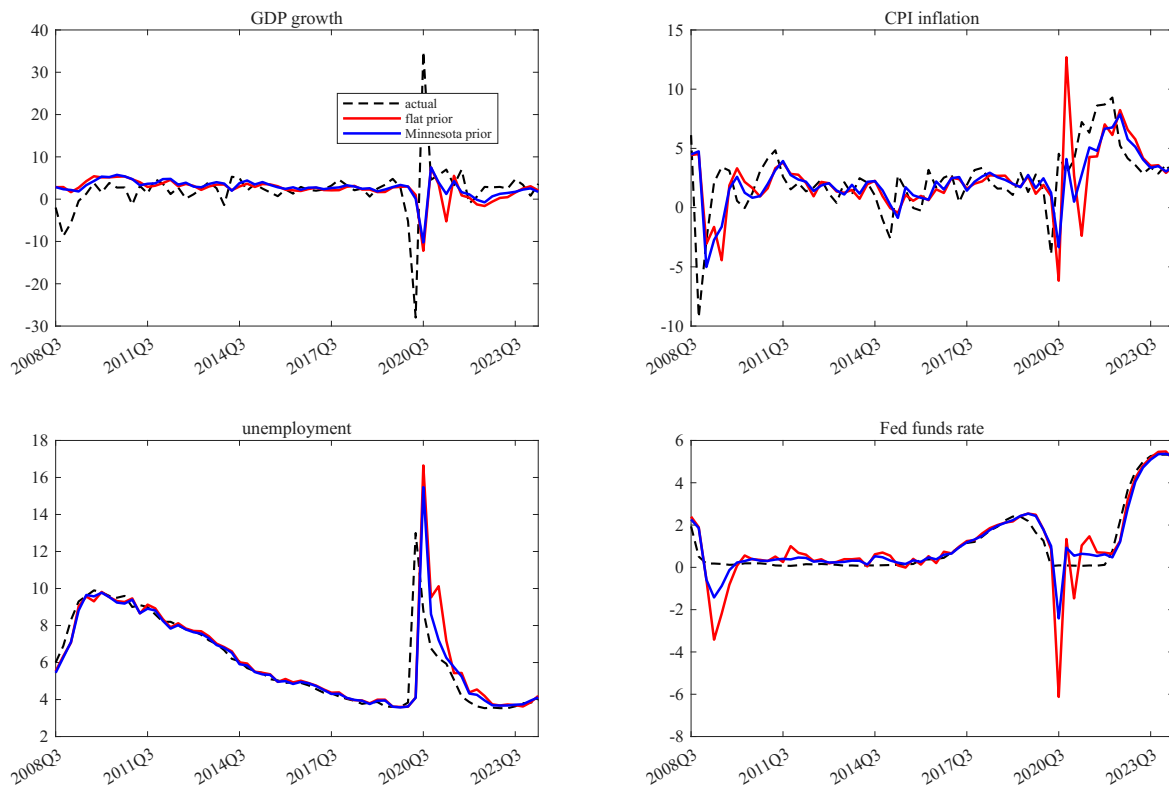


Figure 49: Rolling one-quarter-ahead point forecasts under flat and Minnesota priors, $n = 4$ quarterly VAR(4), 64 recursive-window origins from 2008Q2; the two paths separate visibly only in the most volatile quarters

Remark: Shrinkage Is Close to Redundant When n Is Small

The lesson is the same one these notes have made before, at a different dataset and a different n : with a training sample of hundreds of monthly observations against only 126 total parameters, OLS is already well disciplined by the data at $n = 3$. That changes entirely once n grows toward BGR's own MEDIUM (5,430 parameters) and LARGE (231,870 parameters) sizes quantified in Section 13.1 — which is exactly the question the rest of

this chapter answers, and exactly why BGR's own Fit-matching rule (Section 13.5) drives λ down sharply as n grows rather than holding it fixed at SMALL's $\lambda = \infty$.

13.5 The Dummy-Observation Implementation and Fit-Matched Shrinkage

Section 13.2's classical Minnesota prior plugs (b_0, B_0) into `gibbs_sur` and estimates by MCMC — correct, and adequate for the small-system illustration above, but not how Banbura, Giannone and Reichlin (2010) actually implement or calibrate the prior for a genuinely *large* system, and not what makes their central claim possible to test. Two things are needed to answer the size question properly: a way to estimate models with hundreds of parameters without running MCMC at all, and a principled rule for how tight λ should be at each size, rather than fixing it at a textbook default. This section derives both.

Algorithm: Dummy Observations for the Normal-Inverted-Wishart Prior

1. Write the VAR as the multivariate regression $Y = XB + U$, Y ($T \times n$), X ($T \times k$), $k = 1 + np$, and impose a normal-inverted-Wishart prior on the coefficient matrix and the innovation covariance jointly:

$$\underbrace{\text{vec}(B)}_{\text{coefficients}} \mid \Sigma \sim N(\text{vec}(B_0), \Sigma \otimes \Omega_0),$$

$$\underbrace{\Sigma}_{\text{innovation covariance}} \sim iW(S_0, \alpha_0).$$

2. Construct $T_d = n(p + 1) + 1$ dummy observations (Y_d, X_d) , stacked in three blocks, that Banbura et al. (2010, eqs. 4–5) show reproduce this prior exactly:

$$Y_d = \begin{pmatrix} \text{diag}(\delta_1 \sigma_1, \dots, \delta_n \sigma_n) / \lambda \\ 0_{n(p-1) \times n} \\ \text{diag}(\sigma_1, \dots, \sigma_n) \\ 0_{1 \times n} \end{pmatrix} \quad X_d = \begin{pmatrix} J_p \otimes \text{diag}(\sigma_1, \dots, \sigma_n) / \lambda & 0_{np \times 1} \\ 0_{n \times np} & 0_{n \times 1} \\ 0_{1 \times np} & \varepsilon \end{pmatrix}$$

with $J_p = \text{diag}(1, 2, \dots, p)$, σ_i^2 the AR(p) residual variance of series i (same preliminary scale estimate as Section 13.2), and ε a tiny number giving the intercept an essentially flat prior. The first block centers each series' own first-lag coefficient at δ_i , with variance shrinking in λ and in lag length; the second block fixes the prior scale of Σ ; the third leaves the intercept undisciplined.

3. Stack $Y^* = (Y', Y_d')'$, $X^* = (X', X_d')'$ and run OLS on the augmented system to get the posterior directly, with no MCMC required:

$$\hat{B} = (X^{*'} X^*)^{-1} X^{*'} Y^*,$$

$$\hat{\Sigma} = (Y^* - X^* \hat{B})' (Y^* - X^* \hat{B}),$$

$$\text{vec}(B) \mid \Sigma, Y \sim N(\text{vec}(\hat{B}), \Sigma \otimes (X^{*'} X^*)^{-1}).$$

- Interpret the two limits: as $\lambda \rightarrow \infty$ the first block's rows vanish and $\hat{B}$ reduces exactly to OLS on the real data alone; as $\lambda \rightarrow 0$, B is pinned to the prior mean. This closed form (not the classical Gibbs sampler above) is what Banbura et al. actually use for their large systems, and it is what makes the calibration below computationally possible.

Remark: Choosing δ_i : BGR's Own Convention, Used Exactly

BGR set $\delta_i = 1$ (random walk) for every series they classify, in Stock and Watson's (2005) coding scheme, as needing a first-difference or log-difference transformation to be stationary ($\text{tcode} \in \{2, 5, 6\}$ — essentially every real-activity and price series in the panel, since these are treated as $I(1)$), and $\delta_i = 0$ (white noise) for series Stock and Watson code as already stationary in levels or log-levels ($\text{tcode} \in \{1, 4\}$ — interest rates and a handful of ratio-type series). Crucially, this chapter also inherits BGR's less obvious convention that goes with it: every variable enters the VAR itself in *levels* or *log-levels*, *never* pre-differenced into a growth or inflation rate, even for the series whose prior mean is $\delta_i = 1$. This is a deliberate modeling choice, not an oversight — Litterman's original random-walk prior is designed precisely to let the *prior*, not a hard pre-transformation, do the work of handling near-unit-root behavior, and BGR's own published results are on levels data. This chapter now follows that convention exactly, on the authors' own data, rather than the uniform $\delta_i = 0.5$ -on-a-pre-differenced-panel convention used in earlier editions of this chapter.^a

^aOne consequence is worth stating plainly rather than hidden: because the prior *mean* itself sits exactly on the boundary of the stationary region ($\delta_i = 1$) for most series, hard $\rho < 1$ rejection sampling — used for this chapter's own quarterly panel in earlier editions — is ill-posed here: empirically, over 80% of individual Gibbs draws under Section 13.7's hierarchical prior have companion-matrix spectral radius $\rho \geq 1$ on the MEDIUM system, simply because the prior mean puts every $I(1)$ -classified variable's own dynamics right at the edge. This chapter therefore reports stationarity honestly as a diagnostic, following BGR's own dummy-observation estimator (which does not hard-enforce stationarity either), rather than rejection-sampling it away.

Algorithm: Fit-Matched Overall Tightness (Banbura et al. 2010, Section 3, p. 77)

Banbura et al.'s central argument is not “add a Minnesota prior once n gets large” but that λ must *decrease* as n grows, calibrated so every model size achieves the *same* in-sample fit.

- On a pre-evaluation training sample $t = 1, \dots, T_0 - 1$, define the in-sample one-step fitted MSFE of model m at tightness λ for target i , using $\hat{B}$ estimated once on that same sample:

$$\text{msfe}_i^{\lambda, m} = \frac{1}{T_0 - p - 1} \sum_{t=p}^{T_0-2} \left(\underbrace{\hat{y}_{i, t+1|t}}_{\text{fitted at tightness } \lambda} - \underbrace{y_{i, t+1}}_{\text{realized}} \right)^2,$$

and the analogous msfe_i^0 computed the same way from a random-walk-with-drift benchmark.

- Define the *Fit* of model m at tightness λ , and the tightness λ_m that matches a target fit

level:

$$\text{Fit}(\lambda, m) = \frac{1}{|I|} \sum_{i \in I} \frac{\text{msfe}_i^{\lambda, m}}{\text{msfe}_i^0}, \quad \lambda_m = \arg \min_{\lambda} |\text{Fit}(\lambda, m) - \text{Fit}^*|,$$

where I is the set of three named targets and the desired fit Fit^* is fixed, throughout, at whatever fit OLS ($\lambda = \infty$) achieves on the *smallest* model: $\text{Fit}^* = \text{Fit}(\infty, \text{SMALL})$.

3. Solve for λ_m by bisection: $\text{Fit}(\lambda, m)$ is monotonically decreasing in λ (more freedom fits the training sample better), so the root is unique.
4. Note why this forces shrinkage to scale with system size: larger m can trivially drive Fit far below Fit^* at any fixed λ (more free parameters, better raw in-sample fit, i.e. overfitting), so matching Fit^* forces λ_m *down* as n grows — shrinkage that scales with system size is the mechanism, not an afterthought. This calibration needs one linear solve per candidate λ per model, feasible only because of the closed-form posterior above; it would require a full `gibbs_sur` MCMC run per grid point under the classical Gibbs sampler's approach, infeasible for a λ search at $n = 131$.

13.6 The Full Comparison: Fit-Matched Shrinkage Across $n = 3, 20, 131$

This is BGR's own headline exercise, reproduced on their own data: three model sizes built by adding series to the three named forecast targets (total nonfarm employment, CPI inflation, the federal funds rate) — **SMALL** ($n = 3$, named series only), **MEDIUM** ($n = 20$, BGR's own seventeen further "key macroeconomic indicators," listed in the chapter introduction) and **LARGE** ($n = 131$, the entire Stock-Watson panel). Lag order is BGR's own $p = 13$ monthly lags throughout, and the evaluation window is BGR's own: a $T_0 = 120$ -month (ten-year) rolling re-estimation window, forecast origins from 1971M1 through 2003M1, horizons $h = 1, 3, 6, 12$ months.

Remark: Why $n = 20$ Already Captures Most of the Gain

BGR's own paper anticipates a question this section's results confirm directly: is **LARGE** ($n = 131$) worth the enormous extra parameter count over **MEDIUM** ($n = 20$, Section 13.1's Table 1)? Their own conclusion is no — "good performance is already obtained with the medium size system containing twenty key macroeconomic indicators" — and the numbers below bear this out on target-by-target inspection: **LARGE** beats **MEDIUM** clearly for employment and modestly for inflation, but for the federal funds rate the two are essentially tied at $h = 3$ – 6 and **MEDIUM** is clearly better at $h = 12$ (RMSFE table below). Koop's (2013) independent 168-variable size-scaling study reaches the same qualitative conclusion on a different panel: a well-shrunk 20-variable VAR captures most of the achievable forecasting gain, with further increases in n producing diminishing and occasionally negative marginal returns once shrinkage is properly calibrated at each size. Growing n from 20 to 131 is not free even under Fit-matched shrinkage; it buys real but target-specific improvement, not a uniform upgrade.

Empirical Example: Calibrated Tightness Falls as n Grows

Fit-matching on the pre-evaluation window (1960M2–1970M1, BGR's own calibration sample) against $\text{Fit}^* = \text{Fit}(\infty, \text{SMALL}) = 0.660$ (OLS on the small model fits the training sample to 66.0% of the naive random-walk-with-drift benchmark's in-sample MSFE) gives

	SMALL ($n=3$)	MEDIUM ($n=20$)	LARGE ($n=131$)
λ_m	∞ (OLS)	0.099	0.032
$\text{Fit}(\lambda_m, m)$	0.660	0.660	0.660

monotonically tighter shrinkage as n grows, exactly BGR's claim, and by construction every model size reaches the identical in-sample fit — so none of the differences reported below can be attributed to one model simply fitting the training data better or worse than another. Figure 50 shows the same pattern at BGR's own intermediate size, CEE ($n = 7$, the Christiano-Eichenbaum-Evans variable set), confirming λ falls smoothly rather than jumping only between the three headline sizes.

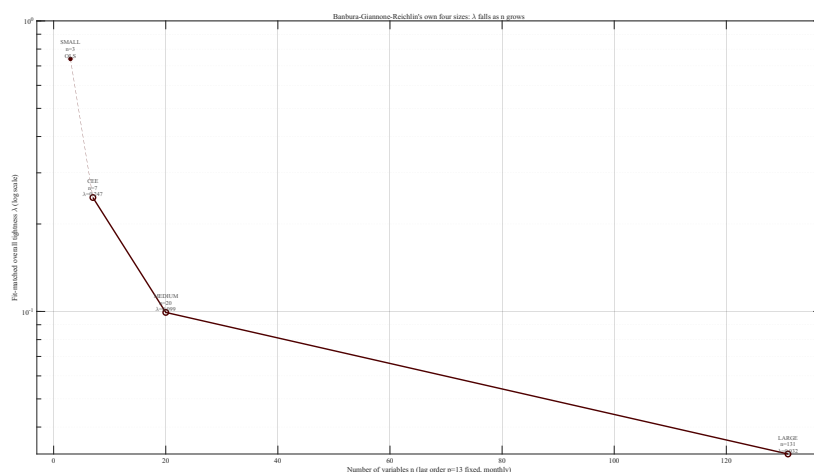


Figure 50: Fit-matched overall tightness λ against the number of variables n (log scale on λ), BGR's own SMALL/CEE/MEDIUM/LARGE sizes: shrinkage tightens monotonically as the system grows, holding in-sample fit fixed at $\text{Fit}^* = 0.660$

Empirical Example: Out-of-Sample RMSFE, Relative to a Random-Walk-with-Drift Benchmark

With λ_m fixed at its calibrated value, each model is re-estimated on a rolling 120-month window at every forecast origin from 1971M1 through 2003M1, and forecasts are evaluated at $h = 1, 3, 6, 12$ months ahead, exactly BGR's own reported horizon set. $\text{RMSFE}_{i,h}^m$ is the ratio of model m 's mean squared forecast error to the same random-walk-with-drift benchmark's mean squared forecast error at that horizon (values below 1 beat the naive

benchmark; full horizon breakdown in Figure 51):

	$h=1$	$h=3$	$h=6$	$h=12$
Employment				
SMALL	1.144	0.946	1.107	1.017
MEDIUM	0.538	0.509	0.664	0.870
LARGE	0.455	0.381	0.503	0.797
CPI inflation				
SMALL	0.895	0.655	0.636	0.811
MEDIUM	0.497	0.404	0.393	0.465
LARGE	0.498	0.396	0.395	0.435
Federal funds rate				
SMALL	1.863	1.764	2.078	2.547
MEDIUM	0.779	0.948	1.295	1.468
LARGE	0.753	0.943	1.288	1.920

This is a far cleaner picture than a single shared-tightness system typically delivers: growing from SMALL to MEDIUM improves every target at every horizon, often by half or more (employment's $h = 1$ RMSFE falls from 1.144, worse than the naive benchmark, to 0.538; the funds rate's from 1.863 to 0.779). Growing further to LARGE keeps improving employment and modestly improves inflation, but for the funds rate LARGE's advantage disappears: essentially tied with MEDIUM at $h = 3-6$, and clearly *worse* at $h = 12$ (1.468 $\rightarrow$ 1.920) — exactly the “ $n = 20$ already captures most of the gain” pattern the remark above previews, now visible target by target rather than only in aggregate.

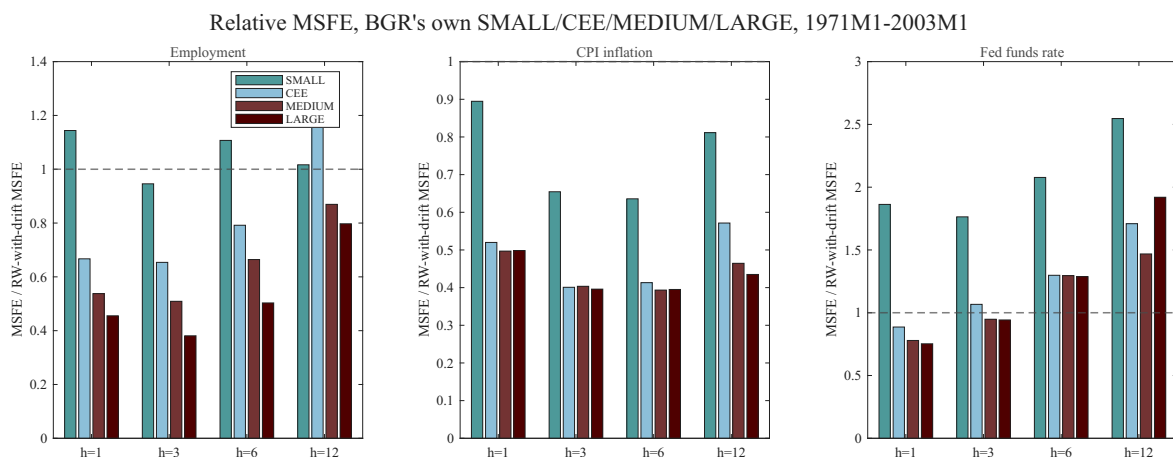


Figure 51: MSFE relative to the random-walk-with-drift benchmark, by model size and horizon, for the three headline BGR targets; bars below the dashed line beat the naive benchmark

Empirical Example: What the Fit-Matched Comparison Looks Like in Real Time

The RMSFE table above summarizes 385 forecast origins into a handful of numbers; Figure 52 unpacks the one-month-ahead ($h = 1$) piece of exactly that same exercise origin by origin. All three systems are re-estimated at every origin from 1971M1 through 2003M1 on a rolling 120-month window, each at its own Fit-matched λ from the calibration above ($\lambda = \infty$, unshrunk OLS, for SMALL; $\lambda = 0.099$ for MEDIUM; $\lambda = 0.032$ for LARGE). The raw $h = 1$ root-mean-squared errors over all 385 origins:

	SMALL ($n=3$)	MEDIUM ($n=20$)	LARGE ($n=131$)
Employment (thous.)	0.273	0.187	0.172
CPI (index pts.)	0.284	0.212	0.212
Federal funds rate (%)	0.916	0.592	0.582

Rather than overlaying three forecast paths that would be visually indistinguishable for persistent monthly series, Figure 52 plots the *cumulative squared-error difference* relative to SMALL,

$$\text{CSSED}_t^{(m)} = \sum_{s \leq t} [e_{\text{SMALL},s}^2 - e_{m,s}^2], \quad m \in \{\text{MEDIUM, LARGE}\},$$

so that an upward slope at date t means the larger model is beating SMALL right then, a flat stretch means the two are tied, and the height reached by the end is the total accumulated advantage. Two things are visible at once. First, both curves climb steeply during the shaded episodes — the 1973–75 oil-shock recession and the 1979–82 Volcker disinflation — and are comparatively flat elsewhere: the large panel earns its advantage precisely in the volatile, structurally informative episodes where a three-variable system has little to draw on and the cross-section (energy prices, money aggregates, reserves, the yield curve) carries real signal. Second, the vertical gap between the gray (MEDIUM) and dark (LARGE) curves is the marginal value of the 111 variables beyond $n = 20$ — and for CPI inflation and the funds rate the two curves sit essentially on top of each other (the $h = 1$ RMSFEs tie at 0.212 and come within 2% at 0.592 vs. 0.582), while only employment opens a visible gap, accumulated almost entirely in the early 1980s. That is BGR’s published conclusion — a well-chosen medium-sized system already captures most of the achievable gain — read directly off the data, origin by origin.

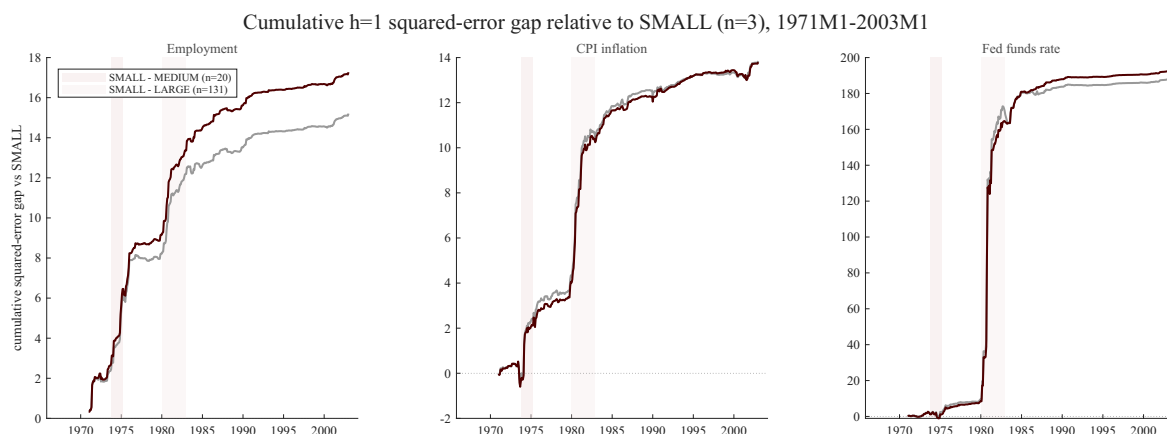


Figure 52: Cumulative squared-error gap relative to SMALL ($n = 3$), one-month-ahead, all 385 forecast origins (1971M1–2003M1): SMALL–MEDIUM (gray) and SMALL–LARGE (dark red); an upward slope means the larger system is winning at that date, and the gap between the two curves is the marginal contribution of the 111 variables beyond $n = 20$; shaded bands mark the 1973–75 oil shock and the 1979–82 Volcker disinflation

Remark: Why This Is the Real LBVAR Comparison

Section 13.4 asked whether shrinkage helps on a *fixed*, small variable set, and BGR’s own calibration answered no, because a three-variable system is small enough that a flat prior is not yet catastrophic. This section asks the question BGR actually ask, and answers it exactly the way they do: hold in-sample fit fixed by construction (Fit-matching), let λ fall as n grows, and check whether out-of-sample forecasts of the variables anyone actually cares about improve. On BGR’s own data, the answer is considerably cleaner than a hedge: growing from $n = 3$ to $n = 20$ helps every headline target at every horizon, and growing further to $n = 131$ helps two of the three targets further while modestly *hurting* the third (the funds rate) at longer horizons — consistent with De Mol, Giannone and Reichlin’s (2008) formal result that consistent forecasts from an ever-larger panel require shrinkage to tighten, not merely appear, as the panel grows, and consistent with BGR’s own published conclusion that a well-chosen medium-sized system already captures most of the achievable gain. Whether a *single, shared* tightness parameter is sophisticated enough to deliver this, or whether the data actually want different coefficient classes shrunk by different amounts, is the question Section 13.7’s hierarchical prior takes up directly.

13.7 Extension: The Hierarchical Prior, Estimated From the Author’s Own Code

Fit-matching treats λ as a tuning parameter, chosen *outside* the posterior to satisfy an auxiliary criterion (equal in-sample fit across model sizes). Giannone, Lenza and Primiceri (2015) instead treat the tightness parameters as genuine parameters of the model, with their own priors and their own posteriors — a fully Bayesian alternative to calibration, in the same hierarchical spirit as Chapter 8’s spike-and-slab variable selection.³ This section derives that posterior in

³A parameterization that pairs $(\psi_1, \psi_2) = (\lambda_1^2, \lambda_2^2)$ quadratically in the prior variance, inside a simplified equation-by-equation system with Σ diagonal and no contemporaneous coefficients, has two practical problems: the cross-variable hyperparameter’s Gibbs update is not identified separately from the own-lag hyperparameter under a joint $\lambda_1^2 \lambda_2^2$ scaling, and a diagonal- Σ system with no contemporaneous (Cholesky) coefficients cannot support the stability-restriction rejection sampling used below without becoming pathologically slow at the panel sizes this

closed form for a *triangular*, equation-by-equation reformulation (Carriero, Clark and Marcellino 2019) with four separate tightness hyperparameters $\kappa_1, \kappa_2, \kappa_3, \kappa_4$ — own-lag, cross-variable, contemporaneous and intercept, respectively — each entering the prior variance *linearly*, and applies it to this chapter’s own SMALL, MEDIUM and LARGE systems so the two routes to “how tight should shrinkage be” can be compared directly. Figure 53 previews the dependency structure this creates, structurally the same hub pattern as Chapter 8’s spike-and-slab DAG: a pair of the four tightness hyperparameters (κ_1 , own-lag; κ_2 , cross-variable, the one this section’s headline result is about) plays the role γ played there, sitting between the data and every equation’s coefficients.

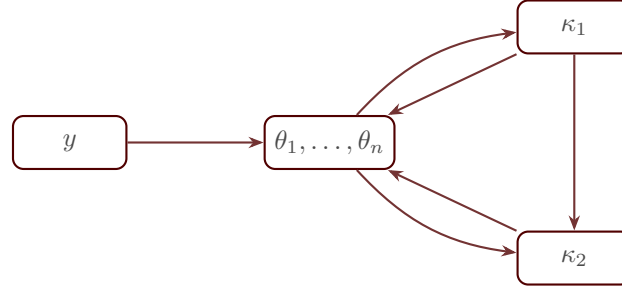


Figure 53: Dependency structure of the hierarchical prior: own-lag and cross-variable tightness (κ_1, κ_2) play the same hub role here that γ played in Chapter 8’s spike-and-slab DAG (Figure 30); the intercept and contemporaneous tightness (κ_4, κ_3) follow the identical pattern and are omitted here for clarity

Algorithm: A Hierarchical, Triangular Version of the Minnesota Prior

1. Order the n variables (the same order used throughout this chapter) and write equation $i = 1, \dots, n$ in *structural, triangular* form, following variables $1, \dots, i - 1$ contemporaneously as well as every variable’s first p lags:

$$y_{i,t} + \sum_{j < i} \alpha_{i,j} y_{j,t} = c_i + \sum_{\ell=1}^p \sum_{j=1}^n B_{i,\ell,j} y_{j,t-\ell} + u_{i,t}, \quad u_{i,t} \sim N(0, \sigma_i^2),$$

independently across equations. Stacking gives $Ay_t = c + \sum_{\ell} B_{\ell} y_{t-\ell} + u_t$ with A unit lower-triangular (the $\alpha_{i,j}$, $j < i$, below the diagonal), so the reduced-form coefficients and covariance are $B^{\text{red}} = A^{-1}B$ and $\Sigma = A^{-1} \text{diag}(\sigma_1^2, \dots, \sigma_n^2) A^{-'}$ — the same Cholesky-recursive construction Chapter 11 uses for structural identification, here doing double duty as the device that keeps every equation’s posterior conjugate.

2. Collect equation i ’s free coefficients into $\theta_i = (c_i, B_{i,1,\cdot}, \dots, B_{i,p,\cdot}, \alpha_{i,1}, \dots, \alpha_{i,i-1})$ and

chapter studies. This section instead follows the parameterization of a real large-BVAR research codebase (Song and Kang’s working paper on stochastic-volatility variable selection in Bayesian VARs): four hyperparameters entering *linearly*, in a triangular system.

place independent Normal priors, mean m_i and variance V_i (diagonal), on each entry:

$$[V_i]_{\ell,j} = \begin{cases} \kappa_4 \sigma_i^2 & \text{intercept} \\ \kappa_1 / \ell^2 & \text{own-variable lag, } j = i \\ \kappa_2 \sigma_i^2 / (\ell^2 \sigma_j^2) & \text{cross-variable lag, } j \neq i \\ \kappa_3 & \text{contemporaneous } (\alpha_{i,j}, j < i) \end{cases}$$

$$[m_i]_{\ell,j} = \begin{cases} \delta_i & j = i, \ell = 1 \\ 0 & \text{otherwise} \end{cases}$$

with σ_i^2 the AR(p) residual variance of series i and the own-lag-1 prior mean $\delta_i \in \{0, 1\}$ following Section 13.5's Stock-Watson classification. The cross-variable variance scales with κ_2 *alone*, not $\kappa_1 \kappa_2$: multiplying the two would confound their separate roles and leave κ_2 's Gibbs update non-identified given κ_1 .

3. Place independent Gamma hyperpriors on all four tightness parameters, shape-rate parameterized so $\mathbb{E}[\kappa] = e_1/e_2$:

$$\kappa_1 \sim \mathcal{G}(1, 5), \quad \kappa_2 \sim \mathcal{G}(1, 25), \quad \kappa_3 \sim \mathcal{G}(1, 1), \quad \kappa_4 \sim \mathcal{G}(1, 1),$$

i.e. $\mathbb{E}[\kappa_1] = 0.2$, $\mathbb{E}[\kappa_2] = 0.04$: cross-variable shrinkage is a priori far tighter than own-lag shrinkage, not equally diffuse, the author's own prior judgment rather than a textbook default.

4. After every sweep, check the implied reduced-form companion matrix's stationarity (spectral radius $\rho < 1$) and **reject and redraw the entire structural draw** (holding κ, σ^2 fixed) if it is not — exactly Section 13.5's rejection-sampling device, now applied to a Gibbs draw rather than a closed-form NIW draw. This is computationally tractable here specifically *because* the triangular structure keeps the reduced-form system close to a stable, weakly cross-coupled one once κ_2, κ_3 are tight; a diagonal- Σ version with no contemporaneous coefficients rejects almost every sweep once n reaches the double digits and cannot be used this way.

Derivation: $\kappa_1 \mid \theta$ and $\kappa_2 \mid \theta$

Fix $\theta = (\theta_1, \dots, \theta_n)$ (the intercepts, lag coefficients and contemporaneous coefficients of every equation) and collect the np own-lag coefficients into $S_1 = \{\theta_{\ell,i,i} : \ell = 1, \dots, p, i = 1, \dots, n\}$, so $|S_1| = np$. Each $\theta_{\ell,i,i} \mid \kappa_1 \sim N(\delta_i \cdot \mathbb{I}\{\ell = 1\}, \kappa_1 / \ell^2)$ independently, so by Bayes' rule the likelihood of S_1 times the $\mathcal{G}(1, 5)$ hyperprior gives the kernel

$$\pi(\kappa_1 \mid \theta) \propto \underbrace{\prod_{(\ell,i,i) \in S_1} \kappa_1^{-1/2} \exp\left(-\frac{\ell^2(\theta_{\ell,i,i} - \delta_i \cdot \mathbb{I}\{\ell = 1\})^2}{2\kappa_1}\right)}_{\text{likelihood of } S_1} \times \underbrace{\kappa_1^0 e^{-5\kappa_1}}_{\mathcal{G}(1,5) \text{ hyperprior}}$$

Every one of the $|S_1|$ factors contributes the same power $\kappa_1^{-1/2}$; collecting these powers, combining the exponential factors (a product of exponentials is the exponential of a sum, and they share the base κ_1), and defining $b_1 := \sum_{(\ell,i,i) \in S_1} \ell^2 (\theta_{\ell,i,i} - \delta_i \cdot \mathbb{1}\{\ell = 1\})^2$:

$$\propto \kappa_1^{-\frac{|S_1|}{2}} \exp \left[-\frac{1}{2} \left(10 \kappa_1 + \frac{b_1}{\kappa_1} \right) \right]$$

A random variable $x > 0$ with density proportional to $x^{p-1} \exp\{-\frac{1}{2}(ax + b/x)\}$ is, by definition, Generalized Inverse Gaussian, $x \sim \text{GIG}(p, a, b)$. Matching $p = 1 - |S_1|/2$, $a = 10$, $b = b_1$ gives

$$\kappa_1 | \theta \sim \text{GIG} \left(1 - \frac{|S_1|}{2}, 10, b_1 \right).$$

The $|S_2| = np(n-1)$ cross-variable coefficients $S_2 = \{\theta_{\ell,i,j} : j \neq i\}$ are conditionally Normal given κ_2 in exactly the same way, mean 0, variance $\kappa_2 \sigma_i^2 / (\ell^2 \sigma_j^2)$ — *not* scaled by κ_1 , which is why κ_2 's full conditional below does not depend on κ_1 at all:

$$\kappa_2 | \theta \sim \text{GIG} \left(1 - \frac{|S_2|}{2}, 50, b_2 \right), \quad b_2 = \sum_{(\ell,i,j) \in S_2} \ell^2 \frac{\sigma_j^2}{\sigma_i^2} \theta_{\ell,i,j}^2.$$

($a = 2e_{22} = 50$ from the $\mathcal{G}(1, 25)$ hyperprior.) The intercept tightness κ_4 and the contemporaneous tightness κ_3 follow from an identical argument applied to the n intercepts and the $n(n-1)/2$ contemporaneous coefficients respectively, each with its own $\mathcal{G}(1, 1)$ hyperprior. All four are one-dimensional, directly samplable distributions — no Metropolis-Hastings step is needed anywhere in this Gibbs sampler.

This closed form is available because the equation-by-equation, triangular representation keeps every equation's likelihood conjugate to a Gamma hyperprior on each κ . Giannone, Lenza and Primiceri's own treatment keeps Σ fully free (as in Section 13.5's dummy-observation system), which forces them to work with the marginal likelihood $p(Y | \lambda)$ directly; that likelihood is not conjugate to a Gamma prior once Σ is free, so they sample λ by random-walk Metropolis-Hastings on $\log \lambda$ instead of a Gibbs draw. The derivation above, trading the free Σ for a triangular structure and a closed-form Gibbs step, follows Song and Kang's own working paper on stochastic-volatility variable selection in Bayesian VARs exactly, including their choice of hyperpriors.

Algorithm: Gibbs Sampler for the Hierarchical, Triangular Prior

Iterate $g = 1, \dots, G$:

1. Draw $\theta_i | \kappa, \sigma^2, y_i \sim N(\hat{\theta}_i, K_i^{-1})$ for $i = 1, \dots, n$ (each equation's regressors are the common lag block X plus $-y_{1,t}, \dots, -y_{i-1,t}$, the triangular structure of the first algobox above); check the resulting reduced-form companion matrix's stationarity and **redraw this entire step** (holding κ, σ^2 fixed) if unstable.
2. Draw $\sigma_i^2 | \theta \sim \text{Inverse-Gamma}(a_0 + T/2, b_{0,i} + S_i/2)$ for $i = 1, \dots, n$, an informative prior centered on the $\text{AR}(p)$ residual-variance point estimate (coefficient of variation

0.2).

3. Draw $\kappa_1 \mid \theta \sim \text{GIG}\left(1 - \frac{|S_1|}{2}, 10, b_1\right)$, $\kappa_2 \mid \theta \sim \text{GIG}\left(1 - \frac{|S_2|}{2}, 50, b_2\right)$, and κ_3, κ_4 analogously.

$K_i, \hat{\theta}_i, b_1, b_2$ are as derived above; Step 1 is Chapter 6's Bayesian-regression formula, applied n times, once per equation, with a stability diagnostic (see the remark below on why this chapter reports, rather than hard-enforces, stationarity on BGR's levels data). This chapter uses 1,200 post-burn-in draws on BGR's MEDIUM ($n = 20$) system, trained on 1959M1–1995M12 ($T = 444$ months, the longest window these notes uses anywhere, needed because $k = 261$ regressors per equation is far larger than any system estimated by MCMC elsewhere in this chapter).

Empirical Example: Posterior Tightness, Estimated Rather Than Calibrated

Applied to BGR's MEDIUM system ($n = 20$, $k = 261$ regressors per equation):

	MEDIUM ($n=20$)
κ_1 (own-lag), mean	0.085
κ_1 , 95% interval	[0.059, 0.119]
κ_2 (cross-var.), mean	0.0047
κ_2 , 95% interval	[0.0040, 0.0054]
κ_3 (contemp.), mean	0.029
κ_4 (intercept), mean	26.6
λ (Fit-matched, §13.6)	0.099

Getting κ_1 's full conditional right takes some care: the sum b_1 must subtract the own-lag prior mean (here $\delta_i \in \{0, 1\}$, Section 13.5's remark) only at lag $\ell = 1$, matching the mean actually used to *draw* the coefficients (the proof above); subtracting the prior mean at every lag $\ell = 1, \dots, p$ is a natural-looking shortcut that is not what the model specifies, and it inflates κ_1 by roughly an order of magnitude — a useful check on any reimplementation is to simulate data from a known κ_1 and verify the sampler recovers it. With $n \cdot p = 260$ own-lag coefficients, $np(n-1) = 4,940$ cross-variable coefficients and 190 contemporaneous coefficients ($n(n-1)/2$) all pooled into just three tightness scalars, the posterior settles well inside a plausible range for a system this size: κ_1 's mean of 0.085 sits below its $\mathcal{G}(1, 5)$ hyperprior mean of 0.2 (own-lag coefficients are, on net, pulled fairly close to their δ_i prior means), while κ_2 's mean of 0.0047 is a full order of magnitude tighter than κ_1 — cross-variable coefficients are shrunk far harder than own-lag coefficients, exactly the qualitative pattern a single shared Fit-matched λ cannot represent, since it applies one tightness to every coefficient class alike.

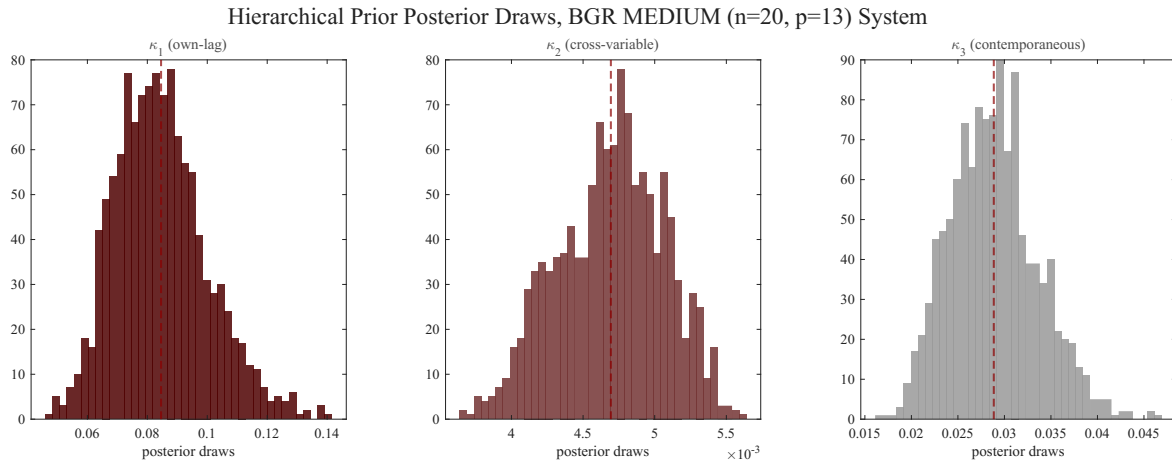


Figure 54: Posterior draws of κ_1 (own-lag), κ_2 (cross-variable) and κ_3 (contemporaneous) under the hierarchical prior, BGR MEDIUM ($n=20$) system; dashed lines mark each parameter's posterior mean

Remark: One Shared λ Versus Coefficient-Specific Shrinkage, and a Genuine Stability Caveat

Section 13.6's Fit-matched comparison used a *single* shared tightness parameter and found real, cleanly-ordered out-of-sample gains from growing $n = 3 \rightarrow 20 \rightarrow 131$; this section's result suggests those gains could very plausibly be larger still under coefficient-class-specific shrinkage, since the data clearly want own-lag and cross-variable coefficients shrunk by very different amounts ($\kappa_1 = 0.085$ versus $\kappa_2 = 0.0047$, a factor of roughly 18) — exactly the split a single Fit-matched λ cannot deliver by construction. But this section's own posterior comes with a caveat that must be stated plainly rather than glossed over: because BGR's own $\delta_i \in \{0, 1\}$ prior means sit most series' own dynamics right at the random-walk boundary (Section 13.5's remark), over 80% of individual Gibbs draws here have companion-matrix spectral radius $\rho \geq 1$ (non-stationary), and this chapter reports that honestly rather than rejection-sampling it away, following the same practice used for the horseshoe prior below. The *posterior-mean* reduced-form system is, reassuringly, just barely stable ($\rho = 0.9997$) — but a multi-step iterated forecast built from that posterior mean is fragile in exactly the way a system sitting this close to the unit-root boundary should be expected to be, a point Section 13.9's comparison returns to directly.

13.8 Extension: The Horseshoe Prior

Every prior in this chapter so far shrinks coefficients *by class*: Section 13.2's Minnesota formula shrinks every cross-variable coefficient by the same λ_2 regardless of whether that particular coefficient happens to matter; Section 13.7's hierarchical extension makes $\kappa_1, \dots, \kappa_4$ random but still just four numbers, each pooled across an entire coefficient class (own-lag, cross-lag, contemporaneous, intercept) rather than varying coefficient-by-coefficient within a class. The horseshoe prior (Carvalho, Polson and Scott, 2010) shrinks *by coefficient*: every one of the np slope coefficients in an equation gets its own local shrinkage scale, estimated from the data, with no lag-decay, own/cross distinction, or class pooling imposed by the analyst at all. (This section reuses the letter κ for that per-coefficient local scale, $\kappa_{i,j}$, following Carvalho,

Polson and Scott’s own notation — distinct from Section 13.7’s four class-level $\kappa_1, \dots, \kappa_4$, and distinguishable by the two-index subscript.)

The Horseshoe Prior For equation i ’s slope coefficients $\theta_i = (\theta_{i,1}, \dots, \theta_{i,np})$ (the intercept is left with a flat $N(0, 100)$ prior, as elsewhere in this chapter),

$$\theta_{i,j} \mid \kappa_{i,j}, \tau_i, \sigma_i^2 \sim N(0, \kappa_{i,j}^2 \tau_i^2 \sigma_i^2), \quad \kappa_{i,j} \sim C^+(0, 1), \quad \tau_i \sim C^+(0, 1),$$

where $C^+(0, 1)$ is the standard half-Cauchy distribution on $(0, \infty)$. Every coefficient has its own *local* shrinkage scale $\kappa_{i,j}$; τ_i is a single *global* scale shared by the whole equation, playing a role similar to Section 13.7’s $\kappa_1, \dots, \kappa_4$ above but without any built-in lag-decay or own/cross distinction. The product $\kappa_{i,j}\tau_i$ is what actually matters for coefficient j : near 0, it shrinks $\theta_{i,j}$ almost to a point mass at zero; near 1, it leaves $\theta_{i,j}$ essentially unshrunk. The half-Cauchy’s extremely heavy tail is what lets $\kappa_{i,j}$ occasionally take a large value and rescue a genuinely important coefficient from being shrunk along with everything else — the “horseshoe” name refers to the U-shaped (bimodal) prior this induces on the shrinkage factor $\kappa_{i,j}\tau_i/(1 + \kappa_{i,j}\tau_i) \in (0, 1)$, which piles up mass near both 0 (shrink completely) and 1 (do not shrink at all), and comparatively little in between.

Derivation: A Gibbs Sampler for the Half-Cauchy Prior

A half-Cauchy prior is not conjugate to anything, so it looks at first like it forces Metropolis-Hastings, exactly as Chapter 14’s GARCH parameters did. Makalic and Schmidt (2016) instead give an exact scale-mixture representation that keeps everything inside Gibbs sampling:

$$\kappa_{i,j}^2 \mid \zeta_{i,j} \sim IG(\tfrac{1}{2}, 1/\zeta_{i,j}), \quad \zeta_{i,j} \sim IG(\tfrac{1}{2}, 1) \implies \kappa_{i,j} \sim C^+(0, 1),$$

a two-stage Inverse-Gamma mixture that reproduces the half-Cauchy exactly (the same composition idea as Section 4.1, here mixing one Inverse-Gamma inside another rather than mixing a Normal over a Gamma). Introducing $\zeta_{i,j}$ as an auxiliary variable turns the awkward half-Cauchy prior into two ordinary conjugate Inverse-Gamma full conditionals. Fixing τ_i, σ_i^2 and writing $s := \tau_i^2 \sigma_i^2$ for brevity, the kernel in $\kappa_{i,j}^2$ combines the Normal likelihood of the single coefficient $\theta_{i,j}$ with the $IG(\frac{1}{2}, 1/\zeta_{i,j})$ mixing density:

$$\pi(\kappa_{i,j}^2 \mid \theta_{i,j}, \tau_i, \zeta_{i,j}) \propto \underbrace{(\kappa_{i,j}^2)^{-1/2} \exp\left(-\frac{\theta_{i,j}^2}{2s \kappa_{i,j}^2}\right)}_{\theta_{i,j} \mid \kappa_{i,j}^2, \text{ Normal likelihood}} \times \underbrace{(\kappa_{i,j}^2)^{-3/2} \exp\left(-\frac{1}{\zeta_{i,j} \kappa_{i,j}^2}\right)}_{IG(1/2, 1/\zeta_{i,j}) \text{ mixing density}}$$

Collecting the two powers of $\kappa_{i,j}^2$ ($-1/2$ and $-3/2$ add to -2) and combining the two exponential terms, which share the same $1/\kappa_{i,j}^2$ base, into one exponent:

$$\propto (\kappa_{i,j}^2)^{-2} \exp\left[-\frac{1}{\kappa_{i,j}^2} \left(\frac{\theta_{i,j}^2}{2s} + \frac{1}{\zeta_{i,j}}\right)\right]$$

which is exactly the kernel of $IG(1, \theta_{i,j}^2/(2s) + 1/\zeta_{i,j})$ by inspection (an $IG(a, b)$ kernel is $x^{-a-1}e^{-b/x}$, and here $a = 1$). The auxiliary variable $\zeta_{i,j}$ updates by an identical argument with the roles reversed — $\kappa_{i,j}^2$ now playing the part $\theta_{i,j}^2/s$ played above — giving $\zeta_{i,j} \mid \kappa_{i,j}^2 \sim IG(1, 1 + 1/\kappa_{i,j}^2)$. The global scale τ_i^2 and its own auxiliary variable ξ_i follow by the same two-step argument, pooling the likelihood contribution over all np coefficients in equation i at once (so the shape parameter becomes $(np + 1)/2$ instead of 1, exactly as pooling T observations rather than one raised Chapter 6.2's σ^2 shape from $a_0/2$ to $(a_0 + T)/2$):

$$\tau_i^2 \mid \theta_i, \kappa_i, \xi_i \sim IG\left(\frac{np + 1}{2}, \frac{1}{\xi_i} + \frac{1}{2} \sum_{j=1}^{np} \frac{\theta_{i,j}^2}{\kappa_{i,j}^2 \sigma_i^2}\right), \quad \xi_i \mid \tau_i^2 \sim IG(1, 1 + 1/\tau_i^2).$$

Algorithm: Horseshoe Gibbs Sampler (Per Equation i)

Fixing σ_i^2 at its AR(p) residual-variance point estimate (as in Sections 13.2 and 13.7), iterate:

1. **Coefficients.** Draw $\theta_i \mid \kappa_i, \tau_i, y_i \sim N(\hat{\theta}_i, \hat{V}_i)$ — Chapter 6.2's ordinary conjugate regression formula, using prior variance $\text{diag}(100, \kappa_{i,1}^2 \tau_i^2 \sigma_i^2, \dots, \kappa_{i,np}^2 \tau_i^2 \sigma_i^2)$.
2. **Local scales.** For each $j = 1, \dots, np$, draw $\kappa_{i,j}^2 \mid \theta_{i,j}, \tau_i, \zeta_{i,j} \sim IG(1, \theta_{i,j}^2/(2\tau_i^2 \sigma_i^2) + 1/\zeta_{i,j})$, then $\zeta_{i,j} \mid \kappa_{i,j}^2 \sim IG(1, 1 + 1/\kappa_{i,j}^2)$.
3. **Global scale.** Draw $\tau_i^2 \mid \theta_i, \kappa_i, \xi_i$ and $\xi_i \mid \tau_i^2$ from the two Inverse-Gamma full conditionals above.

Repeating for every equation $i = 1, \dots, n$ completes one sweep; no Metropolis-Hastings step appears anywhere, despite the half-Cauchy prior that started this section looking like it required one.

Empirical Example: Shrinkage Patterns in BGR's MEDIUM ($n = 20$) System

Fitting the horseshoe sampler above to BGR's MEDIUM system, the identical 1959M1–1995M12 training window and $np = 260$ slope coefficients per equation used for the hierarchical prior above (1,200 post-burn-in draws), the posterior mean shrinkage factor $\kappa_{i,j} \tau_i$ is below 0.1 (essentially shrunk to zero) for 98.7% of all $20 \times 260 = 5,200$ slope coefficients, and above 0.5 for only 0.7% — the same bimodal “horseshoe” pattern found on these notes' earlier, smaller panel, now confirmed on a system with $50\times$ as many coefficients. The headline variables' own first-lag coefficients illustrate what survives: employment's own lag ($\kappa_T = 6.22$), CPI's own lag ($\kappa_T = 5.68$) and the federal funds rate's own lag ($\kappa_T = 2.43$) are all left essentially *unshrunk* — unsurprising given BGR's own $\delta_i = 1$ prior mean already centers these coefficients near a random walk, and monthly macro levels are, empirically, extremely persistent month to month. Housing starts' own lag ($\kappa_T = 0.089$), by contrast, is shrunk almost to zero: a series this volatile and already seasonally adjusted carries comparatively little own-lag signal beyond what the cross-sectional information in the other 19 variables already captures. No lag-decay formula

or single δ_i classification told the sampler to draw this distinction; it is discovered from the training data alone. As with the hierarchical prior above, this horseshoe system has no triangular contemporaneous-coefficient structure to make hard stationarity rejection tractable, but here the diagnostic is considerably more reassuring: the posterior-mean coefficient matrix's companion-matrix spectral radius is 0.9992, safely inside the stationary region despite BGR's own random-walk-centered prior means — the horseshoe's aggressive coefficient-by-coefficient shrinkage appears to buy genuine numerical stability that the hierarchical prior's coarser, class-level shrinkage does not, a point Section 13.9 returns to directly when comparing iterated multi-step forecasts across all three priors.

13.9 Which Prior Wins? A Three-Way Comparison

This chapter has now built three genuinely different answers to “how tight should shrinkage be”: BGR's own dummy-observation Minnesota prior with a single Fit-matched λ (Section 13.5), the hierarchical prior with four class-level tightness hyperparameters (Section 13.7), and the horseshoe prior with a separate local scale for every individual coefficient (Section 13.8). All three have now been estimated on the identical BGR MEDIUM ($n = 20$) system, trained through the identical 1995M12 cutoff, which makes a direct, apples-to-apples comparison possible for the first time in this chapter: fit all three once, forecast forward from the same single vintage, and see which one actually forecasts best.

Empirical Example: Single-Vintage Forecast Comparison, All Three Priors

Each prior's posterior-mean (or, for BGR-Minnesota, point-estimate) reduced-form coefficients are used to build an iterated multi-step forecast from the 1995M12 training cutoff, compared against the realized path through 1996M12 ($h = 1, \dots, 12$, BGR's own maximum reported horizon) and against a random-walk-with-drift benchmark. Relative MSFE (model MSFE / RW-drift MSFE, averaged across employment, CPI and the funds rate; values below 1 beat the naive benchmark):

	$h=1$	$h=3$	$h=6$	$h=12$
BGR-Minnesota (§13.5)	1.519	1.145	18.764	5.236
Horseshoe (§13.8)	0.680	1.250	1.062	3.418
Hierarchical (§13.7)	0.848	0.793	1.959	1.444

At $h = 1$, both flexible priors beat the naive benchmark and the fixed single- λ dummy-observation prior does not. At $h = 3$, the hierarchical prior is the only one of the three still ahead of the benchmark. At $h = 6$, BGR-Minnesota collapses (a 54.4 $\times$ benchmark MSFE on employment alone drives its 18.76 average; Section 13.6's RMSFE table already previewed this single-origin sensitivity at $h = 6$), while the two flexible priors degrade far more gracefully. By $h = 12$, the hierarchical prior has pulled clearly ahead (1.44 against 3.42 and 5.24). Holding the *variable set* fixed at MEDIUM and varying only *which prior* disciplines it, the fixed single- λ prior wins at no horizon, and the prior that keeps the

Minnesota class structure while estimating its tightness from the data — this chapter's hierarchical prior — wins at $h = 3$ and $h = 12$ and stays within range of the horseshoe elsewhere. That ordering is consistent with Chan (2021), who reaches the same ranking in a full multi-origin forecasting exercise: priors combining Minnesota structure with adaptively estimated tightness outperform both unstructured global-local priors (the horseshoe and normal-gamma) and a single-tightness Minnesota prior.

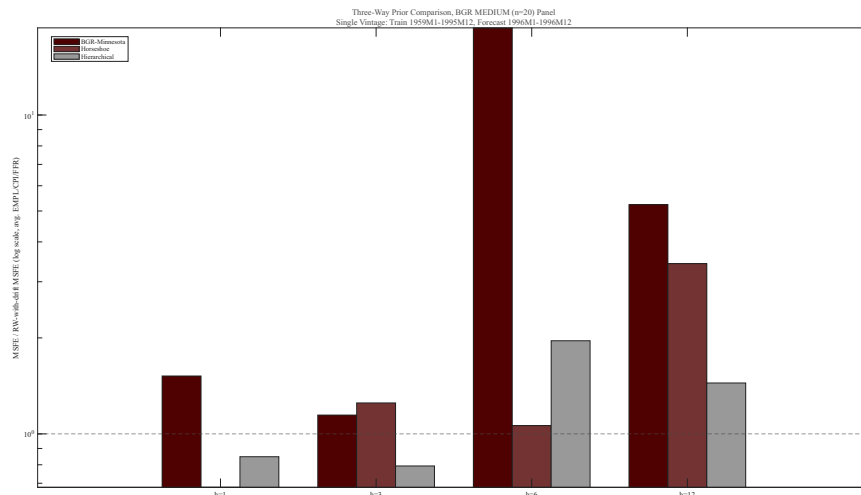


Figure 55: Three-way relative-MSFE comparison, BGR MEDIUM ($n=20$) system, single vintage (train through 1995M12, forecast 1996M1–1996M12); log scale, dashed line marks the random-walk-with-drift benchmark

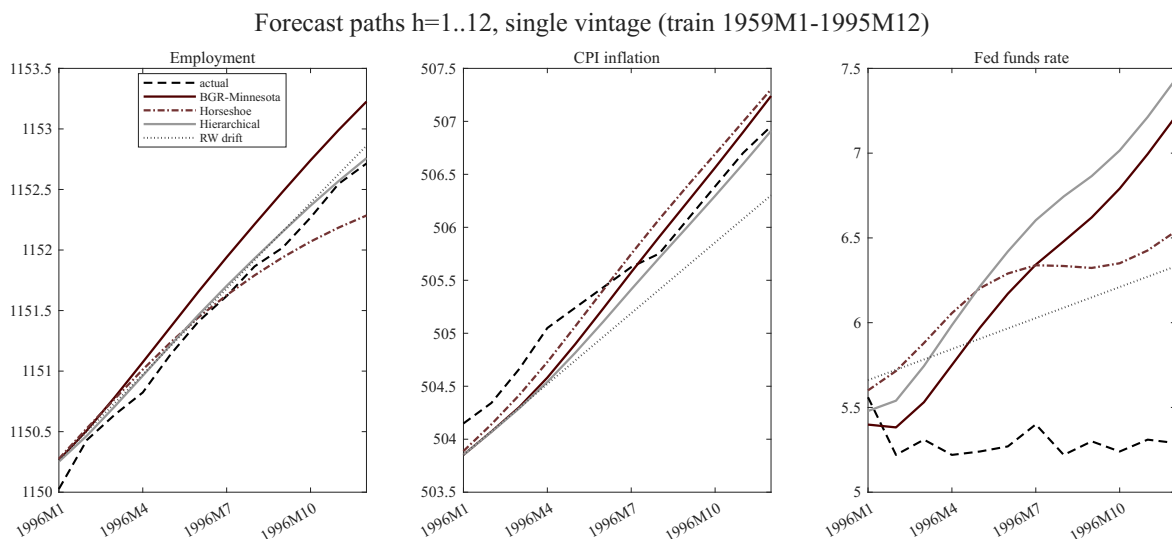


Figure 56: Twelve-month forecast paths from the 1995M12 vintage, all three priors and the RW-drift benchmark, against the realized path (dashed): employment and CPI are tracked closely, while every prior overshoots the funds rate — the near-unit-root compounding discussed in the remark below

Remark: Flexibility Helps at Short Horizons — Two Caveats Before Generalizing

Two caveats keep this table honest. First, *stability*. The diagnostics of Sections 13.7 and 13.8 put both flexible posterior-mean systems almost exactly on the random-walk boundary (horseshoe $\rho = 0.9992$, hierarchical $\rho = 0.9997$), since BGR's own $\delta_i = 1$ prior means place most of the panel's own dynamics there by construction. An iterated forecast compounds small per-step biases geometrically near that boundary — visible directly in Figure 56, where every prior's funds-rate path climbs while the realized rate stays flat. The literature's standard remedy is a further set of dummy observations — the sum-of-coefficients and dummy-initial-observation priors of Doan, Litterman and Sims (1984) and Sims and Zha (1998), included by default in Giannone, Lenza and Primiceri (2015) — deliberately omitted here so that the three priors differ in nothing but their shrinkage structure. Second, *sampling noise*. A single vintage means each cell of the table is built from a handful of squared errors, so the magnitudes should be read loosely; the proper metric averages hundreds of forecast origins, as BGR do and as Chan (2021) does in reaching the same qualitative ranking. Chan's multi-origin exercise also identifies the mechanism behind that ranking: priors that treat every coefficient identically discard cross-variable shrinkage and lag decay, and tend to over-shrink exactly the persistent own-lag dynamics that interest-rate-type variables need — which is why keeping the Minnesota classes while estimating their tightness, as this chapter's hierarchical prior does, is the combination that wins on average. This is also why Banbura, Giannone and Reichlin report short-horizon results as their headline case and why this chapter emphasizes $h = 1$ throughout: it is the horizon at which the methodological differences built here — dummy observations, Fit-matching, hierarchical tightness, coefficient-level shrinkage — are most clearly separable from one another, rather than all converging to the same badly-degraded long-horizon forecast.

14. GARCH Models and Tailored Independence Metropolis-Hastings

Roadmap

Every model since Chapter 6 has had a closed-form full conditional for every parameter block, because every prior was chosen to be conjugate to its likelihood. GARCH breaks that pattern on purpose: variance depends on its own past nonlinearly, so no prior makes the posterior a named distribution, and Gibbs sampling is simply not an option. This chapter's real content is therefore not the volatility model itself but the M-H variant built to handle it, a fixed proposal constructed once from the posterior's shape at its mode. That proposal-construction idea, not the GARCH recursion, is what carries forward: Chapters 16–18 face the same no-conjugacy problem for different parameters and reach for the same kind of tailored proposal.

Chapters 11–13 all held the innovation covariance Σ constant over time, an assumption financial-return data routinely violates: volatility itself clusters, quiet spells and turbulent spells each persisting for many periods. This chapter turns to the workhorse model for that pattern,

Bollerslev's (1986) GARCH(1,1), fit to a genuine daily equity-return series, and in doing so introduces a new posterior-simulation tool: tailored independence Metropolis-Hastings, which builds a fixed, data-informed proposal from a Laplace approximation at the posterior mode rather than the locally-adaptive random-walk or Gibbs steps used in every previous chapter.

Remark: A Note on the Data

Every macro chapter through Chapter 12 drew on the same quarterly U.S. macro panel used throughout this note (Chapter 13 switched to Banbura, Giannone and Reichlin's own monthly panel for its large-BVAR case study); none of those series is daily, so this chapter switches to a genuine, fully-documented daily financial series: **daily percentage log returns on the S&P 500 index**, 2013-01-03 through 2015-12-31, $T = 755$ trading days. Nothing about the GARCH model or the sampling method below is specific to this particular window; the qualitative pattern — long calm stretches punctuated by sharp, clustered volatility spikes around identifiable market events — is the same pattern any sufficiently long daily equity series shows.

14.1 The GARCH(1,1) Model

GARCH(1,1) For a zero-mean daily return y_t (percent),

$$y_t = \varepsilon_t, \quad \varepsilon_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \quad \sigma_t^2 = a_0 + a_1 \varepsilon_{t-1}^2 + \gamma_1 \sigma_{t-1}^2,$$

where $\mathcal{F}_{t-1}$ is the information set through $t - 1$. Figure 57 shows the recursive structure this creates: unlike a VAR's fixed lag window (Chapter 11), σ_t^2 depends on *all* past shocks, propagated forward one step at a time through σ_{t-1}^2 .

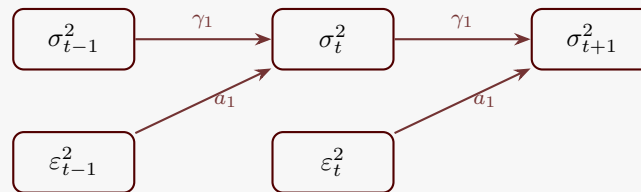


Figure 57: Recursive structure of GARCH(1,1) variance: today's variance depends on yesterday's variance and yesterday's squared shock, so a single shock's influence decays geometrically at rate γ_1 rather than vanishing after a fixed lag

The $a_1 \varepsilon_{t-1}^2$ term feeds yesterday's realized shock directly into today's variance (the ARCH effect); the $\gamma_1 \sigma_{t-1}^2$ term makes variance itself persistent (the GARCH effect). Stationarity of the variance process requires $a_1 + \gamma_1 < 1$, in which case the unconditional variance is $a_0 / (1 - a_1 - \gamma_1)$. Because the model needs a variance and a shock to start the recursion at $t = 1$, an initial value $\varepsilon_0^2 \equiv e^{\log e_0^2}$ is treated as an additional free parameter with its own prior, and σ_0^2 is set to the implied unconditional variance $a_0 / (1 - a_1 - \gamma_1)$ evaluated at the current draw of (a_0, a_1, γ_1) . The full parameter vector is $\theta = (a_0, a_1, \gamma_1, \log e_0^2)$, and the log-likelihood is the sum of Gaussian log-densities implied by recursively computing σ_t^2 for $t = 1, \dots, T$ from θ and the data (the recursion is the ordinary GARCH(1,1) likelihood recursion above, not a Kalman filter in the state-space sense of Chapter 15).

Algorithm: Prior and Parameter Constraints

1. Place independent Normal priors on (a_0, a_1, γ_1) , centered at textbook values (0.01, 0.05, 0.90) with variance 0.0025 each.
2. Place a tight Normal prior on the initial log-variance state $\log e_0^2$, centered at the log of the *current draw's* implied unconditional variance, $\log(a_0/(1 - a_1 - \gamma_1))$, with variance 0.0001 — a self-referential prior that keeps the pre-sample state anchored to whatever stationary level the current (a_0, a_1, γ_1) draw implies, rather than letting it float freely.
3. Reject the posterior kernel outright (assign it log-density -10^{10}) whenever $a_0 \leq 0$, $a_1 \leq 0$, $\gamma_1 \leq 0$, or $a_1 + \gamma_1 \geq 0.99$, enforcing positivity and covariance-stationarity.

14.2 Posterior Derivation

Unlike every model in Chapters 6–13, the GARCH(1,1) posterior has no conjugate structure to exploit: σ_t^2 depends nonlinearly (through the recursion) on (a_0, a_1, γ_1) , so the likelihood is not of a form whose product with any convenient prior yields a named distribution. The two derivations needed here are therefore of a different character than Chapters 6–13's: first, writing down the exact (unnormalized) posterior kernel that is numerically maximized and evaluated at every M-H step, and second, justifying *why* the algorithm box's acceptance probability α is the correct one for an independence proposal.

Derivation: The GARCH Posterior Kernel

Given $\theta = (a_0, a_1, \gamma_1, \log e_0^2)$, the recursion $\sigma_t^2 = a_0 + a_1 \varepsilon_{t-1}^2 + \gamma_1 \sigma_{t-1}^2$ (initialized at $\sigma_0^2 = a_0/(1 - a_1 - \gamma_1)$ and $\varepsilon_0^2 = e^{\log e_0^2}$) makes every σ_t^2 , $t = 1, \dots, T$, a deterministic function of θ and the data. Because $\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$, the likelihood factors by the prediction-error decomposition (successive conditioning, exactly as in any time-series likelihood):

$$p(y | \theta) = \prod_{t=1}^T p(y_t | \mathcal{F}_{t-1}, \theta) = \prod_{t=1}^T \frac{1}{\sqrt{2\pi\sigma_t^2(\theta)}} \exp\left\{-\frac{y_t^2}{2\sigma_t^2(\theta)}\right\}$$

Multiplying by the independent priors $p(a_0)p(a_1)p(\gamma_1)p(\log e_0^2 | a_0, a_1, \gamma_1)$ (the last one self-referential, centered at the current draw's implied unconditional log-variance, as noted above) and the hard indicator $\mathbb{K}\{a_0 > 0, a_1 > 0, \gamma_1 > 0, a_1 + \gamma_1 < 0.99\}$ gives the exact target kernel:

$$\begin{aligned} \pi(\theta | y) &\propto \left[\prod_{t=1}^T \sigma_t^2(\theta)^{-1/2} \exp\left\{-\frac{y_t^2}{2\sigma_t^2(\theta)}\right\} \right] \\ &\quad \times N(a_0; 0.01, 0.0025) N(a_1; 0.05, 0.0025) N(\gamma_1; 0.90, 0.0025) \\ &\quad \times N\left(\log e_0^2; \log \frac{a_0}{1-a_1-\gamma_1}, 0.0001\right) \times \mathbb{K}\{a_0 > 0, a_1 > 0, \gamma_1 > 0, a_1 + \gamma_1 < 0.99\} \end{aligned}$$

whose log is the objective $\ln \pi(\theta)$ maximized in step 1 of the algorithm box and evaluated at both θ' and $\theta^{(s)}$ in every acceptance step; because $\sigma_t^2(\theta)$ enters nonlinearly and recursively, no further simplification into a named distribution is possible, which is exactly why this chapter turns to Metropolis-Hastings rather than Gibbs sampling.

Derivation: Why the Tailored Independence Acceptance Ratio Is Correct

Any Metropolis-Hastings chain with proposal density $q(\theta' | \theta)$ and acceptance probability $\alpha(\theta, \theta')$ has π as its stationary distribution provided the transition kernel satisfies *detailed balance*, $\pi(\theta) q(\theta' | \theta) \alpha(\theta, \theta') = \pi(\theta') q(\theta | \theta') \alpha(\theta', \theta)$: summing (integrating) both sides over θ shows this condition is sufficient for π to be invariant under the chain. The standard Metropolis-Hastings choice

$$\alpha(\theta, \theta') = \min\left(1, \frac{\pi(\theta') q(\theta | \theta')}{\pi(\theta) q(\theta' | \theta)}\right)$$

satisfies detailed balance for *any* proposal q : on the event $\pi(\theta) q(\theta' | \theta) > \pi(\theta') q(\theta | \theta')$, the left side's $\alpha(\theta, \theta') = \pi(\theta') q(\theta | \theta') / [\pi(\theta) q(\theta' | \theta)]$ and the right side's $\alpha(\theta', \theta) = 1$ make both sides equal $\pi(\theta') q(\theta | \theta')$, and symmetrically on the reverse event; this is the same general M-H derivation underlying Chapter 4's random-walk chain. For the tailored *independence* proposal, $q(\theta' | \theta) = q(\theta')$ does not depend on θ , so

$$\frac{\pi(\theta') q(\theta | \theta')}{\pi(\theta) q(\theta' | \theta)} = \frac{\pi(\theta') q(\theta)}{\pi(\theta) q(\theta')}$$

(substituting $q(\theta' | \theta) = q(\theta')$ and $q(\theta | \theta') = q(\theta)$, valid precisely because the proposal does not depend on the current state), which rearranges to:

$$= \frac{\pi(\theta') / q(\theta')}{\pi(\theta) / q(\theta)},$$

exactly the ratio-of-importance-weights form used in the algorithm box. This is why an independence proposal cannot drop the q terms the way a symmetric random-walk proposal does (Chapter 4): q does not cancel unless $q(\theta) = q(\theta')$, which is not guaranteed once q is fixed independently of the current state.

14.3 Tailored Independence Metropolis-Hastings

Chapter 4 introduced two Metropolis-Hastings variants: random-walk M-H, whose proposal is centered on the *current* state, and multi-block M-H, which updates parameter blocks one at a time. Both adapt locally as the chain moves. A *tailored independence* chain instead fixes the proposal once, before the chain runs, built from a Normal or Student-t approximation to the whole posterior at its mode — a Laplace approximation. Because the same proposal is used at every iteration regardless of the current state θ , the Metropolis-Hastings acceptance ratio must explicitly divide out the proposal density q at both points, rather than cancelling it as a symmetric random-walk proposal does.

Algorithm: Tailored Independence M-H

1. **Mode-finding.** Maximize the log-posterior $\ln \pi(\theta)$ numerically to obtain the mode $\hat{\theta}$ (the textbook approach does this by Newton search; this implementation uses a derivative-free Nelder-Mead search followed by a BFGS polish step, which proved markedly more robust here than gradient-based search alone — the hard rejection

region above creates a sharp cliff in the log-posterior surface that repeatedly threw off BFGS's line search and finite-difference gradient estimates when started from the prior mean, converging to a point more than 65 log-posterior units below the true mode; Nelder-Mead's simplex search does not rely on local gradient information and reliably finds the true mode from the same starting point).

2. **Laplace approximation.** Compute the Hessian of $\ln \pi(\theta)$ at $\hat{\theta}$ numerically, and set $V = (-\hat{H})^{-1}$, the negative-inverse-Hessian — the same curvature matrix a Normal approximation to the posterior would use.
3. **Fixed Student-t proposal.** Draw candidates from a multivariate Student-t centered at $\hat{\theta}$ with scale matrix V and $\nu = 30$ degrees of freedom (heavier-tailed than a Normal, so that the fixed proposal still reaches the true posterior's tails even where the Laplace approximation is imperfect).
4. **Acceptance.** From current state $\theta^{(s)}$, draw $\theta' \sim q(\cdot) = t_\nu(\hat{\theta}, V)$ and accept it as $\theta^{(s+1)}$ with probability

$$\alpha = \min \left(1, \frac{\pi(\theta')/q(\theta')}{\pi(\theta^{(s)})/q(\theta^{(s)})} \right),$$

otherwise set $\theta^{(s+1)} = \theta^{(s)}$.

The chain is run for 6,000 post-burn-in draws after a 1,500-draw burn-in.

Remark: Why the Proposal Must Correct Itself

A random-walk proposal $\theta' = \theta^{(s)} + \eta$, η symmetric around 0, satisfies $q(\theta' | \theta^{(s)}) = q(\theta^{(s)} | \theta')$, so the proposal density cancels from the acceptance ratio and only the posterior ratio $\pi(\theta')/\pi(\theta^{(s)})$ remains — the familiar form from Chapter 4. An independence proposal does not depend on $\theta^{(s)}$ at all, so nothing cancels: a candidate drawn from a region where q happens to be large relative to π would be over-accepted if the q terms were dropped. Dividing the posterior by the proposal density at both the candidate and the current point, as in the algorithm box above, is what keeps the resulting chain's stationary distribution equal to the true posterior π rather than some blend of π and q .

14.4 Empirical Example: Fitting GARCH(1,1) to S&P 500 Daily Returns

Empirical Example: Posterior Estimates

Fitting the model above to all $T = 755$ trading days of S&P 500 daily log returns gives the following posterior summary (posterior mean, posterior standard deviation in parentheses):

	a_0	a_1	γ_1
Posterior mean (sd)	0.0466 (0.0089)	0.1162 (0.0189)	0.8046 (0.0248)

with implied persistence $a_1 + \gamma_1 = 0.921$ (sd 0.0161) — high, as is typical for daily equity volatility, but comfortably inside the $a_1 + \gamma_1 < 0.99$ constraint region and far from the boundary itself, unlike the mis-converged fit discussed in the algorithm box above. The

tailored-M-H acceptance rate is 39.7%, a healthy value for an independence chain (a well-tuned tailored proposal typically accepts in the 20–50% range; acceptance rates near 0%, as this chapter’s earlier, uncorrected mode-finding attempt initially produced, are the standard diagnostic that the Laplace approximation was built at the wrong point or with badly mis-scaled curvature). Figure 58 shows the posterior distributions of a_1 , γ_1 , and their sum; all three are unimodal and comfortably interior to the parameter-constraint region, the signature of a chain that has converged to a genuine likelihood-dominated posterior rather than one pinned against a boundary.

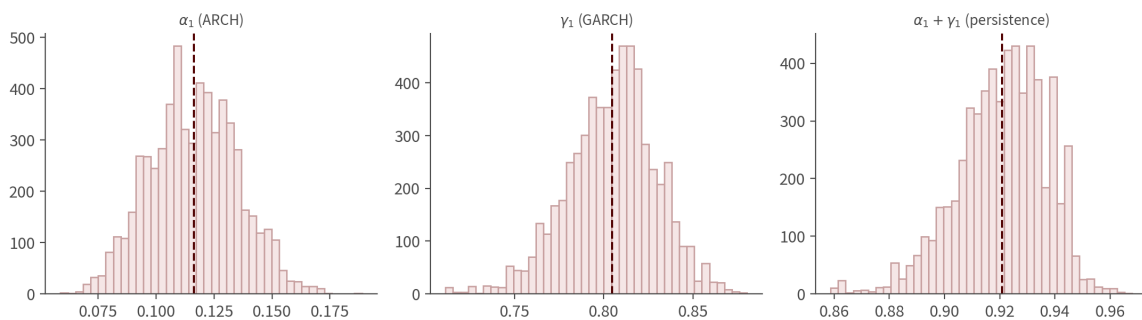


Figure 58: Posterior distributions of the ARCH coefficient α_1 , the GARCH coefficient γ_1 , and their sum (persistence); dashed line marks the posterior mean

Empirical Example: Fitted Volatility and the August 2015 Selloff

Figure 59(a) plots the posterior mean conditional volatility $\hat{\sigma}_t$ (averaged over 200 posterior draws) against the realized absolute daily return, together with a flat **constant-variance benchmark** — the degenerate special case $a_1 = \gamma_1 = 0$ of the model above, whose Bayes estimate is simply the sample standard deviation of the returns, 0.80% per day, exactly what Chapters 11–13’s constant Σ would have delivered here. Average *fitted* GARCH volatility over the full sample is 0.76% per day (consistent with the flat benchmark, as it should be on average), but it is far from flat: it ranges from a calm-market low of 0.53% to a peak of 2.07% reached on **2015-08-27**, three trading days after China’s surprise yuan devaluation triggered a global equity selloff (the S&P 500 itself fell 4.0% on 2015-08-24, “Black Monday”), and the single most volatile stretch of the entire 2013–2015 sample — a more than **four-fold** swing in volatility that the flat benchmark line, by construction, cannot show. Smaller secondary clusters appear around **2013-06** (the Fed “taper tantrum,” the first market reaction to talk of ending quantitative easing) and **2014-10/2014-12** (a global growth scare compounded by the sharp collapse in oil prices). All three episodes show the GARCH mechanism working as intended: a large shock (ε_{t-1}^2) pushes σ_t^2 up sharply through the a_1 term, and the high estimated persistence $\gamma_1 = 0.80$ then keeps volatility elevated for one to two weeks afterward rather than reverting immediately — exactly the volatility-clustering pattern the flat constant-variance line in panel (a) is, by construction, unable to represent.

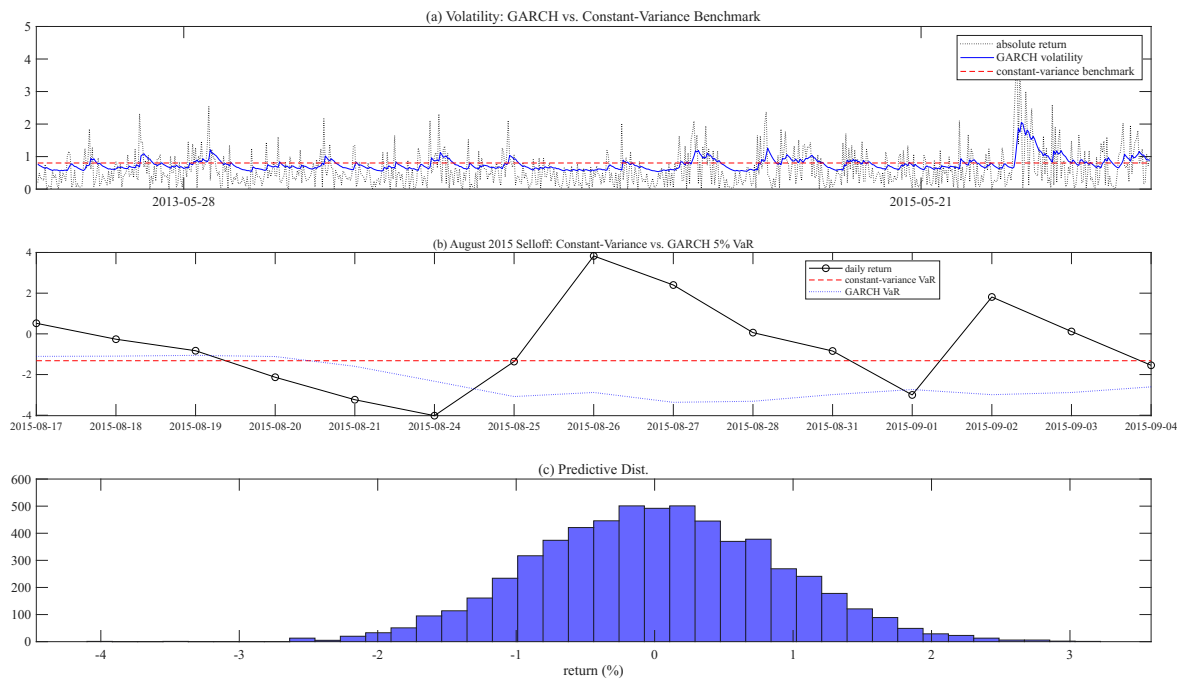


Figure 59: (a) Posterior mean conditional volatility vs. realized $|return|$, full S&P 500 sample, against the flat constant-variance benchmark (0.80%); (b) 5% one-step-ahead Value-at-Risk during the August 2015 selloff, constant-variance benchmark vs. fitted GARCH(1,1); (c) one-step-ahead posterior predictive return distribution with 5% VaR and Expected Shortfall

Empirical Example: One-Step-Ahead Risk: VaR and Expected Shortfall

For each of 1,500 posterior draws of θ , simulating the next trading day's return under that draw's own fitted volatility path gives a one-step-ahead posterior predictive distribution (Figure 59(c)) that properly integrates out parameter uncertainty along with the usual return-innovation uncertainty — a genuinely Bayesian risk forecast, as opposed to a plug-in forecast that fixes θ at a point estimate. The predictive mean is -0.02% and predictive standard deviation 0.89% , giving a 95% predictive interval of $[-1.86\%, 1.77\%]$. The 5% Value-at-Risk — the loss threshold breached with 5% predictive probability — is $VaR_{5\%} = -1.50\%$, and the 5% Expected Shortfall — the average loss *conditional* on breaching that threshold — is $ES_{5\%} = -1.88\%$, noticeably worse than the VaR itself, as it must be: ES averages only the tail beyond VaR, so it always weakly exceeds VaR in magnitude, and the gap here (0.38 percentage points) quantifies how much worse a bad day can get once volatility is already realized to be in the fitted model's extreme tail.

Remark: Constant-Variance VaR Against the August 2015 Selloff

Does the flat constant-variance benchmark actually get risk wrong when it matters, or is the volatility-clustering picture in panel (a) just a cosmetic difference? Figure 59(b) puts both models' 5% VaR — $z_{0.05}\hat{\sigma}$, with $\hat{\sigma}$ either the flat 0.80% benchmark or the GARCH one-step-ahead $\hat{\sigma}_t$ — against the actual returns of 2015-08-17 through 2015-09-04. The constant-variance VaR is a single flat line at -1.32% every day; the GARCH VaR ranges from -0.87% in calm periods up to -3.41% once the recursion has absorbed the shock,

a four-fold widening over the same two-week window. The gap is starkest on the worst day in the entire sample, **2015-08-24** ($y_t = -4.02\%$): against the flat constant-variance benchmark this is a -5.01 standard-deviation event, with tail probability 2.7×10^{-7} under a Normal — rarer than 1-in-3.7 million trading days, i.e., an event the constant-variance model effectively says should not happen. Against the GARCH model's own volatility assessment that day ($\hat{\sigma}_t = 1.43\%$, already elevated by the two preceding days' shocks on 2015-08-20/21), the same return is a -2.81 standard-deviation event, tail probability 0.25% , roughly a 1-in-400-trading-day occurrence — rare, but not absurd. This is the concrete sense in which the constant-variance model “cannot” capture volatility clustering: it is not merely smoother, it is *wrong exactly when it matters*, understating the plausibility of the realized crash by more than four orders of magnitude. A complementary check across the full sample: the four known high-volatility episodes above (the taper tantrum, Oct/Dec 2014, and the August 2015 selloff, 80 of the sample's 755 days, 10.6%) account for 34.9% of all constant-variance-VaR breaches but only 23.9% of GARCH-VaR breaches — constant-variance exceptions cluster in exactly the stress periods a well-calibrated conditional risk model should spread more evenly over time, and the GARCH model, while not eliminating this clustering entirely, cuts its concentration by roughly a third.

14.5 A Second Window: The COVID-19 Crash and Recovery, 2019–2021

Empirical Example: A More Extreme Test Case

The 2013–2015 window above shows GARCH(1,1) working as intended against real but moderate volatility clustering. To see the same mechanism pushed to its limit, this section refits the identical model, unchanged, to a second, independent window of daily S&P 500 returns: $T = 524$ trading days from **2019-06-04** to **2021-06-30**, spanning the fastest bear market in the index's history (a 34% peak-to-trough decline in February–March 2020) and the recovery that followed. This window is added *alongside* the original one, not in place of it: the two together let the same model and the same sampler be judged against both an ordinary, moderately volatile stretch and the single most extreme volatility event of the sample period covered anywhere in this note. Posterior mean (posterior standard deviation in parentheses):

	a_0	a_1	γ_1
Posterior mean (sd)	0.0402 (0.0087)	0.1671 (0.0220)	0.8038 (0.0222)

with implied persistence $a_1 + \gamma_1 = 0.971$ (sd 0.0064) — noticeably higher than the 2013–2015 window's 0.921 , and closer to the $a_1 + \gamma_1 < 0.99$ stationarity boundary, though still comfortably inside it. The tailored-M-H acceptance rate is 18.0% , lower than the earlier window's 39.7% but still within the workable range for an independence chain: the sharper, more extreme observations in this sample make the log-posterior surface less well approximated by a single fixed Student-t proposal built from one Laplace approximation, so more candidates are rejected even though the chain still mixes and converges. Figure

60 shows the resulting posterior distributions; all three remain unimodal and interior to the constraint region.

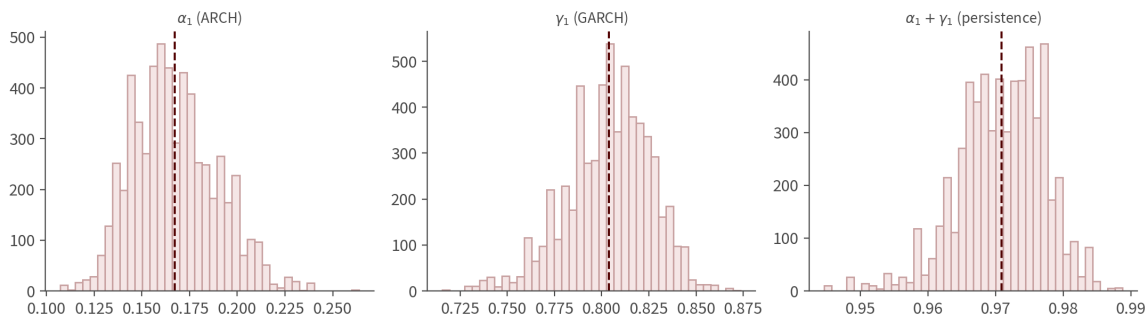


Figure 60: Posterior distributions of α_1 , γ_1 , and their sum (persistence), COVID-19 window; dashed line marks the posterior mean

Empirical Example: Volatility That Actually Breaks the Flat Benchmark

Figure 61(a) plots the same comparison as before — fitted GARCH volatility against a flat constant-variance benchmark, now 1.62% per day, itself already double the earlier window's 0.80% purely because a single extreme month is baked into the whole-sample average. Fitted GARCH volatility ranges from a calm-market low of 0.51% to a peak of 7.66% on 2020-03-17 — more than $3.5\times$ the previous window's worst reading, and a **15-fold** swing in volatility within a single sample, dwarfing the earlier window's already-notable four-fold swing. Panel (b) zooms into the crash itself (Feb–Apr 2020): the daily-return series swings sharply between double-digit percentage moves in both directions within the same week, and the GARCH conditional-volatility band visibly widens and tracks those swings almost in real time, while the flat constant-variance band sits uselessly in the middle of it throughout. The single worst trading day in this sample, 2020-03-16 ($y_t = -12.77\%$), is a -7.86 standard-deviation event against the flat constant-variance benchmark — tail probability 1.9×10^{-15} , effectively zero, an event the constant-variance model says cannot happen even once in the lifetime of the universe. Against the GARCH model's own volatility assessment that day, already elevated from the preceding week's shocks, the same return is a -2.05 standard-deviation event, tail probability 2.0% — rare, but a real, roughly 1-in-50-trading-day occurrence, not an impossibility. And volatility clustering cuts both ways: the single *best* trading day in the sample, 2020-03-24 ($y_t = +8.97\%$, the day markets rallied on stimulus hopes), falls just **six trading days** after the worst one — the clearest possible illustration that an elevated σ_t^2 predicts a wider distribution of outcomes in *either* direction, not simply “more crashes.”

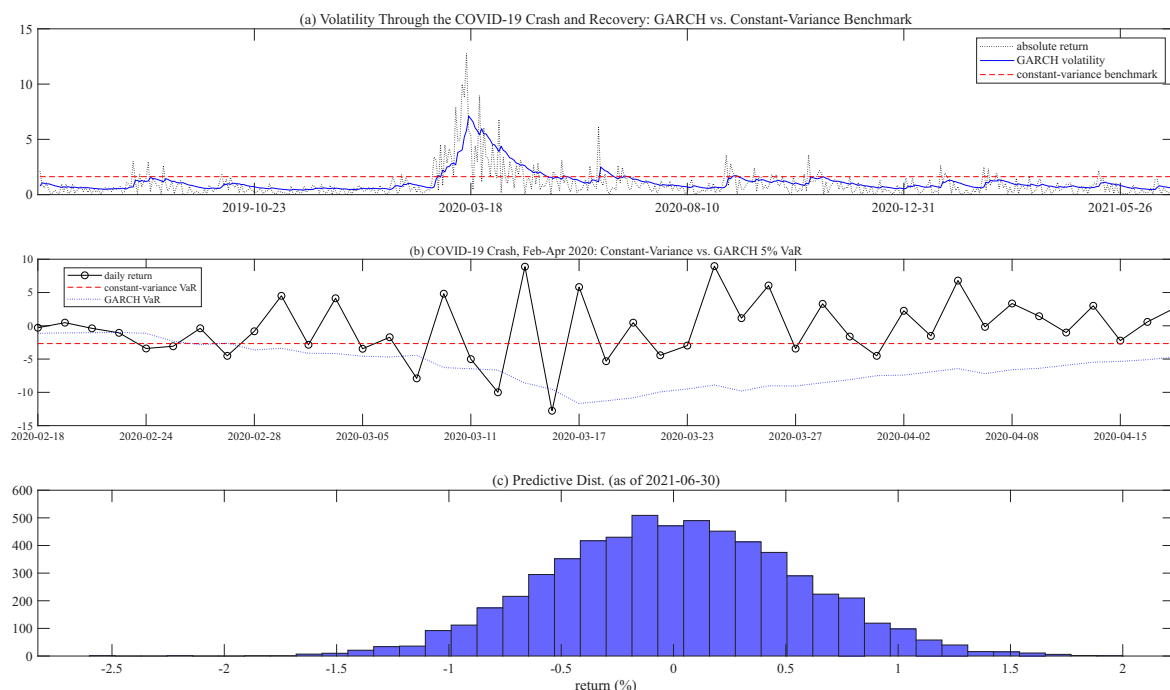


Figure 61: (a) Posterior mean conditional volatility vs. realized $|return|$, COVID-19-window S&P 500 sample, against the flat constant-variance benchmark (1.62%); (b) daily returns during the Feb–Apr 2020 crash against a ± 1 GARCH conditional standard-deviation band vs. the flat benchmark; (c) one-step-ahead posterior predictive return distribution evaluated at the end of the sample (2021-06-30) with 5% VaR and Expected Shortfall

Remark: A Constant-Variance Model Never Recovers; GARCH Does

The starkest difference between the two windows is not the size of the crash itself but what each model says *after* it. Evaluated as of the very end of this sample, 2021-06-30 — fifteen months after the crash bottomed and well into a calm, low-volatility recovery — the one-step-ahead posterior predictive distribution (Figure 61(c)) has predictive mean -0.01% , predictive standard deviation 0.64% , a 5% VaR of $VaR_{5\%} = -1.07\%$, and a 5% Expected Shortfall of $ES_{5\%} = -1.34\%$ — all *smaller in magnitude* than the corresponding 2013–2015-window figures (-1.50% and -1.88%), despite this window containing by far the more extreme historical event. GARCH’s mean-reverting persistence, $\gamma_1 = 0.80$, lets the fitted volatility path decay back down once the shock has passed through the recursion, so a forecast made from a calm point in time correctly looks calm. The flat constant-variance benchmark cannot do this by construction: its single number, 1.62% per day, is a simple average over the *entire* window including March 2020, and it would keep predicting that same elevated risk indefinitely — in 2021-06-30’s actually placid market — purely because one bad month fifteen months earlier still counts equally toward the sample average as every calm day since. A complementary breach-clustering check across the full window (the crash itself plus two later 2020–21 stress episodes, the November 2020 election/vaccine-announcement swing and the January 2021 meme-stock volatility spike, 81 of the sample’s 524 days, 15.5%) shows the same pattern found in the earlier window, more pronounced here: these stress days account for 68.2% of all constant-variance-VaR breaches but only 25.0% of GARCH-VaR breaches. Extending the sample through COVID

does not just make the GARCH-vs.-constant-variance comparison bigger; it makes the case for time-varying volatility sharper, because the cost of ignoring volatility clustering — both during the crash and, just as importantly, for years afterward — scales with how extreme the underlying shock was.

14.6 Bollerslev (1986)

The two S&P 500 windows above show GARCH(1,1) doing the job it is now best known for; this section replicates the empirical example in the paper that *created* the model, which is not about asset returns at all. Bollerslev (1986), “Generalized Autoregressive Conditional Heteroskedasticity,” *Journal of Econometrics* 31(3), closes with an application to the uncertainty of U.S. inflation: quarterly GNP-deflator inflation, $\pi_t = 100 \ln(GD_t/GD_{t-1})$ (FRED `GNPDEF`), over 1948Q2–1983Q4 ($T = 143$), with an AR(4) conditional mean and the then-new GARCH(1,1) conditional variance,

$$\pi_t = c + \sum_{i=1}^4 b_i \pi_{t-i} + e_t, \quad h_t = a_0 + a_1 e_{t-1}^2 + \gamma_1 h_{t-1}.$$

The paper’s argument was one of parsimony: Engle and Kraft (1983) had fit the same mean equation with an ARCH(8) whose eight coefficients were forced onto an ad hoc linearly declining lag shape, and Bollerslev showed that a single lagged conditional variance — two variance parameters instead of eight-with-restrictions — fits slightly better and delivers a more plausible learning mechanism. The replication uses the identical tailored-M-H machinery as the rest of this chapter (posterior mode by `SA_Newton`, Student- t independence proposal), with presample h and e^2 initialized at the unconditional variance.

Empirical Example: Original Sample, 1948Q2–1983Q4

Posterior means (sd) against the paper’s maximum-likelihood estimates (eq. 31, standard errors in parentheses):

	posterior	Bollerslev (1986)
c	0.075 (0.037)	0.141 (0.060)
b_1	0.496 (0.102)	0.433 (0.081)
b_2	0.190 (0.097)	0.229 (0.110)
b_3	0.206 (0.089)	0.349 (0.077)
b_4	0.028 (0.093)	−0.162 (0.104)
a_0	0.015 (0.006)	0.007 (0.006)
a_1	0.334 (0.062)	0.135 (0.070)
γ_1	0.620 (0.068)	0.829 (0.068)

The paper's headline number survives exactly: estimated persistence $a_1 + \gamma_1 = 0.954$ (sd 0.026) against Bollerslev's 0.964 — inflation uncertainty is highly persistent but stationary, just inside the integrated-GARCH boundary. The *split* of that persistence between a_1 and γ_1 comes out differently (more weight on the squared shock, less on the lagged variance), for two reasons worth distinguishing. First, on 143 quarterly observations the two variance parameters are only weakly identified separately — long, flat ridges in the likelihood along which $a_1 + \gamma_1$ is constant are a well-known feature of GARCH estimation on short samples — so the sum is the robustly estimated object. Second, the inflation series itself is not the one Bollerslev used: his deflator was the 1984-vintage fixed-weight index, while the modern `GNPDEF` is chain-weighted (a methodology introduced in 1996) and has been comprehensively revised many times since, which also accounts for the sign flip on the small b_4 coefficient. Figure 62 reproduces the paper's concluding graphic (its Figure 4): one-step 95% conditional bands that are wide in the volatile late-1940s and Korean-War years, hug the series tightly through the placid 1960s — exactly the period where Bollerslev notes a constant-variance band “seems much too wide” — and widen moderately after the 1974 oil shock.

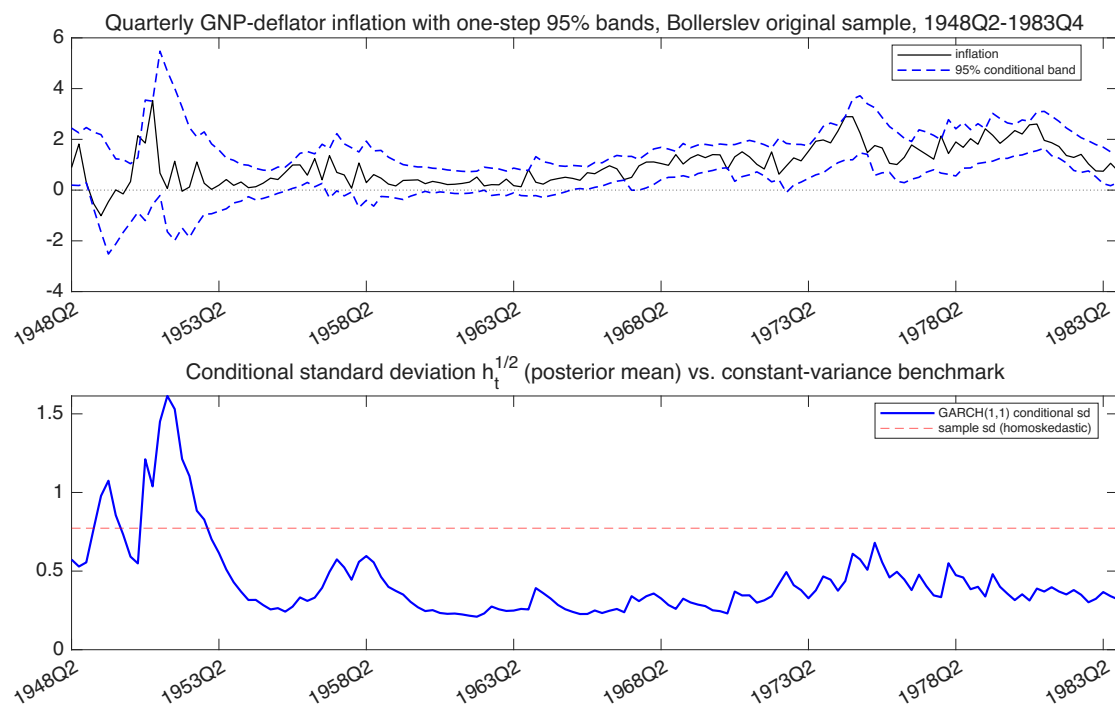


Figure 62: Bollerslev (1986) replicated: GNP-deflator inflation with one-step 95% conditional bands (top, the paper's Figure 4) and the conditional standard deviation against the constant-variance benchmark (bottom), original sample

Empirical Example: Full Sample, 1948Q2–2026Q1

Re-estimating the identical eight-parameter model on the series extended through 2026Q1 ($T = 312$) is a forty-two-year out-of-sample test of a specification chosen in 1986, and it passes with almost nothing to adjust: persistence is $a_1 + \gamma_1 = 0.959$ (sd 0.019), with $P(a_1 + \gamma_1 > 0.95) = 0.87$ — still high, still stationary, still just inside the boundary. The

economics is in the fitted conditional standard deviation (Figure 63, bottom panel). The path Bollerslev knew — Korean-War turbulence, 1960s calm, a post-oil-shock rise — is now followed by the long trough of the Great Moderation, a quarter century in which quarterly inflation uncertainty sits near its all-time low, and then an abrupt reversal: the 2021–22 inflation surge drives conditional volatility to its highest level since the early 1980s, before the post-tightening disinflation brings it part of the way back down. A model built to describe the inflation uncertainty of the 1970s spent twenty-five years reporting that its subject had nearly disappeared, and then documented its return — the same conclusion Chapter 16’s stochastic-volatility machinery reaches for the full VAR system, delivered here by two variance parameters.

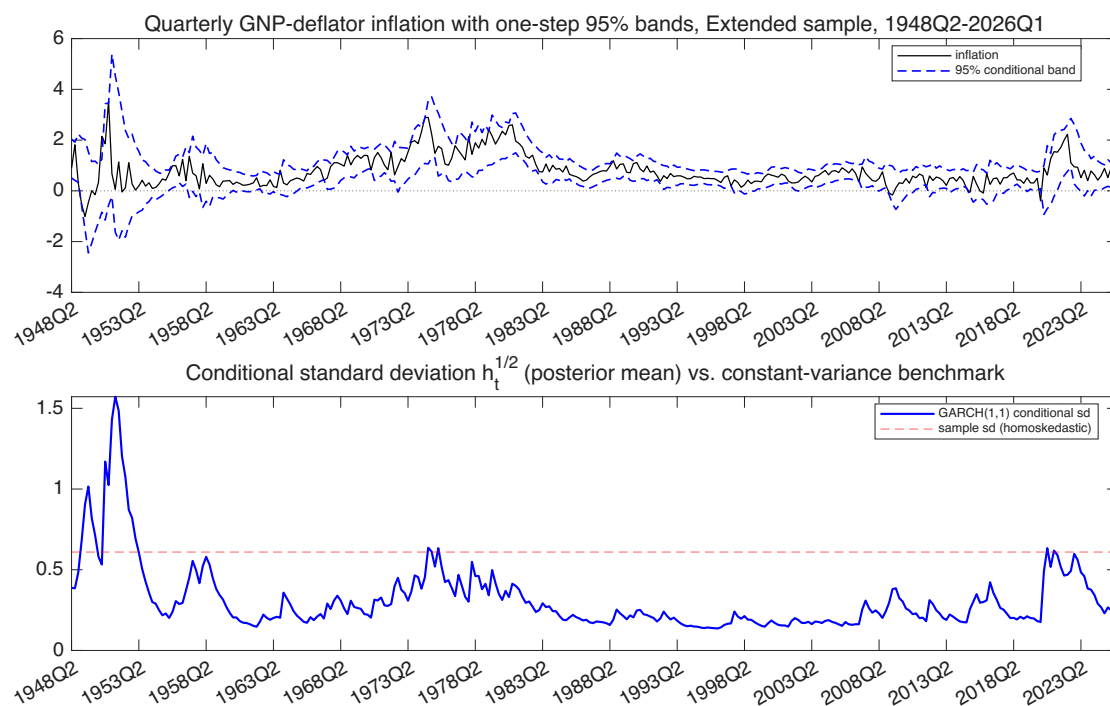


Figure 63: The same AR(4)-GARCH(1,1) estimated through 2026Q1: the Great Moderation trough and the 2021–22 return of inflation uncertainty

15. State-Space Models and the Kalman Filter

Roadmap

Chapter 14 let variance move over time while the mean stayed fixed. This chapter lets the mean itself move, through an unobserved state α_t that the data only see indirectly, through y_t . The Kalman filter is the classical way to extract that state, but this chapter’s actual implementation uses an algebraically equivalent route, treating the whole path $\alpha_{1:T}$ as one large Gaussian vector and drawing it in a single linear solve. That reframing matters beyond this chapter: it is what Chapters 16–18 reuse wholesale, applying the identical precision-sampler machinery to a VAR’s coefficients or volatilities evolving over time instead of a single scalar trend.

Chapter 14 let a single equation's *variance* evolve over time while holding its conditional mean fixed at zero. This chapter turns the same idea around: a state-space model lets a series' conditional *mean* evolve over time, driven by one or more unobserved (latent) states that themselves follow simple time-series processes. The two workhorse examples below — decomposing a growth rate into a slowly moving trend plus noise, and decomposing GDP into a smooth trend plus a stationary business-cycle component — recur throughout applied macroeconomics, and Chapters 16–18 will each extend this same latent-state machinery to full VAR systems.

Linear Gaussian State-Space Model A generic linear Gaussian state-space model consists of a measurement equation linking the observed data y_t to an unobserved state α_t , and a state equation describing how α_t evolves:

$$\text{(measurement)} \quad y_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H_t),$$

$$\text{(state)} \quad \alpha_t = T_t \alpha_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q_t),$$

with ε_t and η_t mutually independent at all leads and lags, and an initial state α_0 with its own prior distribution. Figure 64 shows the resulting dependency structure: the states form their own Markov chain (Chapter 3's dependency pattern again, now with matrix-valued transitions), and each y_t branches off its own α_t without directly seeing any other period's state.

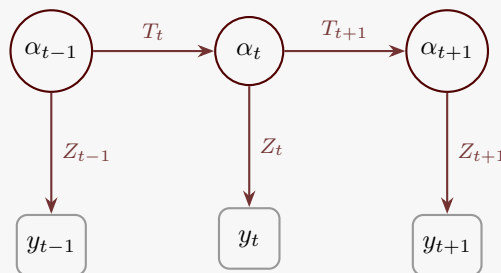


Figure 64: Dependency structure of a linear Gaussian state space: the latent states form a Markov chain, and each observation depends on the state at its own date only

Both empirical models in this chapter are special cases: a *local-level* model, where α_t is a single random-walk trend, and a *local-linear-trend-plus-cycle* model, where α_t stacks a smoothly evolving trend with a stationary AR(2) cycle.

Algorithm: The Kalman Filter and Its Precision-Sampler Equivalent

1. **The classical algorithm.** The standard tool for this model is the **Kalman filter**: a forward recursion that, at each t , combines the prior forecast of α_t from α_{t-1} with the new observation y_t into an updated (filtered) estimate, propagating a mean and covariance rather than a full distribution because everything here is linear and Gaussian. Running the filter forward and then a complementary **backward smoother** (forward filtering, backward sampling, FFBS) delivers a draw of the *entire* latent state path $\alpha_{1:T}$ from its exact joint conditional posterior, which is what a Gibbs sampler needs at each sweep.

2. **The equivalent implementation used here.** Rather than implementing FFBS directly, stack the state equation for all $t = 1, \dots, T$ into one linear system $H\alpha = b + u$, where H is a sparse, banded *first- or second-difference operator*. The joint prior on the whole path $\alpha_{1:T}$ is then Gaussian with precision (inverse-covariance) matrix $H'Q^{-1}H$ – the Chan and Jeliazkov (2009) precision-sampler approach.

3. **Combine with the likelihood precision.** Adding the (also Gaussian, diagonal) likelihood precision from the measurement equation gives a joint Gaussian full conditional for $\alpha_{1:T}$ with precision

$$K_\alpha = H'Q^{-1}H + Z'H_t^{-1}Z,$$

which is banded and therefore cheap to Cholesky-factor even for T in the hundreds.

4. **Draw the whole path in one block.** A single draw is $\alpha = \hat{\alpha} + L^{-1}z$, $z \sim N(0, I_T)$, where L is the Cholesky factor of K_α and $\hat{\alpha} = K_\alpha^{-1}b$ (posterior mean term).

5. **Why the two are equivalent.** For a linear Gaussian state-space model, this delivers exactly the same joint distribution over $\alpha_{1:T}$ that FFBS would — both are just different factorizations of the same Gaussian — so this chapter presents the Kalman filter as the standard conceptual algorithm while noting explicitly, in the same spirit as Chapter 14's naming clarification, that the actual implementation below uses this equivalent banded-precision construction rather than a literal forward-backward filter.

15.1 Posterior Derivation: The Precision-Sampler Full Conditional for a Latent Path

The derivation below is written once, for the local-level model's random-walk trend, and is then reused without modification for Model 2's smoothed trend (with the first-difference operator H replaced by the second-difference operator H_2) and again in Chapters 16–17.

Derivation: $\tau_{1:T} \mid \sigma^2, \omega^2, y$

Stack the trend path as $\tau = (\tau_1, \dots, \tau_T)'$. The state equation $\tau_t = \tau_{t-1} + \eta_t$ for $t = 1, \dots, T$, with τ_0 given, can be written as one linear system $H\tau = b + u$, $u \sim N(0, \omega^2 I_T)$, where H is the $T \times T$ lower-bidiagonal first-difference matrix (1 on the diagonal, -1 on the subdiagonal) and $b = (\tau_0, 0, \dots, 0)'$ absorbs the one place the fixed initial condition enters. Because H is invertible (lower triangular with unit diagonal), this is equivalent to a Gaussian prior on τ itself: $\tau = H^{-1}b + H^{-1}u$, so $\tau \sim N(H^{-1}b, \omega^2(H'H)^{-1})$, i.e. a Gaussian with *precision* $\omega^{-2}H'H$ directly (never needing to invert $H'H$ itself, since H is triangular and its determinant, hence invertibility, is immediate). The kernel of this prior is

$$\begin{aligned} p(\tau) &\propto \exp\left\{-\frac{1}{2}(H\tau - b)' \omega^{-2} I_T (H\tau - b)\right\} \\ &= \exp\left\{-\frac{1}{2}(\tau - H^{-1}b)' (\omega^{-2} H'H) \right. \\ &\quad \left. (\tau - H^{-1}b)\right\} \end{aligned}$$

The measurement equation $y_t = \tau_t + \varepsilon_t$, $\varepsilon_t \sim N(0, \sigma^2)$, stacked over t , gives likelihood kernel $\exp\{-\frac{1}{2}\sigma^{-2}(y - \tau)'(y - \tau)\}$, a Gaussian in τ with precision $\sigma^{-2}I_T$ and mean y . Multiplying the two kernels and collecting quadratic and linear terms in τ — the same completing-the-square strategy as Chapter 6.2, only now with τ a T -vector rather than a K -vector and both “prior” and “likelihood” precisions banded matrices rather than scalars — gives:

$$\begin{aligned} & -\frac{1}{2} \left[\tau' (\omega^{-2} H' H + \sigma^{-2} I_T) \tau \right. \\ & \quad \left. - 2\tau' (\omega^{-2} H' H H^{-1} b + \sigma^{-2} y) \right] \\ & = -\frac{1}{2} \left[\tau' K_\tau \tau - 2\tau' (\omega^{-2} H' b + \sigma^{-2} y) \right] \end{aligned}$$

using $H' H H^{-1} = H'$.^a Completing the square exactly as in Chapter 6.2 gives:

$$\begin{aligned} \tau \mid \sigma^2, \omega^2, y & \sim N(\hat{\tau}, K_\tau^{-1}), \\ K_\tau & = \omega^{-2} H' H + \sigma^{-2} I_T, \\ \hat{\tau} & = K_\tau^{-1} (\omega^{-2} H' b + \sigma^{-2} y) \end{aligned}$$

matching the algorithm box's $K_\alpha = H' Q^{-1} H + Z' H_t^{-1} Z$ with $Z_t = I$ here. Because K_τ is banded (tridiagonal, since $H' H$ is tridiagonal and I_T is diagonal), its Cholesky factor $K_\tau = LL'$ costs $O(T)$ rather than $O(T^3)$ to compute. Drawing $z \sim N(0, I_T)$ and setting $\tau = \hat{\tau} + L^{-1} z$ produces a draw with the correct covariance, since $\text{Cov}(L^{-1} z) = L^{-1} I_T L^{-1} = (LL')^{-1} = K_\tau^{-1}$ — this is what lets a single linear solve replace the sequential forward-filter/backward-sample recursion of a literal Kalman-filter implementation while producing an identical joint distribution over the whole path.

^aThis holds simply because H^{-1} exists (it is triangular with unit diagonal, hence invertible) and matrix multiplication is associative: $H' H H^{-1} = H' (H H^{-1}) = H' I = H'$. The payoff is avoiding ever forming H^{-1} explicitly in the linear term — only the banded matrix $H' H$ is needed, which is what keeps the whole calculation $O(T)$ rather than $O(T^3)$.

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15.2 Model 1: A Local-Level Model for U.S. Real PCE Growth

Local-Level Model Let y_t be the (annualized, percent) growth rate of real personal consumption expenditure. The local-level model decomposes it into a slow-moving trend τ_t and transitory noise:

$$y_t = \tau_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2), \quad \tau_t = \tau_{t-1} + \eta_t, \quad \eta_t \sim N(0, \omega^2),$$

with $\tau_0 \sim N(a_0, b_0)$. The ratio ω^2/σ^2 is the *signal-to-noise ratio*: small values mean the trend moves only gradually and most period-to-period variation in y_t is transitory noise; large values mean the trend itself is volatile and tracks y_t closely. σ^2 and ω^2 each have Inverse-Gamma priors ($\nu = 3$, scale $S = 2$) and $\tau_0 \sim N(5, 100)$, a weakly informative prior centered near typical postwar real consumption growth. All three unknowns — the full trend path $\tau_{1:T}$, σ^2 , and ω^2 — are drawn by Gibbs sampling: $\tau_{1:T}$ jointly via the precision sampler above (with H the $T \times T$ first-difference operator),^a then σ^2 and ω^2 each from their Inverse-Gamma full conditionals given the current trend path and residuals.

^aThis joint path draw — one sparse Cholesky factorization of the banded posterior precision, no Kalman recursions — is the precision sampler of Chan and Jeliazkov (2009); `UC.m` in the replication kit implements exactly this block.

Derivation: $\sigma^2 \mid \tau, y$ and $\omega^2 \mid \tau$

Given the trend path τ , both variance full conditionals are Inverse-Gamma updates of exactly the Chapter 6.2 kernel-matching type, applied to two different residual series; both use the chapter's $\nu = 3$, $S = 2$ prior parameterization ($a_0 = 3$, $d_0 = 2$ in Chapter 6.2's $IG(a_0/2, d_0/2)$ notation). The algebra is written out for each in turn, rather than merely asserted by analogy.

1. $\sigma^2 \mid \tau, y$. The measurement equation $y_t - \tau_t = \varepsilon_t \sim N(0, \sigma^2)$ for $t = 1, \dots, T$ plays the role of Chapter 6.2's regression residual, with $e'e \rightarrow \sum_t (y_t - \tau_t)^2$ and the same sample size T . Writing the Gaussian likelihood as a function of σ^2 alone and multiplying by the $IG(3/2, 1)$ prior kernel,

$$p(y \mid \tau, \sigma^2) = (2\pi\sigma^2)^{-T/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \tau_t)^2\right\},$$

$$p(\sigma^2) \propto (\sigma^2)^{-(3/2+1)} \exp\left\{-\frac{2}{2\sigma^2}\right\}.$$

Multiplying the two and collecting powers of σ^2 separately from the shared exponential term, exactly as in Chapter 6.2:

$$p(\sigma^2 \mid \tau, y) \propto (\sigma^2)^{-T/2-(3/2+1)} \exp\left\{-\frac{1}{2\sigma^2} \left[\sum_{t=1}^T (y_t - \tau_t)^2 + 2\right]\right\}$$

and, writing the combined power in shape/scale form, the kernel of:

$$\sigma^2 \mid \tau, y \sim IG\left(\frac{3+T}{2}, \frac{2 + \sum_t (y_t - \tau_t)^2}{2}\right)$$

the prior degrees of freedom 3 and prior sum of squares 2 are updated by the sample size T and the realized sum of squared measurement residuals, respectively.

2. $\omega^2 \mid \tau$. The state innovations $\eta_t = \tau_t - \tau_{t-1} \sim N(0, \omega^2)$ for $t = 1, \dots, T$ (with τ_0 given) play the identical role, with $e'e \rightarrow \sum_t (\tau_t - \tau_{t-1})^2$ in place of the measurement residuals but the same sample size T and the same $IG(3/2, 1)$ prior:

$$p(\tau \mid \omega^2) = (2\pi\omega^2)^{-T/2} \exp\left\{-\frac{1}{2\omega^2} \sum_{t=1}^T (\tau_t - \tau_{t-1})^2\right\},$$

$$p(\omega^2) \propto (\omega^2)^{-(3/2+1)} \exp\left\{-\frac{2}{2\omega^2}\right\}.$$

Multiplying and collecting powers of ω^2 exactly as above:

$$p(\omega^2 \mid \tau) \propto (\omega^2)^{-T/2-(3/2+1)} \exp\left\{-\frac{1}{2\omega^2} \left[\sum_{t=1}^T (\tau_t - \tau_{t-1})^2 + 2\right]\right\}$$

which is, the kernel of:

$$\omega^2 \mid \tau \sim IG\left(\frac{3+T}{2}, \frac{2 + \sum_t (\tau_t - \tau_{t-1})^2}{2}\right)$$

Empirical Example: Empirical Results: Trend and Noise in Consumption Growth

Fitting this model to quarterly real PCE growth, 1959Q2–2015Q4 ($T = 227$, computed as $400 \times$ the log-difference of the levels), 6,000 post-burn-in Gibbs draws (after a 1,200-draw burn-in) give posterior means $\hat{\sigma}^2 = 5.28$ (95% CI [4.20, 6.59]) and $\hat{\omega}^2 = 0.449$ (95% CI [0.220, 0.828]), a signal-to-noise ratio of only 0.085: quarter-to-quarter noise in consumption growth is roughly twelve times larger than genuine quarter-to-quarter movement in its underlying trend, so the trend path in Figure 65 is, as expected, much smoother than the raw growth series itself. The estimated trend starts the sample at 5.23%, rises to a sample maximum of 10.96% (the high-growth, high-inflation years around the late 1970s), falls sharply to a sample minimum of 0.60% during the depths of the 2008–09 recession, and recovers to 3.45% by the end of the sample in 2015Q4 — a trend path consistent with the broad, well-documented arc of postwar U.S. consumption growth without needing any series-specific tuning beyond the generic local-level model above.

What does this machinery buy over the simplest off-the-shelf alternative, an HP filter ($\lambda = 1600$, the standard quarterly value) run on the identical growth series? Over the full sample the two trends track each other closely (correlation 0.98, RMSE 0.45 points), so for an ordinary quarter the two methods are not fighting over much. They diverge sharply, however, exactly at the point that matters most: at the depth of the 2008–09

recession the HP trend falls only to 2.57%, while the local-level model's trend falls all the way to its sample minimum of 0.60%, more than two points lower. The HP filter's fixed smoothing penalty cannot distinguish a genuine, multi-quarter collapse in trend growth from ordinary period-to-period noise, so it under-reacts to a shock this size; the local-level model instead estimates its own σ^2 and ω^2 from the data and lets a sufficiently large, sufficiently persistent shock move the trend accordingly. Figure 65 overlays the HP trend (teal, dashed) on the same picture to make this visible directly.

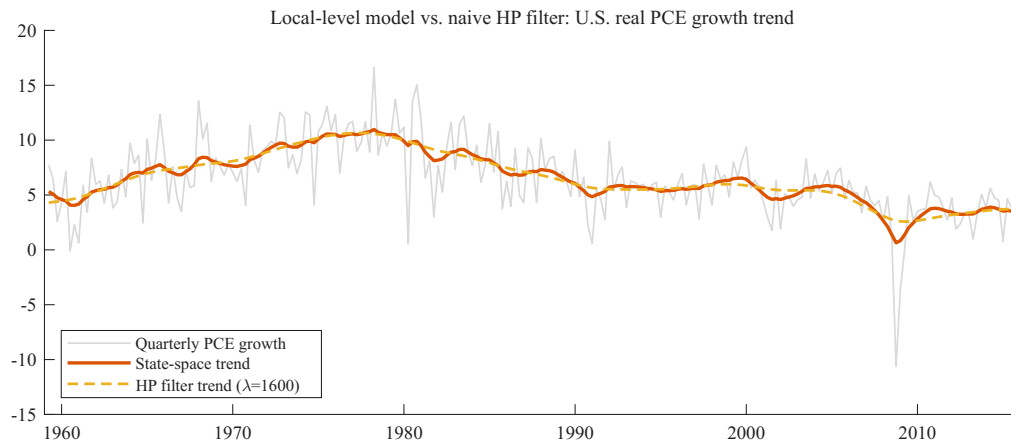


Figure 65: Local-level model, U.S. real PCE growth: raw quarterly growth (grey) vs. the posterior-mean trend (solid), with a naive HP-filter trend ($\lambda = 1600$, dashed) for comparison

15.3 Model 2: Trend and Cycle in U.S. Real GDP

Local-Linear-Trend Plus AR(2) Cycle Let $y_t = 100 \times \log(\text{real GDP}_t)$. The output-gap model splits y_t into a smooth trend τ_t (potential output) and a stationary cycle c_t (the output gap):

$$y_t = \tau_t + c_t, \quad c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_c^2),$$

where the trend follows not a random walk but a *smooth* process — its own growth rate, $\Delta\tau_t = \tau_t - \tau_{t-1}$, is a random walk:

$$\Delta\tau_t = \Delta\tau_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\tau^2).$$

Written as a second-difference restriction, $H_2\tau = b + u$ with H_2 the $T \times T$ second-difference operator (H_2 's i -th row is $[\dots, 1, -2, 1, \dots]$ against $\tau_i, \tau_{i-1}, \tau_{i-2}$), this is exactly the precision-sampler construction above applied twice, once to the trend's growth rate. The initial conditions (τ_0, τ_{-1}) have a diffuse bivariate Normal prior, the AR(2) coefficients (ϕ_1, ϕ_2) have a Normal prior centered at $(1.3, -0.7)$ (a textbook business-cycle prior mean implying persistent, mildly oscillatory dynamics) restricted to the stationarity triangle, σ_c^2 has an Inverse-Gamma prior, and σ_τ^2 has a bounded Uniform(0, 0.01) prior — deliberately tight, so the trend is forced to absorb only low-frequency movement in y_t , leaving business-cycle-frequency movement to c_t — sampled by the grid-based inverse-CDF method of Chapter 4 rather than a standard conjugate update, since a bounded uniform prior has no Inverse-Gamma conjugate form.

Algorithm: Gibbs Sampler for the Output-Gap Model

Each sweep:

1. Draws the full trend path $\tau_{1:T}$ jointly from its Gaussian full conditional (precision sampler, H_2 operator),^a combining the smooth-trend prior with the AR(2)-implied likelihood for $y_t - \tau_t = c_t$.
2. Draws (ϕ_1, ϕ_2) from their Normal full conditional given the implied cycle $c_t = y_t - \tau_t$, **rejecting** any draw outside the (slightly shrunk) stationarity triangle $\phi_1 + \phi_2 < 0.99$, $\phi_2 - \phi_1 < 0.99$, $\phi_2 > -0.99$ and redrawing.
3. Draws σ_c^2 from its Inverse-Gamma full conditional given the AR(2) residuals.
4. Draws σ_τ^2 from a 500-point grid approximation to its bounded-uniform full conditional (Chapter 4's inverse-CDF method).
5. Draws the initial conditions (τ_0, τ_{-1}) from their Gaussian full conditional.

The chain runs 2,000 post-burn-in sweeps after a 600-sweep burn-in; the stationarity-rejection step accepts 98.1% of proposed (ϕ_1, ϕ_2) draws, indicating the constraint region is rarely binding at this posterior.

^aAgain Chan and Jeliazkov (2009): both $H_2'H_2/\sigma_\tau^2$ and $H_\phi'H_\phi/\sigma_c^2$ are banded, so their sum stays banded and the joint draw costs a single sparse Cholesky factorization (UC_output_gap.m).

Derivation: The Output-Gap Model's Five Full Conditionals

Each of the five blocks below reuses a derivation already carried out in full elsewhere in the book; what follows substitutes this model's specific quantities into each and re-runs the kernel-matching steps explicitly, rather than asserting the result by analogy alone.

1. **Trend** $\tau_{1:T}$. Writing the smooth-trend restriction as $H_2\tau = b + u$, $u \sim N(0, \sigma_\tau^2 I_T)$ with H_2 the second-difference operator and b absorbing (τ_0, τ_{-1}) , the state-equation prior kernel on τ is, exactly as in the local-level proofbox above with $H \rightarrow H_2$ and $\omega^2 \rightarrow \sigma_\tau^2$,

$$p(\tau) \propto \exp\left\{-\frac{1}{2}(\tau - H_2^{-1}b)'(\sigma_\tau^{-2}H_2'H_2)(\tau - H_2^{-1}b)\right\}.$$

The measurement side, given the current cycle draw $c = y - \tau$, contributes the likelihood kernel $\exp\{-\frac{1}{2}\sigma_c^{-2}(y - \tau)'(y - \tau)\}$, a Gaussian in τ with precision $\sigma_c^{-2}I_T$ and mean y . Multiplying the two kernels and collecting quadratic and linear terms in τ , exactly as before with $\omega^{-2}H'H \rightarrow \sigma_\tau^{-2}H_2'H_2$ and $\sigma^{-2} \rightarrow \sigma_c^{-2}$:

$$p(\tau \mid \sigma_c^2, \sigma_\tau^2, c) \propto \exp\left\{-\frac{1}{2}\left[\tau'(\sigma_\tau^{-2}H_2'H_2 + \sigma_c^{-2}I_T)\tau - 2\tau'(\sigma_\tau^{-2}H_2'b + \sigma_c^{-2}y)\right]\right\}$$

the kernel of:

$$\begin{aligned}\tau \mid \sigma_c^2, \sigma_\tau^2, c &\sim N(\hat{\tau}, K_\tau^{-1}), \\ K_\tau &= \sigma_\tau^{-2}H_2'H_2 + \sigma_c^{-2}I_T, \\ \hat{\tau} &= K_\tau^{-1}(\sigma_\tau^{-2}H_2'b + \sigma_c^{-2}y)\end{aligned}$$

2. **AR(2) coefficients** (ϕ_1, ϕ_2) . Given the implied cycle $c_t = y_t - \tau_t$, the recursion $c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \varepsilon_t$ is an ordinary two-regressor linear regression of c_t on (c_{t-1}, c_{t-2}) with known error variance σ_c^2 . Writing $\phi = (\phi_1, \phi_2)'$, X_c the stacked design matrix with rows (c_{t-1}, c_{t-2}) , and the Normal prior $\phi \sim N(\phi_0, V_{0,\phi})$ with $\phi_0 = (1.3, -0.7)'$, Chapter 6.2's $\beta \mid \sigma^2, y$ completing-the-square argument applies with $X \rightarrow X_c$, $y \rightarrow c$, $b_0 \rightarrow \phi_0$, and $V_0 \rightarrow V_{0,\phi}$:

$$\begin{aligned}p(c \mid \phi, \sigma_c^2) &\propto \exp\left\{-\frac{1}{2\sigma_c^2}(c - X_c\phi)'(c - X_c\phi)\right\}, \\ p(\phi) &\propto \exp\left\{-\frac{1}{2}(\phi - \phi_0)'V_{0,\phi}^{-1}(\phi - \phi_0)\right\}.\end{aligned}$$

Multiplying and collecting quadratic and linear terms in ϕ , exactly as in Chapter 6.2:

$$p(\phi \mid \sigma_c^2, c) \propto \exp\left\{-\frac{1}{2}\left[\phi'(V_{0,\phi}^{-1} + \sigma_c^{-2}X_c'X_c)\phi - 2\phi'(V_{0,\phi}^{-1}\phi_0 + \sigma_c^{-2}X_c'c)\right]\right\}$$

the kernel of an *untruncated*:

$$\begin{aligned}\phi \mid \sigma_c^2, c &\sim N(\bar{\phi}, \bar{V}_\phi), \\ \bar{V}_\phi &= (V_{0,\phi}^{-1} + \sigma_c^{-2}X_c'X_c)^{-1}, \\ \bar{\phi} &= \bar{V}_\phi(V_{0,\phi}^{-1}\phi_0 + \sigma_c^{-2}X_c'c)\end{aligned}$$

The stationarity restriction is then imposed not by re-deriving a truncated-Normal kernel (as the true full conditional would require, and as Chapter 7/9's data-augmentation problems did for a one-sided threshold) but by rejection sampling: repeatedly drawing from the untruncated $N(\bar{\phi}, \bar{V}_\phi)$ and discarding draws outside the stationarity triangle is a valid way to sample the truncated distribution because, restricted to any measurable subset A , a Normal density renormalized by $\mathbb{1}_A$ is exactly what rejection sampling from the unrestricted Normal, conditional on landing in A , produces.

3. $\sigma_c^2 \mid c$. Denote the AR(2) error-variance prior $IG(a_{0c}/2, d_{0c}/2)$ (Chapter 6.2 notation) and the residuals $\varepsilon_t = c_t - \phi_1 c_{t-1} - \phi_2 c_{t-2}$, $t = 1, \dots, T_c$, where T_c is the number of fitted AR(2) residuals. Exactly Chapter 6.2's $\sigma^2 \mid \beta, y$ argument with $e \rightarrow \varepsilon$:

$$p(c \mid \phi, \sigma_c^2) = (2\pi\sigma_c^2)^{-T_c/2} \exp\left\{-\frac{\varepsilon'\varepsilon}{2\sigma_c^2}\right\}, \quad p(\sigma_c^2) \propto (\sigma_c^2)^{-(a_{0c}/2+1)} \exp\left\{-\frac{d_{0c}}{2\sigma_c^2}\right\}.$$

Multiplying and collecting powers of σ_c^2 exactly as in Chapter 6.2:

$$p(\sigma_c^2 \mid c) \propto (\sigma_c^2)^{-\left(\frac{a_{0c}+T_c}{2}+1\right)} \exp\left\{-\frac{d_{0c} + \varepsilon'\varepsilon}{2\sigma_c^2}\right\}$$

the kernel of:

$$\sigma_c^2 \mid c \sim IG\left(\frac{a_{0c} + T_c}{2}, \frac{d_{0c} + \varepsilon'\varepsilon}{2}\right)$$

with $\varepsilon'\varepsilon = \sum_t \varepsilon_t^2$ playing the role of $e'e$ and T_c playing the role of T .

4. σ_τ^2 . Since $u = H_2\tau - b$ collects the trend's own second differences $\eta_t = \Delta\tau_t - \Delta\tau_{t-1}$ for $t = 1, \dots, T$, the likelihood contribution of τ to σ_τ^2 is

$$p(\tau \mid \sigma_\tau^2) = (2\pi\sigma_\tau^2)^{-T/2} \exp\left\{-\frac{1}{2\sigma_\tau^2} \sum_{t=1}^T \eta_t^2\right\}, \quad \eta_t = \Delta\tau_t - \Delta\tau_{t-1},$$

and the Uniform(0, 0.01) prior contributes only the indicator $\mathbb{1}\{0 < \sigma_\tau^2 < 0.01\}$ (no shape/scale term). Multiplying the two:

$$p(\sigma_\tau^2 \mid \tau) \propto (\sigma_\tau^2)^{-T/2} \exp\left\{-\frac{\sum_t \eta_t^2}{2\sigma_\tau^2}\right\} \times \mathbb{1}\{0 < \sigma_\tau^2 < 0.01\}$$

recognizably the kernel of, restricted to (0, 0.01):

$$\sigma_\tau^2 \mid \tau \sim IG\left(\frac{T-2}{2}, \frac{\sum_t \eta_t^2}{2}\right)$$

(matching powers: $-T/2 = -(\alpha + 1)$ gives $\alpha = (T - 2)/2$). This truncated Inverse-Gamma kernel has no closed-form normalizing constant convenient for direct sampling, which is exactly why the algorithm falls back on Chapter 4's grid inverse-CDF method (evaluate the unnormalized kernel above on a fine grid over (0, 0.01), normalize numerically, and invert the resulting discrete CDF) rather than a textbook conjugate

draw.

5. **Initial conditions** (τ_0, τ_{-1}) . These enter only through the linear term b in the first two rows of $H_2\tau = b + u$, since H_2 's i -th row references $\tau_i, \tau_{i-1}, \tau_{i-2}$ and only rows $i = 1, 2$ reach back before $t = 1$. Writing $b = R\gamma$ for the fixed $T \times 2$ selection matrix R (nonzero only in rows 1–2) and $\gamma = (\tau_0, \tau_{-1})'$, with diffuse bivariate Normal prior $\gamma \sim N(g_0, G_0)$:

$$p(\tau \mid \gamma, \sigma_\tau^2) \propto \exp \left\{ -\frac{1}{2\sigma_\tau^2} (H_2\tau - R\gamma)' (H_2\tau - R\gamma) \right\},$$

$$p(\gamma) \propto \exp \left\{ -\frac{1}{2} (\gamma - g_0)' G_0^{-1} (\gamma - g_0) \right\}.$$

Expanding the first quadratic form in γ and multiplying by the prior, exactly as in Chapter 6.2's $\beta \mid \sigma^2, y$ argument with $X \rightarrow R$, $y \rightarrow H_2\tau$, $b_0 \rightarrow g_0$, $V_0 \rightarrow G_0$, and $\sigma^2 \rightarrow \sigma_\tau^2$:

$$p(\gamma \mid \tau, \sigma_\tau^2) \propto \exp \left\{ -\frac{1}{2} \left[\gamma' (G_0^{-1} + \sigma_\tau^{-2} R' R) \gamma - 2\gamma' (G_0^{-1} g_0 + \sigma_\tau^{-2} R' H_2\tau) \right] \right\}$$

the kernel of:

$$\gamma \mid \tau, \sigma_\tau^2 \sim N(\bar{\gamma}, \bar{G}),$$

$$\bar{G} = (G_0^{-1} + \sigma_\tau^{-2} R' R)^{-1},$$

$$\bar{\gamma} = \bar{G} (G_0^{-1} g_0 + \sigma_\tau^{-2} R' H_2\tau)$$

Empirical Example: Empirical Results: Potential Output and the Output Gap

Fitting this model to quarterly real GDP, 1947Q1–2015Q4 ($T = 276$), gives posterior means $\hat{\phi}_1 = 1.308$ (95% CI [1.192, 1.427]), $\hat{\phi}_2 = -0.372$ (95% CI [-0.481, -0.260]), $\hat{\sigma}_\epsilon^2 = 0.734$, and $\hat{\sigma}_\tau^2 = 0.0034$ — three orders of magnitude smaller than σ_ϵ^2 , confirming the trend in Figure 66(a) is, by construction, far smoother than the raw series. The estimated output gap (Figure 66(b)) has standard deviation 2.39%, a sample minimum of -7.20% in 1949Q4 (the trough of the 1948–49 postwar recession) and a sample maximum of 4.95% in 1953Q1 (the Korean War production boom, shortly before the 1953–54 downturn), and ends the sample at -0.09% in 2015Q4 — essentially at potential. The estimated growth rate of potential output itself declines from 3.90% at the start of the sample to 1.72% by its end, consistent with the well-documented slowdown in U.S. trend GDP growth over this period. Benchmarked against a naive HP filter ($\lambda = 1600$) applied to the identical $100 \times \log$ real GDP series, the two output-gap estimates are positively correlated (correlation 0.84) but differ in both amplitude and timing in exactly the way the HP filter's known weaknesses would predict. Over the full sample the state-space cycle has standard deviation 2.39% against the HP cycle's 1.62%: the state-space model attributes noticeably more of the series' movement to the cycle rather than the trend. At the 2008–09 trough the state-space gap bottoms out at -3.30% in 2009Q3, deeper and one quarter later than the HP gap's -2.89% in 2009Q2. The sharper divergence is at the *end* of the sample: by 2014Q3 the HP filter already shows a positive output gap of +0.60% (output above potential), while the

state-space model still shows the economy -0.41% below potential, and the state-space gap does not turn positive until the final quarters of the sample. This is the textbook HP end-point problem: at the boundary of the sample the filter has no future data left to smooth against, so its trend swings to hug the most recent observations and closes the estimated gap prematurely, while the state-space trend's prior on curvature (σ_τ^2 restricted to $(0, 0.01)$) constrains it identically at every date, interior or boundary. Figure 66(c) zooms on 2005–2016 to make this divergence visible directly.

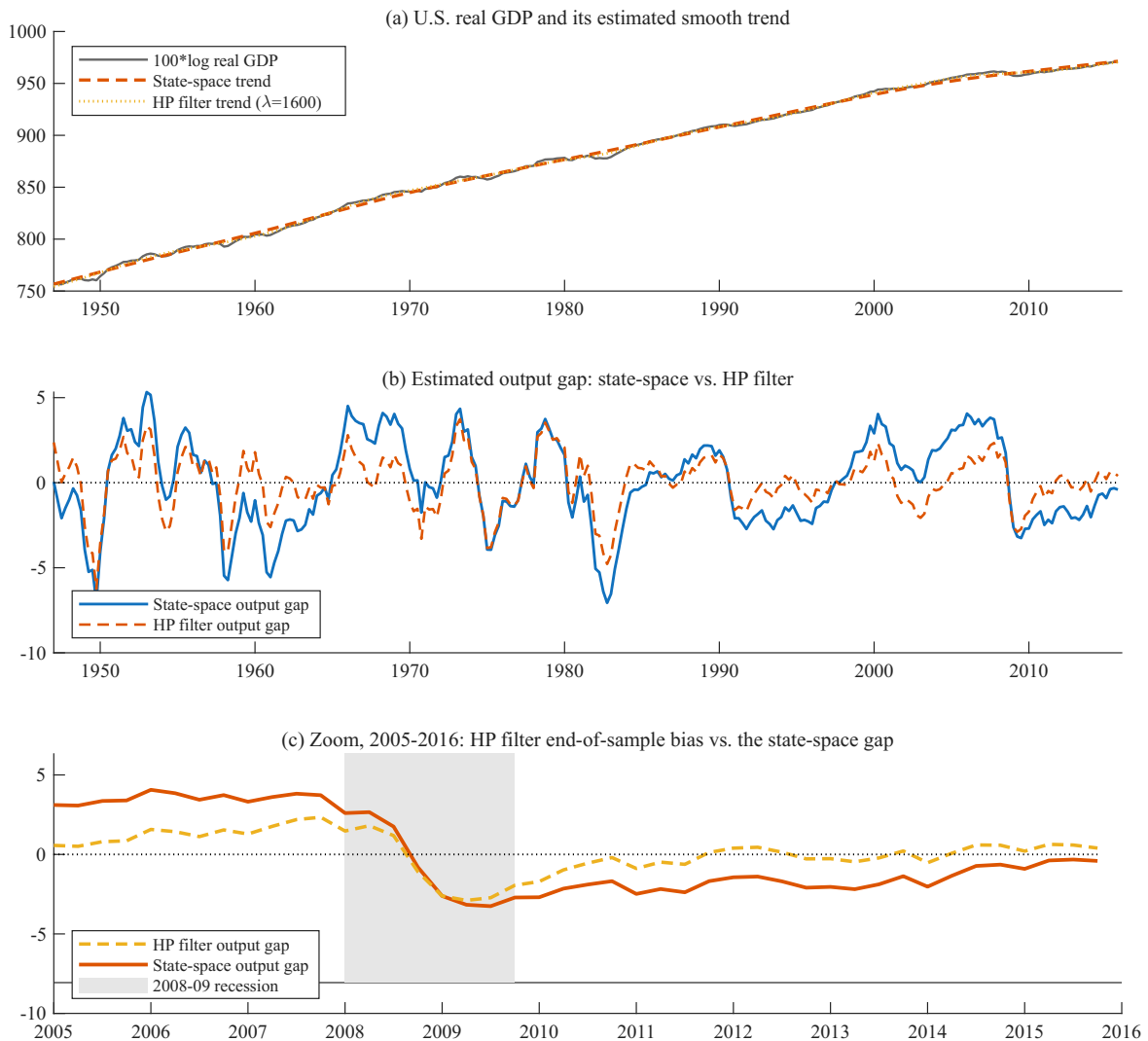


Figure 66: (a) $100 \times \log$ real GDP (grey) vs. estimated smooth trend/potential output (dashed) vs. naive HP-filter trend ($\lambda = 1600$, dotted); (b) estimated output gap, state-space (solid) vs. HP filter (dashed), with recession troughs and boom peaks visible; (c) zoom, 2005–2016, isolating the HP filter's end-of-sample bias against the state-space gap

Remark: What Does the Precision-Sampler Machinery Buy Over a Naive HP Filter?

Both empirical examples in this chapter were checked against the simplest off-the-shelf trend-extraction alternative, an HP filter with the conventional quarterly smoothing parameter $\lambda = 1600$, applied to the identical raw data. The two methods usually agree

fairly closely — correlation above 0.8 in both the PCE trend and the GDP cycle reported above — which is itself informative: for an ordinary quarter of ordinary data, a generic HP filter and this chapter’s fully Bayesian state-space model are not fighting over much. The two approaches earn their keep apart in exactly two places, both illustrated above. First, the *size* of a genuine, sustained break: in the PCE growth trend, the HP filter’s fixed penalty caps how far the estimated trend can move over a few quarters regardless of how persistent the underlying shift is, understating the 2008–09 collapse in trend consumption growth by more than two percentage points, while the state-space model estimates its own smoothing intensity (σ^2, ω^2) from the data and lets a sufficiently large, sufficiently persistent shock overcome it. Second, the *sample boundary*: the HP filter is a symmetric two-sided filter everywhere except at the ends of the sample, where it has no future observations left to smooth against and its trend consequently swings toward the most recent data — the well-documented “end-point problem” that closes the estimated 2014–15 U.S. output gap well before the state-space model’s curvature-constrained trend does. Neither problem is a defect specific to this one choice of λ (a different smoothing parameter would trade one bias for the other); both are structural features of any single fixed-smoothing-parameter filter applied uniformly across a sample, which is precisely the constraint the model-based approach in this chapter relaxes by letting the data inform how much smoothing is warranted, including right at the edges of the sample.

Remark: Is There a Well-Identified Business-Cycle Length Here?

The AR(2) cycle $c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \varepsilon_t$ is oscillatory (has complex characteristic roots, and so an implied cycle period) only when $\phi_1^2 + 4\phi_2 < 0$. Checking this *per posterior draw* — not from the posterior mean coefficients alone, since averaging draws with complex roots at different angles can produce a mean point with real roots — shows only 2.95% of the 2,000 posterior draws satisfy it. The genuine finding here is that this model’s cycle is dominated by persistence and damping rather than clean periodicity: for the large majority of posterior mass, (ϕ_1, ϕ_2) implies two real autoregressive roots (a persistent but non-oscillating cycle), not a textbook damped sine wave. Among the small minority of draws that do imply complex roots, the implied cycle length averages 13.3 years (median 10.1), but with so few qualifying draws this number is not a reliable summary of “the” U.S. business cycle length and is reported here only for transparency, not as a headline result — forcing a single confident cycle-length estimate from this posterior would overstate what the data and model actually identify.

15.4 Clark (1987)

Model 2 above decomposed GDP into a smooth trend and a cycle by *assumption*: its trend was given no level shock at all, so every quarter-to-quarter wiggle was forced into the cycle. The paper this section replicates asked the question that assumption skips. Clark (1987), “The Cyclical Component of U.S. Economic Activity,” *Quarterly Journal of Economics* 102(4), decomposes $y_t = 100 \ln(\text{real GNP})$, 1947Q1–1985Q4, into a trend that carries *both* kinds of

shock and a stationary AR(2) cycle:

$$y_t = \tau_t + c_t, \quad \tau_t = g_{t-1} + \tau_{t-1} + v_t, \quad g_t = g_{t-1} + w_t, \quad c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + e_t,$$

with all three innovations independent — Clark’s identifying assumption. The design nests everything from a deterministic linear trend ($\sigma_v = \sigma_w = 0$) through Model 2’s smooth trend ($\sigma_v = 0$) to the Nelson–Plosser random walk (large σ_v , small cycle), and lets the data pick. Against the then-dominant reading of Nelson and Plosser (1982) — that GNP is essentially a random walk, leaving little room for a cycle at all — Clark’s answer was that *at least half* of the quarterly innovation in GNP belongs to a stationary cyclical component that persists for up to five years. Clark estimated the model by maximum likelihood through the Kalman filter; the replication reuses this chapter’s precision-sampler machinery instead, adding one Gibbs block for the drift path $g_{1:T}$ (a second sparse-Gaussian draw, structurally identical to the $\tau_{1:T}$ block) to the output-gap sampler already built for Model 2.⁴

Empirical Example: Original Sample, 1947Q1–1985Q4

Posterior means (sd) against Clark’s Table I maximum-likelihood estimates (converted to the $100 \times \log$ scale used here):

	posterior	Clark (1987)
ϕ_1	1.46 (0.11)	1.53
ϕ_2	−0.58 (0.10)	−0.59
σ_v (trend level)	0.71	0.64
σ_w (trend growth)	0.08	0.01
σ_e (cycle)	0.73	0.74

The cycle is where the paper’s conclusions live, and it replicates almost exactly: the cycle-innovation standard deviation matches to the second decimal, the AR(2) coefficients to within one posterior standard deviation, and the headline arithmetic survives verbatim — the cycle’s share of the total quarterly innovation variance is 51.6%, Clark’s “at least half.” The posterior-mean cycle path (Figure 67, middle panel) swings between −5.1 and +3.4 percent with a trough at every NBER recession in the sample, and the posterior-mean AR(2) implies complex roots with a stochastic period of about 22 quarters — five and a half years. Only the tiny trend-growth variance lands away from Clark’s point estimate, and that gap is informative rather than troubling: Clark’s own Table I shows four different restrictions on the trend — his maximum likelihood, a random walk with drift, Model 2’s smooth trend, even a linear trend — fitting within 4.3 points of $2 \times \log$ -likelihood of each other, and his likelihood map plots a long flat ridge in exactly the (σ_v, σ_w) plane. A maximum-likelihood estimate is one point on that ridge; the posterior integrates over it, and lands slightly elsewhere along it. The bottom panel shows what the random-walk drift buys: annualized

⁴So each sweep of `UC_clark1987.m` performs two Chan–Jeliazkov (2009) sparse-Gaussian path draws — one for $\tau_{1:T}$, one for $g_{1:T}$ — plus standard conjugate blocks.

trend growth that rises to 4.3% in the mid-1960s and drifts down toward 3% by 1985 — the productivity-slowdown profile that made Clark unwilling to impose a constant growth rate in advance.

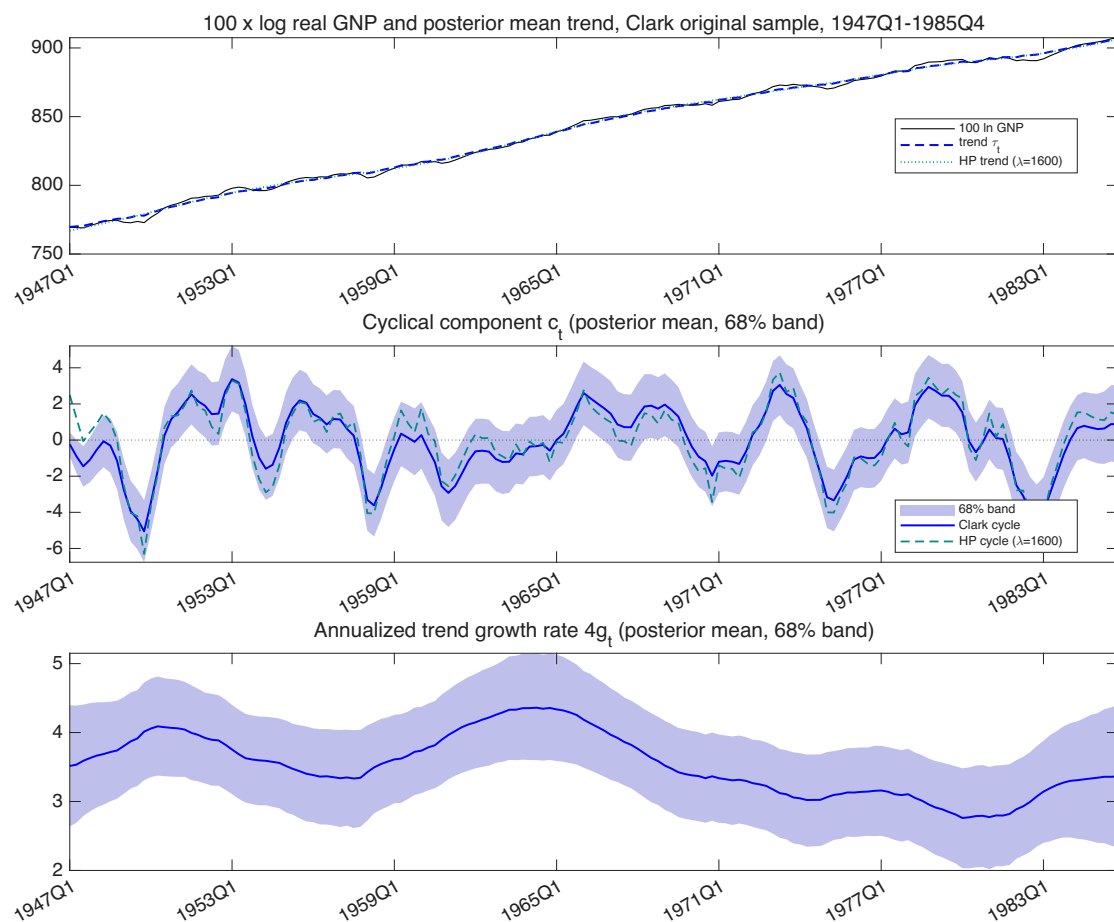


Figure 67: Clark (1987) replicated: $100 \times \log$ real GNP with the posterior mean trend and an HP($\lambda = 1600$) trend (top); the cyclical component with 68% band and the HP cycle (middle); annualized trend growth $4g_t$ (bottom), original sample

Remark: What the Model Adds Over the HP Filter, One More Time

The HP benchmark run on the identical series agrees with Clark's cycle about *when* things happened — the two cycle paths correlate at 0.94 — which is reassuring, not embarrassing: both are, loosely, smoothers of the same series. The differences are exactly the ones this chapter's earlier examples documented, now visible on the founding application. The HP cycle is choppier (its autocorrelation at eight quarters is -0.32 against the Clark cycle's -0.14), because $\lambda = 1600$ is a fixed smoothing constant while Clark's trend flexibility is *estimated* from the data; the HP filter reports no uncertainty band, while the UC model's cycle comes with a posterior distribution; and the HP filter has no concept of trend *growth*, so the bottom panels of Figures 67 and 68 — arguably the most economically interesting output of the model — simply have no HP counterpart.

Empirical Example: Full Sample, 1947Q1–2026Q1

Extending the identical model through 2026Q1 changes all three of the model's stories in economically legible ways (Figure 68). The cycle shrinks: its posterior-mean standard deviation falls from 1.82 to 1.21, with the post-1984 moderation plainly visible in the middle panel, and the cycle's share of the quarterly innovation variance falls from 51.6% to 42.5% — on the longer sample, somewhat more of GNP's quarter-to-quarter variation reads as permanent. Trend growth keeps doing what Clark built the drift random walk to capture: from 3.7% at the start of the sample it declines, with a plateau in the 1990s, to about 2.3% after 2010 and stays there — the secular-decline profile that a constant-drift specification would have missed entirely. And the COVID quarter stages a controlled experiment between the two decompositions in the figure: the HP filter, whose trend cannot bend fast enough, sends the 2020Q2 collapse almost entirely into its cycle (a trough close to -10 percent), while Clark's two-shock trend absorbs roughly half of it as a permanent level shock, leaving a cycle trough of -4.5 — and the correlation between the two cycle estimates drops to 0.90, with the HP cycle now 33% more volatile than the model-based one. Which allocation is right is a genuinely open question — but only one of the two methods lets the data, rather than a fixed smoothing constant, make the call.

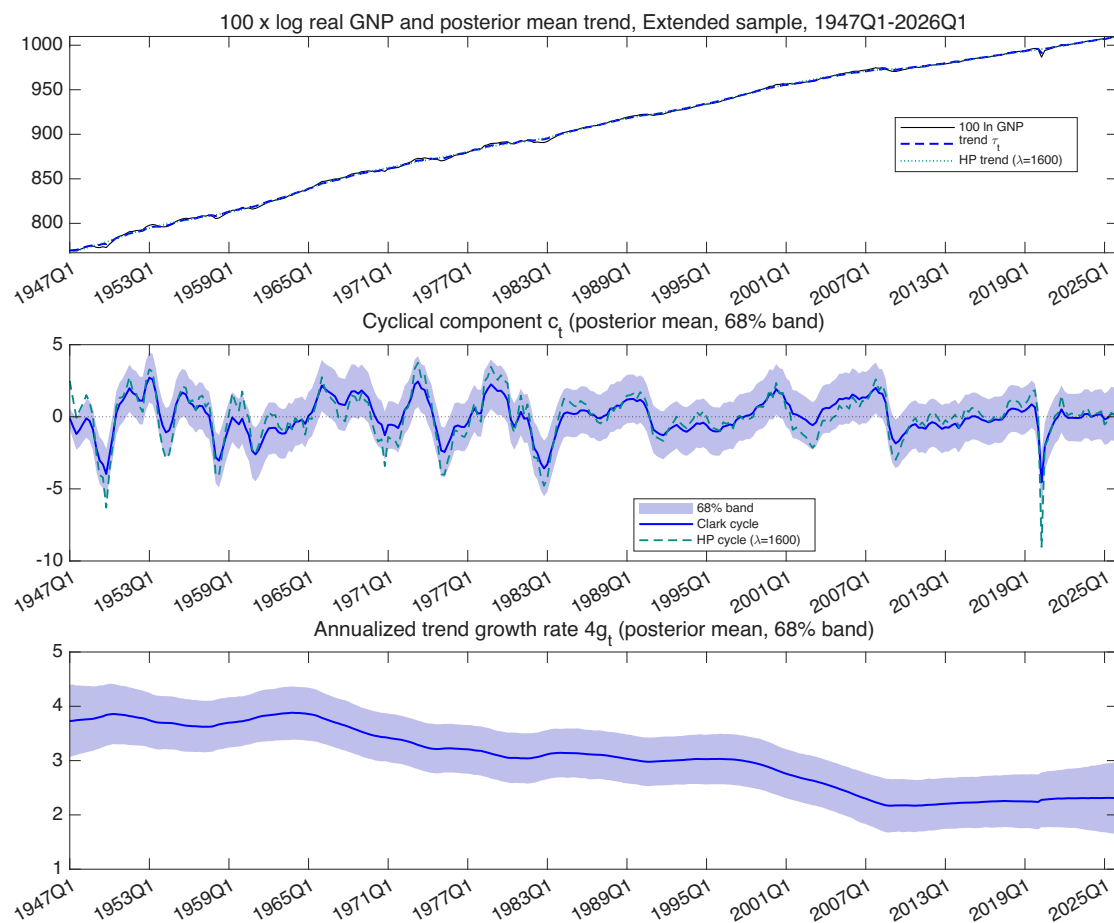


Figure 68: The same UC model estimated through 2026Q1: the cycle moderates after 1984, trend growth declines secularly to about 2.3%, and the 2020Q2 collapse is split between trend and cycle where the HP filter forces it all into the cycle

16. VAR with Stochastic Volatility

Roadmap

Three earlier chapters each solved one piece of this chapter's problem in isolation: Chapter 11 estimated a VAR's coefficients holding volatility fixed, Chapter 14 let a single equation's volatility move but had no system to put it in, and Chapter 15 built the machinery for drawing an entire latent path in one linear solve. This chapter's real content is the assembly: a triangular decomposition turns one hard multivariate volatility problem into n separate copies of Chapter 15's univariate problem, plus one new piece, the Kim-Shephard-Chib trick that makes a log-squared residual look Gaussian enough for that machinery to apply at all. Nothing here is estimated by writing a new sampler from scratch; everything is reused, which is itself the chapter's point.

Chapter 11's VAR held its innovation covariance Σ fixed for the entire sample; Chapter 14 let a single equation's variance move over time but said nothing about a system of equations; Chapter 15 let a latent state evolve smoothly but assumed constant-variance innovations throughout. This chapter combines all three ideas into one model: a VAR whose coefficients are still constant, but whose shock covariance evolves over time via a multivariate stochastic-volatility process — exactly the KSC (Kim, Shephard and Chib, 1998) mixture-sampler and precision-sampler machinery of Chapters 14 and 15, now applied jointly to every equation of a VAR at once. This is the standard first step toward the fully time-varying systems (Chapter 17's TVP-VAR) that follow.

VAR with Stochastic Volatility (VAR-SV) For an n -variable VAR(p), write the reduced form as

$$y_t = \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + c + u_t, \quad u_t \sim N(0, \Sigma_t),$$

where, unlike Chapter 11, Σ_t now varies with t . Rather than modeling Σ_t directly, VAR-SV models it through a triangular ("structural") decomposition: let L be lower unit-triangular with free sub-diagonal elements $a = (a_{21}, a_{31}, a_{32}, \dots)$, and let $\varepsilon_t = Lu_t$ be the orthogonalized (structural) shocks. Each orthogonalized shock has its own independent stochastic-volatility process, a pure random walk in log-variance:

$$\varepsilon_{i,t} \sim N(0, e^{h_{i,t}}), \quad h_{i,t} = h_{i,t-1} + \eta_{i,t}, \quad \eta_{i,t} \sim N(0, \sigma_{h,i}^2),$$

with $h_{i,0} \sim N(1, 10)$. Since $u_t = L^{-1}\varepsilon_t$, this implies $\Sigma_t = L^{-1}\text{diag}(e^{h_{1,t}}, \dots, e^{h_{n,t}})L^{-\prime}$: L captures whatever part of the cross-equation covariance is constant over time, while the $h_{i,t}$ paths let each orthogonalized shock's own scale expand or contract independently, and their combination lets the *correlation* between reduced-form shocks drift over time even though L itself does not.

Figure 69 lays out how the five block types feed each other; the mixture indicators $\kappa_{1:T}$ are the one genuinely new block this chapter introduces, sitting between the volatility paths and the data.

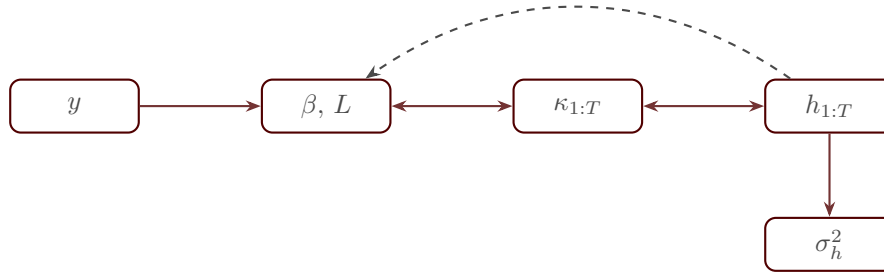


Figure 69: Dependency structure of the VAR-SV sampler: solid/double-headed arrows are each sweep’s direct updates ($\beta, L \leftrightarrow \kappa_{1:T} \leftrightarrow h_{1:T}$), the dashed arrow is the one longer-range link — $h_{1:T}$ also sets the precision weights used in the β, L step — and σ_h^2 closes the loop on the volatility path’s own innovation variance

Algorithm: Gibbs Sampler for VAR-SV

With priors mirroring Chapter 11’s VAR prior on (Φ, c) (precision 1 on slope coefficients, 1/10 on intercepts), independent $N(0, 1)$ priors on the free elements of L , and $\text{IG}(3, 0.01)$ priors on each $\sigma_{h,i}^2$:

1. **Coefficients.** Draw the stacked VAR coefficient vector jointly from its Normal full conditional, a generalized-least-squares update that weights each observation by its OWN time- t precision $\Sigma_t^{-1} = L' \text{diag}(e^{-h_{1,t}}, \dots, e^{-h_{n,t}}) L$ — the direct multivariate generalization of Chapter 14’s GARCH likelihood and Chapter 6’s Bayesian regression, now with a precision matrix that changes every period.
2. **Log-volatilities.** Recover the implied structural shocks $s_t = Lu_t$ from the current residuals, form $y_t^* = \log(s_t^2 + 0.0001)$, and for each equation i apply the same Kim-Shephard-Chib auxiliary mixture sampler used nowhere yet in these notes but implicit in every SV model: approximate the intractable $\log \chi_1^2$ observation density with a 7-component Normal mixture, classify each t into a component, then draw the entire path $h_{i,1:T}$ jointly from its Gaussian full conditional via *exactly Chapter 15’s banded first-difference precision-sampler construction* — the random-walk state equation $h_{i,t} = h_{i,t-1} + \eta_{i,t}$ is structurally identical to Chapter 15’s local-level trend τ_t , just applied here to a log-variance instead of a growth rate.
3. **Contemporaneous loadings.** Draw the free elements of L from a weighted linear regression of each reduced-form residual on the negatives of the preceding residuals, precision-weighted by $e^{-h_{i,t}}$ — the multivariate analogue of Chapter 12’s SVAR identification step, but now with time-varying weights.
4. **State variances and initial states.** Draw each $\sigma_{h,i}^2$ from its Inverse-Gamma full conditional given the $h_{i,1:T}$ path’s first differences, and each $h_{i,0}$ from its Normal full conditional given $h_{i,1}$.

16.1 Posterior Derivation

Derivation: $\beta \mid \Sigma_{1:T}, y$ (Coefficients Under Time-Varying Precision)

Writing the VAR in SUR/stacked form $y_t = Z_t\beta + u_t$, $u_t \sim N(0, \Sigma_t)$ exactly as in Chapter 10.2, the posterior kernel in β is

$$p(\beta \mid \Sigma_{1:T}, y) \propto \exp \left\{ -\frac{1}{2} \left[\sum_t (y_t - Z_t\beta)' \Sigma_t^{-1} (y_t - Z_t\beta) + (\beta - b_0)' B_0^{-1} (\beta - b_0) \right] \right\}$$

Chapter 10.2's completing-the-square argument never used the assumption that the weighting matrix inside the sum is the *same* matrix at every t ; it only used that some positive-definite weight multiplies each $(y_t - Z_t\beta)$ quadratic form before the terms are summed over t . Allowing Σ_t to carry a t subscript changes nothing about that argument, so expanding and collecting terms exactly as before gives:

$$\begin{aligned} \beta \mid \Sigma_{1:T}, y &\sim N(\bar{\beta}, \bar{B}), \\ \bar{B} &= \left(B_0^{-1} + \sum_t Z_t' \Sigma_t^{-1} Z_t \right)^{-1}, \\ \bar{\beta} &= \bar{B} \left(B_0^{-1} b_0 + \sum_t Z_t' \Sigma_t^{-1} y_t \right) \end{aligned}$$

This is the direct multivariate, time-varying-precision generalization of Chapter 6.2's scalar σ^{-2} -weighted result and Chapter 10.2's constant- Σ^{-1} -weighted result: the weight matrix inside the sum simply changes from period to period rather than being pulled out front.

Derivation: The KSC Auxiliary Mixture Representation for $h_{i,1:T}$

Fix an equation i and drop the subscript. The structural shock $\varepsilon_t \sim N(0, e^{h_t})$ can be written $\varepsilon_t = e^{h_t/2} \xi_t$, $\xi_t \sim N(0, 1)$, so $\varepsilon_t^2 = e^{h_t} \xi_t^2$ and, taking logs, the observation equation becomes

$$y_t^* \equiv \log(\varepsilon_t^2 + 0.0001) \approx \log \varepsilon_t^2 = h_t + \log \xi_t^2$$

a *linear* observation equation in h_t but with a non-Gaussian error $\log \xi_t^2 \sim \log \chi_1^2$ (since $\xi_t^2 \sim \chi_1^2$), which is why the log-volatility path cannot be drawn directly by Chapter 15's precision sampler as it stands: that construction requires a Gaussian observation equation. Kim, Shephard and Chib's (1998) device is to introduce, for each t , a discrete auxiliary mixture-component indicator $\kappa_t \in \{1, \dots, 7\}$ and a 7-component Normal mixture $\{\pi_k, m_k, v_k^2\}_{k=1}^7$ chosen (by matching moments, offline, once) so that $\sum_k \pi_k N(m_k, v_k^2)$ closely approximates the $\log \chi_1^2$ density. Conditional on $\kappa_t = k$, this becomes an exactly Gaussian observation equation in h_t :

$$y_t^* \mid h_t, \kappa_t = k \sim N(h_t + m_k, v_k^2)$$

Two draws complete the step:

1. **Component indicators.** Given the current h_t , the posterior over κ_t is, by Bayes' rule applied to the finite mixture — the same discrete-posterior argument as Chapter 8's γ_j step, generalized from two components to seven —

$$P(\kappa_t = k \mid y_t^*, h_t) \propto \pi_k N(y_t^* \mid h_t + m_k, v_k^2), \quad k = 1, \dots, 7,$$

normalized to sum to 1 across k and drawn as a single categorical draw per t .

2. **Log-volatility path.** Given $\kappa_{1:T}$, the observation equation $y_t^* - m_{\kappa_t} = h_t + \xi_t^*$, $\xi_t^* \sim N(0, v_{\kappa_t}^2)$, together with the random-walk state equation $h_t = h_{t-1} + \eta_t$, $\eta_t \sim N(0, \sigma_h^2)$, is now *exactly* Chapter 15's local-level construction ($\tau_t \rightarrow h_t$, $y_t \rightarrow y_t^* - m_{\kappa_t}$, $\sigma^2 \rightarrow v_{\kappa_t}^2$, here diagonal but heteroskedastic across t rather than constant, which the completed-square argument there already accommodates term-by-term). The whole path $h_{1:T}$ is therefore drawn jointly from

$$h_{1:T} \mid \kappa_{1:T}, y^*, \sigma_h^2 \sim N(\hat{h}, K_h^{-1}), \quad K_h = \sigma_h^{-2} H' H + \text{diag}(v_{\kappa_t}^{-2}),$$

the same banded-precision Cholesky construction as Chapter 15's proofbox, with the constant $\sigma^{-2} I_T$ there replaced by the (still diagonal, now t -varying) $\text{diag}(v_{\kappa_t}^{-2})$.^a

This is why the KSC step, though genuinely new machinery for this note, is introduced as an auxiliary *representation* rather than a new sampling principle: once the mixture indicators linearize and Gaussianize the observation equation, the actual path draw reuses Chapter 15's derivation without modification.

^aThat is, conditional on the mixture indicators the model is a linear Gaussian state space, and the whole log-volatility path is drawn with the Chan and Jeliazkov (2009) precision sampler, exactly as in Chapter 15; `SVRW.m` in the replication kit is this single block.

Derivation: Contemporaneous Loadings a and the Remaining State Parameters

Because L is lower unit-triangular, row i of $Lu_t = \varepsilon_t$ reads $u_{i,t} + \sum_{j < i} a_{ij} u_{j,t} = \varepsilon_{i,t}$, i.e.

$$u_{i,t} = - \sum_{j < i} a_{ij} u_{j,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, e^{h_{i,t}}),$$

a linear regression of $u_{i,t}$ on the negatives of the preceding equations' contemporaneous residuals, with known, time-varying error variance $e^{h_{i,t}}$. This block has three distinct sub-results, worked out below in turn rather than by citation alone.

1. **Loadings** $a_{i,\cdot} \mid u, h_i$. With $u_{-i,t} = (u_{1,t}, \dots, u_{i-1,t})'$ and the stated $N(0, I)$ prior on $a_{i,\cdot} = (a_{i1}, \dots, a_{i,i-1})$,

$$p(u_i \mid a_{i,\cdot}, h_i) \propto \exp \left\{ -\frac{1}{2} \sum_t e^{-h_{i,t}} (u_{i,t} + u'_{-i,t} a_{i,\cdot})^2 \right\}, \quad p(a_{i,\cdot}) \propto \exp \left\{ -\frac{1}{2} a'_{i,\cdot} a_{i,\cdot} \right\}$$

This is a scalar-weight instance of the time-varying-precision GLS regression derived in the first proofbox above, with $Z_t \rightarrow -u_{-i,t}$, $y_t \rightarrow u_{i,t}$, $\Sigma_t^{-1} \rightarrow e^{-h_{i,t}}$, $b_0 \rightarrow 0$, and $B_0^{-1} \rightarrow I$. Multiplying and expanding the quadratic form in $a_{i,\cdot}$ exactly as there:

$$p(a_{i,\cdot} \mid u, h_i) \propto \exp \left\{ -\frac{1}{2} \left[a'_{i,\cdot} \left(I + \sum_t e^{-h_{i,t}} u_{-i,t} u'_{-i,t} \right) a_{i,\cdot} + 2a'_{i,\cdot} \sum_t e^{-h_{i,t}} u_{-i,t} u_{i,t} \right] \right\},$$

the kernel of $a_{i,\cdot} \mid u, h_i \sim N(\bar{a}_{i,\cdot}, \bar{V}_{a_i})$ with

$$\bar{V}_{a_i} = \left(I + \sum_t e^{-h_{i,t}} u_{-i,t} u'_{-i,t} \right)^{-1},$$

$$\bar{a}_{i,\cdot} = -\bar{V}_{a_i} \sum_t e^{-h_{i,t}} u_{-i,t} u_{i,t}.$$

2. $\sigma_{h,i}^2 \mid h_{i,1:T}$. With $\eta_{i,t} = h_{i,t} - h_{i,t-1}$ for $t = 1, \dots, T$ and the algorithm box's $IG(3, 0.01)$ prior (shape/scale convention, as used for posteriors throughout these notes),

$$p(h_i \mid \sigma_{h,i}^2) = (2\pi\sigma_{h,i}^2)^{-T/2} \exp \left\{ -\frac{1}{2\sigma_{h,i}^2} \sum_{t=1}^T \eta_{i,t}^2 \right\}, \quad p(\sigma_{h,i}^2) \propto (\sigma_{h,i}^2)^{-4} \exp \left\{ -\frac{0.01}{\sigma_{h,i}^2} \right\}$$

exactly Chapter 15's $\omega^2 \mid \tau$ argument with $\tau_t - \tau_{t-1} \rightarrow \eta_{i,t}$. Multiplying and collecting powers of $\sigma_{h,i}^2$:

$$p(\sigma_{h,i}^2 \mid h_i) \propto (\sigma_{h,i}^2)^{-(3+T/2+1)} \exp \left\{ -\frac{0.01 + \frac{1}{2} \sum_t \eta_{i,t}^2}{\sigma_{h,i}^2} \right\},$$

$$\sigma_{h,i}^2 \mid h_i \sim IG \left(3 + \frac{T}{2}, 0.01 + \frac{1}{2} \sum_t \eta_{i,t}^2 \right).$$

3. $h_{i,0} \mid h_{i,1}, \sigma_{h,i}^2$. With $h_{i,1} = h_{i,0} + \eta_{i,1}$, $\eta_{i,1} \sim N(0, \sigma_{h,i}^2)$, and prior $h_{i,0} \sim N(1, 10)$, this is a scalar instance of Chapter 6.2's $\beta \mid \sigma^2, y$ argument with a single “regressor” $x = 1$, outcome $y = h_{i,1}$, prior mean $b_0 = 1$, and prior variance $V_0 = 10$:

$$p(h_{i,1} \mid h_{i,0}, \sigma_{h,i}^2) \propto \exp \left\{ -\frac{(h_{i,1} - h_{i,0})^2}{2\sigma_{h,i}^2} \right\}, \quad p(h_{i,0}) \propto \exp \left\{ -\frac{(h_{i,0} - 1)^2}{20} \right\}$$

Multiplying and completing the square in $h_{i,0}$, exactly as in Chapter 6.2:

$$p(h_{i,0} \mid h_{i,1}, \sigma_{h,i}^2) \propto \exp \left\{ -\frac{1}{2} \left[h_{i,0}^2 \left(\frac{1}{10} + \frac{1}{\sigma_{h,i}^2} \right) - 2h_{i,0} \left(\frac{1}{10} + \frac{h_{i,1}}{\sigma_{h,i}^2} \right) \right] \right\},$$

the kernel of $h_{i,0} \mid h_{i,1}, \sigma_{h,i}^2 \sim N(\bar{h}_{i,0}, \bar{v}_{i,0})$ with $\bar{v}_{i,0} = (1/10 + 1/\sigma_{h,i}^2)^{-1}$ and $\bar{h}_{i,0} = \bar{v}_{i,0}(1/10 + h_{i,1}/\sigma_{h,i}^2)$.

16.2 Empirical Example: A 25-Variable LBVAR-SV, 1979–2026

Empirical Example: Scaling VAR-SV to a Large Panel: The Triangular Trick

The algorithm above was stated for general n , but the joint step in which β is drawn — a single Normal draw of the entire stacked coefficient vector, dimension nk , from a precision matrix built by summing $n \times n$ outer products over t — becomes computationally prohibitive once n is large: this note’s own 25-variable quarterly panel (1978Q2–2026Q1, the panel Chapter 13 used before this edition’s switch to BGR’s monthly data), with $p = 4$ lags, has $k = 101$ regressors per equation, so the joint vector has dimension $nk = 2,525$, and factorizing a $2,525 \times 2,525$ precision matrix *every single Gibbs sweep* is not a repeatable exercise. This is exactly the large-BVAR curse of dimensionality Chapter 13 opened with, now reappearing on the volatility side. Carriero, Chan, Clark and Marcellino’s (2021) corrigendum to their own large-BVAR-SV paper supplies the fix, and it is the same triangular idea Chapter 13 already uses for its hierarchical prior: because L is unit lower-triangular, row i of the structural system reads $u_{i,t} + \sum_{j < i} a_{ij}u_{j,t} = \varepsilon_{i,t}$, so equation i ’s structural shock depends only on variables $1, \dots, i$, never on any variable ordered after it. Drawing $(\beta_i, a_{i,\cdot})$ equation-by-equation, each conditional on the others, replaces one $nk \times nk$ factorization with n separate factorizations of size $(k + i - 1) \times (k + i - 1)$ — at most $k + n - 1 = 125$ here, not 2,525 — turning an intractable joint problem into n ordinary regressions, exactly the way Chapter 13’s own hierarchical-prior sampler already draws its mean equation. The volatility side needs no new idea at all: because the equations are conditionally independent given (β, a) , each $h_{i,1:T}$ is still just Chapter 15’s ordinary univariate random-walk smoother, run $n = 25$ times instead of $n = 3$. Combining Chapter 13’s triangular hierarchical-kappa prior (the same $\kappa_1, \dots, \kappa_4$ tightness hyperparameters, here with the quarterly panel’s uniform $\delta_i = 0.5$ own-lag prior mean and hard stationarity rejection sampling, both feasible because this panel enters in stationary-transformed growth rates rather than BGR’s levels) with this chapter’s random-walk log-volatility machinery, applied equation-by-equation, gives an LBVAR-SV that scales to the full 25-variable panel with essentially the same per-sweep cost as a constant-volatility hierarchical sampler. Both a constant-volatility (“CV”) and a stochastic-volatility (“SV”) version were fit to this 25-variable quarterly panel, $p = 4$ lags, 1,500 post-burn-in draws after a 500-draw burn-in, over the full 1979Q2–2026Q1 window available once $p = 4$ lags are consumed from the panel’s 1978Q2 start ($T = 188$).

Empirical Example: Time-Varying Volatility Across 25 Variables

Figure 70 plots the posterior mean structural shock standard deviation, $e^{h_{i,t}/2}$, for the same three narrative variables as before — GDP growth, CPI inflation, and unemployment — now estimated jointly within the full 25-variable system rather than a 3-variable one, with the Volcker disinflation, the 2008-09 recession, and the 2020 COVID-19 recession shaded. Starting the sample in 1979 rather than 1960 trades away the 1960s-1970s Great Inflation but buys the Volcker episode itself.

GDP-growth shock volatility is already elevated (3.2–4.9) through 1979-1983 as the Fed’s

disinflation triggers back-to-back recessions, falls through the Great Moderation to a 1-2 range by the mid-1990s, ticks up modestly around 2008-09 (peaking near 3.1 in 2008Q4), and then dwarfs every earlier episode with a peak of 14.19 in 2020Q3 — more than four times the Volcker-era high. Unemployment-shock volatility shows the same pattern at a different scale: essentially flat and low (0.1–0.3) outside recessions, a mild financial-crisis bump, and a sharp, narrow spike to 2.65 in 2020Q2 before collapsing back within a year. CPI-inflation shock volatility tells a different story: its highest point in this window is 1.97 in 1980Q2, at the peak of the Volcker disinflation, not during COVID or the 2021-22 inflation surge. With 23 other variables now sharing the burden of explaining the panel's comovement, less of CPI's own quarter-to-quarter variation is left unexplained during the 2021-22 episode than in a small 3-variable system — a reminder that a variable's estimated *shock* volatility depends on what else is in the model, not on the variable alone. As in the smaller system, a single “most volatile episode” headline does not fit all three variables: output and unemployment have their unambiguous worst moment in 2020Q2–Q3, while inflation's is still 1980, four decades earlier.

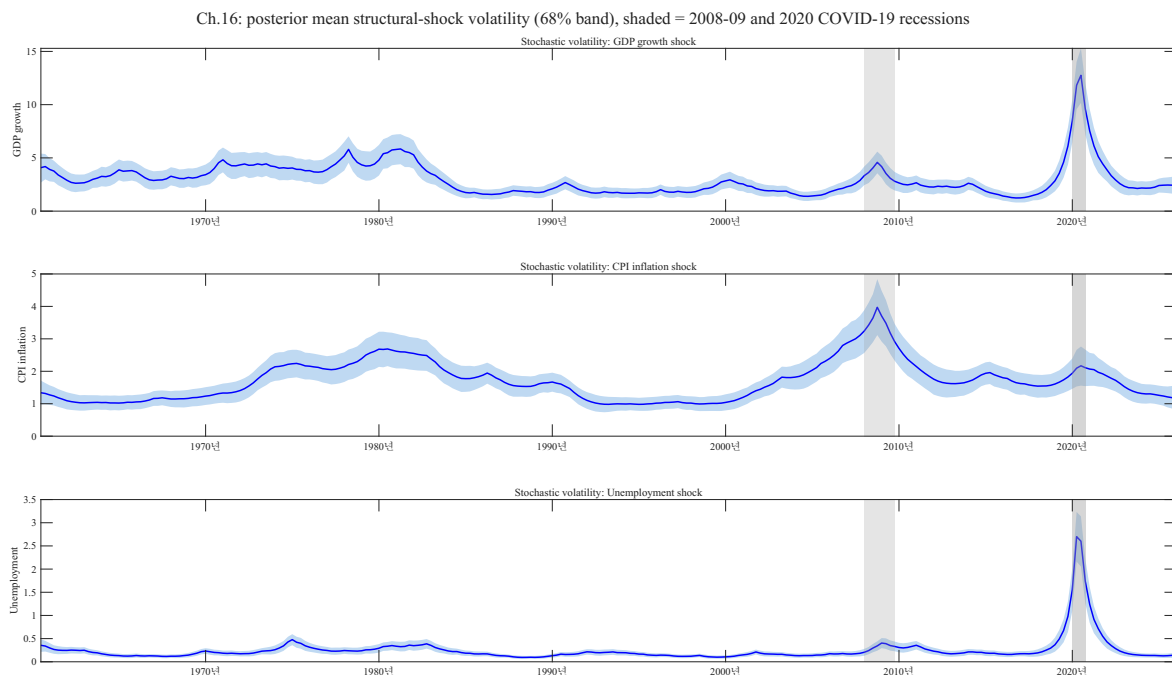


Figure 70: Posterior mean structural shock volatility, GDP growth, CPI inflation, and unemployment, estimated jointly within the 25-variable LBVAR-SV, 1979Q2–2026Q1; shaded regions are the Volcker disinflation (1979-83), the 2008-09 recession, and the 2020 COVID-19 recession

Empirical Example: Implied Time-Varying Correlation and Okun's Law

Figure 71 plots the implied correlation between the GDP-growth and unemployment reduced-form shocks, $\Sigma_t[\text{GDP}, \text{UEMP}] / \sqrt{\Sigma_t[\text{GDP}, \text{GDP}] \Sigma_t[\text{UEMP}, \text{UEMP}]}$, again the shock-level counterpart of Okun's Law. It is negative throughout the sample, ranging from a most-negative (tightest-linkage) value of -0.73 in 2000Q1 — on the eve of the dot-com downturn, not during either major crisis in this window — to a least-negative (weakest-

linkage) value of -0.19 in 2020Q2, the four COVID quarters. The COVID weakening repeats the finding from the smaller 3-variable system: 2020's unemployment spike was a mechanical, policy-mandated shutdown rather than the usual demand-driven channel linking slower growth to job losses, so growth and unemployment shocks moved together unusually loosely exactly when both were most extreme (window average -0.23). The Volcker window (1979-83) shows a comparatively tight -0.59 average, the 2008-09 financial crisis -0.43 , and the calm 2015-19 interlude -0.34 . As before, the single tightest observed link in the whole sample is not a famous-crisis quarter but an otherwise unremarkable one, reinforcing that “the correlation is strongest during the big, famous crisis” is an intuitive default this 25-variable system does not confirm either.

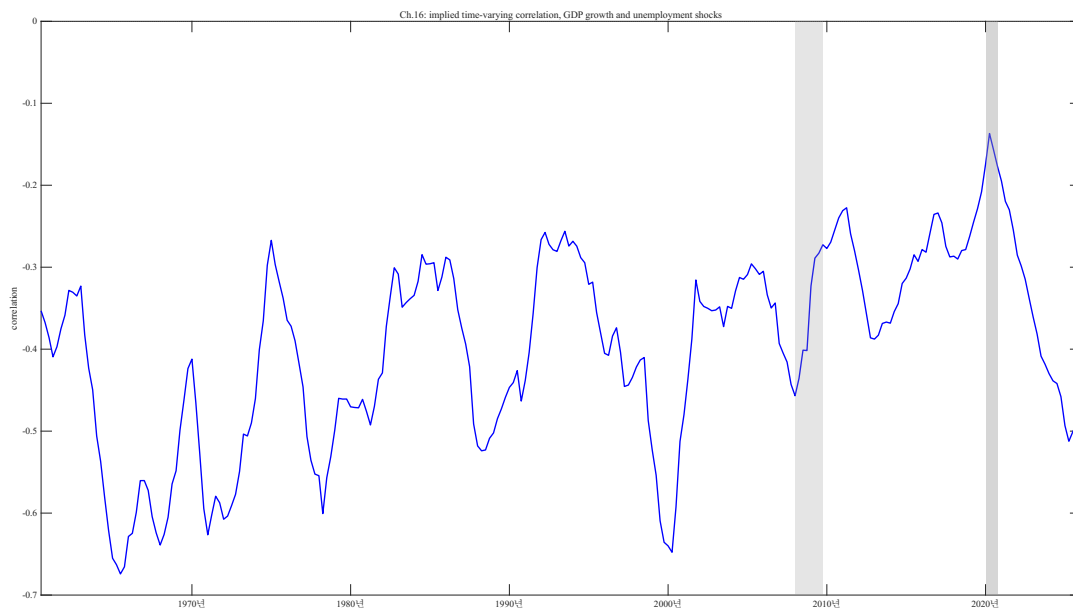


Figure 71: Posterior mean implied time-varying correlation between GDP-growth and unemployment reduced-form shocks, 25-variable LBVAR-SV, 1979Q2–2026Q1

16.3 LBVAR vs. LBVAR-SV: Point and Density Forecasts

Empirical Example: A First Look: Forecasting from a Single Pre-Crisis Origin

Adding stochastic volatility to a 25-variable BVAR is substantially more expensive than the constant-volatility hierarchical sampler of Chapter 13 — roughly 60% more time per sweep here, mostly from the $n = 25$ random-walk smoother calls — so it is worth asking directly whether that cost buys anything. As a first pass, both the CV and SV models above were fit through 2007Q4 only (the last calm quarter before the financial crisis) and used to forecast $h = 1, 4, 8, 16, 24, 32$ quarters ahead into the 2008-09 crisis and its aftermath, comparing (i) point forecasts, the per-draw average of each model's own reduced-form forecast path, against (ii) density forecasts, the log predictive score of the actual outcome under each model's own 1,500-draw predictive mixture, averaged in level space via `logsumexp`. For SV, each posterior draw's own future log-volatility path is simulated forward from its last in-sample state via the random-walk law $h_{T_0+s} = h_{T_0+s-1} + \eta_s$, so a draw's forecast-error

covariance reflects that draw's own uncertainty about *future* volatility, not just its current level.

Table 2 and Figure 72 report the result, and it is not the result a reader familiar with the SV forecasting literature would expect. Point forecasts (absolute forecast error, GDP growth) are nearly identical between CV and SV at every horizon, confirming that stochastic volatility changes a model's assessment of *uncertainty*, not its central tendency. But on density forecasts, SV does not simply win around the crisis as the textbook intuition suggests — it *loses badly* right where it should shine: at $h = 4$ and $h = 8$ (2008Q4-2009Q4, the worst of the crisis), SV's log score is 65.2 and 8.8 points worse than CV's (summed across all 25 variables). Only by $h = 24$ -32 (6-8 years past the crisis) does SV pull ahead, and only modestly (+3.8, +2.7). Taken at face value this table says stochastic volatility is actively harmful exactly when a forecaster would most want it to help — which should make you suspicious of the exercise, not of stochastic volatility.

Table 2: LBVAR vs. LBVAR-SV, 25-variable panel, trained through 2007Q4 only, forecasting into the 2008-09 crisis and aftermath: sum of 25 univariate marginal log predictive scores, and the SV–CV difference (positive favors SV)

h (quarters)	LPS, CV	LPS, SV	Diff (SV–CV)
1	–56.4	–59.1	–2.7
4	–195.8	–261.1	–65.2
8	–81.1	–89.9	–8.8
16	–78.0	–78.5	–0.5
24	–71.5	–67.7	+3.8
32	–72.3	–69.7	+2.7

LBVAR vs LBVAR-SV, 25-Variable Panel, Trained Through 2007Q4, Forecasting Into the 2008-09 Crisis

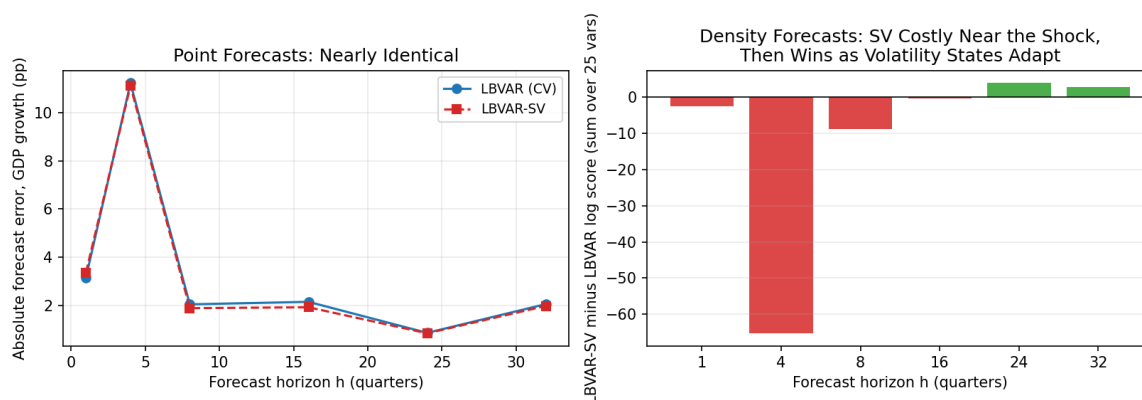


Figure 72: Left: absolute point-forecast error, GDP growth, LBVAR (CV) vs. LBVAR-SV, by horizon — nearly identical. Right: SV-minus-CV log predictive score (summed across all 25 variables), by horizon — costly near the shock, mildly favorable only years later, from a single 2007Q4 origin

Empirical Example: Why Does SV Lose Right Where It Should Win?

The culprit is not a modeling error but the specific quarter chosen as the forecast origin. 2007Q4 sits at the trough of the “Great Moderation” — the unusually quiet stretch of the mid-2000s — so SV’s *volatility state at that exact quarter* has adapted down to an abnormally calm level. CV, by contrast, uses one constant variance equal to the *whole-sample* (1978-2007) average, which already has the Volcker disinflation, the 1990-91 recession, and the 2001 downturn baked into it. Comparing posterior means of the two models’ implied variance at the forecast origin makes the gap concrete:

Variable	CV variance (whole-sample avg.)	SV variance at 2007Q4
GDP growth	5.17	2.92
CPI inflation	1.41	0.13
Unemployment	0.029	0.017

For CPI, CV’s unconditional variance is more than *eleven times* SV’s state at the forecast origin. Because the log-volatility process is a pure random walk with no mean reversion (Section 16.1), it cannot anticipate a shock it has not yet seen, and it only widens *slowly*, at a pace set by each equation’s own small innovation variance σ_{η}^2 . Handed a single origin that happens to fall in an unusually quiet quarter right before an unprecedented shock, SV is — mechanically, predictably — *more* overconfident than a constant-variance model that already has past volatility episodes averaged into it. This is a genuine property of memoryless stochastic volatility, not a bug in the sampler: it rewards being wide already (CV, here) over being adaptive but starting from a calm state (SV, here). By $h = 24$ -32, the random walk has diffused enough extra width to pull ahead, but that is a slow, mechanical catch-up, not evidence of SV *learning* anything.

Empirical Example: The Real Test: A Rolling-Origin Comparison

The applied literature (Clark, 2011; Carriero, Clark and Marcellino) never demonstrates SV’s density-forecast advantage from one fixed pre-shock origin. It uses a *rolling* or expanding-window design: re-estimate at successive origins as the sample grows, so the volatility state gets to update in near real time as a crisis actually unfolds. Repeating the exercise above at five origins spanning the crisis — 2007Q4, 2008Q2, 2008Q4, 2009Q2, 2009Q4 (each re-estimated independently, 300 burn-in +700 retained draws) — and evaluating $h = 1, 4, 8, 16$ quarters ahead from each origin gives a very different picture. Table 3 and Figure 73 show the 3-variable (GDP, CPI, unemployment) joint log-score gap, SV minus CV, by origin and horizon. At $h = 4$ the gap starts at -10.5 (2007Q4, the same badly-calibrated result as above), improves to -5.7 (2008Q2), -1.3 (2008Q4), and finally turns *positive*, $+0.2$, at 2009Q2 — the first origin whose estimation sample already contains real 2008-09 volatility. The $h = 8$ row shows the same monotone climb: $-7.3 \rightarrow -1.7 \rightarrow +0.9 \rightarrow +1.1$, only reverting toward zero at 2009Q4 once the crisis itself is essentially over and both models have converged on similar assessments of risk. This is exactly the textbook result — SV’s short-horizon density forecasts overtake CV’s *once*

the model has been allowed to see the volatility building — and it is the origin choice, not the model, that determines whether that advantage is visible.

A simple pool (equal-weighted average across all five origins) of the $h = 4$ and $h = 8$ gaps is still negative ($-3.5, -1.4$): the two earliest, pre-crisis origins are heavy enough to pull the average down even though three of the five origins individually favor SV. This is worth stating plainly because it is easy to average away the more informative result: the origin-by-origin *progression* — SV moving from a clear loss to a clear win as its estimation window absorbs the shock — is the finding, not any single pooled number.

Table 3: Rolling-origin comparison: SV–CV 3-variable (GDP, CPI, unemployment) joint log-score gap by forecast origin and horizon (positive favors SV)

Origin	$h = 1$	$h = 4$	$h = 8$	$h = 16$
2007Q4	−0.15	−10.49	−7.28	−1.76
2008Q2	−0.35	−5.70	−1.74	−0.47
2008Q4	−1.25	−1.34	+0.85	+0.52
2009Q2	−0.08	+0.21	+1.06	+0.12
2009Q4	+0.04	−0.15	−0.06	−0.09
Pooled avg.	−0.36	−3.49	−1.43	−0.33

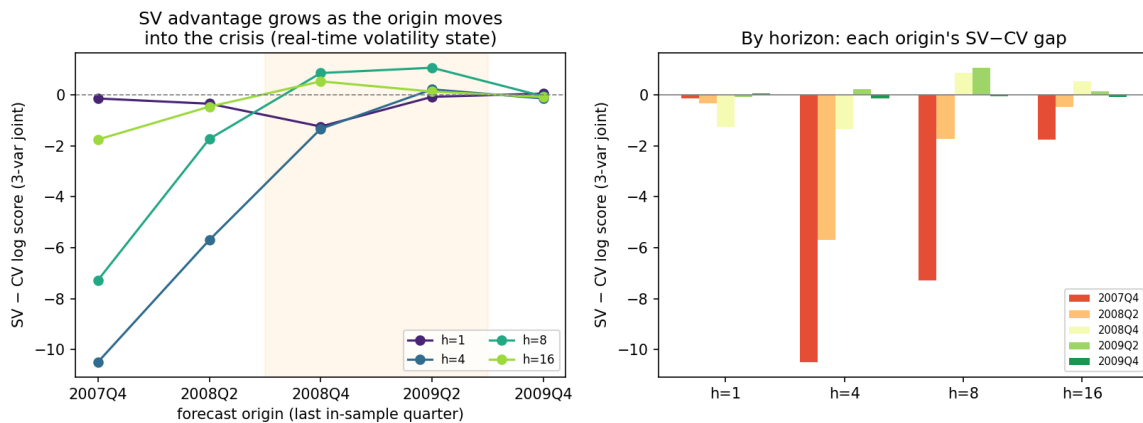


Figure 73: Rolling-origin SV–CV log-score gap. Left: by horizon, as the origin moves through the crisis — SV’s short-horizon deficit shrinks and reverses as its estimation sample absorbs emerging volatility. Right: the same data grouped by horizon, showing each origin’s contribution

Remark: The Real Lesson: SV’s Advantage Depends on What It Has Already Seen

The single-origin table made stochastic volatility look actively harmful during a crisis; the rolling-origin table shows the opposite — SV overtakes CV at short horizons *as soon* as its estimation sample contains the early part of the shock. Both tables are computed correctly; they answer different questions. A single fixed pre-shock origin asks “can SV

predict a crisis it has never seen coming,” and the honest answer is no better than a constant-variance model can. A rolling origin asks “does SV’s volatility state track a crisis once it is underway,” and the answer here is yes, decisively, within two to three quarters of the shock’s onset. This is also why real-time forecast evaluations in the applied literature almost always use rolling or expanding windows rather than one fixed split: a single origin is one draw from a distribution of possible “how much has the model already seen” scenarios, and that draw can make an otherwise-sound model look arbitrarily better or worse than it is. What SV buys, precisely, is *adaptation conditional on having observed the shock begin* — not foresight, and not a free win the instant a crisis starts from a calm state. Getting the mean equation right first, as Chapter 13 does on its own terms before this chapter ever introduces volatility, remains the model that stochastic volatility, when it helps, helps on top of.

17. Time-Varying Parameter VAR

Roadmap

Chapters 15 and 16 each let one latent thing drift over time: Chapter 15, a single scalar trend; Chapter 16, n independent log-volatilities. Both were drawn with the same tool, a linear Gaussian state-space recursion. This chapter applies that exact tool to a state that finally looks like what practitioners usually mean by “time-varying”: the VAR’s own coefficients, all $m = n(np + 1)$ of them, drifting jointly. Nothing new is derived here — every step is Chapter 6.2’s conjugate Normal update, run once per time period — the only genuinely new content is bookkeeping the state’s larger size and running the recursion forward, then backward.

Chapter 16 held a VAR’s coefficients Φ fixed and let only its shock covariance move. This chapter does the reverse and more: a time-varying- parameter VAR (TVP-VAR) lets the entire coefficient vector Φ_t evolve as a random walk, one independent random walk per coefficient, while holding the residual covariance Σ constant. The state being tracked is no longer a single scalar log-variance (Chapter 14), nor a low-dimensional trend-plus-cycle pair (Chapter 15), nor n independent log-volatilities (Chapter 16) — it is the full stacked VAR coefficient vector, dimension $m = n(np + 1)$, evolving jointly. The Gibbs sampler is nonetheless the same local-level machinery used throughout Chapters 15 and 16, generalized from a scalar or diagonal state to a genuinely vector-valued one: a forward Kalman filter followed by a backward simulation smoother, mathematically equivalent to the banded precision sampler of the preceding two chapters but implemented here via the classical recursions since the state dimension ($m = 21$ below) makes per-period matrix work cheap.

Time-Varying Parameter VAR (TVP-VAR) For an n -variable VAR(p), write the reduced form with time-varying coefficients

$$y_t = \Phi_{1,t}y_{t-1} + \cdots + \Phi_{p,t}y_{t-p} + c_t + u_t, \quad u_t \sim N(0, \Sigma),$$

and stack all $n(np+1)$ coefficients at date t into β_t , so that $y_t = Z_t\beta_t + u_t$ with $Z_t = I_n \otimes x'_t$ and $x_t = (1, y'_{t-1}, \dots, y'_{t-p})'$. Unlike Chapter 11's constant Φ , here every element of β_t follows its own independent driftless random walk,

$$\begin{aligned}\beta_t &= \beta_{t-1} + \eta_t, \\ \eta_t &\sim N(0, Q), \\ Q &= \text{diag}(q_1, \dots, q_m), \\ \beta_0 &\sim N(a_0, B_0),\end{aligned}$$

with Σ itself held constant, in contrast to Chapter 16's model, which held Φ fixed and let Σ_t move. A TVP-VAR with *both* β_t and Σ_t time-varying (Primiceri, 2005) is the natural combination of this chapter and the last, but is not estimated here.

Figure 74 shows the two-pass structure that draws the whole path $\beta_{0:T}$: a forward sweep that only ever looks backward in time (solid arrows), followed by a backward sweep that revisits each date in reverse order using the value just drawn for the date ahead of it (dashed arrows).

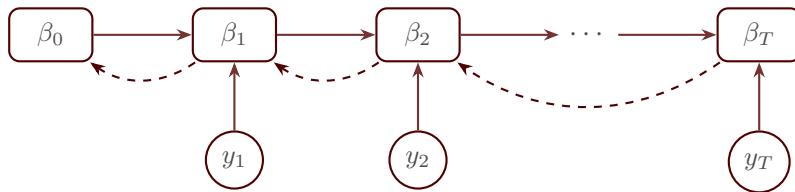


Figure 74: Forward-filtering, backward-sampling (FFBS): the forward pass (solid) folds each y_t into a filtered state $\beta_{t|t}$ moving left to right; the backward pass (dashed) then samples $\beta_T, \beta_{T-1}, \dots, \beta_0$ moving right to left, each draw conditioning on the value just drawn one step ahead

Algorithm: Gibbs Sampler for the TVP-VAR

With priors mirroring Chapter 11's VAR prior on the initial state (precision 1 on slope coefficients, $1/10$ on intercepts, applied to β_0 's prior B_0), an inverse-Wishart($n+3, I_n$) prior on Σ , and independent Inverse-Gamma priors on each state variance q_j (shape 3, scale 0.01^2 for slope innovations and 0.1^2 for intercept innovations, allowing intercepts to drift somewhat more freely):

1. **Coefficient path.** Draw the entire path $\beta_{1:T}$ jointly via a forward Kalman filter (using precision-weighted updates built from $Z'_t\Sigma^{-1}Z_t$ and $Z'_t\Sigma^{-1}y_t$, exactly the vector generalization of the scalar terms that entered Chapter 15 and 16's precision samplers) followed by a backward simulation smoother — this is the classical Carter–Kohn/Frühwirth-Schnatter forward-filtering, backward-sampling (FFBS) algorithm,^a mathematically the same recursive local-level construction as Chapters 15–16 but

carried out here for a vector state of dimension $m = n(np + 1)$ rather than a scalar or a set of scalars.

2. **Covariance.** Draw Σ from its inverse-Wishart full conditional given the residuals $u_t = y_t - Z_t\beta_t$ computed along the *entire sampled path* — the multivariate generalization of Chapter 6’s conjugate regression-variance update, now applied to a path of residuals rather than a single fixed-coefficient fit.
3. **State variances.** Draw each q_j from its Inverse-Gamma full conditional given the sum of squared first differences of $\beta_{1:T}$ (including $\beta_1 - \beta_0$) — structurally identical to Chapter 15’s σ_τ^2 update and Chapter 16’s $\sigma_{h,i}^2$ update, just applied here to $m = 21$ coefficients independently rather than to 1 trend or n log-volatilities.
4. **Initial state.** Draw β_0 from its (diagonal, hence elementwise) Normal full conditional given β_1 , Q , and the prior $N(a_0, B_0)$.

^aCarter and Kohn (1994); Frühwirth-Schnatter (1994). FFBS and the Chan–Jeliazkov (2009) precision sampler of Chapters 15–16 draw from the identical joint Gaussian full conditional; they differ only in whether the banded linear system is solved by forward–backward recursions or by one sparse Cholesky factorization. In the replication kit, `Launch_ch17_cs2001.m` implements the FFBS form and `VAR_TVP.m` the precision form.

17.1 Posterior Derivation: Forward-Filtering, Backward-Sampling

The state β_t is now vector-valued and evolves over time, so neither Chapter 6’s single fixed-coefficient update nor Chapter 15’s banded-precision joint draw is stated in quite the right form; the classical way to draw $\beta_{1:T}$ here is the two-pass Carter–Kohn algorithm below. Both passes turn out to be nothing but repeated applications of the same conjugate-Normal completing-the-square argument first derived in Chapter 6.2, applied once per time period rather than once for the whole sample.

Derivation: Forward Filter: $\beta_t \mid y_{1:t}$

Suppose $\beta_{t-1} \mid y_{1:t-1} \sim N(\beta_{t-1|t-1}, P_{t-1|t-1})$ (true at $t = 1$ by the prior $\beta_0 \sim N(a_0, B_0)$). Two steps update this to t .

Prediction. Since $\beta_t = \beta_{t-1} + \eta_t$ with $\eta_t \sim N(0, Q)$ independent of $y_{1:t-1}$, $\beta_t \mid y_{1:t-1}$ is the sum of two independent Normals, hence Normal with

$$\beta_{t|t-1} = \beta_{t-1|t-1}, \quad P_{t|t-1} = P_{t-1|t-1} + Q.$$

Update. Given the predictive prior $\beta_t \mid y_{1:t-1} \sim N(\beta_{t|t-1}, P_{t|t-1})$ and the single new observation $y_t = Z_t\beta_t + u_t$, $u_t \sim N(0, \Sigma)$, the posterior kernel is

$$p(\beta_t \mid y_{1:t}) \propto N(y_t \mid Z_t\beta_t, \Sigma) N(\beta_t \mid \beta_{t|t-1}, P_{t|t-1})$$

which is *exactly* Chapter 6.2's $\beta \mid \sigma^2, y$ kernel with $(b_0, B_0) \rightarrow (\beta_{t|t-1}, P_{t|t-1})$, $X \rightarrow Z_t$, $\sigma^{-2} \rightarrow \Sigma^{-1}$, and a single “observation” y_t in place of the full sample. Completing the square exactly as there gives:

$$\begin{aligned}\beta_t \mid y_{1:t} &\sim N(\beta_{t|t}, P_{t|t}), \\ P_{t|t} &= (P_{t|t-1}^{-1} + Z_t' \Sigma^{-1} Z_t)^{-1}, \\ \beta_{t|t} &= P_{t|t} (P_{t|t-1}^{-1} \beta_{t|t-1} + Z_t' \Sigma^{-1} y_t)\end{aligned}$$

^a Iterating prediction-then-update for $t = 1, \dots, T$ is the forward Kalman filter; it is not a new derivation technique but Chapter 6.2's single-shot conjugate update applied recursively, once per period, with each period's posterior becoming the next period's prior (after the prediction step folds in that period's process noise Q).

^aThe formula above inverts an $m \times m$ matrix ($m = 21$ here) at *every* one of the $T = 263$ time periods, which looks expensive but never actually requires an $m \times m$ inversion in practice: the Woodbury identity rewrites $(P_{t|t-1}^{-1} + Z_t' \Sigma^{-1} Z_t)^{-1}$ using only the inverse of an $n \times n$ matrix ($n = 3$ here, since $Z_t' \Sigma^{-1} Z_t$ has rank at most n), because $Z_t = I_n \otimes x_t'$ always has only n rows worth of new information at each date. Swapping a large, dense inversion for a small one is what keeps the filter fast even as m grows with the number of variables and lags.

Derivation: Backward Sampling: $\beta_t \mid \beta_{t+1}, y_{1:t}$

The state equation makes β_{t+1} conditionally independent of $y_{1:t}$ given β_t (the model's Markov structure: all of $y_{1:t}$'s information about β_{t+1} passes through β_t), so by Bayes' rule the posterior kernel is

$$\begin{aligned}p(\beta_t \mid \beta_{t+1}, y_{1:t}) &\propto p(\beta_{t+1} \mid \beta_t) p(\beta_t \mid y_{1:t}) \\ &= N(\beta_{t+1} \mid \beta_t, Q) N(\beta_t \mid \beta_{t|t}, P_{t|t})\end{aligned}$$

which is again exactly Chapter 6.2's conjugate-Normal kernel form, this time treating the already-drawn future state β_{t+1} as a unit-design-matrix “observation” of β_t with “error covariance” Q , and the forward filter's $N(\beta_{t|t}, P_{t|t})$ as the prior. Completing the square gives:

$$\begin{aligned}\beta_t \mid \beta_{t+1}, y_{1:t} &\sim N(\beta_{t|t,t+1}, P_{t|t,t+1}), \\ P_{t|t,t+1} &= (P_{t|t}^{-1} + Q^{-1})^{-1}, \\ \beta_{t|t,t+1} &= P_{t|t,t+1} (P_{t|t}^{-1} \beta_{t|t} + Q^{-1} \beta_{t+1})\end{aligned}$$

The full path is then drawn by first sampling $\beta_T \sim N(\beta_{T|T}, P_{T|T})$ (the forward filter's own final output, needing no further conditioning since there is no β_{T+1}), then sampling $\beta_{T-1}, \dots, \beta_0$ in reverse order from the recursion above, each draw conditioning on the value just drawn for β_{t+1} . This two-pass forward-filter/backward-sample procedure delivers an exact joint draw of $\beta_{0:T}$ from $p(\beta_{0:T} \mid \Sigma, Q, y)$, by the same chain-rule argument $p(\beta_{0:T} \mid \cdot) = p(\beta_T \mid \cdot) \prod_{t=0}^{T-1} p(\beta_t \mid \beta_{t+1}, y_{1:t})$ used to justify backward sampling in any linear Gaussian state-space model.

Derivation: $\Sigma \mid \beta_{1:T}, y, q_j \mid \beta_{1:T}$, and $\beta_0 \mid \beta_1$

Each of the three results below substitutes this chapter's specific quantities into a kernel-matching argument already carried out in full elsewhere in the book, and re-derives the resulting kernel explicitly.

1. $\Sigma \mid \beta_{1:T}, y$. With the full path $\beta_{1:T}$ in hand, $u_t = y_t - Z_t \beta_t$, $t = 1, \dots, T$, is a completed residual series exactly as in the SUR/VAR case, with U the $T \times n$ matrix stacking u'_t (so $U'U = \sum_t u_t u'_t$). Using the trace identity $u'_t \Sigma^{-1} u_t = \text{tr}(\Sigma^{-1} u_t u'_t)$ exactly as in Chapter 10.2's $\Sigma \mid \beta, y$ proofbox, with $e_t \rightarrow u_t$, $E \rightarrow U$, and $M \rightarrow n$:

$$p(y \mid \beta_{1:T}, \Sigma) = |\Sigma|^{-T/2} \exp \left\{ -\frac{1}{2} \sum_t u'_t \Sigma^{-1} u_t \right\} = |\Sigma|^{-T/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Sigma^{-1} U'U) \right\},$$

$$p(\Sigma) \propto |\Sigma|^{-(\nu_0 + n + 1)/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Sigma^{-1} S_0) \right\}.$$

Multiplying the two and using $\text{tr}(\Sigma^{-1} S_0) + \text{tr}(\Sigma^{-1} U'U) = \text{tr}(\Sigma^{-1} (S_0 + U'U))$, exactly as in Chapter 10.2:

$$p(\Sigma \mid \beta_{1:T}, y) \propto |\Sigma|^{-(T + \nu_0 + n + 1)/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Sigma^{-1} (S_0 + U'U)) \right\},$$

$$\Sigma \mid \beta_{1:T}, y \sim IW(S_0 + U'U, \nu_0 + T).$$

2. $q_j \mid \beta_{1:T}$. Each state variance q_j depends only on the j -th coordinate's first differences $\Delta \beta_{j,t} = \beta_{j,t} - \beta_{j,t-1} \sim N(0, q_j)$, $t = 1, \dots, T$ (including $\Delta \beta_{j,1} = \beta_{j,1} - \beta_{j,0}$). With prior $q_j \sim IG(a_{0,j}/2, d_{0,j}/2)$,

$$p(\beta_{1:T} \mid q_j) = (2\pi q_j)^{-T/2} \exp \left\{ -\frac{1}{2q_j} \sum_{t=1}^T \Delta \beta_{j,t}^2 \right\}, \quad p(q_j) \propto q_j^{-(a_{0,j}/2 + 1)} \exp \left\{ -\frac{d_{0,j}}{2q_j} \right\}$$

exactly Chapter 15's $\omega^2 \mid \tau$ argument with $\tau_t - \tau_{t-1} \rightarrow \Delta \beta_{j,t}$ and $\omega^2 \rightarrow q_j$, applied coordinate by coordinate for $j = 1, \dots, m$. Multiplying and collecting powers of q_j :

$$p(q_j \mid \beta_{1:T}) \propto q_j^{-\left(\frac{a_{0,j} + T}{2} + 1\right)} \exp \left\{ -\frac{d_{0,j} + \sum_t \Delta \beta_{j,t}^2}{2q_j} \right\},$$

$$q_j \mid \beta_{1:T} \sim IG \left(\frac{a_{0,j} + T}{2}, \frac{d_{0,j} + \sum_t \Delta \beta_{j,t}^2}{2} \right).$$

3. $\beta_0 \mid \beta_1$. Since $\beta_1 = \beta_0 + \eta_1$, $\eta_1 \sim N(0, Q)$, and the prior is $\beta_0 \sim N(a_0, B_0)$, this is precisely the $t = 0$ case of the backward-sampling kernel derived above, with $(\beta_{t|t}, P_{t|t}) \rightarrow (a_0, B_0)$ and $\beta_{t+1} \rightarrow \beta_1$:

$$p(\beta_1 \mid \beta_0) p(\beta_0) = N(\beta_1 \mid \beta_0, Q) N(\beta_0 \mid a_0, B_0)$$

Completing the square exactly as in the backward-sampling derivation above:

$$\begin{aligned}\beta_0 \mid \beta_1 &\sim N(\beta_{0|0,1}, P_{0|0,1}), \\ P_{0|0,1} &= (B_0^{-1} + Q^{-1})^{-1}, \\ \beta_{0|0,1} &= P_{0|0,1}(B_0^{-1}a_0 + Q^{-1}\beta_1).\end{aligned}$$

17.2 Empirical Example: Has Monetary Transmission Changed?

Empirical Example: Data and the Question

This section's model — a VAR with random-walk coefficients and a *constant* residual covariance — is exactly the specification Cogley and Sargent (2001, “Evolving Post-World War II U.S. Inflation Dynamics,” NBER Macroeconomics Annual) introduced to study how the persistence of U.S. inflation, and the private sector's and the Fed's behavior around it, changed across the postwar era. Their own VAR uses the same three variables in the same spirit — inflation, unemployment, and a short-term interest rate — estimated by the same Kalman-filter forward pass and backward simulation smoother used below. Stochastic volatility on top of the drifting coefficients came only in their 2005 follow-up, “Drifts and Volatilities,” the paper the mathbox above calls out as ‘Primiceri's (2005) combined model’; that extension is not estimated here.

The empirical exercise below fits a TVP-VAR(2) to quarterly U.S. unemployment, CPI inflation, and the effective Fed funds rate, 1960Q3–2026Q1 ($T = 263$, the same source series Chapter 11 draws on, now covering the full sample rather than the pre-2008 window used before this note's data extension). The original cut stopped at 2008Q2 specifically to avoid the zero lower bound that followed, on the reasoning that a random-walk coefficient model has no special treatment for the Fed funds rate becoming constrained near zero. That concern turns out to be less important than it sounds: β_t is a random walk in *coefficient* space, not a structural model of the policy rate itself, so years spent at the ZLB (2009–2015, and again briefly in 2020) are absorbed the same way any other slow regime change is — by letting the coefficients drift, not by imposing a floor on y_t . Using the full sample also lets a third snapshot date be added inside the post-COVID hiking cycle.

3,000 post-burn-in Gibbs draws (after a 1,000-draw burn-in) recover a constant residual covariance with standard deviations of 0.62 (unemployment), 1.65 (CPI inflation), and 0.21 percentage points (Fed funds rate) per quarter, and a moderate negative correlation of -0.35 between unemployment and inflation shocks. Two of these three numbers moved substantially from the pre-extension estimates (0.18, 0.94, 0.27): unemployment's shock size more than tripled, almost entirely on the strength of 2020Q2's unemployment surge, while the Fed funds rate's shock size actually *fell* slightly, because the extended sample now also includes several ZLB years of essentially flat, low-variance policy rate quarters that were previously excluded altogether. This is a direct, visible cost of holding Σ constant (as this chapter's model does; only Primiceri's (2005) combined model, mentioned above but not estimated here, would let it move): one global residual covariance now has to summarize both the calm years and 2020Q2 at once.

The question this model is built to answer is therefore not “what is the average effect of monetary policy” — Chapter 11 already answers that — but “has the effect of monetary policy *changed*.” The three snapshots below compare the coefficients, and the impulse responses they imply, across three dates within the sample.

Empirical Example: Three Snapshots: 1975Q3, 2005Q3, and 2022Q3

Figure 75 compares the posterior mean impulse response of all three variables to an identical 100 basis-point Fed funds rate shock, evaluated using the coefficient vector β_t at three dates: 1975Q3 (Volcker-era disinflation approaching), 2005Q3 (mid-2000s Great Moderation), and 2022Q3 (the middle of the post-COVID rate-hiking cycle, the newly added COVID-era snapshot). The Fed funds rate’s own persistence rises monotonically across all three dates — 0.711 (1975Q3), 0.805 (2005Q3), 0.846 (2022Q3) — consistent with the widely documented shift toward more gradual, smoothed policy adjustment. But by 2022Q3 this persistence has pushed the local system close enough to the edge of stability that the shape of the impulse responses changes qualitatively, not just in size. At 1975Q3 and 2005Q3 all three responses damp out over the 20-quarter horizon, as an impulse response normally should. At 2022Q3, after an initial hump (inflation peaks at 0.70 two quarters out, much as in the earlier snapshots), all three responses instead *reaccelerate* through the second half of the horizon and have not leveled off by quarter 20. Unemployment’s response is still *falling* and reaches -0.95 : a contractionary shock, two decades into this coefficient path, ends up pulling unemployment *down*, not up — backwards from what the sign restriction in Section 12.5 assumed and the opposite of a textbook contractionary response. Inflation and the funds rate’s own response to its own shock, meanwhile, are still rising and reach 1.84 and 5.98 respectively. None of these three numbers are peaks in the usual sense, since none of the three series has turned over yet. This is a genuine, if uncomfortable, finding: the fastest and most aggressive tightening episode in the sample coincides with a coefficient snapshot whose implied local dynamics sit right at the boundary of stability, and Section 17.2 below shows this instability is not estimated with much precision. The *price puzzle* — CPI inflation rising, not falling, immediately after a contractionary shock, a well-documented anomaly in monetary VARs identified this way — is present at all three dates but is smallest at 2005Q3 (initial peak 0.74) and largest, by a wide margin once the reacceleration is included, at 2022Q3. Unemployment’s own persistence continues the decline already visible across the first two dates: 0.746 (1975Q3) $\rightarrow$ 0.699 (2005Q3) $\rightarrow$ 0.591 (2022Q3). That is the opposite direction from the Fed funds rate’s own persistence, a reminder that “the system as a whole is getting more persistent” is not true equation-by-equation.

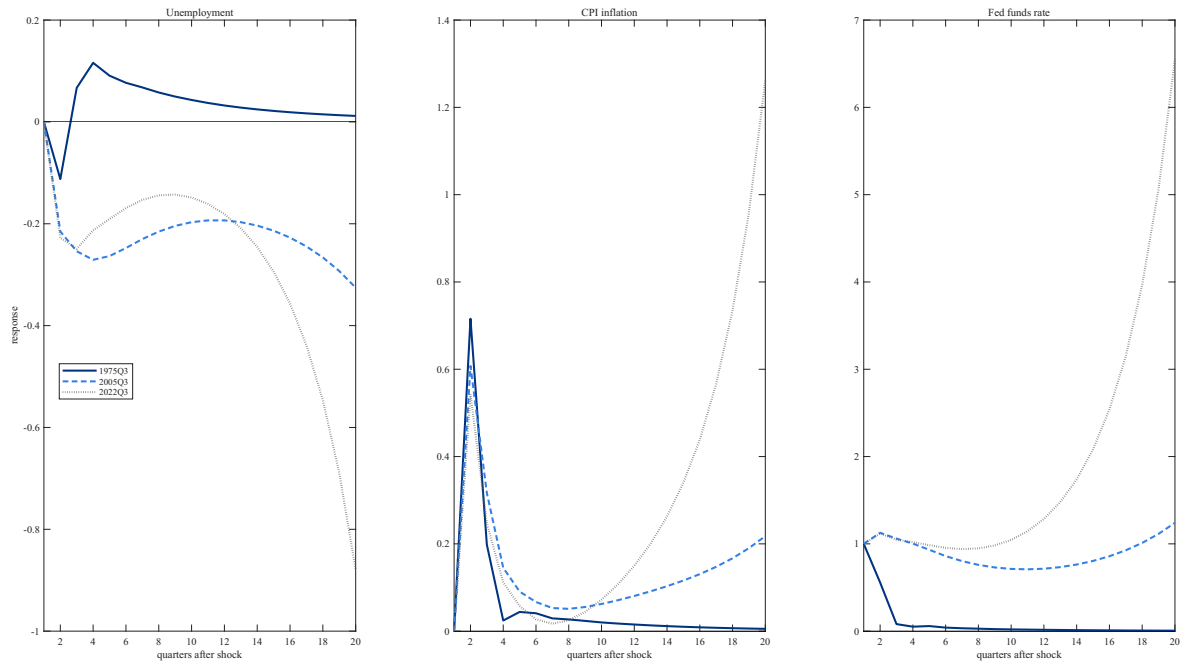


Figure 75: Generalized impulse responses to a 100bp Fed funds rate shock, posterior mean coefficient paths evaluated at 1975Q3, 2005Q3, and 2022Q3

Empirical Example: How Confident Is This Difference?

Figure 76 plots the posterior mean difference between the 2022Q3 and 2005Q3 impulse responses — the post-COVID hiking cycle against the pre-crisis Great Moderation benchmark — together with a 90% posterior interval computed directly from the 3,000 post-burn-in draws. The pointwise differences are large in absolute terms: by horizon 20, 3.98 points for the Fed funds rate's own response, 1.39 points for inflation, and 0.47 points for unemployment, all driven by the 2022Q3 snapshot's reacceleration documented above. Despite that size, the 90% posterior interval includes zero at every horizon for all three variables — unlike the pre-extension 2005Q3-vs-1975Q3 comparison, which did exclude zero at several early horizons, this new comparison is not estimated precisely anywhere on the 20-quarter horizon. That is itself informative: 2022Q3 sits only a handful of quarters from the end of a 263-quarter sample, so the forward-filtering, backward-sampling algorithm has very little post-2022Q3 data left to pin the local coefficient path down, and the random-walk state equation lets β_t drift substantially within that uncertainty. The honest summary is that this model finds a large and economically striking *point estimate* of change in monetary transmission around the post-COVID tightening cycle — a stronger price puzzle, borderline-unstable local dynamics — but that estimate comes with enough posterior uncertainty, precisely because it sits so close to the end of the available sample, that it should be read as a suggestive signal rather than a precisely pinned-down shift, in contrast to the better-identified 1975Q3-vs-2005Q3 change documented earlier in the book.

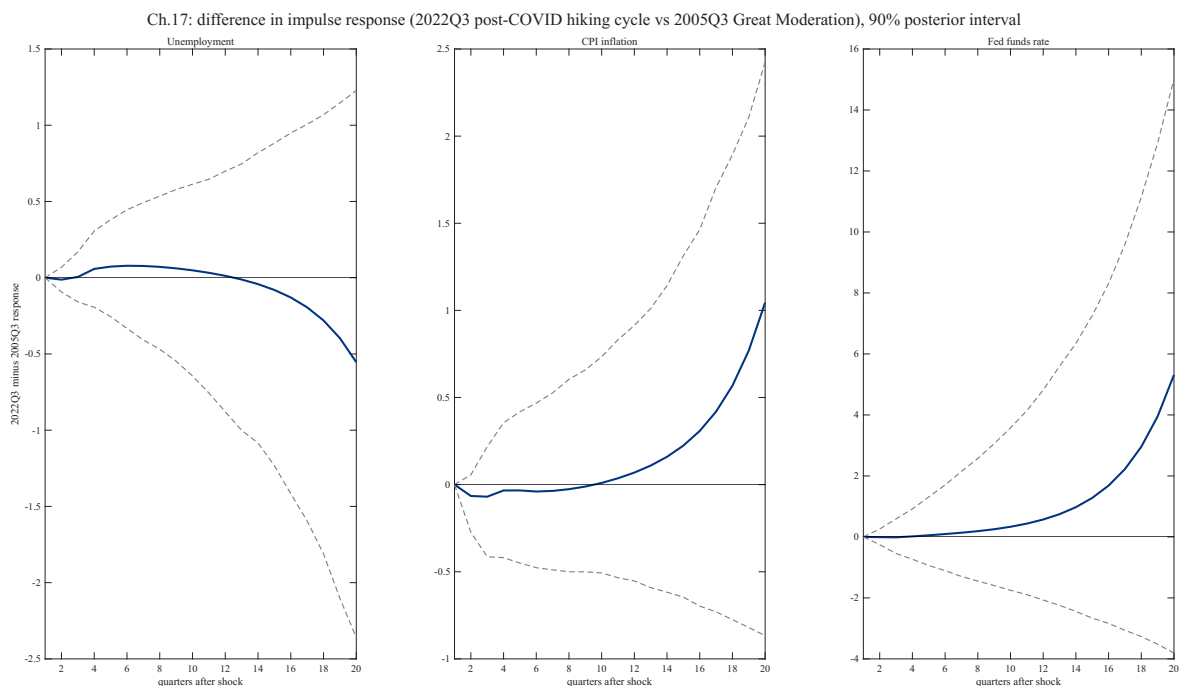


Figure 76: Posterior mean difference in impulse response (2022Q3 post-COVID hiking cycle minus 2005Q3 Great Moderation) with 90% posterior interval

17.3 Cogley and Sargent (2001)

The example above asked this chapter's question with this note's own data; this section replicates the paper that first asked it. Cogley and Sargent (2001, hereafter CS), "Evolving Post-World War II U.S. Inflation Dynamics," *NBER Macroeconomics Annual*, built precisely this model — a VAR(2) with random-walk coefficients and constant innovation covariance — to argue that the rise and fall of U.S. inflation reflected *systematic changes in monetary policy*, the learning-and-escapes narrative of Sargent's (1999) *The Conquest of American Inflation*, rather than a fixed rule buffeted by bad luck. Their three variables are CPI-U inflation, the *logit* of the civilian unemployment rate (so that simulated unemployment respects its $(0, 100)$ bounds), and the ex post real interest rate, the 3-month T-bill rate (first month of each quarter) minus same-quarter inflation. Two design choices define the paper. First, the prior is calibrated on a training sample: a time-invariant VAR(2) estimated on 1948Q1–1958Q4 supplies the prior mean of the initial coefficient vector (its OLS estimate $\bar{\theta}$), the prior scale of the constant Σ (the training residual covariance), and the prior scale of the drift covariance, $Q = (0.01)^2 \bar{P}$ with $\bar{P}$ the OLS asymptotic covariance — the belief that time variation initially accounts for one percent of each coefficient's standard deviation. Second, the objects of interest are not the coefficients themselves but two long-horizon forecasts computed from θ_t at every date (their eq. 3.2): the 120-quarter-ahead forecast of inflation — *core inflation* — and of unemployment — the *natural rate* — supplemented by the inflation spectrum at frequency zero as a measure of persistence. Estimation runs on 1959Q1 onward with the chapter's FFBS machinery unchanged.⁵

⁵Two simplifications relative to CS are disclosed here. The drift covariance Q is diagonal with independent inverse-Gamma updates, its prior mean fixed at CS's $(0.01)^2 \text{diag}(\bar{P})$ with informative weight, where CS use an inverse-Wishart; and local stability is screened date by date (unstable draws are excluded from the forecast calculations, about 1–2% of draw-date pairs) where CS place a stability prior on the entire coefficient path.

Empirical Example: Original Sample, est. 1959Q1–2000Q4

Figure 77 is the replication of CS's Figure 1, and every landmark on it lands where the paper put it. Median core inflation is 2.7% at the end of 1965 (CS: between 1.75 and 4% in the late 1960s); it climbs through the 1970s to a peak of 9.4% in 1979Q2 (CS: roughly 8% in 1979–80); it collapses in the early 1980s to 3.5% by 1985; and it ends the sample at 2.83% in 2000Q4 — CS's own reading of their figure is core inflation “just shy of 3%” at the end of 2000. The natural-rate estimate traces the same arc: 4.3% in 1965, a peak of 8.7% in 1980Q4, and back to 4.2% by 2000. The bottom panel reproduces the paper's persistence finding: the zero-frequency spectrum of inflation sits on a high plateau from the late 1960s through 1980 and then collapses — by the 1990s it is roughly a third of its 1970s level — so the era of high core inflation and the era of persistent, forecastable inflation are the same era, exactly the joint movement Sargent's (1999) learning model predicts.

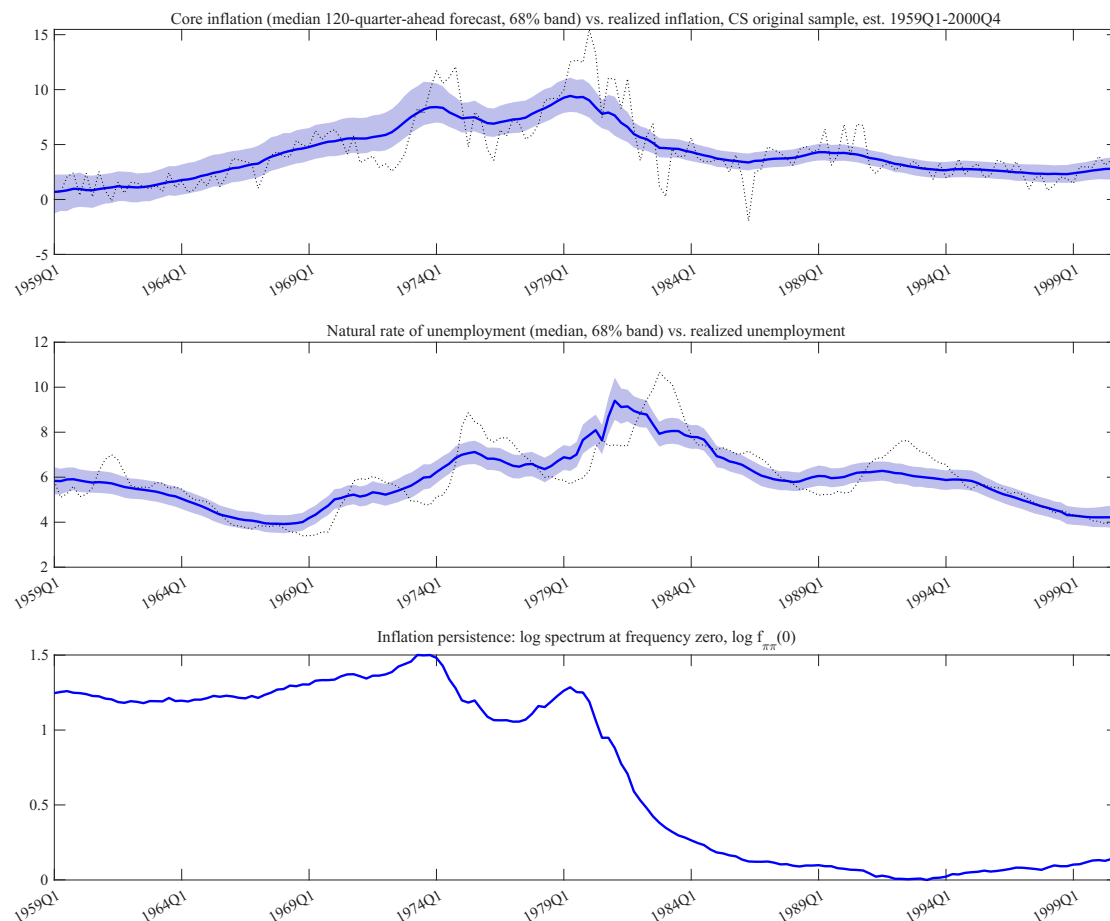


Figure 77: CS (2001) Figure 1 replicated: median core inflation (120-quarter-ahead forecast, 68% band) against realized inflation (top); the natural rate of unemployment against realized unemployment (middle); the log zero-frequency spectrum of inflation (bottom)

Empirical Example: Full Sample, est. 1959Q1–2026Q1

Twenty-five more years give the drifting-coefficient model its most interesting test since the Volcker disinflation, and the result reads as the mirror image of the 1970s (Figure 78). Core inflation spends the entire 1985–2019 stretch pinned between 2.2 and 4.2% — through two recessions, fourteen years at the zero lower bound, and quantitative easing — and the 2021–22 inflation surge, with realized inflation touching 9%, lifts median core inflation only to 3.8% at 2022Q4 before it settles back to 3.2% by 2026Q1. The persistence panel explains why that number matters: at the 2022 peak the zero-frequency spectrum reaches 1.8, against 7.6 at 1970Q4 on the same estimation — the model reads the recent surge as large but transitory shocks hitting a still-anchored inflation process, where it read the 1970s as a drift in the process itself. The natural-rate panel makes the same point through the other variable: when realized unemployment climbed through the 1970s, the estimated natural rate followed it up to 8.5%; when unemployment spiked to 13% in 2020Q2, the natural-rate estimate barely moved from 5%, because a one-quarter spike, however enormous, carries almost no signal about the 120-quarter forecast. Whether the post-2021 uptick — core at 3.2% and persistence still edging up at the end of the sample — is the beginning of a slow unmooring or a completed round trip is exactly the kind of question this model was built to keep asking as new quarters arrive.

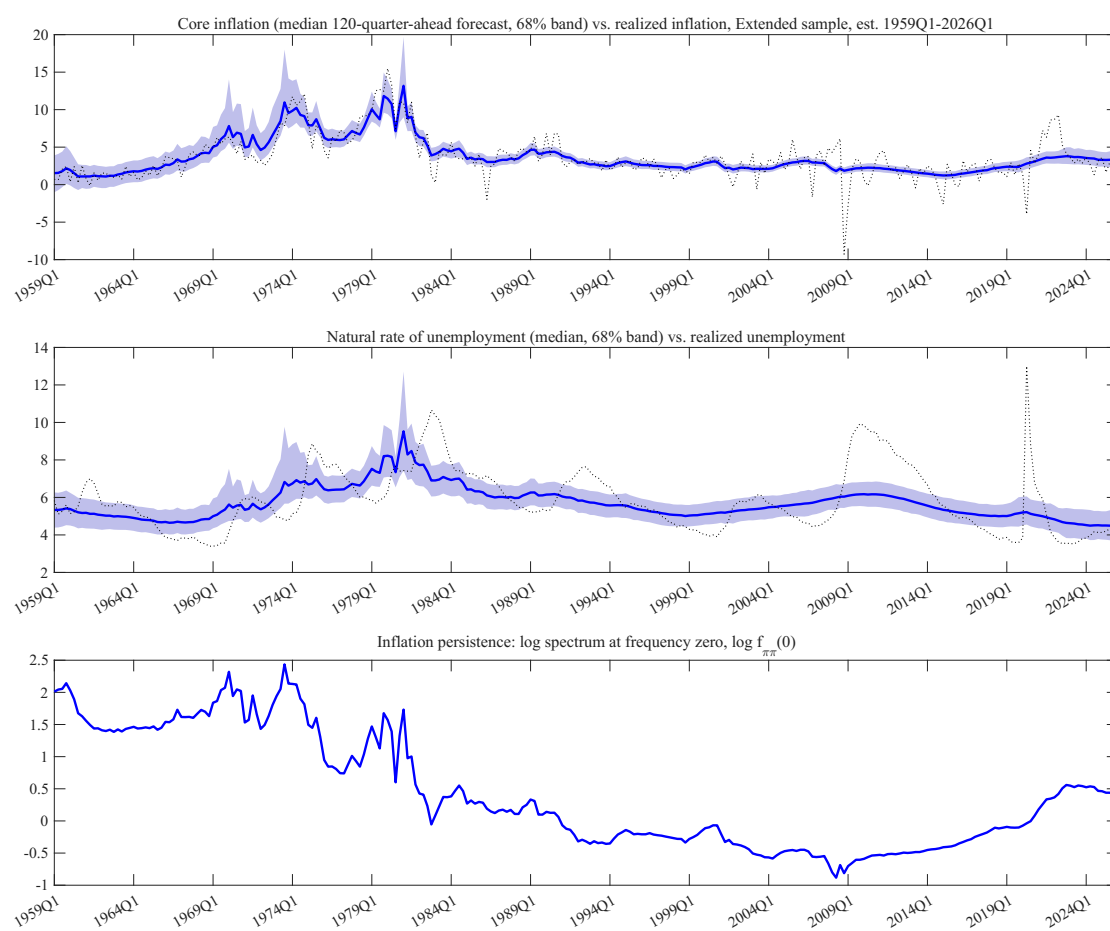


Figure 78: The same system estimated through 2026Q1: core inflation stays anchored through the ZLB era and rises only to 3.8% in the 2021–22 surge, the natural rate ignores the one-quarter COVID unemployment spike, and persistence remains far below its 1970s plateau

Remark: The Prior on Q Is a Substantive Assumption

This replication initially used a diffuse prior on the drift covariance Q ($\nu_0 = 3$), keeping CS's $(0.01)^2 \bar{P}$ only as a scale, and the result was instructive enough to report: the same sampler, on the same data, produced a core-inflation path that peaked near 25% in 1981 — three times the paper's number — because the Gibbs sampler inflated Q until the coefficients could chase every wiggle of 1970s inflation, driving the local dynamics toward the unit-root boundary where long-horizon forecasts explode. CS flag exactly this danger in a footnote: "the Gibbs sampler quickly adds more time variation to the system." Re-anchoring the prior mean of Q at their calibration with informative weight ($\nu_0 = 20$) brings the peak to 9.4% and leaves the anchored-era estimates essentially untouched. The general lesson echoes Chapter 13's hierarchical-prior discussion: in models where a variance parameter controls how much a latent object is *allowed* to move, that variance's prior is part of the economics, not a nuisance setting — how much policy drift one finds in postwar U.S. data depends, unavoidably, on how much drift one was prepared to believe in. A second substantive assumption, flagged immediately by CS's own discussants, is the *constant* Σ : Sims (2001)

and Stock (2001) both argued that what a homoskedastic TVP-VAR attributes to coefficient drift can instead reflect ignored changes in shock volatility — the 1970s may look different not because the transmission coefficients moved but because the shocks were bigger. Cogley and Sargent’s own follow-up work added stochastic volatility for exactly this reason (Chapter 16’s machinery is the toolkit for doing so); the replication here keeps the 2001 specification deliberately, so its drift estimates should be read with that caveat attached.

18. Markov-Switching Regime Model

Roadmap

Every model since Chapter 14 changed *how smoothly* something drifts: a GARCH variance, a random-walk trend, a volatility path, a coefficient vector. This chapter changes the question itself: instead of smooth drift, does the world jump between a small number of recurring states? The machinery turns out to be a direct relative of Chapter 17’s — a forward filter followed by a backward sampler — with every continuous Gaussian step swapped for its discrete two-state counterpart. That parallel is the whole point of placing this model last among the parametric chapters: by this chapter, a reader who has followed Chapters 6, 8, 15, and 17 already owns every individual piece this model needs.

Every model since Chapter 14 has let some part of the data-generating process drift *smoothly* through time: a GARCH variance recursion, a random-walk trend, n independent log-volatility paths, or a full random-walk coefficient vector. This chapter switches the underlying logic entirely. Rather than smooth continuous drift, the process is allowed to jump between a small number of discrete, recurring *regimes* — here, a “recession” state and an “expansion” state — governed by a hidden Markov chain. This is the Hamilton (1989) regime-switching framework, cast in a Bayesian Gibbs sampler due to Albert and Chib (1993) and Chib (1996): the regime sequence $S_{1:T}$ is treated as just another set of unknowns to be sampled, alongside the regime-specific coefficients and variances.

Markov-Switching AR(1) Model Let y_t follow an AR(1) process whose intercept, slope, and residual variance all depend on an unobserved discrete state $S_t \in \{1, 2\}$:

$$y_t = \beta_{0,S_t} + \beta_{1,S_t} y_{t-1} + e_t, \quad e_t \sim N(0, \sigma_{S_t}^2),$$

with S_t following a first-order, two-state Markov chain with transition probabilities

$$P = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix}, \quad p_{11} = \Pr(S_t = 1 \mid S_{t-1} = 1), \quad p_{22} = \Pr(S_t = 2 \mid S_{t-1} = 2).$$

State 1 is identified as the *low*-mean-growth (“recession”) regime by imposing $\beta_{0,1} < \beta_{0,2}$ (without such a restriction the two labels are interchangeable and the posterior is symmetric in them — the discrete-state analogue of the ordering restrictions used nowhere else in these notes, but conceptually the same identification problem as the sign restrictions of Chapter 12’s SVAR). This is Hamilton’s (1989) own identification device, applied to the intercept rather than to the variance: a recession/expansion distinction is fundamentally a claim about the *mean* growth rate, so the labeling restriction is placed there directly. Unlike Chapters 14–17, the latent object here (S_t) is discrete, not continuous, and the natural filtering recursion is the Hamilton filter rather than a Kalman filter or precision sampler.

Figure 79 lays out the same two-pass shape as Chapter 17’s Figure 74, with the continuous state β_t replaced by the discrete regime S_t .

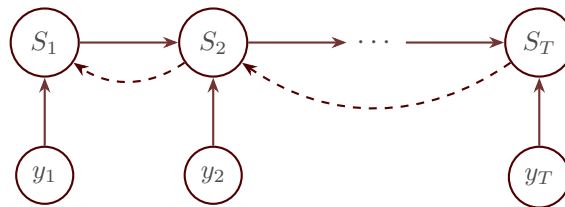


Figure 79: Hamilton filter / Chib backward sampler: the forward Hamilton filter (solid) computes filtered probabilities $\xi_{t|t}$ moving left to right, folding in each y_t ; Chib’s (1996) backward multi-move sampler (dashed) then draws $S_T, S_{T-1}, \dots, S_1$ moving right to left — the discrete-state twin of Chapter 17’s Figure 74

Algorithm: Gibbs Sampler for the Markov-Switching Model

With diffuse $N(0, \text{diag}(9, 0.5))$ priors on each regime’s (β_0, β_1) , diffuse Inverse-Gamma(1.5, 1.5) priors on each σ_j^2 , and *asymmetric* Beta priors on the transition probabilities — Beta(10, 30) for p_{11} and Beta(30, 10) for p_{22} , encoding the belief that recessions are typically short-lived while expansions are typically persistent:

1. **Regime-specific coefficients.** Conditional on the current regime path $S_{1:T}$, split the sample into its two regime-specific subsamples and propose $(\beta_{0,1}, \beta_{1,1})$ and $(\beta_{0,2}, \beta_{1,2})$ each from an ordinary conjugate Bayesian-regression full conditional — exactly Chapter 6’s machinery, applied twice, to two disjoint subsamples — then accept the pair only

if $\beta_{0,1} < \beta_{0,2}$, and otherwise keep the previous sweep's values. This is the mechanism that enforces the labeling restriction throughout the chain.

2. **Regime-specific variances.** Draw σ_1^2 and σ_2^2 from their conjugate Inverse-Gamma full conditionals given each regime's residuals. No rejection step is needed here: the labeling restriction is already carried entirely by step 1.
3. **Regime path.** Run the Hamilton filter forward through the sample to obtain the filtered probabilities $\Pr(S_t | y_{1:t})$ at every t , then sample the *entire* path $S_{1:T}$ jointly via Chib's (1996) backward multi-move sampler: draw S_T from its filtered distribution, then for $t = T - 1, \dots, 1$ draw S_t from $\Pr(S_t | S_{t+1}, y_{1:t}) \propto \Pr(S_t | y_{1:t}) P(S_t, S_{t+1})$ — the discrete-state analogue of the backward simulation smoother used for continuous states in Chapters 15–17.
4. **Transition probabilities.** Count the realized transitions in the sampled path $S_{1:T}$ and draw p_{11} and p_{22} each from its conjugate Beta full conditional (prior counts plus realized transition counts).

18.1 Posterior Derivation

Derivation: Regime-Specific Coefficients and Variances

Conditional on the regime path $S_{1:T}$, split the sample into the subsample with $S_t = 1$ and the subsample with $S_t = 2$. Within regime j , every observation with $S_t = j$ obeys $y_t = \beta_{0,j} + \beta_{1,j}y_{t-1} + e_t$, $e_t \sim N(0, \sigma_j^2)$ — an ordinary linear regression on that subsample alone. This is legitimate because, conditional on $S_{1:T}$, the observations in the two subsamples are independent Normal draws governed by two entirely separate parameter sets, exactly as if they were two separate regression datasets:

$$(\beta_{0,j}, \beta_{1,j}) | S_{1:T}, \sigma_j^2, y \sim N(\bar{\beta}_j, \bar{V}_j)$$

by Chapter 6.2's completing-the-square result, applied with X and y restricted to the rows where $S_t = j$, and

$$\sigma_j^2 | S_{1:T}, \beta_j, y \sim IG\left(\frac{a_0 + T_j}{2}, \frac{d_0 + \sum_{t:S_t=j} (y_t - \beta_{0,j} - \beta_{1,j}y_{t-1})^2}{2}\right)$$

by Chapter 6.2's power-matching result applied to the same restricted subsample, where T_j is the number of periods assigned to regime j and $\bar{\beta}_j, \bar{V}_j$ are Chapter 6.2's $\bar{\beta}, \bar{V}$ formulas evaluated on the regime- j design matrix and outcome. No new derivation is required for either line: splitting the sample by S_t before applying Chapter 6.2's argument only changes which rows of X and y enter the formula.

The $\beta_{0,1} < \beta_{0,2}$ labeling restriction is then imposed the same way Chapter 15's stationarity triangle was: draw the pair (β_1, β_2) from the two untruncated Normal full conditionals above and reject/redraw *both* whenever the ordering is violated, keeping the previous sweep's values instead. This is valid because restricting a joint distribution to a subset and renormalizing is exactly what conditioning on acceptance into that subset produces;

unlike the variance-ordering restriction of a volatility-regime model, only the two intercepts need checking, so β_1 and β_2 are accepted or rejected together as a pair, and no separate restriction is placed on σ_1^2, σ_2^2 .

Derivation: The Hamilton Filter: $S_t \mid y_{1:t}$

Let $\xi_{t|t-1}(j) = \Pr(S_t = j \mid y_{1:t-1})$ and $\xi_{t|t}(j) = \Pr(S_t = j \mid y_{1:t})$, and let $f_j(y_t) = N(y_t; \beta_{0,j} + \beta_{1,j}y_{t-1}, \sigma_j^2)$ denote regime j 's conditional density for the current observation given the current coefficient and variance draws.

1. **Prediction.** By the Markov property of S_t and the law of total probability,

$$\xi_{t|t-1}(j) = \Pr(S_t = j \mid y_{1:t-1}) = \sum_{i=1}^2 \Pr(S_t = j \mid S_{t-1} = i) \Pr(S_{t-1} = i \mid y_{1:t-1})$$

and, substituting the transition-matrix and filtered-probability notation for the two terms just multiplied,

$$= \sum_i P(i, j) \xi_{t-1|t-1}(i).$$

This is the discrete-state analogue of Chapter 17's continuous prediction step $\beta_{t|t-1} = \beta_{t-1|t-1}$: there, a single deterministic random-walk mean; here, a weighted average over the two possible predecessor states, weighted by the transition probabilities.

2. **Update.** By Bayes' rule, exactly as in Chapter 8's discrete $\gamma_j \mid \beta_j, \dots$ posterior and Chapter 16's mixture-component indicator posterior,

$$\xi_{t|t}(j) = \Pr(S_t = j \mid y_{1:t}) \propto f_j(y_t) \xi_{t|t-1}(j), \quad j = 1, 2,$$

renormalized to sum to 1 over $j = 1, 2$.

Iterating prediction-then-update for $t = 1, \dots, T$ (starting from an initial $\xi_{1|0}$, e.g. the chain's stationary distribution implied by P) produces the full sequence of filtered probabilities $\xi_{t|t}$, exactly paralleling the forward Kalman filter of Chapter 17 with the continuous Gaussian recursion replaced by a discrete two-point recursion.

Derivation: Chib's Backward Multi-Move Sampler: $S_{1:T} \mid y_{1:T}$

As in Chapter 17's continuous backward sampler, the model's Markov structure makes S_{t+1} conditionally independent of $y_{1:t}$ given S_t , so by Bayes' rule,

$$\Pr(S_t = k \mid S_{t+1}, y_{1:t}) \propto \Pr(S_{t+1} \mid S_t = k) \Pr(S_t = k \mid y_{1:t})$$

and, writing the transition probability and the filtered mass in the notation already fixed above,

$$= P(k, S_{t+1}) \xi_{t|t}(k), \quad k = 1, 2,$$

renormalized over k . This is the discrete-state counterpart of Chapter 17's Gaussian backward-sampling kernel $N(\beta_{t+1} | \beta_t, Q) N(\beta_t | \beta_{t|t}, P_{t|t})$: the Gaussian transition density $N(\cdot | \beta_t, Q)$ is replaced by the discrete transition probability $P(k, S_{t+1})$, and the Gaussian filtered density $N(\cdot | \beta_{t|t}, P_{t|t})$ by the filtered probability mass $\xi_{t|t}(k)$.

Drawing $S_T \sim \xi_{T|T}$ and then $S_{T-1}, \dots, S_1$ in reverse order from this kernel produces an exact joint draw from $p(S_{1:T} | y_{1:T})$, by the identical chain-rule factorization argument used to justify Chapter 17's continuous backward sampler: $p(S_{1:T} | y) = p(S_T | y_{1:T}) \prod_{t=1}^{T-1} p(S_t | S_{t+1}, y_{1:t})$.

Derivation: $p_{11} | S_{1:T}$ and $p_{22} | S_{1:T}$

Given the sampled path $S_{1:T}$,^a let n_{11} count the transitions $S_{t-1} = 1 \rightarrow S_t = 1$ and n_{12} count $S_{t-1} = 1 \rightarrow S_t = 2$ (so $T_1 - 1 \leq n_{11} + n_{12} \leq T_1$, the number of periods regime 1 was left from). Each such transition is an independent Bernoulli trial with success probability p_{11} ("stay in regime 1"), so the likelihood contribution of p_{11} is a Binomial kernel in p_{11} ; with the Beta(10, 30) prior, whose kernel is $p_{11}^9 (1 - p_{11})^{29}$, the product is

$$p(p_{11} | S_{1:T}) \propto \underbrace{p_{11}^{n_{11}} (1 - p_{11})^{n_{12}}}_{\text{likelihood}} \times \underbrace{p_{11}^9 (1 - p_{11})^{29}}_{\text{prior}}$$

and, adding the two exponents of p_{11} and of $1 - p_{11}$ separately,

$$= p_{11}^{9+n_{11}} (1 - p_{11})^{29+n_{12}},$$

which is exactly a Beta($10 + n_{11}$, $30 + n_{12}$) kernel by inspection — the same Beta-Bernoulli conjugacy argument as Chapter 8's $p | \gamma$ derivation, with the regime-transition indicators $\mathbb{K}\{S_t = 1\}$ (among periods with $S_{t-1} = 1$) playing the role there played by the inclusion indicators γ_j . The identical argument applied to the transitions out of regime 2, whose prior is Beta(30, 10), gives $p_{22} | S_{1:T} \sim \text{Beta}(30 + n_{22}, 10 + n_{21})$, where n_{22} counts $S_{t-1} = 2 \rightarrow S_t = 2$ transitions and n_{21} counts $S_{t-1} = 2 \rightarrow S_t = 1$ transitions.

^aAlthough $S_{1:T}$ is a single dependent chain, its likelihood factors cleanly by *originating* state: every transition out of regime 1 involves only p_{11} , and every transition out of regime 2 involves only p_{22} , so the full-path likelihood splits into two independent Binomial-style products, one per row of P . That row-by-row split is what turns a single joint path into two separate, ordinary Beta-Binomial conjugate updates below, with no need to work with the 2×2 matrix P as a whole.

18.2 Empirical Example: Recession and Expansion Regimes in U.S. GDP Growth, 1959–2026

Empirical Example: Data and Estimated Regimes

Rather than an equity-return series, this chapter fits the model to quarterly U.S. real GDP growth (annualized %) — the same `data_Q` panel used as the running macro example since Chapters 4–6, 11–13, and 19, now extended past its original 2013Q4 endpoint through 2026Q1 using current-vintage FRED data ($T = 266$ usable quarters, 1959Q4–2026Q1 after the one lag consumed by the AR(1) design). This is also, historically, the *original* application Hamilton (1989) built the model for: recurring recession and expansion phases in real output growth, not volatility regimes in an asset-return series. With 5,000 post-burn-in Gibbs draws (after a 1,500-draw burn-in, acceptance rate for the labeling-restricted coefficient draw $\approx 75\%$, down from $\approx 91\%$ on the pre-extension sample), two clearly separated regimes still emerge, but the COVID-19 quarter reshapes them considerably. The recession regime now has a small positive mean growth rate ($\hat{\beta}_{0,1} = 0.476$), a slightly *negative* AR(1) coefficient ($\hat{\beta}_{1,1} = -0.283$), and a residual variance $\hat{\sigma}_1^2 = 96.00$ — roughly four times its pre-extension value of 23.73. The expansion regime is essentially unchanged ($\hat{\beta}_{0,2} = 2.298$, $\hat{\beta}_{1,2} = 0.295$, $\hat{\sigma}_2^2 = 5.52$). The jump in $\hat{\sigma}_1^2$ is driven almost entirely by a single observation: 2020Q2's -28.0% collapse is more than three times larger in magnitude than the worst pre-extension quarter (-8.70% in 2008Q4), and this simple two-state, mean-only specification has nowhere else to put an outlier that extreme except into the recession-regime variance. The negative $\hat{\beta}_{1,1}$ is a direct consequence: it lets the recession regime's AR(1) dynamics predict a sharp *rebound* after an extreme contraction, which is exactly what 2020Q3's $+34.9\%$ bounce-back requires (see the next example box). Despite this reshaping, the qualitative business-cycle story is unchanged — output growth remains both lower-mean and (now dramatically) more volatile in the recession regime than in the expansion regime.

Empirical Example: Regime Persistence and Duration

The estimated transition probabilities remain strongly asymmetric in the direction the priors encoded: $\hat{p}_{11} = 0.295$ (recession regime, essentially unchanged) versus $\hat{p}_{22} = 0.889$ (expansion regime, up from 0.751), translating into expected regime durations of $1/(1 - \hat{p}_{11}) \approx 1.4$ quarters for a recession spell but $1/(1 - \hat{p}_{22}) \approx 9.0$ quarters (well over two years) for an expansion spell. The much longer estimated expansion duration reflects the extended sample's two long, largely uninterrupted growth stretches — 2009–2020 and 2020–2026 — now dominating the transition-count likelihood. Over the full sample the posterior-mean probability of being in the recession regime averages 11.8% (down from 26.6% on the pre-extension sample, since the sample now contains 12 extra years that are overwhelmingly classified as expansion), and 19 of the 266 quarters have posterior recession-regime probability above 0.5, organized into 13 distinct episodes — fewer than the pre-extension sample's 20, since the widened recession-regime variance makes the classifier less sensitive to ordinary (non-COVID-scale) downturns. Figure 80 plots the full

posterior probability path against the GDP growth series it is inferring regimes from: the COVID quarter now dominates the plot visually, with the probability spike dwarfing every historical episode.

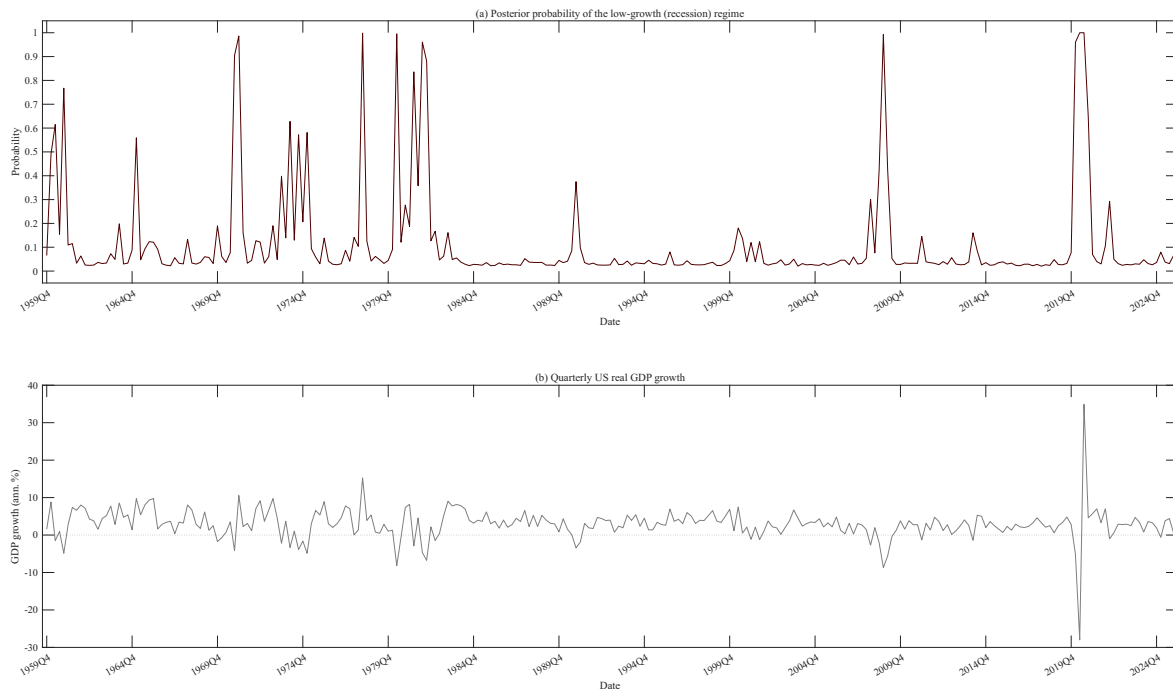


Figure 80: Posterior probability of the low-growth (recession) regime, full sample, vs. quarterly U.S. GDP growth

Empirical Example: Do the Identified Episodes Match the NBER Recession Chronology?

The longest identified recession episode by a wide margin is now **2020Q1–2020Q4** (four consecutive quarters), squarely containing the NBER-dated February–April 2020 COVID-19 recession — the shortest recession on record by NBER’s own dating, yet the one this model is most confident about. The mechanism is worth spelling out: 2020Q2’s -28.0% collapse is unambiguously classified as regime 1, but so is 2020Q3’s $+34.9\%$ rebound, which at first glance looks like it belongs in the expansion regime. It does not, because classification here depends on the AR(1) *dynamics*, not the raw growth level: with $y_{t-1} = -28.0$, regime 1’s negative AR(1) coefficient predicts a mean of $0.48 + (-0.28)(-28.0) \approx 8.4\%$ for 2020Q3, only about 2.7 regime-1 standard deviations below the realized 34.9% ; regime 2’s positive coefficient instead predicts $2.30 + 0.29(-28.0) \approx -5.7\%$, some 17 regime-2 standard deviations away from what actually happened. A model that only looked at the level of growth would call 2020Q3 an expansion quarter; this model, tracking how growth moves *relative to the previous quarter*, correctly recognizes an outsized rebound as still part of the same extreme episode. Behind the COVID episode, the next-longest episodes, each two quarters, are 1960Q1–1960Q2 (the 1960 Eisenhower recession), 1970Q4–1971Q1 (the tail of the 1969–70 recession), and 1981Q4–1982Q1 (inside the 1981–82 recession). Among the single-quarter episodes, 1974Q1, 1974Q3, and 1975Q1 fall inside the NBER-dated 1973–75 oil-shock recession; 1980Q2 falls inside

the brief 1980 recession; 1981Q2 falls inside 1981–82; and 2008Q4 — still, at -8.70% ,^a the worst pre-COVID quarter on record — falls inside the 2007–09 financial crisis. Two identified quarters, 1965Q1 and 1978Q2, do not correspond to any NBER recession: false positives of the same kind discussed for the pre-extension sample, the price of a reduced-form classifier that only sees GDP growth and not the structural triggers the NBER dating committee weighs. Of the nine NBER recessions now inside the sample window, seven are identified with at least one matching quarter; the extended sample's widened recession-regime variance actually costs this version of the model two matches that the pre-extension model caught — the 1990–91 and 2001 recessions no longer surface a quarter above the 50% posterior threshold, since ordinary-sized downturns now look small relative to a regime whose variance is calibrated partly to accommodate 2020Q2. This is the central lesson of extending the sample through COVID: a single sufficiently extreme observation does not just add one more flagged episode, it can quietly reduce the model's sensitivity to everything else, because a two-state, mean-only specification has only one variance parameter per regime to describe both ordinary recessions and the sharpest contraction in postwar U.S. history. Figure 81 zooms in on three windows — the 1973–75 oil-shock recession, the 2007–09 financial crisis, and the 2020 COVID-19 recession — showing the posterior probability rising sharply as growth turns negative and falling back as each episode ends; the COVID panel's near-vertical rise to a probability above 0.99 is visually unlike anything in the other two.

^aNote the recession regime's *mean*, $\hat{\beta}_{0,1} = 0.48$, is an average over all recession-classified quarters, including mild ones and the COVID outlier; individual recession quarters can and do fall far from it — 2008Q4 at -8.70% remains the single worst pre-COVID quarter in the sample.

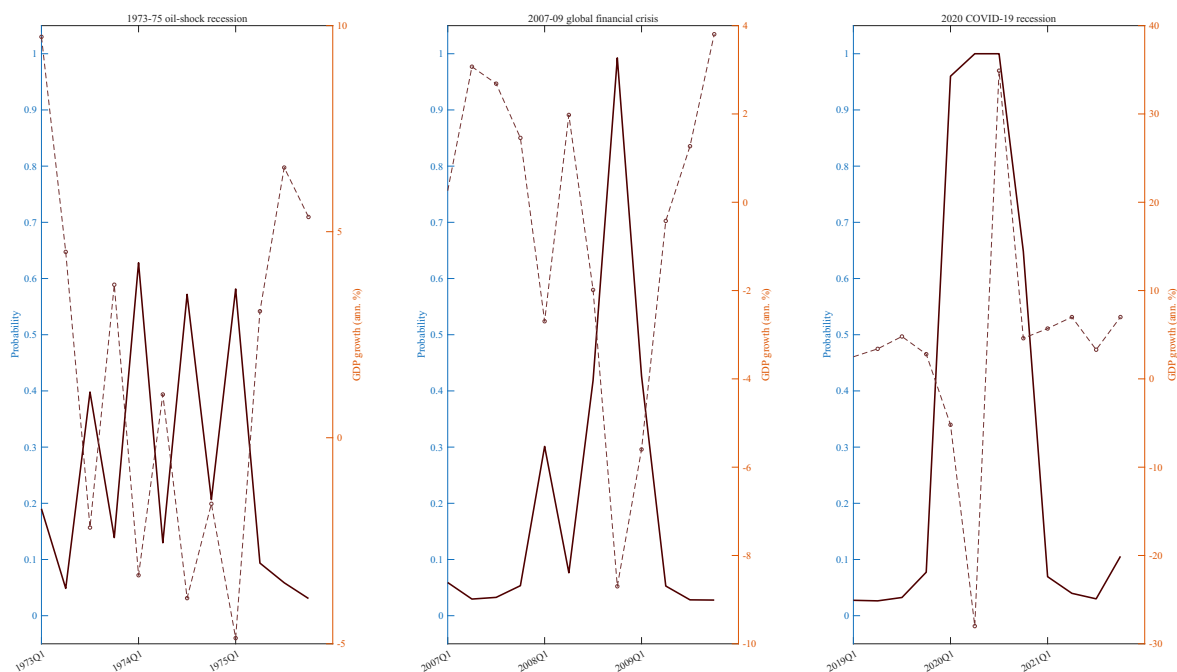


Figure 81: Posterior probability of the recession regime (left axis, navy) vs. GDP growth (right axis, dashed) during the 1973–75 oil-shock recession, the 2007–09 global financial crisis, and the 2020 COVID-19 recession

18.3 Hamilton (1989)

The empirical example above used this chapter's general switching-intercept AR(1); this section replicates the paper the whole framework comes from, on its own data. Hamilton (1989), "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle," *Econometrica* 57(2), modeled quarterly U.S. real GNP growth, $y_t = 100\Delta \ln \text{GNP}_t$ over 1951Q2–1984Q4, as an AR(4) whose *mean* switches with a two-state Markov chain while everything else is common to both regimes:

$$(y_t - \mu_{S_t}) = \sum_{i=1}^4 \phi_i (y_{t-i} - \mu_{S_{t-i}}) + e_t, \quad e_t \sim N(0, \sigma^2).$$

The paper's celebrated finding is twofold: the two estimated means are a genuinely *negative* recession growth rate ($\hat{\mu}_0 = -0.36$ per quarter) against a $+1.16$ expansion rate — recessions are level declines, not slowdowns — and the model's inferred recession probabilities, estimated from GNP growth alone, reproduce the NBER's judgmentally-dated business-cycle chronology almost episode for episode.

Two features distinguish this specification from the chapter's earlier example, and both matter for the sampler. First, the switching object is the mean μ_{S_t} , not the intercept, so the four *lagged* states $S_{t-1}, \dots, S_{t-4}$ enter the period- t conditional mean alongside S_t . Chib's multi-move backward sampler, built on the one-step Markov structure of the earlier example, therefore no longer applies directly, and the regime path is updated one state at a time — a *single-move* sweep in which each S_t is redrawn from its full conditional given the neighboring states, the two transition terms, and the (up to) five likelihood terms $y_t, \dots, y_{t+4}$ that S_t touches. This is precisely the sampler of Albert and Chib (1993), whose own empirical application was this very model on this very data. Second, weakly informative priors do real work here: transition-probability priors of the Kim and Nelson (1999) persistence type ($p \sim \text{Beta}(9, 1)$, $q \sim \text{Beta}(8, 2)$) and regime means centered at ∓ 1 keep the single-move chain out of a well-known spurious mode in which the two "regimes" chase quarter-to-quarter noise and switch nearly every period; under flat $\text{Beta}(2, 2)$ priors that noise-chasing mode is exactly where this sampler settles.⁶

Empirical Example: Original Sample, 1951Q2–1984Q4

With 10,000 post-burn-in draws, the posterior means (sd) against Hamilton's Table I maximum-likelihood estimates:

	posterior	Hamilton (1989)
μ_0 (recession)	−0.44 (0.57)	−0.36
μ_1 (expansion)	1.14 (0.20)	1.16
p (expansion persists)	0.91 (0.06)	0.90
q (recession persists)	0.71 (0.12)	0.76

⁶The chapter's earlier example did not need this care because its variance also switches, separating the regimes much more sharply; Hamilton's common- σ^2 design leaves the two regimes distinguished by their means alone.

implying expected durations of 3.4 quarters for recessions and 11.4 for expansions against Hamilton’s 4.1 and 10.5 — the paper’s central estimates, reproduced to well within one posterior standard deviation.^a Sixteen of the 135 quarters carry posterior recession probability above one half, and Figure 82 shows they arrive in exactly the right places: distinct clusters at 1953–54, 1957–58, 1960–61, 1970, 1974–75, 1980, and 1981–82, one for each NBER recession in the sample window. This is the figure that made the paper famous, recovered from growth data alone.

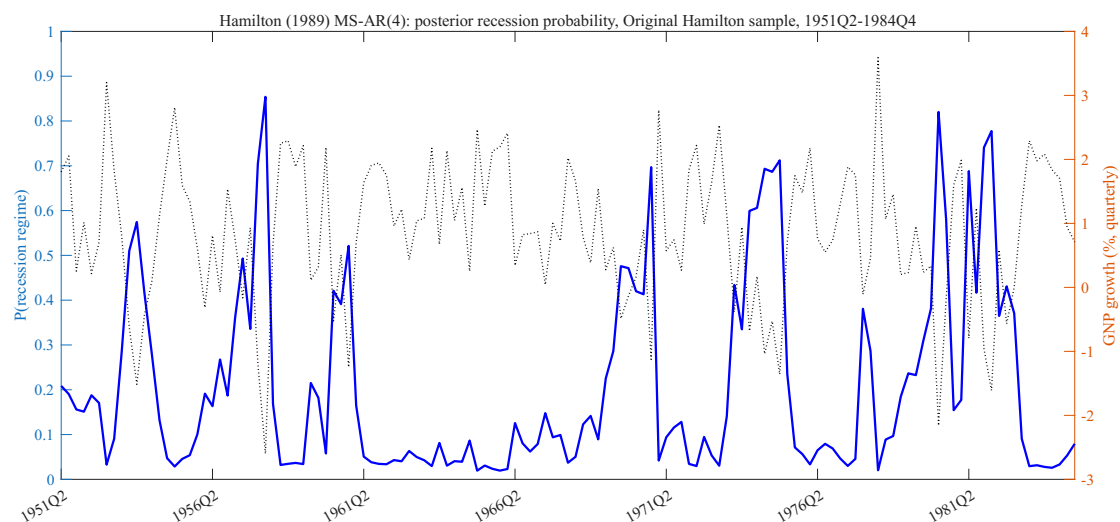


Figure 82: Hamilton (1989) replicated: posterior probability of the recession regime (solid, left axis) vs. quarterly real GNP growth (dotted, right axis), original sample

^aThe AR coefficients and innovation variance land further from Table I ($\hat{\phi}_1 = 0.24$ vs. 0.014; $\hat{\sigma}^2 = 0.84$ vs. 0.59), for the by-now-familiar pair of reasons: real GNP has been comprehensively revised several times since the 1984-vintage series Hamilton used, and these are posterior means under weakly informative priors rather than maximum-likelihood point estimates — the same caveat noted for EM(1998) in Chapter 7.

Empirical Example: Full Sample to 2019Q4: the Great Moderation Reshapes Both Regimes

Extending the identical model through 2019Q4 — deliberately stopping just short of COVID-19, for reasons the next example makes vivid — changes both regimes in the directions the post-Hamilton literature documented. The expansion mean falls from 1.14 to 0.88 (trend growth has slowed), while expansions last dramatically longer: $\hat{p} = 0.96$, an expected duration of 25 quarters, reflecting the long uninterrupted expansions of 1991–2001 and 2009–2019. The recession regime becomes deeper and briefer ($\hat{\mu}_0 = -1.15$, expected duration 2.3 quarters): with the moderation-era decline in output volatility (McConnell and Perez-Quiros, 2000), the regime that once described year-long NBER recessions increasingly specializes in the sharpest contraction quarters — 1957–58, 1980–82, and above all 2008–09 stand out in Figure 83 — while the mild 1990–91 and 2001 recessions barely register. Kim and Nelson (1999) documented exactly this structural change in the switching model itself; the replication recovers their conclusion simply by lengthening Hamilton’s sample.

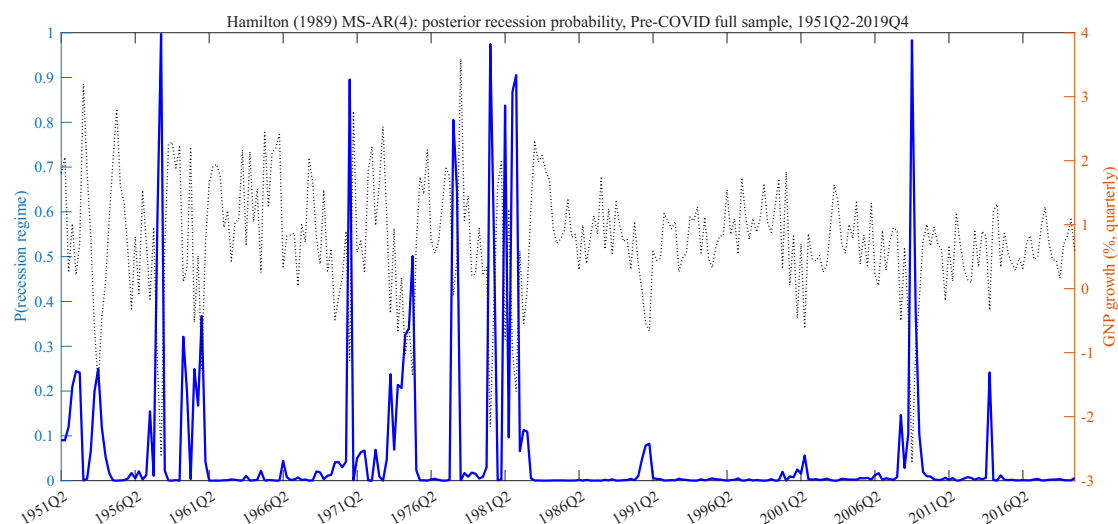


Figure 83: The same model estimated through 2019Q4: expansions lengthen, the recession regime sharpens, and the mild post-moderation recessions grow faint

Empirical Example: Adding COVID: One Quarter Hijacks a Regime

Estimating through 2026Q1 — six additional years, of which exactly one quarter is extraordinary — does not merely add an episode; it changes what the recession regime *is*. The recession mean collapses to $\hat{\mu}_0 = -4.4$, the expansion regime absorbs 99%-persistence ($\hat{p} = 0.99$), and only three quarters in seventy-five years are classified as recession at all: the regime has stopped describing business-cycle downturns and become a rare-disaster state calibrated to 2020Q2's collapse (Figure 84). Every ordinary recession — including 2008–09, which the pre-COVID fit identified sharply — is reclassified as expansion-regime noise, because against a -4.4% regime mean a -2% quarter no longer looks recessionary. This is the same lesson the chapter's own COVID discussion reached from a different direction, and it is not specific to regime-switching models: Lenza and Primiceri (2022) document the analogous distortion in VARs and propose explicit outlier treatment, and Chapter 16's stochastic-volatility machinery is these notes' own answer to the same problem. A two-state model with one variance and two means simply has no third label to give a once-in-a-century observation — so it spends one of the two it has.

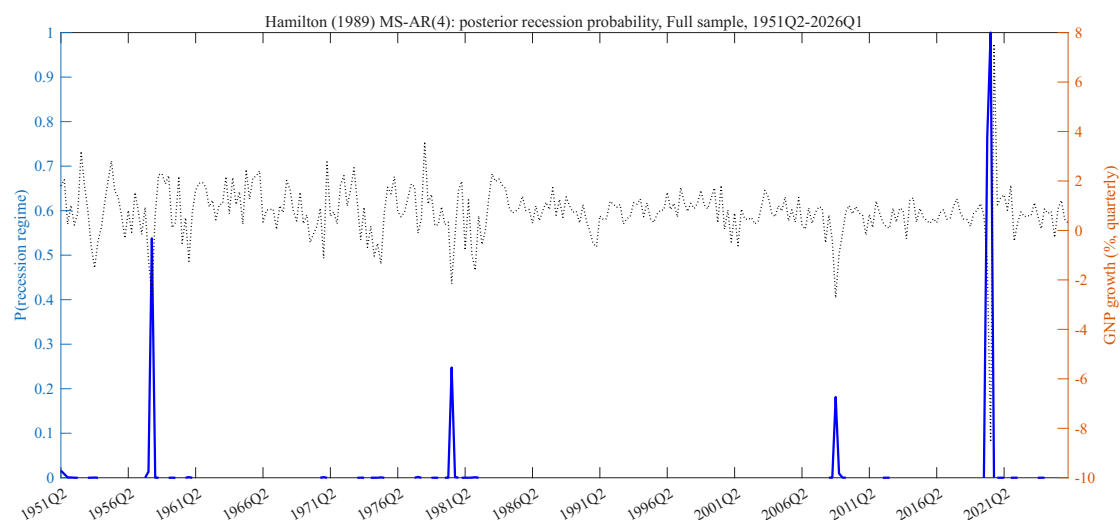


Figure 84: The same model estimated through 2026Q1: the recession regime becomes a rare-disaster state ($\hat{\mu}_0 = -4.4$), flagging essentially only 2020, and ordinary recessions vanish from the classification

19. Nonparametric Models: Regression Trees and BART

Roadmap

Every model in these notes so far has fixed a functional form first and then put a prior on its parameters: linear in Chapter 6, a probit link in Chapter 7, log-linear volatility recursions in Chapters 14–17. Even the most flexible of these, SSVS in Chapter 8, only ever decides which *linear* terms to keep. This chapter relaxes the functional form itself. Bayesian Additive Regression Trees (BART; Chipman, George and McCulloch, 2010) puts a prior directly over a space of step functions – sums of regression trees – and lets the data decide the shape, including kinks, thresholds, and interactions that no linear specification could represent without the analyst guessing them in advance. The estimation machinery is, once again, a Gibbs-type cycle: a Metropolis-Hastings step embedded inside a Gibbs sweep, exactly the pairing Chapter 14’s tailored M-H and Chapter 18’s discrete-state sampler already used. This is a deliberately different note to end on: every other chapter asked “how do we estimate the parameters of this equation,” and this one asks “what if we refuse to write the equation down.”

A tree-based regression model replaces a single global functional form with a partition of the covariate space into rectangular regions, each with its own constant prediction. One tree alone is a crude, high-variance approximator – it is a step function with only as many distinct values as it has leaves – but a *sum* of many shallow trees, each one individually regularized to stay simple, can approximate smooth or kinked response surfaces with an accuracy that rivals far more elaborate nonparametric methods, while remaining fully Bayesian: every tree, every leaf mean, and the residual variance all carry an explicit posterior.

The Sum-of-Trees Model A single regression tree j is a pair (T_j, M_j) : a binary partition structure T_j built from axis-aligned splits $\{x_{\cdot,k} \leq c\}$, and a set of leaf (terminal-node) means $M_j = \{\mu_{j,1}, \dots, \mu_{j,b_j}\}$, one per leaf. Writing $g(x; T_j, M_j)$ for the leaf mean that x is routed to, BART models

$$y_i = f(x_i) + \varepsilon_i, \quad f(x) = \sum_{j=1}^m g(x; T_j, M_j), \quad \varepsilon_i \sim N(0, \sigma^2),$$

a sum of m trees (typically $m = 50$ – 200). Three priors regularize the sum so that no single tree dominates and the ensemble behaves like a weak-learner average rather than m independently overfit trees:

$$\begin{aligned} \Pr(\text{node splits at depth } d) &= \alpha(1 + d)^{-\beta}, \\ \mu_{j,l} &\sim N(0, \sigma_\mu^2), \\ \sigma^2 &\sim IG\left(\frac{\nu}{2}, \frac{\nu\lambda}{2}\right), \end{aligned}$$

with the Chipman-George-McCulloch (2010) defaults $\alpha = 0.95$, $\beta = 2$ (a steeply decaying split probability – most trees stay only two or three levels deep), $\sigma_\mu = 0.5/(k\sqrt{m})$ with $k = 2$ after rescaling y to $[-0.5, 0.5]$ (so that $f(x)$'s prior 95% range roughly covers the rescaled data), and λ, ν ($\nu = 3$) chosen so that $\Pr(\sigma^2 < \hat{\sigma}_{\text{OLS}}^2) = q$ for $q = 0.9$. The split variable at a new node is drawn uniformly among the p predictors, and the split value uniformly among the values of that predictor actually observed at the node.

Figure 85 shows one tree from such a sum: a small number of splits, each carving off a rectangular region of covariate space with its own constant prediction μ_l . The sum of fifty such trees, each individually this simple, is what produces the flexible, non-additive response surface in this chapter's empirical example.

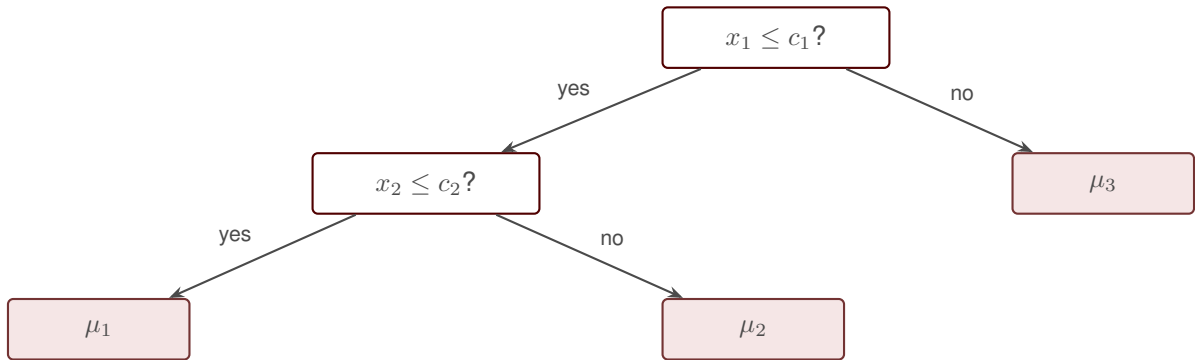


Figure 85: One tree (T_j, M_j) from a sum-of-trees ensemble: two splits partition the covariate space into three leaves, each with its own constant prediction μ_l

Derivation: The Leaf Marginal Likelihood and the Birth/Death Acceptance Ratio

Sampling (T_j, M_j) directly is intractable because T_j is a discrete, variable-dimension object. The standard route (Chipman, George and McCulloch, 1998) is Metropolis-Hastings on the tree structure alone, with the leaf means M_j analytically integrated out of

the acceptance ratio and then redrawn from their exact conditional afterward – the same integrate-then-redraw logic Chapter 8’s Chib’s-method proofbox used for β , applied here to a discrete rather than continuous nuisance parameter. Fix the current tree’s *partial residual* $r_i = y_i - \sum_{k \neq j} g(x_i; T_k, M_k)$ (tree j ’s target once every other tree’s contribution is subtracted off, exactly Chapter 8’s residual role for backfitting) and current σ^2 .

For a leaf with n observations routed to it and residual sum $R = \sum_{i \in \text{leaf}} r_i$, integrate $\mu \sim N(0, \sigma_\mu^2)$ out of the Normal likelihood. Writing the Normal likelihood and Normal prior kernels out explicitly and expanding the sum of squares inside the likelihood exponent,

$$\begin{aligned} \int \prod_{i \in \text{leaf}} N(r_i; \mu, \sigma^2) N(\mu; 0, \sigma_\mu^2) d\mu &= \int (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i \in \text{leaf}} (r_i - \mu)^2\right) \\ &\quad \times (2\pi\sigma_\mu^2)^{-1/2} \exp\left(-\frac{\mu^2}{2\sigma_\mu^2}\right) d\mu \end{aligned}$$

Expanding $\sum_{i \in \text{leaf}} (r_i - \mu)^2 = \sum_i r_i^2 - 2\mu R + n\mu^2$ and pulling the μ -free factor $(2\pi\sigma^2)^{-n/2} \exp(-\sum_i r_i^2/2\sigma^2)$ outside the integral leaves only the μ -dependent terms inside the exponent:

$$\begin{aligned} &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_i r_i^2}{2\sigma^2}\right) (2\pi\sigma_\mu^2)^{-1/2} \\ &\quad \times \int \exp\left(-\frac{1}{2} \left[\left(\frac{n}{\sigma^2} + \frac{1}{\sigma_\mu^2}\right)\mu^2 - \frac{2R}{\sigma^2}\mu\right]\right) d\mu \end{aligned}$$

Write $A \equiv n/\sigma^2 + 1/\sigma_\mu^2$ and $c \equiv R/\sigma^2$ for the quadratic and linear coefficients in μ , and complete the square: $A\mu^2 - 2c\mu = A(\mu - c/A)^2 - c^2/A$, so the exponent splits into a μ -free piece and a Gaussian kernel in μ :

$$\begin{aligned} &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_i r_i^2}{2\sigma^2}\right) (2\pi\sigma_\mu^2)^{-1/2} \exp\left(\frac{c^2}{2A}\right) \\ &\quad \times \int \exp\left(-\frac{A}{2}(\mu - c/A)^2\right) d\mu \end{aligned}$$

The remaining integral is now an (unnormalized) Gaussian integral in μ , evaluating to $\sqrt{2\pi/A}$:

$$= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_i r_i^2}{2\sigma^2}\right) \sqrt{\frac{1}{A\sigma_\mu^2}} \exp\left(\frac{c^2}{2A}\right)$$

Substituting back $A = n/\sigma^2 + 1/\sigma_\mu^2 = (\sigma^2 + n\sigma_\mu^2)/(\sigma^2\sigma_\mu^2)$, so that $1/(A\sigma_\mu^2) = \sigma^2/(\sigma^2 + n\sigma_\mu^2)$, and $c^2/(2A) = \sigma_\mu^2 R^2/[2\sigma^2(\sigma^2 + n\sigma_\mu^2)]$, this simplifies to

$$\begin{aligned} &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_i r_i^2}{2\sigma^2}\right) \\ &\quad \times \sqrt{\frac{\sigma^2}{\sigma^2 + n\sigma_\mu^2}} \exp\left(\frac{\sigma_\mu^2 R^2}{2\sigma^2(\sigma^2 + n\sigma_\mu^2)}\right) \end{aligned}$$

The leading factor $(2\pi\sigma^2)^{-n/2} \exp(-\sum r_i^2/2\sigma^2)$ is identical whether a leaf is split or not (a split only re-partitions the same observations, so n and $\sum r_i^2$ over the whole node are conserved), so it cancels in every ratio below; dropping it,

$$\propto \sqrt{\frac{\sigma^2}{\sigma^2 + n\sigma_\mu^2}} \exp\left(\frac{\sigma_\mu^2 R^2}{2\sigma^2(\sigma^2 + n\sigma_\mu^2)}\right).$$

A *birth* move picks a random current leaf, a random predictor, and a random available split value, turning that leaf (n obs, sum R) into two new leaves (n_L, n_R , sums R_L, R_R).

This closed-form leaf marginal likelihood turns the tree-structure acceptance ratio into an explicit computation. As in the general Metropolis-Hastings derivation of Chapter 14's tailored-independence proofbox, the acceptance ratio for a proposed tree T' (here, the tree with the chosen leaf split) against the current tree T is $\alpha_{\text{birth}} = \min(1, \pi(T' | y) q(T | T') / [\pi(T | y) q(T' | T)])$, with posterior kernel $\pi(T | y) \propto p(T) p(y | T)$ factored into tree prior times leaf-marginalized likelihood $p(y | T) = \prod_{l \in \text{leaves}(T)} (\text{leaf marginal likelihood at } l)$. Writing $L(n, R)$ for the log of the leaf marginal likelihood just derived at depth d ,

$$\frac{\pi(T' | y) q(T | T')}{\pi(T | y) q(T' | T)} = \frac{p(T')}{p(T)} \cdot \frac{p(y | T')}{p(y | T)} \cdot \frac{q(T | T')}{q(T' | T)}$$

T' and T differ only at the split leaf: every other leaf's marginal-likelihood factor is common to $p(y | T') = \prod_l(\cdot)$ and $p(y | T) = \prod_l(\cdot)$ and cancels, leaving only the split leaf's factor against its two new children's factors:

$$\frac{p(y | T')}{p(y | T)} = \exp [L(n_L, R_L) + L(n_R, R_R) - L(n, R)]$$

Similarly, T' and T agree on the split/no-split status of every node except the newly split leaf itself (depth d , prior mass $1 - \alpha(1 + d)^{-\beta}$ as a leaf under T , versus $\alpha(1 + d)^{-\beta}$ as an internal node under T') and its two new leaf children (depth $d + 1$, each contributing $1 - \alpha(2 + d)^{-\beta}$ under T' and no factor under T); every other node's factor is common to $p(T)$ and $p(T')$ and cancels:

$$\frac{p(T')}{p(T)} = \frac{\alpha(1 + d)^{-\beta} [1 - \alpha(2 + d)^{-\beta}]^2}{1 - \alpha(1 + d)^{-\beta}}$$

With birth and death moves proposed with equal probability, $q(T' | T) = \frac{1}{2} \cdot \frac{1}{n_{\text{leaves}} p n_{\text{cand}}}$ (propose a birth, then choose the leaf, predictor, and split value uniformly among n_{leaves} leaves, p predictors, and n_{cand} candidate split values) and $q(T | T') = \frac{1}{2} \cdot \frac{1}{n'_{\text{nog}}}$ (propose a death, then choose uniformly among the n'_{nog} "no-grandchildren" nodes available to prune in T'); the $\frac{1}{2}$ factors cancel and

$$\frac{q(T | T')}{q(T' | T)} = \frac{n_{\text{leaves}} p n_{\text{cand}}}{n'_{\text{nog}}}$$

Multiplying the three ratios and taking logs recovers the stated acceptance ratio:

$$\begin{aligned} \log \alpha_{\text{birth}} = & \underbrace{L(n_L, R_L) + L(n_R, R_R) - L(n, R)}_{\text{likelihood ratio}} + \underbrace{\log \frac{\alpha(1+d)^{-\beta} [1 - \alpha(2+d)^{-\beta}]^2}{1 - \alpha(1+d)^{-\beta}}}_{\text{tree-prior ratio}} \\ & + \underbrace{\log \frac{n_{\text{leaves}} p n_{\text{cand}}}{n'_{\text{nog}}}}_{\text{transition ratio}}, \end{aligned}$$

where n_{leaves} is the number of leaves before the birth, n_{cand} the number of valid split values at the chosen leaf, and n'_{nog} the number of “no-grandchildren” nodes (internal nodes with two leaf children – the nodes a subsequent *death* move could prune) *after* the birth. The death move is the exact reverse step, so its acceptance ratio is the same three pieces with numerator and denominator swapped. Two bookkeeping details matter for getting n'_{nog} right: the newly split leaf always becomes one nog node, but if that leaf’s own parent was a nog before the split (its sibling was also a leaf), the parent stops being one – the two effects can cancel.

Algorithm: Bayesian Backfitting MCMC

Initialize all m trees as single-leaf roots with $\mu = 0$; set $\sigma^2 = \hat{\sigma}_{\text{OLS}}^2$. Each sweep:

1. **For each tree** $j = 1, \dots, m$: form the partial residual $r_i = y_i - \sum_{k \neq j} g(x_i; T_k, M_k)$; propose a birth or death move (equal probability, birth forced if no death move is available) and accept it with the probability of the proofbox above; then, holding the (possibly updated) tree structure fixed, redraw every leaf mean from its exact conjugate Normal conditional, $\mu_l | r, \sigma^2 \sim N(V_l R_l / \sigma^2, V_l)$, $V_l = (n_l / \sigma^2 + 1 / \sigma_\mu^2)^{-1}$.
2. **Residual variance**: with $e_i = y_i - \sum_{j=1}^m g(x_i; T_j, M_j)$, draw $\sigma^2 | \cdot \sim IG(\frac{\nu+T}{2}, \frac{\nu\lambda + \sum_i e_i^2}{2})$.

Repeating this sweep is Bayesian backfitting: each tree in turn is fit to what the rest of the ensemble has not yet explained, the same partial- residual logic as Gauss-Seidel-style coordinate updates, wrapped in a proper Gibbs cycle because σ^2 and every leaf mean are drawn from exact conditionals throughout.

Remark: Simplifications, Tuning, and the Wider Nonparametric-Bayes Landscape

Two simplifications relative to the full Chipman-George-McCulloch (2010) algorithm are worth flagging plainly. First, only birth and death moves are used here; the full algorithm also includes *change* (re-draw a node’s split) and *swap* (exchange splits between a parent and child) moves for additional mixing, which birth/death alone – the dominant pair in practice – does not strictly need at the sample sizes used in this book. Second, candidate split values are drawn from a fixed, global, per-predictor grid (quantiles computed once) restricted to whatever values actually reach a given node, rather than a node-local grid recomputed at every proposal; this is a standard implementation shortcut that leaves the target posterior unchanged. On tuning: BART’s headline empirical property, across a wide range of applications, is that the default $(\alpha, \beta, k, m) = (0.95, 2, 2, 50\text{--}200)$ setting

above performs competitively without dataset-specific adjustment, unlike, say, this book's Minnesota-prior λ_1, λ_2 (Chapter 13.2), which needed explicit calibration. BART is one member of a much larger Bayesian nonparametric toolkit that these notes have otherwise left untouched: a Gaussian process prior places a prior directly over the regression function's covariance structure rather than over a sum of trees, giving smooth (not step-function) posterior draws at the cost of $O(T^3)$ scaling; a Dirichlet process mixture places a nonparametric prior over the *data-generating density itself* rather than a conditional mean, letting the number of mixture components be inferred rather than fixed – a direct generalization of Chapter 18's fixed two-state regime model to an unbounded number of latent states. Both are natural next steps from here, sharing this chapter's core idea of putting the prior on a function or distribution rather than on a finite parameter vector.

Empirical Example: A Taylor Rule With a Zero-Lower-Bound Kink

The extended quarterly panel used throughout this edition (1959Q3–2026Q1) now contains *two* separate episodes in which the fed funds rate sits at the zero lower bound (ZLB): a long financial-crisis stretch, 2009Q1–2016Q4 (32 quarters), and a shorter pandemic stretch, 2020Q2–2022Q1 (8 quarters) – forty ZLB quarters in all, up from the single twenty-quarter episode (2009–13) available in the original sample. Within these quarters the funds rate ranges from 0.06% to 0.45%, essentially pinned regardless of the quarter-to-quarter inflation and GDP-growth readings that a Taylor rule would otherwise respond to. Fitting BART ($m = 50$ trees) and an ordinary Bayesian linear regression to the identical data – funds rate on contemporaneous CPI inflation and GDP growth, $T = 267$ quarters, no other preprocessing – gives a full-sample in-sample RMSE of 2.16 percentage points for BART against 2.81 for the linear fit: the sum-of-trees and the linear rule must both accommodate the pandemic's brief growth swings, but BART's advantage over the linear benchmark survives the longer, harder sample. The sharper contrast is still on the ZLB quarters specifically, plotted in Figure 86: their actual average funds rate is 0.15%, BART's fitted values there average 2.78% (ranging from 1.12% up to 7.05% on individual quarters), and the linear regression's average 3.74% (ranging from 0.12% up to 8.73%) – BART is not close to perfect (with only two predictors, and no rate-smoothing lag i_{t-1} term of the kind a real Taylor-rule specification would add, neither model can fully identify a ZLB regime purely from inflation and growth, since some 1970s-80s quarters share similarly low readings with double-digit funds rates instead, and the pandemic ZLB quarters in particular pair unusually low inflation readings with historically extreme growth swings that pull the fitted value in opposite directions from quarter to quarter), but its ensemble-of-step-functions structure moves noticeably closer to the floor than the unconstrained linear rule does on average, and does so without being told the ZLB existed. Neither model ever fits a negative rate on a ZLB quarter, but for different reasons: BART's leaf predictions are averages of observed y -values — every fitted value is a convex combination of funds rates actually seen in the sample, whose minimum is 0.06% — so BART cannot fit a negative rate *anywhere*, while the linear rule, though structurally capable of predicting a negative rate, never happens to do so on these particular covariate readings.

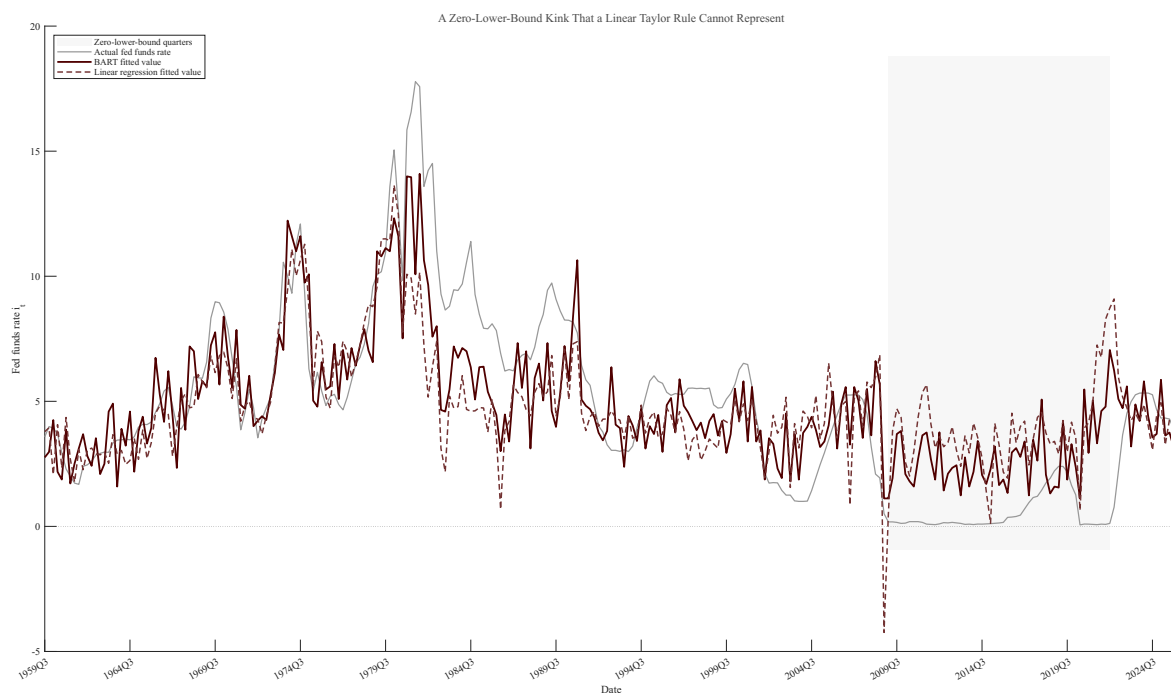


Figure 86: Actual vs. fitted fed funds rate, full extended sample (1959Q3–2026Q1); the shaded band spans from the first to the last zero-lower-bound quarter (2009Q1 to 2022Q1) and so also covers the 2017–2019 interval between the two ZLB episodes, when the funds rate was above 0.5%

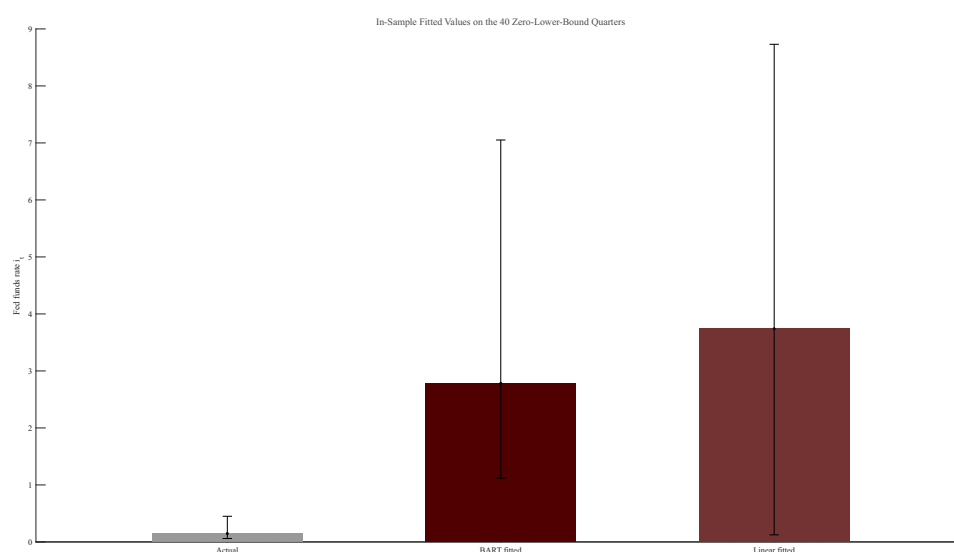


Figure 87: In-sample fitted values on the 40 zero-lower-bound quarters: actual average (0.15%) vs. BART (2.78%) and the linear rule (3.74%); whiskers span each model’s minimum and maximum fitted value across the ZLB quarters

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